AAPP011-4-2 Capstone Project – Cover Sheet
ASSIGNMENT

Instruction:

- Marks will be awarded for good presentation and thoroughness in your approach.
- Referencing Code: If you use some code, or ideas for code, that are taken or adapted from another source (book, magazine, internet, discussion forum, etc), then this **must** be cited and referenced using the APA convention within your source code. Failure to reference code properly is considered plagiarism.
- Complete this cover sheet and attach it to your project's first page.
- This project is to be attempted by a group of 6 students.

Student declaration:

I declare that:

- *We understand what is meant by plagiarism*
- *The implications of plagiarism have been explained to us by our lecturer*
- *This project is all our work and we have acknowledged any use of the published or unpublished works of other people.*

Group Leader's Signature: *ling* Date: 15th July 2025

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1.0 Acknowledgement

We would like to thank everyone who helped us with our RUNTIME FITNESS diet plan project.

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We also thank each team member for working hard and helping each other throughout the project.

A big thank you to the people who shared their opinions and past experiences when developing an application or a website. Your feedback helped us make the application or website better.

Lastly, thank you to our families and friends for encouraging us and supporting us during this project.

2.0 Abstract

This project investigates the development of a fitness management system with personalized diet plans that caters to the demand for personalized health and wellness. The study addresses the challenges of managing user interactions within an intuitive interface, including workout tracking, diet planning. The study aims to design, develop, and assess the RUNTIME FITNESS Management System that supports members and administrators with tailored features based on their roles.

The RUNTIME FITNESS Management System was developed using the Spiral Model, a risk-driven system development methodology. Functional requirements were gathered and refined through stakeholder engagement and translated into modular components. Tools such as HTML, CSS, PHP, MySQL, and JavaScript were used to implement core functionalities, including registration and login, workout progress tracking, diet planning, chat functionality, and administrative controls. The implementation's outcome indicates that the system achieves its intended goals. User testing ensures that features such as progress tracking and profile management are easy to use.

In conclusion, the RUNTIME FITNESS Management System enhances user experience with personalized health tracking. The project demonstrates how iterative development and user-centered design can improve the accessibility of online fitness platforms.

3.0 Workload Matrix

ASIA PACIFIC UNIVERSITY OF TECHNOLOGY AND INNOVATON

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Student Coursework Workload Matrix - Grades and Feedback Attachment

INTAKE: UCDF2308ICT(SE)	STUDENT NAME	SERENE LOH ZI TING	SIEW ZHEN LYNN	SIM TIAN	YAP XIN LING	YEO PEI WEN	YONG XUAN LYN
System Name: RUNTIME FITNESS	TP NO.	TP075920	TP076386	TP077056	TP077224	TP077057	TP076797

A. Group Component

CLO	ASSIGNMENT COMPONENT	CONTRIBUTION PERCENTAGE	CONTRIBUTION PERCENTAGE	CONTRIBUTION PERCENTAGE	CONTRIBUTION PERCENTAGE	CONTRIBUTION PERCENTAGE	CONTRIBUTION PERCENTAGE	TOTAL %
1.0	Acknowledgement	100.00	-	-	-	-	-	100
2.0	Abstract	-	-	-	100.00	-	-	100
3.0	Introduction							
3.1	Project Background	-	-	-	-	100.00	-	100
3.2	Problem Statement	-	-	-	-	100.00	-	100
3.3	Project Aim	-	100.00	-	-	-	-	100
3.4	Project Objectives	-	100.00	-	-	-	-	100
3.5	Proposed Solutions/Scope	-	-	100.00	-	-	-	100
3.6	Functionalities/Deliverables	-	50.00	-	-	50.00	-	100
4.0	Project Plan							
4.1	System Development Methodology	-	-	-	100.00	-	-	100
4.2	Project Gantt Chart	-	-	100.00	-	-	-	100
5.0	System Hierarchy Chart	-	-	-	-	100.00	-	100
6.0	System Design							
6.1	Context Diagram	-	-	-	100.00	-	-	100
6.2	Data Flow Diagrams & Data Dictionary	16.67	16.67	16.67	16.67	16.67	16.67	100
6.3	DFD Level-0	-	-	-	-	100.00	-	100
6.4	DFD Level-1	16.67	16.67	16.67	16.67	16.67	16.67	100
6.5	Program Flowcharts	16.67	16.67	16.67	16.67	16.67	16.67	100
6.6	Database Design						100.00	100
6.7	Interface Design	-	-	100.00	-	-	-	100
6.8	Report Design	-	-	-	-	-	100.00	100
6.9	Test Cases Preparation	100.00						100
7.0	Execution/Implementation							
7.1	Screenshots	-	100.00	-	-	-	-	100
7.2	Significant Source Codes	-	100.00	-	-	-	-	100
8.0	Outcomes of Testing							
8.1	Unit Testing	100.00	-	-	-	-	-	100
8.2	User Acceptance	100.00	-	-	-	-	-	100
9.0	User Manual	-	-	100.00	-	-	-	100
10.0	Conclusion	-	-	-	-	-	100.00	100
11.0	Enhancements	-	-	-	100.00	-	-	100
12.0	Meeting Minutes	-	-	-	-	-	100.00	100
13.0	Progress Sheet	-	-	-	100.00	-	-	100
	Total Marks and Contribution	16.67%	16.67%	16.67%	16.67%	16.67%	16.67%	
	Signature	SERENE	celine	simtian	ling	PeiWen	XuanLyn	

Figure 1: Workload Matrix

4.0 Introduction

4.1 Project Background

This paper will document the proposal of a project for the pre-development of personalised diet plan and workout application. In today's fast-paced digital age, health and fitness have become increasingly important, yet many individuals struggle to maintain consistent and personalized diet plans. With the rise in health awareness and the growing demand for convenient wellness solutions, RUNTIME FITNESS was conceptualized to bridge this gap.

The idea behind RUNTIME FITNESS came from the observation that people don't have enough time, knowledge or motivation to follow a healthy diet. Most diet plans are the same for everyone and don't match people's different needs, like their body type, health goals, daily routine or food choices.

To address these challenges, RUNTIME FITNESS is developed as a user-friendly application that offers personalized diet plans tailored to users' goals, whether it's weight loss, muscle gain, or overall health improvement. By combining technology with expert dietary guidelines, the app aims to support users in making sustainable lifestyle changes through customized meal suggestions, nutritional tracking, and progress monitoring.

This project not only contributes to individual well-being but also reflects a broader commitment to leveraging digital innovation for healthier communities.

4.2 Problem Statement

Throughout these years, many individuals have chosen to exercise at home rather than attend a gym centre or hire a personal coach. This trend was significantly accelerated by the Covid-19 pandemic in 2019, which led to global lockdowns that forced people to stay at home. Nevertheless, many fitness apps are not user-friendly and do not offer personalised guidance or fully guided workout plans (QuickPose.ai, 2023). As a result, users, especially beginners, often feel confused about how to do their workouts in a proper way, which will lead to poor workout experiences and increase the risk of injury. Furthermore, a personalised diet plan is one of the essential features that is often overlooked (Sharma, 2024). Most fitness applications focus solely on doing workouts without offering personalised diet plans that align with users' fitness goals.

Besides, most applications do not provide enough motivation to encourage users to stay consistent with their exercise routines (Yates, 2022). Without interactive features, users may quickly lose interest and struggle with long-term commitment. Moreover, the cost of using fitness applications can also be a major barrier. Some applications require users to pay for the monthly or annual subscription fees to use some important features, which further limits accessibility for a wider range of users.

Another common problem is the poor user interface and user experience (UI/UX) design found in many fitness applications (American Sports & Fitness Association, n.d.). Users may struggle with complicated navigation and unclear workout instructions. Apart from that, most fitness applications do not provide real-time feedback during workouts. This has limited their ability to correct their posture instantly and support users in correcting their form instantly.

4.3 Project Aim

The primary aim of this project is to develop RUNTIME FITNESS, a web-based application that provides personalized diet planning and structured workout support for individuals seeking healthier lifestyles. The system emphasizes user-friendliness, accessibility, and personalization, enabling users to follow tailored health programs aligned with their specific goals, routines, and preferences.

4.4 Project Objectives

Key objectives include delivering an application with a seamless onboarding experience, guided workout videos, and real-time support through chats with coaches and nutritionists. The app encourages user consistency through motivational tools such as streak tracking, goal setting, and reminder notifications, all while maintaining a freemium model for wide accessibility.

The project also aims to ensure high usability, security, and performance, backed by intuitive interface design, responsive layouts, encrypted data handling, and regular backups. Guided by the Spiral Model development methodology, the system evolves iteratively based on continuous risk assessment and user feedback to ensure it meets practical user needs and offers a refined digital fitness experience.

4.5 Proposed Solutions/Scope

RUNTIME FITNESS is a fitness management system providing a structured platform to manage member data, track fitness progress, and offer personalized plans such as diet plans and workout plans. It is designed to assist members in achieving their health goals through consistent logging and tailored guidance. Admin has the authority to access the system and view reports, which are automatically generated by the system for monitoring and evaluation purposes.

4.5.1 Member Registration and Login

The system enables new users to register as members by providing their details through a registration form. Once registered, users can securely log in to access their dashboard. This function ensures that only authorised users can view and manage their data. It also maintains a complete list of all registered users.

4.5.2 Personalised Diet and Workout Plan

This feature is implemented through a chatbot that interacts with users to collect relevant personal details such as fitness goals and preferences. Based on the input, the system will generate customised diet and workout plans tailored to individual needs, whether it is for weight loss, muscle gain, or maintaining weight. The chatbot ensures a user-friendly experience while promoting guided and sustainable fitness progress.

4.5.3 Nutritional Journal

Members can record their daily meals, including breakfast, lunch, dinner, and snacks. This helps them become more aware of their habits and stay consistent with their dietary goals. The journal also allows users to log water intake, mood, vitamins or supplements consumed. Additionally, a dedicated text area is provided for users to write a short personal diary entry. This feature encourages holistic wellness tracking by integrating nutritional and emotional reflections.

4.5.4 Workout Progress

Members can log their completed workouts, including types of exercises, durations, and intensity levels. This function helps members track how consistently they are following their workout plans. It also allows for progress evaluation over time, making fitness tracking more structured and goal-oriented.

4.5.5 Progress Tracking

This feature tracks key health metrics such as weight, body measurements (waist, hip, thigh), water intake and sleep hours. Progress is visually displayed through three charts for evaluation and continuous improvement. Upon member log out, a notification prompts the user to confirm whether they have logged their progress for the day. If not, they are encouraged to update it; otherwise, they may choose to log out directly. This feature promotes consistency and accountability in fitness tracking.

4.5.4 Reporting and Monitoring

Admin can view multiple reports, including member profiles, registration reports, and activity reports. These reports offer valuable insights into system usage and member progress, supporting better evaluation and decision making. By centralising this information, the system enhances administrative oversight and operational efficiency.

4.6 Functionalities/Deliverables

Target Users

- Members
- Administrators

4.6.1 Functional Requirements

Members will be able:

- To sign up or log in
- To view profile
- To edit the profile
- To upload the workout progress
- To create personalised diet plan
- To create workout plan
- To add nutritionist journal
- To track their progress

- To share their thoughts in the community section

Administrators will be able:

- To log in
- To view profile
- To edit member profile
- To generate the report

4.6.2 Non-Functional Requirements

- **Technical/Implementation Constraints**
  - The system must be developed in HTML, CSS, JavaScript, MySQL, and PHP
  - The system must use MySQL for managing user and fitness-related data
  - The system must support responsive web design
- **Performance**
  - The system's annual downtime must be less than 50 minutes
  - The system must be user-friendly and intuitive for all user types
  - The system must be free from critical bugs in the production environment
  - The system's response time must be less than 3 seconds
- **Security**
  - All users' health and personal data must be encrypted both at rest and in transit
  - The system must implement secure authentication
- **Usability**
  - The app must feature a responsive web design
  - Interfaces must be easy to navigate for members, coaches, and nutritionists

- Accessibility best practices should be followed

- **Backup and Recovery**

- The system must have automatic recovery procedures in case of data loss or server failure

- Daily backups of user and application data must be performed and stored securely

5.0 Project Plan

5.1 System Development Methodology

5.1.1 Introduction to Spiral Model

The project development team has selected the Spiral Model as the system development methodology for the RUNTIME FITNESS Management System. The Spiral Model is a risk-driven software development framework introduced by Barry W. Boehm in the late 1980s (Gupta et al., 2021). It integrates key principles of the Waterfall Model and Iterative Development, facilitating the constant enhancement of the system through repeated cycles (Stepanov, 2021).

A major strength of the Spiral Model lies in its strong emphasis on risk analysis, early client feedback, and progressive refinement of system requirements. By incorporating client feedback at an early stage, the Spiral Model enables the project development team to develop a highly personalized and user-centric product (Maryani et al., 2022). Each phase within the Spiral Model is characterized by a distinct design goal, reviewed and evaluated by the client upon completion. As Leong et al. (2023) highlight, the development starts with a limited set of requirements and progressively expands the functionality and specifications with each successive spiral cycle. This approach enables the system to evolve systematically while continuously addressing risks and user needs.

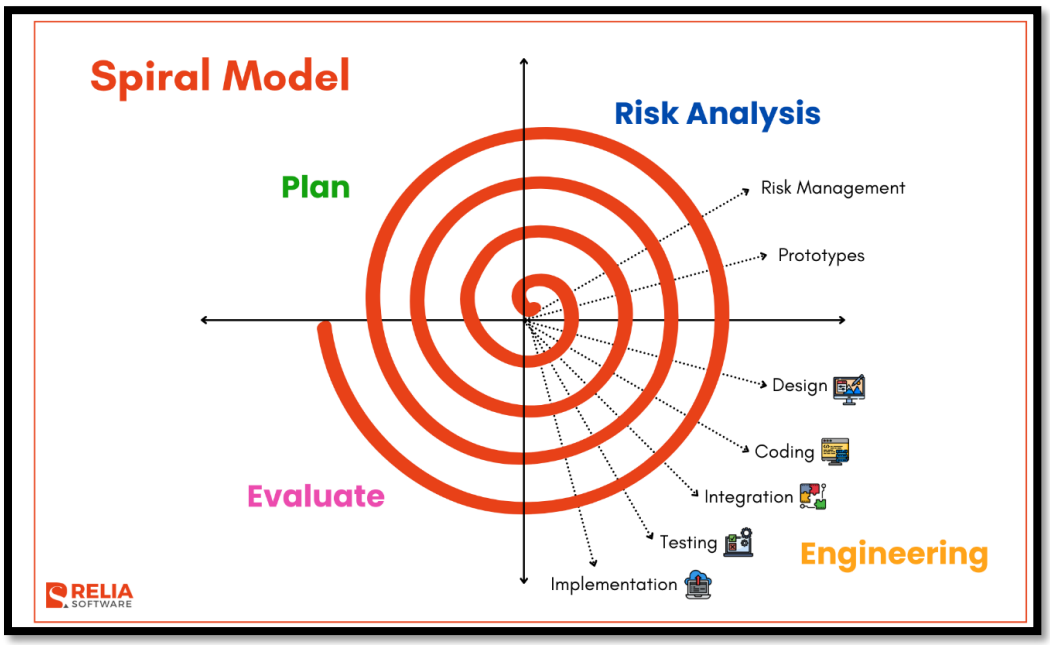


Figure 2: Illustration of the Spiral Model

(Note: The Spiral Model is well-suited for large and complex projects with high-risk factors and need for early user feedback. Adapted from "8 Software Development Life Cycle (SDLC) Methodologies", by Truong (2024), Relia Software.)

Figure 1 illustrates the structure of the Spiral Model, where each loop represents a distinct phase of the software development process. The Spiral Model organizes the project into a series of iterations, each comprising four main phases: Planning, Risk Analysis, Engineering, and Evaluation (Hossain, 2023). With each iteration, new functionalities are added, or existing features are refined. This iterative approach ensures the system evolves through continuous feedback and risk management, making it highly suitable for complex, user-focused projects like the RUNTIME FITNESS Management System.

5.1.2 Phases in the Spiral Model

As the development of the RUNTIME FITNESS Management System progresses, the Spiral Model guides the process through four iterative and structured phases. Each phase is vital in ensuring continuous refinement and risk-managed evolution of the system.

1. Planning Phase

The initial phase of the Spiral Model begins with the planning phase, and this phase plays a critical role in setting the foundation for the entire system development process. During this phase, the objectives of the system and its requirements, including functional and non-functional requirements, are gathered and defined in detail. The project development team works closely with key stakeholders such as members, coaches, nutritionists, and administrators in order to fully understand their expectations and define goals and the scope of the system (Hossain, 2023). Through interviews, surveys, and requirement gathering sessions, the project development team identifies core features of the RUNTIME FITNESS Management System, including user registration, personalized diet plans, real-time coach consultation, and member progress dashboards. A comprehensive project plan is developed to ensure a structured approach. This includes a clear breakdown of tasks, timelines, milestones, and deliverables, often visualized through tools like Gantt charts. These charts help map out the entire development cycle and track progress. The planning phase also involves estimating required resources like budget, manpower, and technical tools and outlining a high-level system architecture and design concept. These early decisions ensure that development activities in later phases are well organized and guided by realistic constraints. The planning

phase aligns technical development with user expectations and business goals, minimizing the chances of miscommunication and costly changes later in the project (Kyeremeh, 2021).

2. Risk Analysis Phase

The second phase of the Spiral Model is the risk analysis phase, an essential step where potential risks are identified, assessed, and mitigated in order to minimize their impact on the project. This phase focuses on technical and business-related risks that could hinder the successful development and deployment of the system. For the RUNTIME FITNESS Management System, several risk categories are considered. Technical risks include integration challenges with third-party fitness devices, system scalability issues, and the potential for inaccuracies in automated diet or fitness recommendations. Security and privacy risks are significant as the system handles sensitive personal health data. User-related risks, like low adoption due to a confusing or unintuitive interface, are also evaluated. These risks are identified through detailed feasibility studies and risk assessment, which help prioritize them based on their likelihood and impact (Hossain, 2023). To mitigate these risks, appropriate strategies are developed. For instance, implementing strong data encryption and secure authentication protocols helps protect user privacy. A modular and scalable system architecture ensures flexibility and performance as the user base grows. Collaborating with certified healthcare professionals can enhance the credibility and accuracy of the system's diet and fitness plans. By addressing these concerns early, the project development team minimizes the chances of costly redesigns or failures later in the project lifecycle (Stepanov, 2021). Additionally, a low-fidelity or functional prototype is often developed at the end of this phase. This prototype enables stakeholders to visualize key system components, validate critical features, and propose alternative solutions before full-scale development begins (Kyeremeh, 2021). In short, this phase ensures that informed decisions are made with a focus on reducing uncertainties and aligning the project with realistic, risk-aware strategies (Hossain, 2023).

3. Engineering Phase

The engineering phase begins after the planning and risk analysis phases are complete, marking the transition from conceptualization to actual system development. This phase focuses on translating the approved system design and requirements into a functioning product through a series of iterative development cycles. The engineering process involves designing the system architecture, developing user interface components, writing code for crucial features like user authentication, personalized diet planning, and fitness tracking, and integrating APIs for

compatibility with third-party fitness devices. Each cycle of the Spiral Model targets the construction and refinement of specific modules, starting from core functionalities and gradually progressing toward more advanced features (Hossain, 2023). For the RUNTIME FITNESS Management System, the initial cycles may involve developing the login system, user registration, and basic profile management. Subsequent iterations focus on implementing personalized features such as workout plan generation, nutrition advice modules, consultation booking systems, and real-time progress tracking dashboards. A key aspect of this phase is the implementation of thorough testing procedures to guarantee software quality and reliability. Various levels of testing, including unit testing, integration testing, and system testing, are carried out at each iteration to identify and fix errors early in the development cycle (Kyeremeh, 2021). Each spiral loop results in a version of the system with incremental improvements based on previous feedback and requirements (Hossain, 2023). This phased approach not only enables continuous validation but also allows the system to evolve in alignment with user expectations and technical constraints. By the end of this phase, a complete version of the RUNTIME FITNESS Management System is available, ready for further refinement and evaluation.

4. Evaluation Phase

The evaluation phase is an integral component of the Spiral Model that occurs at the end of each development cycle. In this phase, the partially developed product is presented to stakeholders, including users, project managers, and developers, for thorough assessment and feedback. This review process typically involves usability testing, user surveys, interviews, and direct observation to evaluate the functionality, performance, and user experience of the system (Stepanov, 2021). The goal is to determine whether the current iteration meets the specified requirements and aligns with user expectations. Stakeholder feedback is critical for identifying technical flaws and areas where the system can be improved. It also helps to uncover new requirements that may not have been previously considered. For the RUNTIME FITNESS Management System, feedback may highlight the need for a more intuitive user interface, clearer data visualization in progress tracking dashboards, or additional wellness features such as mental health tips and hydration reminders. Based on the insights gained during this phase, necessary adjustments are made to the system's design, functionalities, or user flow to better meet user needs and business goals (Hossain, 2023). This iterative feedback loop ensures that the system is continuously refined and remains relevant to real-world use cases. By incorporating feedback into subsequent development cycles, the system evolves in a

controlled, user-centered manner that enhances overall satisfaction and success (Kyeremeh, 2021).

5.1.3 Justification

The reason why the project team chose this methodology for the development of the RUNTIME FITNESS Management System is due to its outstanding support for flexibility, risk management, iterative improvement, and client involvement. The Spiral Model's high flexibility enables the team to accommodate evolving requirements and integrate continuous feedback (Hossain, 2023). Given that RUNTIME FITNESS aims to provide personalized diet and workout plans catering to users with diverse fitness goals, this adaptability is essential.

The strong emphasis on client involvement at every iteration ensures that the system remains aligned with user expectations. Frequent engagement with coaches, fitness experts, and end users promotes high user satisfaction (Hossain, 2023). The Spiral Model's prototyping approach enables early visualization of key system components, such as diet recommendation engines. Early prototyping allows requirements to be refined before full-scale development begins, minimizing rework (Gupta et al., 2021).

The iterative nature of the model, with multiple cycles of enhancement, supports continuous improvement. This is highly beneficial for RUNTIME FITNESS, where new features such as additional workout plans or dietary guidelines can be added over time in response to trends and feedback (Stepanov, 2021). Risk Management is a major advantage, allowing potential issues such as data security or system scalability to be addressed proactively, reducing the chance of major failures later. (Gupta et al., 2021).

The Spiral Model encourages comprehensive documentation and traceability at each stage. This ensures that future updates, audits, and system maintenance can be efficiently managed (Hossain, 2023). A structured approach to resource allocation and project monitoring promotes effective utilization of time and labor, which benefits project managers (Kyeremeh, 2021).

All in all, the Spiral Model is the most suitable methodology for ensuring the reliable development of the RUNTIME FITNESS Management System because of its advantages, which include early error detection, higher productivity, component reusability, systematic development, and the early production of working prototypes as all of which are critical for the successful and reliable development of the RUNTIME FITNESS Management System.

5.2 Project Gantt Chart

Activity/Date	April				MAY				JUNE					JULY	
	W1	W2	W3	W4	W1	W2	W3	W4	W1	W2	W3	W4	W5	W1	W2
1. Project Proposal															
1.1 Introduction															
1.1.1 Project Background															
1.1.2 Problem Context															
1.1.3 Proposed Solution															
1.1.4 Project Objectives															
1.1.5 Project Scope															
1.2 Project Plan															
1.2.1 System Development Methodology															
1.3 System Hierarchy Chart															
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3.0 System Implementation															
3.1 Login/ Registration															
3.2 Diet Plan															
3.3 Workout Plan															
3.4 User Profile/ Homepage															
3.5 Admin															

6.0 System Hierarchy Chart

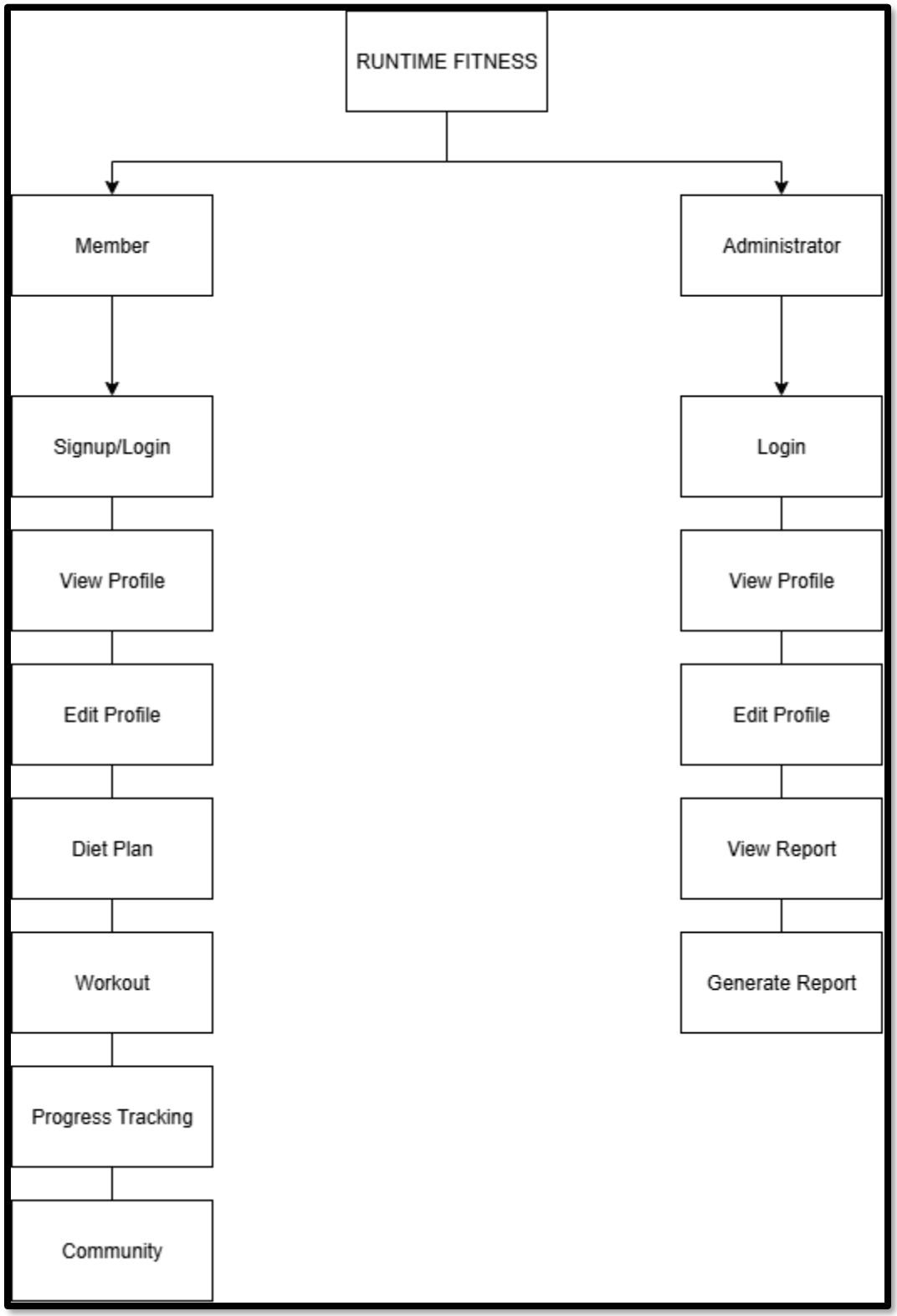


Figure 3: Hierarchy Chart of RUNTIME FITNESS

7.0 System Design

7.1 Context Diagram

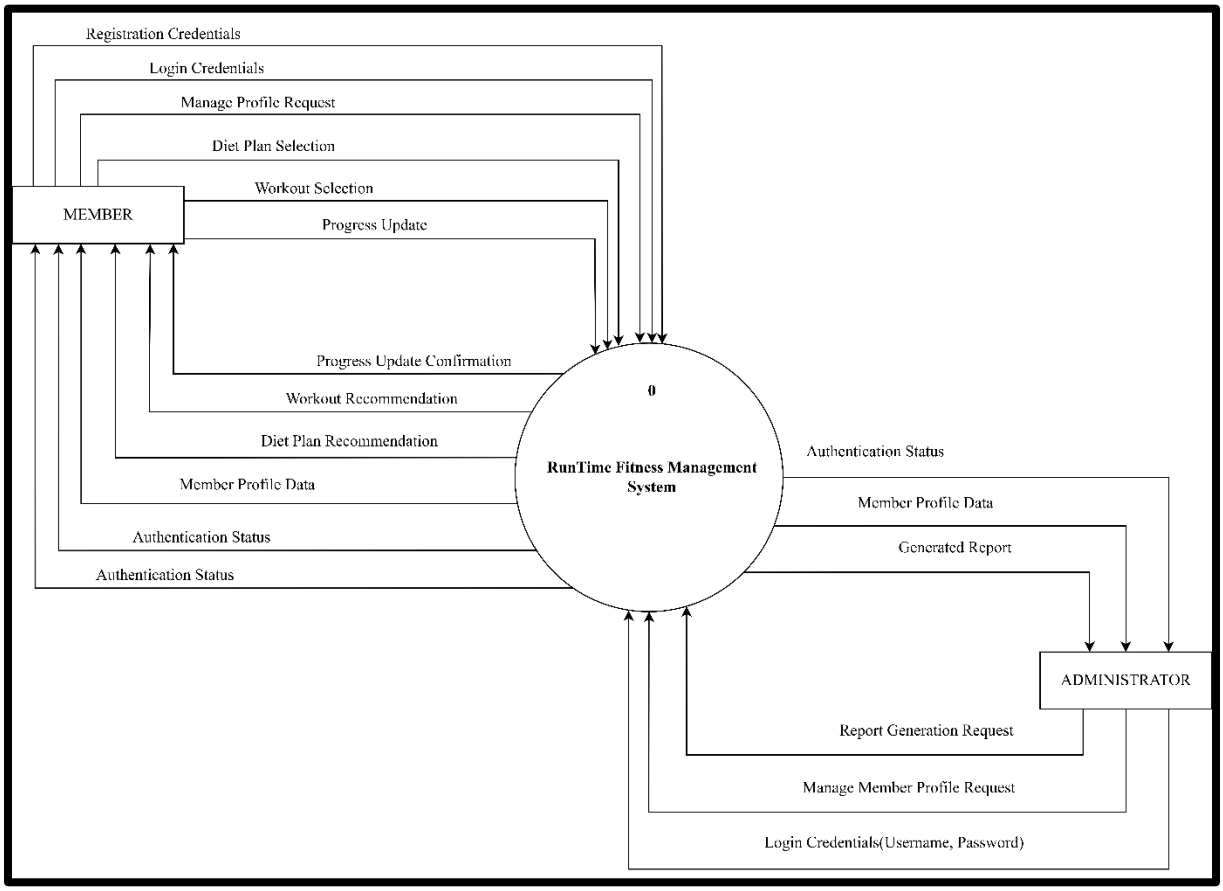


Figure 4: Context Diagram of RUNTIME FITNESS

7.2 Data Flow Diagrams & Data Dictionary

Data Dictionary – Process 1 Registration (YAP XIN LING TP077224)

External Entity

1)

Name: MEMBER

Description: An individual who registers, logs in, and resets their password by using the system.

Input Data Flows: Success Message, Error Message

Output Data Flows: Registration Details, Login Details, Reset Password Details

2)

Name: ADMINISTRATOR

Description: A system administrator who logs in to an account and resets the password by using the system.

Input Data Flows: Success Message, Error Message

Output Data Flows: Login Details, Reset Password Details

Process

1)

Name: 1.1 Enter Registration Information

Description: This process receives registration information from the user to register an account.

Input Data Flows: Registration Details

Output Data Flows: Registration Details

Process:

START

DISPLAY “Please enter your registration details.”

INPUT username, fullName, email, phone, password, confirmPassword, securityKeyword, dob, gender, height, weight, goal

END

2)

Name: 1.2 Validate Registration Information

Description: This process validates the user's registration information to register an account, including required field check, email format, password confirmation, and uniqueness.

Input Data Flows: Registration Details

Output Data Flows: Registration Details, Error Message, Success Message

Process:

START

IF username IS EMPTY OR fullName IS EMPTY OR email IS EMPTY OR phone IS EMPTY OR password IS EMPTY OR confirmPassword IS EMPTY OR securityKeyword IS EMPTY OR dob IS EMPTY OR gender IS EMPTY OR height IS EMPTY OR weight IS EMPTY OR goal IS EMPTY THEN

DISPLAY "All fields are required."

RETURN to registration form

END IF

IF password ≠ confirmPassword THEN

DISPLAY "Passwords do not match."

RETURN to registration form

END IF

IF NOT isValidEmail(email) THEN

DISPLAY "Invalid email format."

RETURN TO registration form

END IF

IF usernameExists(username) THEN

DISPLAY “Username already taken.”

RETURN TO registration form

END IF

IF emailExists(email) THEN

DISPLAY “Email already registered.”

RETURN TO registration form

END IF

END

3)

Name: 1.3 Store Registration Information

Description: This process stores the validated registration information of the user in the USERDATA_T data store.

Input Data Flows: Registration Details

Output Data Flows: Registration Details

Process:

START

STORE Registration Details INTO USERDATA_T

DISPLAY “Registration Successful”

END

4)

Name: 1.4 Enter Login Credentials

Description: This process accepts the user’s login information from the user.

Input Data Flows: Login Details

Output Data Flows: Login Details

Process:

START

DISPLAY “Enter your username and password.”

INPUT username, password

END

5)

Name: 1.5 Validate Login Credentials

Description: This process validates login information from the user against USERDATA_T, and required field check

Input Data Flows: Login Details, Login Credentials

Output Data Flows: Success Message, Error Message

Process:

START

IF username IS EMPTY OR password IS EMPTY THEN

DISPLAY “All fields are required.”

RETURN to login form

END IF

IF credentials MATCH USERDATA_T THEN

DISPLAY “Login Successful.”

ELSE

DISPLAY “Invalid username or password.”

END IF

END

6)

Name: 1.6 Enter Reset Password Information

Description: This process collects reset password credentials from the user for password resetting.

Input Data Flows: Reset Password Details

Output Data Flows: Reset Password Details

Process:

START

DISPLAY “Enter your username, security keyword, and new password.”

INPUT username, securityKeyword, newPassword

END

7)

Name: 1.7 Validate Reset Password Information

Description: This process authenticates identity and validates the user’s input before allowing password update.

Input Data Flows: Password Credentials

Output Data Flows: Success Message, Error Message

Process:

START

IF username IS EMPTY OR securityKeyword IS EMPTY OR newPassword IS EMPTY THEN

DISPLAY “All fields are required.”

RETURN to reset password form

END IF

IF username AND securityKeyword MATCH USERDATA_T THEN

DISPLAY “Password Reset Successful.”

ELSE

DISPLAY “Invalid reset credentials.”

END IF

END

8)

Name: 1.8 Store New Password

Description: This process stores the new password of the user in the USERDATA_T data store.

Input Data Flows: New Password

Output Data Flows: New Password

Process:

START

STORE newPassword INTO USERDATA_T

END

Data Flow

1)

Name: Registration Details

Description: This data flow carries the registration details from the user during the registration process.

Origin/Source: MEMBER external entity, Enter Registration Information process, Validate Registration Information process, Store Registration Information process, USER_DATA.T data store

Destination/Sink: Enter Registration Information process, Validate Registration Information process, Store Registration Information process, USERDATA_T data store

Data Structure: username, fullName, email, phone, password, confirmPassword, securityKeyword, dob, gender, height, weight, goal, role

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user

Element Characteristics:

-Type: Varchar

-Length: 50

-Output Format: xxxxxxxxxx

Data Element Name: fullName

Description: Full name of the user

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: email

Description: Unique email of the user

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: phone

Description: Unique phone number of the user

Element Characteristics:

-Type: Varchar

-Length: 15

-Output Format: xxxxxxxxxx

Data Element Name: password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: confirmPassword

Description: Re-entered password to confirm password

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: securityKeyword

Description: A unique keyword that is used for reset password

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: dob

Description: Date of birth of the user

Element Characteristics:

-Type: date

-Length: 20

-Output Format: YYYY-MM-DD

Data Element Name: gender

Description: Gender of the user

Element Characteristics:

-Type: Varchar

-Length: 10

-Output Format: Male/Female

Data Element Name: height

Description: Height of the user in cm

Element Characteristics:

-Type: Float

-Length: 10

-Output Format: xxxxxxxxxx

Data Element Name: weight

Description: Weight of the user in kg

Element Characteristics:

-Type: Float

-Length: 10

-Output Format: xxxxxxxxxx

Data Element Name: goal

Description: The purpose of the user for using the system

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: role

Description: The role of the user

Element Characteristics:

-Type: Varchar

-Length: 10

-Output Format: xxxxxxxxxx

2)

Name: Login Details

Description: This data flow carries the login details from the user

Origin/Source: MEMBER external entity, ADMINISTRATOR external entity, Enter Login Credentials process

Destination/Sink: Enter Login Credentials process, Validate Login Credentials process

Data Structure: username, password

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user

Element Characteristics:

-Type: Varchar

-Length: 50

-Output Format: xxxxxxxxxx

Data Element Name: password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

3)

Name: Login Credentials

Description: This is a flow that carries the login credentials from the USERDATA_T

Origin/Source: USERDATA_T data store

Destination/Sink: Validate Login Credentials

Data Structure: username, password

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user.

Element Characteristics:

-Type: Varchar

-Length: 50

-Output Format: xxxxxxxxxx

Data Element Name: password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

4)

Name: Reset Password Details

Description: This is a flow that carries the input of the user for password reset.

Origin/Source: MEMBER external entity, ADMINISTRATOR external entity, Enter Reset Password Information process

Destination/Sink: Enter Reset Password Information process, Validate Reset Password Information process

Data Structure: username, securityKeyword, newPassword

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user

Element Characteristics:

-Type: Varchar

-Length: 50

-Output Format: xxxxxxxxxx

Data Element Name: securityKeyword

Description: A unique keyword that is used for reset password.

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: newPassword

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

5)

Name: Password Credentials

Description: This data flow carries data in the USERDATA_T data store to validate reset password information.

Origin/Source: USERDATA_T data store

Destination/Sink: Validate Reset Password Information process

Data Structure: username, securityKeyword, newPassword

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user.

Element Characteristics:

-Type: Varchar

-Length: 50

-Output Format: xxxxxxxxxx

Data Element Name: securityKeyword

Description: A unique keyword that is used for reset password.

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: newPassword

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

6)

Name: New Password

Description: The new password set by the user during the password reset process.

Origin/Source: Validate Reset Password Information process, Store New Password process

Destination/Sink: Store New Password process, USERDATA_T data store

Data Structure: newPassword

Data Elements:

Data Element Name: newPassword

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

7)

Name: Success Message

Description: This data flow carries the success message after the user successfully registers, logs in, and resets the password.

Origin/Source: Validate Registration Information process, Validate Login Credentials process, Validate Reset Password Information process

Destination/Sink: MEMBER external entity, ADMINISTRATOR external entity

Data Structure: Success message details

Data Elements:

Data Element Name: Success message details

Description: A short message that indicates the user has register, login, and reset password.

Element Characteristics:

-Type: Varchar

-Length: 500

-Output Format: xxxxxxxxxx

8)

Name: Error Message

Description: This data flow carries the error message after the user enters invalid credentials to register, login, and reset password.

Origin/Source: Validate Registration Information process, Validate Login Credentials process, Validate Reset Password Information process

Destination/Sink: MEMBER external entity, ADMINISTRATOR external entity

Data Structure: Invalid field details

Data Elements:

Data Element Name: Invalid field details

Description: Error returned if the input is invalid

Element Characteristics:

-Type: Varchar

-Length: 500

-Output Format: xxxxxxxxxx

Data Store

1)

Name: USERDATA_T

Description: This data store saves the user data

Input Data Flows: Registration Details, New Password Credentials

Output Data Flows: Registration Details, Login Credentials, Password Credentials

Data Structure: username, fullName, email, phone, password, securityKeyword, dob, gender, height, weight, goal, role

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user

Element Characteristics:

-Type: Varchar

-Length: 50

-Output Format: xxxxxxxxxx

Data Element Name: fullName

Description: Full name of the user

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: email

Description: Unique email of the user

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: phone

Description: Unique phone number of the user

Element Characteristics:

-Type: Varchar

-Length: 15

-Output Format: xxxxxxxxxx

Data Element Name: password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: securityKeyword

Description: A unique keyword that is used for reset password

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: dob

Description: Date of birth of the user

Element Characteristics:

-Type: date

-Length: 20

-Output Format: YYYY-MM-DD

Data Element Name: gender

Description: Gender of the user

Element Characteristics:

-Type: Varchar

-Length: 10

-Output Format: Male/Female

Data Element Name: height

Description: Height of the user in cm

Element Characteristics:

-Type: Float

-Length: 10

-Output Format: xxxxxxxxxx

Data Element Name: weight

Description: Weight of the user in kg

Element Characteristics:

-Type: Float

-Length: 10

-Output Format: xxxxxxxxxx

Data Element Name: goal

Description: The purpose of the user for using the system

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: role

Description: The role of the user

Element Characteristics:

-Type: Varchar

-Length: 10

-Output Format: xxxxxxxxxx

Data Dictionary - Process 2 Manage Profile (SERENE LOH ZI TING TP075920)

EXTERNAL ENTITY

1)

Name: Member

Description: Registered user of the system. Can view/edit profile, delete account, and change password.

Input data flows: Personalized Data, Deletion Confirmation, New Password

Output Data Flows: User Data

2)

Name: Admin

Description: System admin with view-only or limited control privileges. Can view user profiles and receive deletion confirmation.

Input data flows: Deletion Confirmation

Output data flows: User Data, Deletion Confirmation

PROCESS: 2.1 Edit Profile

1)

Name: 2.1 Edit Profile

Description: Allows member to edit their own personal information

Input Data flows: Personalized Data

Output Data flows: User Data (to UserData_T)

Process Description:

START

IF member clicks “Edit Profile”

SHOW editable profile fields

WAIT for member to submit changes

VALIDATE data format (e.g., email, phone)

IF data is valid

SAVE changes to UserData_T

DISPLAY success message

ELSE

SHOW error message

END

2)

Name: 2.2 View Profile

Description: Retrieves the member’s profile for display

Input data flows: Username from the system session or internal memory

Output data flows: username from the system session or internal memory

Process Description:

START

IF member clicks “Profile” icon

FETCH Username from session

RETRIEVE profile data from UserData_T

Display profile fields (read-only)

END

3)

Name: 2.3 View Member Profile

Description: Allows member to edit their own personal information

Input Data Flows: Personalized Data

Output Date Flows: User Data (to UserData_T)

Process Description:

START

IF admin selects “view member profile”

SELECT member from list

FETCH selected username

RETRIEVE user data from UserData_T

DISPLAY profile fields (read-only)

END

4)

Name: 2.4 Delete Account

Description: Processes account deletion request from member or admin

Input data flows: Deletion Request (username or Target username)

Output data flows: Deletion Confirmation (to Member, Admin)

Process Description:

START

IF user initiates account deletion

ASK for confirmation

IF confirmed

DELETE user data in UserData_T

SEND deletion confirmation to user

ELSE IF admin initiates deletion

SELECT username from list

DELETE user data in UserData_T

SEND deletion confirmation to admin

END

5)

Name: 2.5 Change Password

Description: Allows the member to update their password securely

Input Data flows: New Password

Output data flows: Updated Password (to Data Store)

Process Description:

START

IF user submits new password

VALIDATE password format (e.g., length, complexity)

IF valid

UPDATE password in UserData_T

DISPLAY success message

ELSE

DISPLAY error message

END

DATA FLOWS

Name: Personalized Data

Description: Edited profile fields from the member

Origin: Member

Destination: Process 2.1 Edit Profile

Data Structure: Username, fullName, Email, Phone number, Gender, DOB, Security keywords, Height, Weight, Fitness Goal

2)

Name: User Data

Description: Retrieved or updated user profile information

Origin: UserData_T

Destination: Processes 2.1, 2.2, 2.3

Data Structure: Username, fullName, Email, Phone number, Password, Security keywords, Role

3)

Name: Deletion Confirmation

Description: Confirmation message after a delete action

Origin: Process 2.4 Delete Account

Destination: Member, Admin

Data Structure: Username, status

4)

Name: New Password

Description: Retrieved or updated user profile information

Origin: Member

Destination: Process 2.5 Change Password

Data Structure: Username, Security keywords, New Password

DATA STORE

Name: D2 (UserData_T)

Description: Stores user profile info, password, and status

Input Data flows: User Data, New Password

Output Data flows: User Data

Data structure: UserID, Name, Email, Password, Role, Security Keywords, etc.

Data Elements:

Data element Name: username

Description: The unique login identifier for the user

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: xxxxxxxxxx

Data Element Name: fullName

Description: Full name of the user

Element Characteristics:

- Type: Varchar
- Length: 100
- Output Format: xxxxxxxxxx

Data Element Name: email

Description: Unique email of the user

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: phone

Description: Unique phone number of the user

Element Characteristics:

-Type: Varchar

-Length: 15

-Output Format: xxxxxxxxxx

Data Element Name: password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: securityKeyword

Description: A unique keyword that is used for reset password

Element Characteristics:

-Type: Varchar

-Length: 100

-Output Format: xxxxxxxxxx

Data Element Name: dob

Description: Date of birth of the user

Element Characteristics:

-Type: date

-Length: 20

-Output Format: YYYY-MM-DD

Data Element Name: gender

Description: Gender of the user

Element Characteristics:

-Type: Varchar

-Length: 10

-Output Format: Male/Female

Data Element Name: height

Description: Height of the user in cm

Element Characteristics:

-Type: Float

-Length: 10

-Output Format: xxxxxxxxxx

Data Element Name: weight

Description: Weight of the user in kg

Element Characteristics:

-Type: Float

-Length: 10

-Output Format: xxxxxxxxxx

Data Element Name: fitness goal

Description: The purpose of the user for using the system

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Element Name: role

Description: The role of the user

Element Characteristics:

-Type: Varchar

-Length: 10

-Output Format: xxxxxxxxxx

Data element Name: status

Description: Indicates if the account is active, deactivated, or deleted

Element Characteristics:

- Type: Varchar
- Length: 20
- Output Format: “Active”, “Deactivated”, “Deleted”

Data Element Name: New password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

-Type: Varchar

-Length: 255

-Output Format: xxxxxxxxxx

Data Dictionary - Process 3 Chat (YONG XUAN LYN – TP076797)

External Entity

Name: MEMBER

Description: The system end user.

Input Data Flows: Comment

Output Data Flows: Comment

Process

1.

Name: 3.1 Define comment metadata

Description: Get comment and set comment metadata.

Input Data Flows: Comment

Output Data Flows: Comment metadata

Process:

START

GET comment from MEMBER

SET variables

cht_user_id = MEMBER id
comment = MEMBER input
comment_date = current_timestamp()

SAVE variables to CHATDATA_T

END

2.

Name: 3.2 Display Comment in Community

Description: Display comment in community chat.

Input Data Flows: Comment

Output Data Flows: Comment

Process:

START

GET comment && cht_user_id from CHATDATA_T

DISPLAY comment && cht_user_id

END

Data Flow

1.

Name: Comment

Description: Input made by users.

Origin/Source: MEMBER

Destination/Sink: 3.1 Define comment metadata

Data Structure: comment

Data Elements:

- Data Element Name: comment
- Description: Input made by users.
- Element Characteristics:
  - Type: varchar
  - Length: 255
  - Output format: xxxxxxxxxx

2.

Name: Comment metadata

Description: Metadata of comment.

Origin/Source: 3.1 Define comment metadata, CHATDATA_T, 3.2 Display Comment in Community

Destination/Sink: CHATDATA_T, 3.2 Display Comment in Community, MEMBER

Data Structure: cht_user_id, comment, comment_date

Data Elements:

- Data Element Name: cht_user_id
- Description: Unique identification code for each user. Can reference usr_user_id.
- Element Characteristics:
    - Type: int
    - Length: 11
    - Output format: xxxxxxxxx

- Data Element Name: comment
- Description: Input made by users.
- Element Characteristics:
    - Type: varchar
    - Length: 255
    - Output format: xxxxxxxxxx

- Data Element Name: comment_date
- Description: Date when comment was created.
- Element Characteristics:
    - Type: datetime
    - Length: -
    - Output format: YYYY-MM-DD hh:mm:ss

Data Dictionary - Process 4 Diet Plan (SIEW ZHEN LYNN TP076386)

External Entity

Name: MEMBER

Description: An individual who interacts with the system to receive a personalized diet plan by providing required details and preferences.

Input Data Flows: Success/Error Message, Personalized Diet Plan

Output Data Flows: Plan Type Selection, Usage Preference, Body Details, Activity Level Selection, Main Goal Selection, Daily Food Budget

Process

Name: 4.1 Choose Plan Type

Description: This process allows the MEMBER to select the type of diet plan they want (e.g., vegan, keto, balanced).

Input Data Flows: Plan Type Selection from MEMBER

Output Data Flows: Plan Type Selection

Process:

```
START
    DISPLAY available plan types to MEMBER
    RECEIVE planType selection from MEMBER
    IF planType IS EMPTY THEN
        DISPLAY "Please select a plan type."
        RETURN to plan type selection
    END IF
    PASS planType to next process
END
```

Name: 4.2 Determine the Use of Recorded Details

Description: Determines whether the MEMBER wants to use previously recorded body details or enter new details.

Input Data Flows: Plan Type Selection

Output Data Flows: Usage Preference

Process:

START

ASK MEMBER if they want to use recorded details or enter new details

RECEIVE usagePreference from MEMBER

IF usagePreference IS EMPTY THEN

DISPLAY "Please select an option."

RETURN to usage preference selection

END IF

PASS usagePreference to next process

END

Name: 4.3 Enter Body Details

Description: Collects the MEMBER's body details if they choose to enter new details.

Input Data Flows: Usage Preference, Body Details from MEMBER

Output Data Flows: Body Details, Recorded Details

Process:

START

IF usagePreference = "NewEntry" THEN

PROMPT MEMBER to enter height, weight, age, gender

RECEIVE bodyDetails from MEMBER

PASS bodyDetails to next process

ELSE

RETRIEVE recorded bodyDetails

PASS recorded bodyDetails to next process

END IF

END

Name: 4.4 Validate Body Details

Description: Validates the body details provided by the MEMBER.

Input Data Flows: Body Details

Output Data Flows: Success/Error Message, Body Details

Process:

START

IF height IS EMPTY OR weight IS EMPTY OR age IS EMPTY OR gender IS EMPTY THEN

DISPLAY "All body details are required."

RETURN to body details entry

END IF

IF height NOT numeric OR weight NOT numeric OR age NOT numeric THEN

DISPLAY "Height, weight, and age must be numbers."

RETURN to body details entry

END IF

IF gender NOT IN ("Male", "Female") THEN

DISPLAY "Invalid gender."

RETURN to body details entry

END IF

DISPLAY "Body details validated successfully."

PASS bodyDetails to next process

END

Name: 4.5 Save Body Details

Description: Saves the validated body details to the data store.

Input Data Flows: Body Details

Output Data Flows: Body Details

Process:

START

RECEIVE validated bodyDetails

STORE bodyDetails in DIETPLAN_T

PASS bodyDetails to next process

END

Name: 4.6 Select Activity Level

Description: Allows the MEMBER to select their activity level.

Input Data Flows: Activity Level Selection from MEMBER

Output Data Flows: Activity Level

Process:

START

DISPLAY activity level options to MEMBER

RECEIVE activityLevel selection from MEMBER

IF activityLevel IS EMPTY THEN

DISPLAY "Please select an activity level."

RETURN to activity level selection

END IF

PASS activityLevel to next process

END

Name: 4.7 Select Main Goal

Description: Allows the MEMBER to select their main goal for the diet plan.

Input Data Flows: Main Goal Selection from MEMBER

Output Data Flows: Main Goal

Process:

START

DISPLAY main goal options to MEMBER

RECEIVE mainGoal selection from MEMBER

IF mainGoal IS EMPTY THEN

DISPLAY "Please select a main goal."

RETURN to main goal selection

END IF

PASS mainGoal to next process

END

Name: 4.8 Enter Daily Food Budget

Description: Collects the MEMBER’s daily food budget (calories or nutrients).

Input Data Flows: Daily Food Budget from MEMBER

Output Data Flows: Food Budget

Process:

START

PROMPT MEMBER to enter daily food budget

RECEIVE foodBudget from MEMBER

IF foodBudget IS EMPTY THEN

DISPLAY "Please enter your daily food budget."

RETURN to food budget entry

END IF

PASS foodBudget to next process

END

Name: 4.9 Validate Daily Food Budget

Description: Validates the daily food budget entered by the MEMBER.

Input Data Flows: Food Budget

Output Data Flows: Success/Error Message, Food Budget

Process:

START

IF foodBudget IS EMPTY OR NOT numeric THEN

DISPLAY "Food budget must be a number."

RETURN to food budget entry

END IF

IF foodBudget < 800 OR foodBudget > 5000 THEN

DISPLAY "Food budget must be between 800 and 5000."

RETURN to food budget entry

END IF

DISPLAY "Food budget validated successfully."

PASS foodBudget to next process

END

Name: 4.10 Save Daily Food Budget

Description: Saves the validated daily food budget to the data store.

Input Data Flows: Food Budget

Output Data Flows: Food Budget

Process:

START

RECEIVE validated foodBudget

STORE foodBudget in DIETPLAN_T

PASS foodBudget to next process

END

Name: 4.11 Generate Personalized Diet Plan

Description: Generates a personalized diet plan for the MEMBER based on all collected and validated data.

Input Data Flows: Plan Type, Body Details, Activity Level, Main Goal, Food Budget

Output Data Flows: Personalized Diet Plan

Process:

START

RETRIEVE planType, bodyDetails, activityLevel, mainGoal, foodBudget from DIETPLAN_T

GENERATE personalized diet plan using all input data

STORE planContent in DIETPLAN_T

SEND personalized diet plan to MEMBER

END

Data Flow

1.

Name: Login Details

Description: This data flow carries the login details from the user.

Origin/Source: MEMBER external entity, ADMINISTRATOR external entity, Enter Login Credentials process

Destination: Enter Login Credentials process, Validate Login Credentials process

Data Structure: username, password

Data Elements:

Data Element Name: username

Description: An identifier that uniquely identifies the user

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: xxxxxxxxxx

Data Element Name: password

Description: A combination of alphabet, number, and special characters to secure the user account.

Element Characteristics:

- Type: Varchar
- Length: 255
- Output Format: xxxxxxxxxx

2.

Name: Plan Type Selection

Description: The type of diet plan selected by the member (e.g., vegan, keto, balanced).

Origin/Source: MEMBER

Destination/Sink: Choose Plan Type process (4.1), Determine the Use of Recorded Details process (4.2)

Data Structure: planType

Data Elements:

Data Element Name: planType

Description: The type of diet plan selected by the member

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: xxxxxxxxxx

3.

Name: Usage Preference

Description: Indicates whether the member wants to use previously recorded details or enter new ones.

Origin/Source: Determine the Use of Recorded Details process (4.2), MEMBER

Destination/Sink: Enter Body Details process (4.3)

Data Structure: usagePreference

Data Elements:

Data Element Name: usagePreference

Description: Member's preference to use recorded details or enter new details

Element Characteristics:

- Type: Varchar
- Length: 20
- Output Format: UseRecorded/NewEntry

4.

Name: Body Details

Description: The physical details of the member required for diet plan generation.

Origin/Source: MEMBER, Enter Body Details process (4.3), Save Body Details process (4.5)

Destination/Sink: Validate Body Details process (4.4), Save Body Details process (4.5), DIETPLAN_T data store

Data Structure: height, weight, age, gender

Data Elements:

Data Element Name: height

Description: Height of the member in cm

Element Characteristics:

- Type: Float
- Length: 10
- Output Format: xxxxxxxxxx

Data Element Name: weight

Description: Weight of the member in kg

Element Characteristics:

- Type: Float
- Length: 10
- Output Format: xxxxxxxxxx

Data Element Name: age

Description: Age of the member

Element Characteristics:

- Type: Integer
- Length: 3
- Output Format: xx

Data Element Name: gender

Description: Gender of the member

Element Characteristics:

- Type: Varchar
- Length: 10
- Output Format: Male/Female

5.

Name: Activity Level

Description: The activity level selected by the member (e.g., sedentary, moderate, active).

Origin/Source: MEMBER, Select Activity Level process (4.6)

Destination/Sink: DIETPLAN_T data store, Enter Daily Food Budget process (4.8)

Data Structure: activityLevel

Data Elements:

Data Element Name: activityLevel

Description: The member's selected activity level

Element Characteristics:

- Type: Varchar
- Length: 20
- Output Format: Sedentary/Moderate/Active

6.

Name: Main Goal

Description: The main goal selected by the member (e.g., weight loss, muscle gain, maintenance).

Origin/Source: MEMBER, Select Main Goal process (4.7)

Destination/Sink: DIETPLAN_T data store, Enter Daily Food Budget process (4.8)

Data Structure: mainGoal

Data Elements:

Data Element Name: mainGoal

Description: The member's main goal for the diet plan

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: WeightLoss/MuscleGain/Maintenance

7.

Name: Daily Food Budget

Description: The daily caloric or nutritional budget set by the member.

Origin/Source: MEMBER, Enter Daily Food Budget process (4.8)

Destination/Sink: Validate Daily Food Budget process (4.9), Save Daily Food Budget process (4.10), DIETPLAN_T data store

Data Structure: foodBudget

Data Elements:

Data Element Name: foodBudget

Description: The daily food budget (calories or nutrients)

Element Characteristics:

- Type: Integer
- Length: 5
- Output Format: xxxxx

8.

Name: Success/Error Message

Description: Message indicating the success or failure of validation or saving operations.

Origin/Source: Validate Body Details process (4.4), Validate Daily Food Budget process (4.9)

Destination/Sink: MEMBER

Data Structure: message

Data Elements:

Data Element Name: message

Description: Success or error message for the member

Element Characteristics:

- Type: Varchar
- Length: 255
- Output Format: xxxxxxxxxx

9.

Name: Personalized Diet Plan

Description: The generated personalized diet plan for the member.

Origin/Source: Generate Personalized Diet Plan process (4.11), DIETPLAN_T data store

Destination/Sink: MEMBER

Data Structure: planType, bodyDetails, activityLevel, mainGoal, foodBudget, planContent

Data Elements:

Data Element Name: planType

Description: The type of diet plan generated

Element Characteristics:

- Type: Varchar
- Length: 50

- Output Format: xxxxxxxxxx

Data Element Name: bodyDetails

Description: The member's body details used for the plan

Element Characteristics:

- Type: JSON/Object
- Output Format: {height: xx, weight: xx, age: xx, gender: xx}

Data Element Name: activityLevel

Description: The member's activity level

Element Characteristics:

- Type: Varchar
- Length: 20
- Output Format: Sedentary/Moderate/Active

Data Element Name: mainGoal

Description: The member's main goal

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: WeightLoss/MuscleGain/Maintenance

Data Element Name: foodBudget

Description: The daily food budget

Element Characteristics:

- Type: Integer
- Length: 5
- Output Format: xxxxx

Data Element Name: planContent

Description: The actual content of the personalized diet plan

Element Characteristics:

- Type: Text
- Length: 2000
- Output Format: xxxxxxxxxx

Data Store

Name: DIETPLAN_T

Description: This data store saves all information related to the personalized diet plan for each member.

Input Data Flows: Body Details, Activity Level, Main Goal, Food Budget

Output Data Flows: Personalized Diet Plan

Data Structure: memberID, planType, bodyDetails, activityLevel, mainGoal, foodBudget, planContent

Data Elements:

1.

Data Element Name: memberID

Description: Unique identifier for the member

Element Characteristics:

- Type: Integer
- Length: 10
- Output Format: xxxxxxxxxx

2.

Data Element Name: planType

Description: The type of diet plan

Element Characteristics:

- Type: Varchar

- Length: 50

- Output Format: xxxxxxxxxx

3.

Data Element Name: bodyDetails

Description: The member’s body details

Element Characteristics:

- Type: JSON/Object

- Output Format: {height: xx, weight: xx, age: xx, gender: xx}

4.

Data Element Name: activityLevel

Description: The member’s activity level

Element Characteristics:

- Type: Varchar

- Length: 20

- Output Format: Sedentary/Moderate/Active

5.

Data Element Name: mainGoal

Description: The member’s main goal

Element Characteristics:

- Type: Varchar

- Length: 50

- Output Format: WeightLoss/MuscleGain/Maintenance

6.

Data Element Name: foodBudget

Description: The daily food budget

Element Characteristics:

- Type: Integer
- Length: 5
- Output Format: xxxxx

7.

Data Element Name: planContent

Description: The actual content of the personalized diet plan

Element Characteristics:

- Type: Text
- Length: 2000
- Output Format: xxxxxxxxxx

Data Dictionary - Process 5 Workout (YEO PEI WEN TP077057)

External Entity

Name: MEMBER

Description: A user who involves in the system has his or her own personalized workout plan by following the BMI category with height and weight.

Input Data Flows: BMI Category, Weekly Workout Routine, Workout Progress, Workout Duration Chart

Output Data Flows: Height and Weight Details

Process

Name: 5.1 Show Profile Data

Description: This process is retrieved the MEMBER profile which are weight and height.

Input Data Flows: MEMBER ID

Output Data Flows: Weight and Height Details

Process:

START

FETCH wop_user_id

RETRIEVE wop_height, wop_weight FROM workout_plan

DISPLAY wop_height, wop_weight TO MEMBER

PROMPT: “These are the most recent details we’ve saved for you. Height: wop_height, Weight: wop_height”

END

Name: 5.2 Confirm or Enter Data

Description: Asks the MEMBER whether to use the profile data or enter new height and weight.

Input Data Flows: Weight and Height Details, Choice from MEMBER

Output Data Flows: Weight and Height Details (for BMI calculation)

Process:

START

PROMPT: “Do you want to use your profile’s height and weight? (yes/no)”

IF YES THEN

    PASS wop_height and wop_weight TO next process

ELSE

    PROPMT member TO enter new height and weight

    PASS new height and weight TO next process

END

Name: 5.3 Calculate BMI

Description: Calculate the Body Mass Index (BMI) based on the selected or entered data.

Input Data Flows: Weight and Height Details

Output Data Flows: Weight and Height Details

Process:

START

INPUT: height, weight

CALCULATE BMI = weight / (height/100) ^2

CLASSIFY BMI into category:

IF BMI < 18.5 → Underweight

ELSE IF BMI < 25 → Normal

ELSE IF BMI < 30 → Overweight

ELSE IF BMI < 35 → Obese1

ELSE IF BMI < 40 → Obese2

ELSE → Obese3

PASS BMI and category TO next process

END

Name: 5.4 Generated Workout Plan

Description: Creates a workout plan based on BMI category

Input Data Flows: Weight and Height Details

Output Data Flows: Workout Plan Details

Process:

START

INPUT: BMI category

SELECT workout set FROM workout_plan based on category

GENERATE structured weekly workout plan (Mon-Sun)

SAVE plan INTO WORKOUTDATA_T under wop_user_id

PASS plan TO next process

END

Name: 5.5 Show Weekly Routine

Description: Displays the weekly workout plan with tick options for completed workouts.

Input Data Flows: Workout Plan Details

Output Data Flows: Weekly Workout Plan Details to MEMBER

Process:

START

RETRIEVE workout plan FROM WORKOUTDATA_T

DISPLAY weekly workout plan TO MEMBER

ALLOW MEMBER TO tick completed workouts

FOR each ticked workout:

    SEND to next process

END

Name: 5.6 Tick Completed Workout

Description: Allows MEMBERS to mark a workout as completed and sends the record for tracking.

Input Data Flows: Weekly Workout Plan Details to MEMBER

Output Data Flows: Completed Weekly Workout Plan Details from MEMBER

Process:

START

INPUT: workout ticked by MEMBER

RECORD completed workout date and duration

UPDATE WORKOUTDATA_T under wop_user_id

PASS data TO next process

END

Name: 5.7 Show Completed Workout and Duration

Description: Displays a table and chart of completed workouts and total workout duration.

Input Data Flows: Completed Weekly Workout Plan Details

Output Data Flows: Completed Weekly Workout Plan Details to MEMBER, Weekly Workout Duration Chart

Process:

START

RETRIEVE all completed workouts FROM WORKOUTDATA_T

DISPLAY as table (Workout Progress)

AGGREGATE duration per day/week

DISPLAY as chart (Workout Duration Chart)

END

Data Flow

1.

Name: Weight and Height Details

Description: This flow contains the user's height and weight values, either retrieved from the profile or entered manually.

Origin/Source: MEMBER external entity, Show Profile Data process, Confirm or Enter Data process

Destination: Calculate BMI process

Data Structure: wos_user_id, wop_height, wop_weight, wop_bmi, wop_category

Data Elements:

Data Elements Name: wos_user_id

Description: The unique ID of the member.

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: xxxxxxx

Data Elements Name: wop_height

Description: The height of the member in centimeters.

Element Characteristics:

- Type: Decimal
- Length: (5, 2)
- Output Format: xxxxx.xx

Data Elements Name: wop_weight

Description: The weight of the member in kilograms.

Element Characteristics:

- Type: Decimal
- Length: (5, 2)
- Output Format: xxxxx.xx

Data Elements Name: wop_bmi

Description: The BMI of the member.

Element Characteristics:

- Type: Decimal
- Length: (5, 2)
- Output Format: xxxxx.xx

Data Elements Name: wop_category

Description: The BMI category of the member.

Element Characteristics:

- Type: Varchar
- Length: 20
- Output Format: xxxxxxxx

2.

Name: Completed Workout Details

Description: This data flow carries data about workouts completed by the member, including date and duration.

Origin/Source: Tick Completed Workout process

Destination: WORKOUTDATA_T data store, Show Completed Workout and Duration process

Data Structure: wos_day, wos_item, wos_checked, wos_date

Data Elements:

Data Elements Name: wos_day

Description: The day on which member completed the workout.

Element Characteristics:

- Type: varchar
- Length: 20
- Output Format: xxxxxxxxx

Data Elements Name: wos_item

Description: The workout that completed by the member.

Element Characteristics:

- Type: varchar
- Length: 225
- Output Format: xxxxxxxxxxx

Data Elements Name: wos_checked

Description: The status of the workout plan to do.

Element Characteristics:

- Type: tinyint
- Length: 1
- Output Format: x

Data Elements Name: wos_date

Description: The date on which member completed the workout.

Element Characteristics:

- Type: date
- Output Format: YYYY-MM-DD

Data Store

Name: WORKOUTDATA_T

Description: This data store saves all information related to the personalized workout plan for each member.

Input Data Flows: Workout Plan Details, Completed Weekly Workout Plan Details

Output Data Flows: Weight and Height Details, Completed Weekly Workout Plan Details

Data Structure: wos_user_id, wop_height, wop_weight, wop_bmi, wop_category, wos_day, wos_item, wos_checked, wos_date

Data Elements:

1.

Data Elements Name: wos_user_id

Description: The unique ID of the member.

Element Characteristics:

- Type: Varchar
- Length: 50
- Output Format: xxxxxxx

2.

Data Element Name: wop_height

Description: The height of the member in centimeters.

Element Characteristics:

- Type: Decimal
- Length: (5,2)
- Output formal: xxxxx.xx

3.

Data Element Name: wop_weight

Description: The weight of the member in kilograms.

Element Characteristics:

- Type: Decimal

- Length: (5,2)
- Output formal: xxxxx.xx

4.

Data Elements Name: wop_bmi

Description: The BMI of the member.

Element Characteristics:

- Type: Decimal
- Length: (5, 2)
- Output Format: xxxxx.xx

5.

Data Elements Name: wop_category

Description: The BMI category of the member.

Element Characteristics:

- Type: Varchar
- Length: 20
- Output Format: xxxxxxxx

6.

Data Element Name: wop_day

Description: The day of the member completed the workout.

Element Characteristics:

- Type: Varchar
- Length: 20
- Output formal: xxxxxxx

7.

Data Element Name: wop_item

Description: The workout completed by the member.

Element Characteristics:

- Type: Varchar
- Length: 255

- Output formal: xxxxxxx

8.

Data Element Name: wop_checked

Description: The status of the workout that need to be done by the member.

Element Characteristics:

- Type: tinyint
- Length: 1
- Output formal: x

9.

Data Element Name: wop_date

Description: The date completed the workout by the member.

Element Characteristics:

- Type: Date
- Output formal: YYYY-MM-DD

Data Dictionary – Process 6 Progress (SIM TIAN TP077056)

EXTERNAL ENTITY

1.

Name: MEMBER

Description: An individual who updates their progress using the system, including weight, body measurements (waist, hip, thigh), and water and sleep logs.

Input Data Flows: Error Message, Success Message

Output Data Flows: Daily Weight Details, Body Measurements Details, Water and Sleep Log Details

PROCESS

1.

Name: 6.1 Enter Daily Weight

Description: This process receives Daily Weight Details from the user.

Input Data Flows: Daily Weight Details

Output Data Flows: Daily Weight Details

Process:

START

INPUT weight

END

2.

Name: 6.2 Validate Daily Weight

Description: This process validates the received Daily Weight Details.

Input Data Flows: Daily Weight Details

Output Data Flows: Daily Weight Details, Error Message, Success Message

Process:

START

IF weight >=10 && weight <=200

THEN RETURN to progress page

ELSE

DISPLAY “Weight must be between 10-200kg.”

END

3.

Name: 6.3 Store Daily Weight

Description: This process stores Daily Weight Details into the database.

Input Data Flows: Daily Weight Details

Output Data Flows: Daily Weight Details

Process:

START

STORE Daily Weight Details INTO PROGRESSDATA_T

OUTPUT "Weight saved successfully"

END

4.

Name: 6.4 Enter Body Measurements

Description: This process receives Body Measurements Details from the user.

Input Data Flows: Body Measurements Details

Output Data Flows: Body Measurements Details

Process:

START

INPUT waist, hip, thigh

END

5.

Name: 6.5 Validate Body Measurements

Description: This process validates the Body Measurements Details.

Input Data Flows: Body Measurements Details

Output Data Flows: Body Measurements Details, Error Message, Success Message

Process:

START

IF waist >=10 && waist <=200

THEN RETURN

ELSE IF hip >=10 && hip <=200

THEN RETURN

ELSE IF thigh >=10 && thigh <=200

THEN RETURN

ELSE DISPLAY “Waist, hip and thigh must be between 10-200cm.”

END

6.

Name: 6.6 Store Body Measurements

Description: This process stores Body Measurements Details into the database.

Input Data Flows: Body Measurements Details

Output Data Flows: Body Measurements Details

Process:

START

STORE Body Measurements Details INTO PROGRESSDATA_T

OUTPUT "Body Measurements saved successfully"

END

7.

Name: 6.7 Enter Water & Sleep Log

Description: This process receives Water & Sleep Log Details from the user.

Input Data Flows: Water & Sleep Log Details

Output Data Flows: Water & Sleep Log Details

Process:

START

INPUT water, sleep

END

8.

Name: 6.8 Validate Water & Sleep Log

Description: This process validates Water & Sleep Log Details.

Input Data Flows: Water & Sleep Log Details

Output Data Flows: Water & Sleep Log Details, Error Message, Success Message

Process:

START

IF (water >=0 && water <=20) && (sleep >= 0 && sleep<=24)

THEN RETURN to progress page

ELSE

DISPLAY “Water must be between 0-20 cups. Sleep must be between 0-24 hours.”

END

9.

Name: 6.9 Store Water & Sleep Log

Description: This process stores Water & Sleep Log Details into the database.

Input Data Flows: Water & Sleep Log Details

Output Data Flows: Water & Sleep Log Details

Process:

START

STORE Water & Sleep Log Details INTO PROGRESSDATA_T

OUTPUT "Water & Sleep Log saved successfully"

END

10.

Name: 6.10 Generate Chart

Description: This process generates weight chart, body measurements chart, and water & sleep log chart for the user.

Input Data Flows: Progress Data Details

Output Data Flows: Progress Data Details

Process:

START

OUTPUT weight chart, body measurements chart, water & sleep log chart

END

DATA FLOW

1)

Name: Daily Weight Details

Description: This data flow carries Daily Weight Details from the user during the progress tracking process.

Origin/Source: MEMBER external entity, Enter Daily Weight process, Validate Daily Weight process, Store Daily Weight process, PROGRESSDATA_T

Destination/Sink: Enter Daily Weight process, Validate Daily Weight process, Store Daily Weight process, PROGRESSDATA_T

Data Structure: weight

Data Elements:

- **Data Element Name:** weight
- **Description:** Weight of the user in cm.
- **Element Characteristics:**
    - Type: DECIMAL
    - Length: 5
    - Output Format: xxxxx

2)

Name: Body Measurements Details

Description: This data flow carries Body Measurements Details from the user during the progress tracking process.

Origin/Source: MEMBER external entity, Enter Body Measurements process, Validate Body Measurements process, Store Body Measurements process, PROGRESSDATA_T

Destination/Sink: Enter Body Measurements process, Validate Body Measurements process, Store Body Measurements process, PROGRESSDATA_T

Data Structure: waist, hip, thigh

Data Elements:

- **Data Element Name:** waist
- **Description:** Waist of the user in cm.

- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx
- **Data Element Name:** hip
- **Description:** Hip of the user in cm.
- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx
- **Data Element Name:** thigh
- **Description:** Thigh of the user in cm.
- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx

3)

Name: Water & Sleep Log Details

Description: This data flow carries Water & Sleep Log Details from the user during the progress tracking process.

Origin/Source: MEMBER external entity, Enter Water & Sleep Log process, Validate Water & Sleep Log process, Store Water & Sleep Log process, PROGRESSDATA_T

Destination/Sink: Enter Water & Sleep Log process, Validate Water & Sleep Log process, Store Water & Sleep Log process, PROGRESSDATA_T

Data Structure: water, sleep

Data Elements:

- **Data Element Name:** water
- **Description:** Cups of water that drink by the user.
- **Element Characteristics:**
  - Type: DECIMAL

  - Length: 5
  - Output Format: xxxxx
- **Data Element Name:** sleep
- **Description:** Sleep hours of the user.
- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx

4)

Name: Progress Data Details

Description: This data flow carries Progress Data Details from the database during the progress tracking process.

Origin/Source: PROGRESSDATA_T

Destination/Sink: Generate Chart process

Data Structure: weight, waist, hip, thigh, water, sleep

Data Elements: Each field (weight, waist, hip, thigh, water, sleep) has the following characteristics:

- Type: DECIMAL
- Length: 5
- Output Format: xxxxx

DATA STORE

1)

Name: D1 PROGRESSDATA_T

Description: This data store saves the progress data of the user.

Input Data Flows: Daily Weight Details, Body Measurements Details, Water & Sleep Log Details

Output Data Flows: Progress Data Details

Data Structure: logID, weight, waist, hip, thigh, water, sleep

Data Elements:

- **Data Element Name:** logID
- **Description:** An identifier that uniquely identifies each log.
- **Element Characteristics:**
    - Type: VARCHAR
    - Length: 10
    - Output Format: xxxxx

- **Data Element Name:** weight
- **Description:** Weight of the user in cm.
- **Element Characteristics:**
    - Type: DECIMAL
    - Length: 5
    - Output Format: xxxxx

- **Data Element Name:** waist
- **Description:** Waist of the user in cm.
- **Element Characteristics:**
    - Type: DECIMAL
    - Length: 5
    - Output Format: xxxxx

- **Data Element Name:** hip
- **Description:** Hip of the user in cm.
- **Element Characteristics:**

  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx

- **Data Element Name:** thigh
- **Description:** Thigh of the user in cm.
- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx

- **Data Element Name:** water
- **Description:** Cups of water that drink by the user.
- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx

- **Data Element Name:** sleep
- **Description:** Sleep hours of the user.
- **Element Characteristics:**
  - Type: DECIMAL
  - Length: 5
  - Output Format: xxxxx

Data Dictionary - Process 7 Report (YONG XUAN LYN TP 076797)

External Entity

Name: ADMIN

Description: The system administrator.

Input Data Flows: Categorized number of new user registration, Categorized number of active members

Output Data Flows: Registration && last login timestamp of members

Process

1.

Name: 7.1 Display Registration Report

Description: Display the number of registered members sorted by week, month, and year.

Input Data Flows: Registration timestamp of members

Output Data Flows: Categorized number of new user registration

Process:

START

GET current_timestamp()

GET ALL usr_registration_date

SORT usr_registration_date BY week, month, year && COUNT

DEFINE variables:

reg_week = this week's registrations
reg_month = this month's registrations
reg_year = this year's registrations

ASSIGN variables

DISPLAY variables

END

2.

Name: 7.2 Display Active Member Report

Description: Display the number of active members sorted by week, month, and year.

Input Data Flows: Last login timestamp of members

Output Data Flows: Categorized number of active members

Process:

START

GET current_timestamp()

GET ALL usr_last_login_date

SORT usr_last_login_date BY week, month, year && COUNT

DEFINE variables

active_week = this week's active members
active_month = this month's active members
active_year = this year's active members

ASSIGN variables

DISPLAY variables

END

Data Flow

3.

Name: Registration timestamp of members

Description: The timestamp recorded when a new member registration is done.

Origin/Source: REPORT_T

Destination/Sink: 7.1 Display Registration Report

Data Structure: registration_datetime

Data Elements:

- Data Element Name: registration_datetime
- Description: The timestamp recorded when a new member registration is done.
- Element Characteristics:
    - Type: datetime
    - Length:
    - Output format: YYYY-MM-DD hh:mm:ss

4.

Name: Last login timestamp of members

Description: The most current timestamp recorded when a member last logs in.

Origin/Source: REPORT_T

Destination/Sink: 7.2 Display Active Member Report

Data Structure: last_login_datetime

Data Elements:

- Data Element Name: last_login_datetime
- Description: The most current timestamp recorded when a member last logs in.
- Element Characteristics:
    - Type: datetime
    - Length:
    - Output format: YYYY-MM-DD hh:mm:ss

5.

Name: Categorized number of new user registration

Description: Numbers of new member registrations sorted and categorized using week, month, and year.

Origin/Source: 7.1 Display Registration Report

Destination/Sink: ADMIN External Entity

Data Structure: reg_week, reg_month, reg_year

Data Elements:

- Data Element Name: reg_week
- Description: This week's number of new registered members.
- Element Characteristics:
    - Type: int
    - Length: 8
    - Output format: xxxxxxxx

- Data Element Name: reg_month
- Description: This month's number of new registered members.
- Element Characteristics:
    - Type: int
    - Length: 8
    - Output format: xxxxxxxx

- Data Element Name: reg_year
- Description: This year's number of new registered members.
- Element Characteristics:
    - Type: int
    - Length: 8
    - Output format: xxxxxxxx

6.

Name: Categorized number of active members

Description: Numbers of logins sorted and categorized using week, month, and year.

Origin/Source: 7.2 Display Active Member Report

Destination/Sink: ADMIN External Entity

Data Structure: active_week, active_month, active_year

Data Elements:

- Data Element Name: active_week
- Description: Active members, sorted by week.
- Element Characteristics:
    - Type: int
    - Length: 8
    - Output format: xxxxxxxx

- Data Element Name: active_month
- Description: Active members, sorted by month.
- Element Characteristics:
    - Type: int
    - Length: 8
    - Output format: xxxxxxxx

- Data Element Name: active_year
- Description: Active members, sorted by year.
- Element Characteristics:
    - Type: int
    - Length: 8
    - Output format: xxxxxxxx

7.

Name: Registration && last login timestamp of members

Description: The current_datetime() collected when a user registers/logs in.

Origin/Source: ADMIN

Destination/Sink: REPORT_T

Data Structure: registration_datetime, last_login_datetime

Data Elements:

- Data Element Name: registration_datetime
- Description: The timestamp recorded when a new member registration is done.
- Element Characteristics:
    - Type: datetime
    - Length:
    - Output format: YYYY-MM-DD hh:mm:ss

- Data Element Name: last_login_datetime
- Description: The most current timestamp recorded when a member last logs in.
- Element Characteristics:
    - Type: datetime
    - Length:
    - Output format: YYYY-MM-DD hh:mm:ss

7.3 DFD Level-0

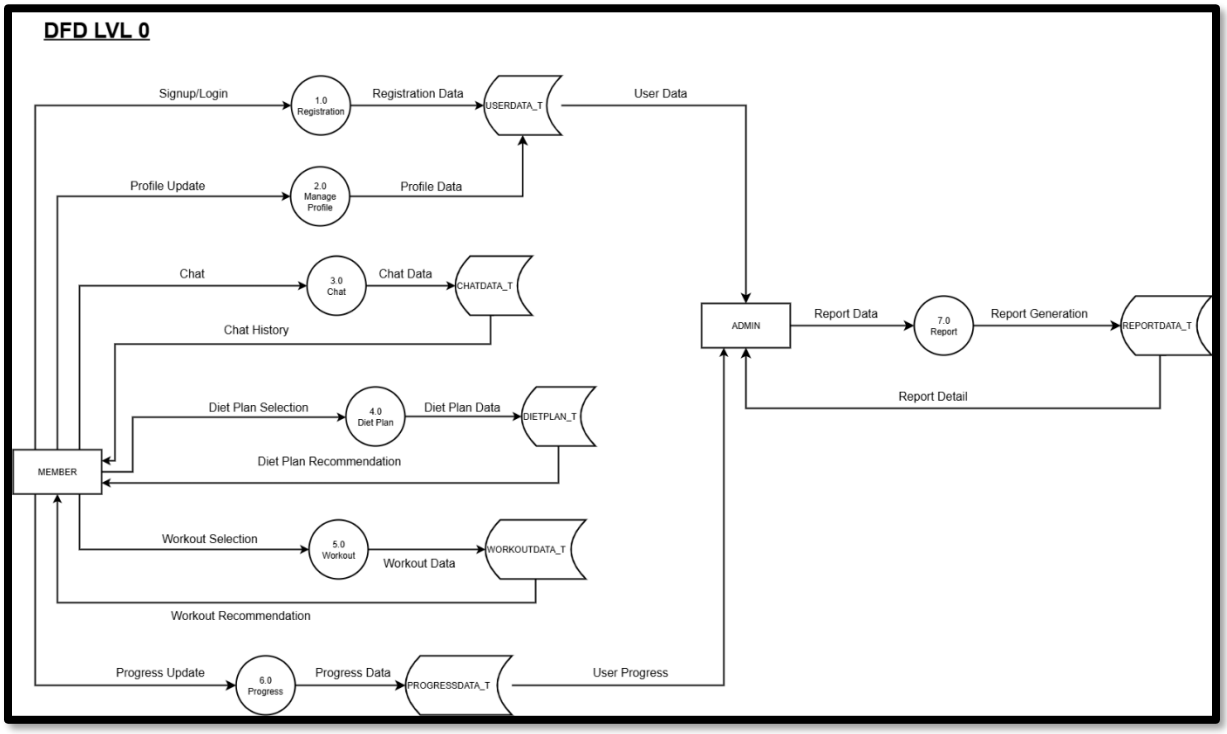


Figure 5: DFD Level-0

7.4 DFD Level-1

7.4.1 DFD Level 1 - Registration (YAP XIN LING – TP077224)

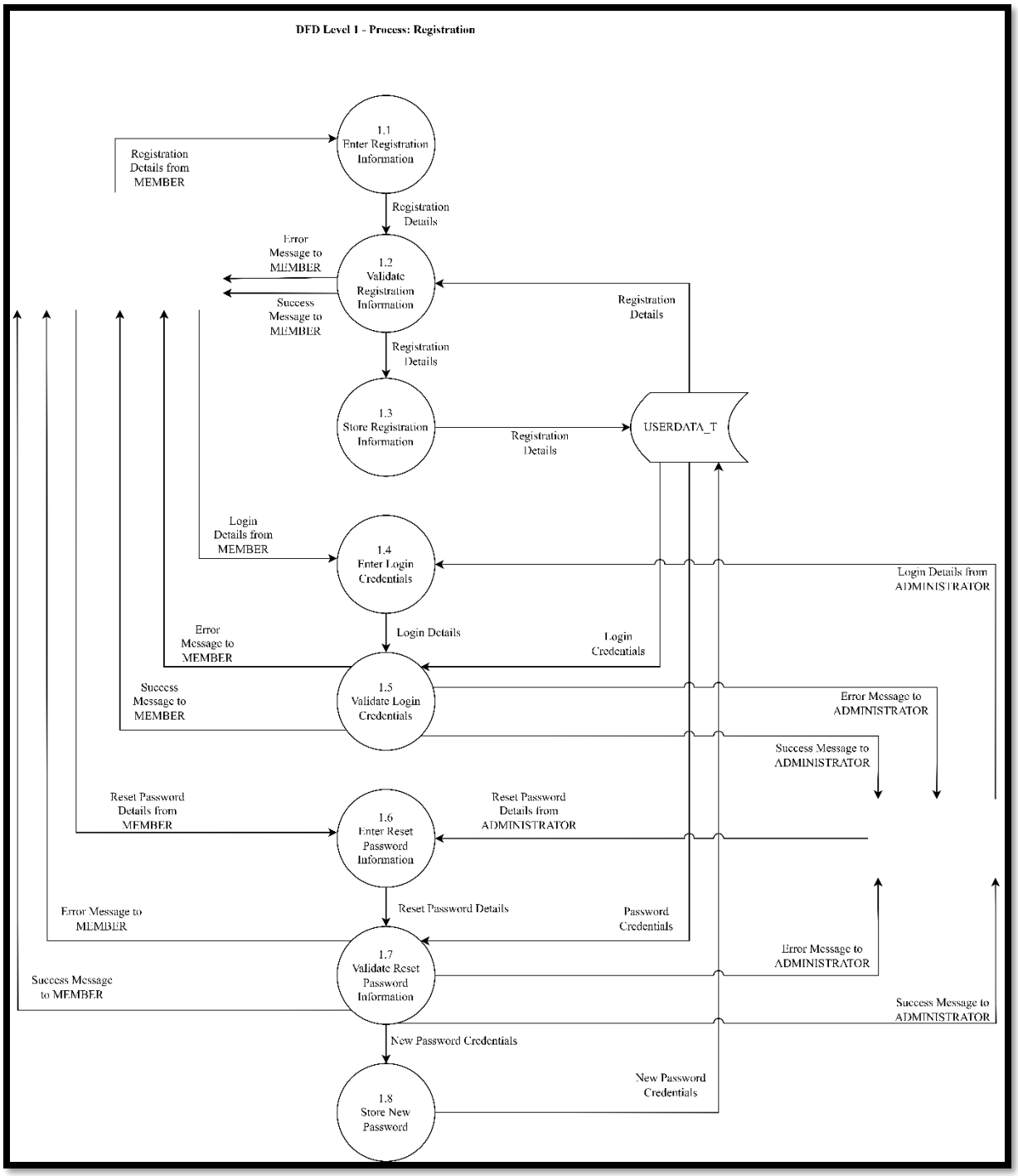


Figure 6: DFD Level 1 (Registration Process)

7.4.2 DFD Level 1 – Manage Profile (SERENE LOH ZI TING – TP075920)

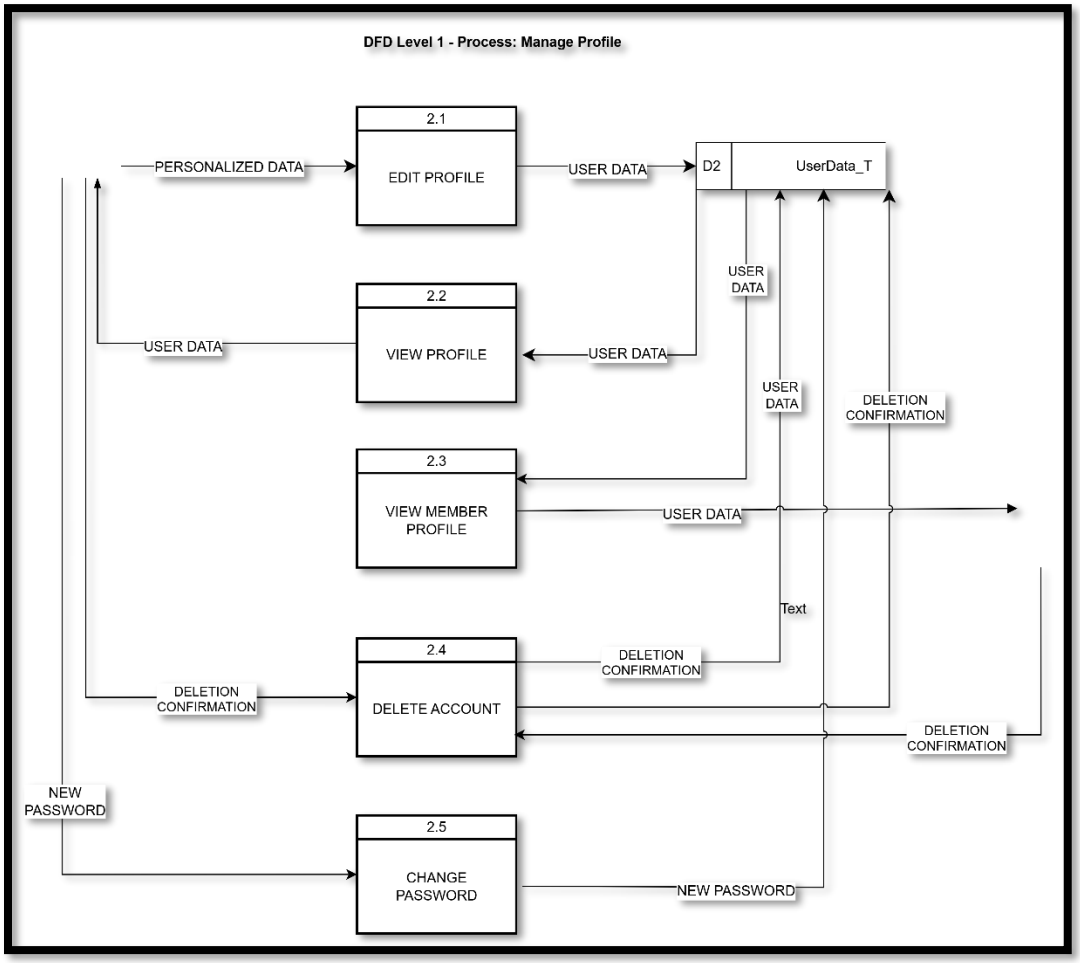


Figure 7: DFD level 1 (manage profile)

7.4.3 DFD Level 1 – Chat (YONG XUAN LYN – TP076797)

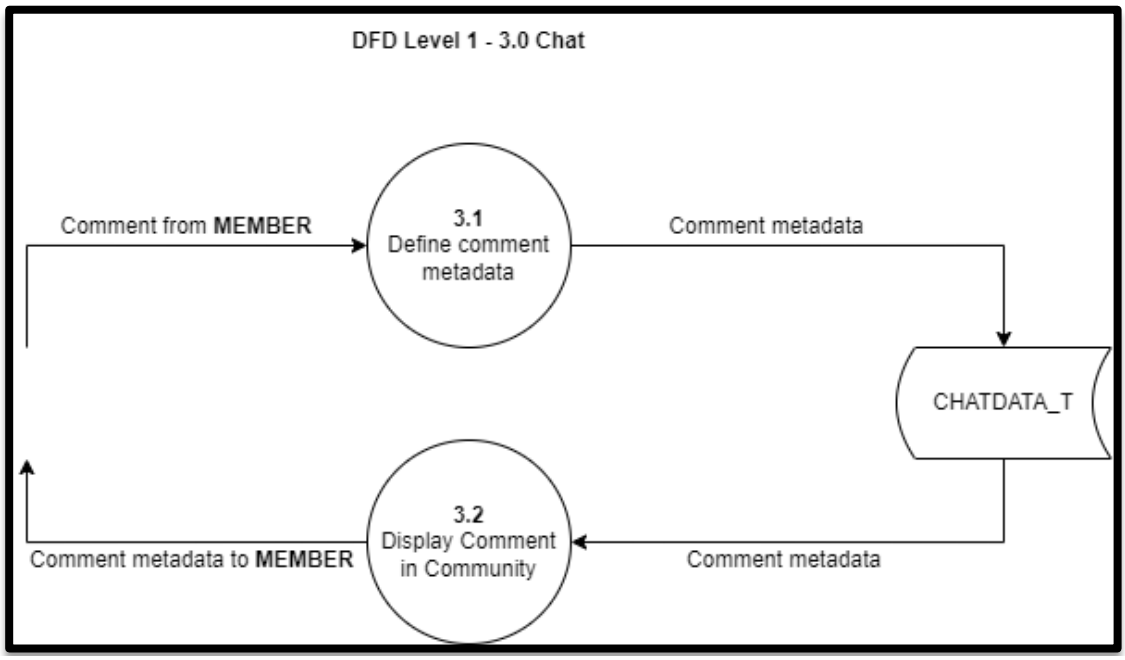


Figure 8: DFD Level 1 (chat)

7.4.4 DFD Level 1 – Diet Plan (SIEW ZHEN LYNN– TP076386)

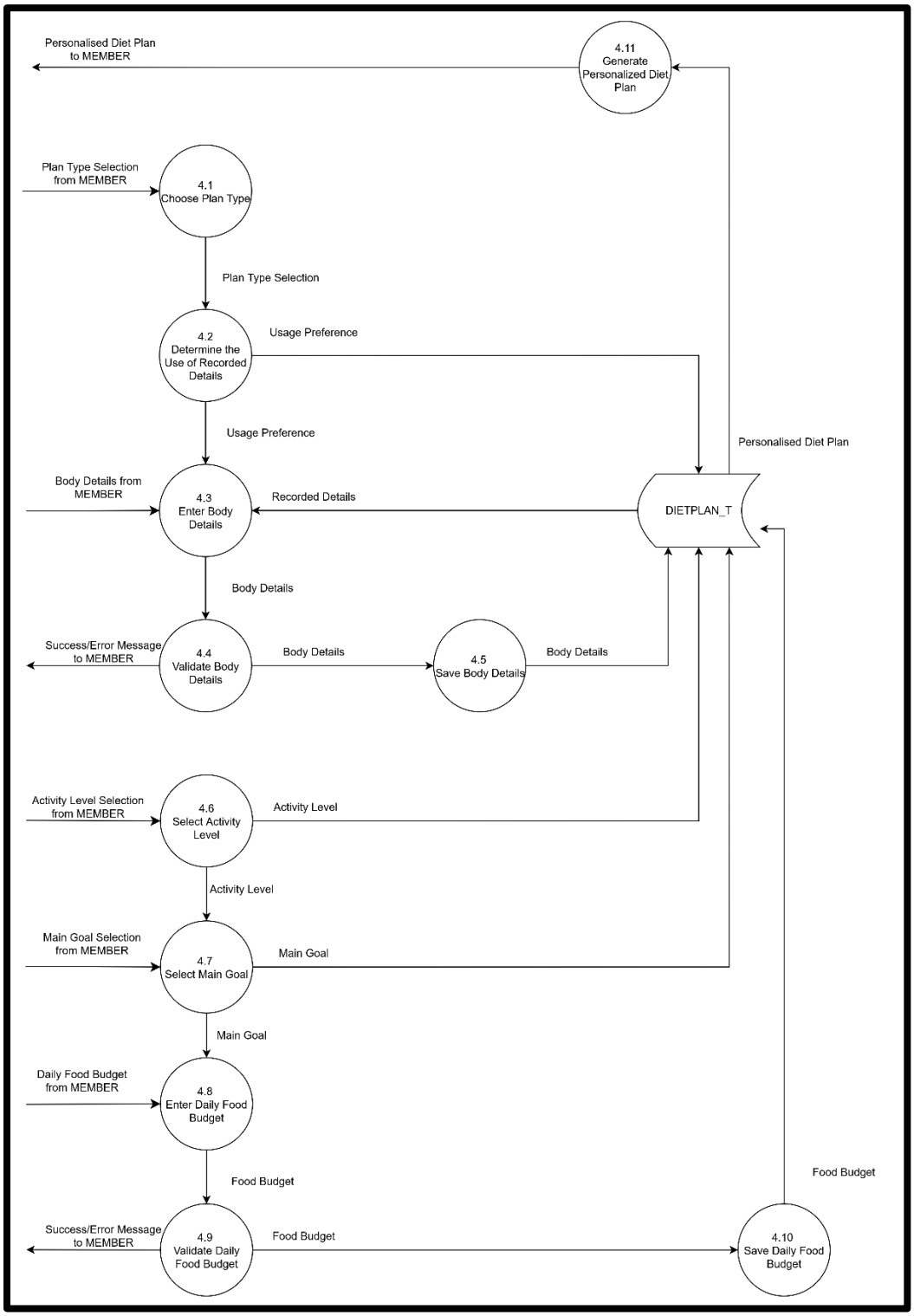


Figure 9: DFD Level 1 (Diet Plan)

7.4.5 DFD Level 1 – Workout (YEO PEI WEN – TP077057)

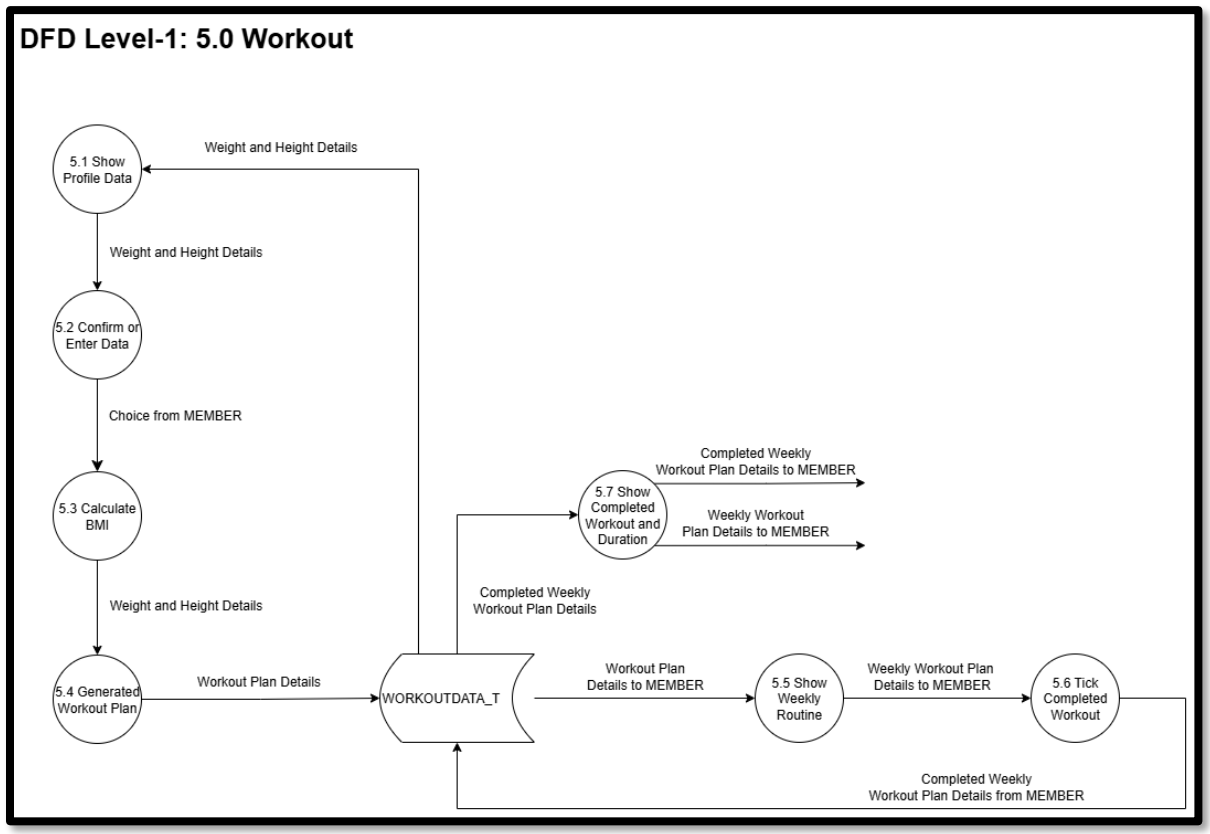


Figure 10: DFD Level-1 (Workout)

7.4.6 DFD Level 1 – Progress (SIM TIAN – TP077056)

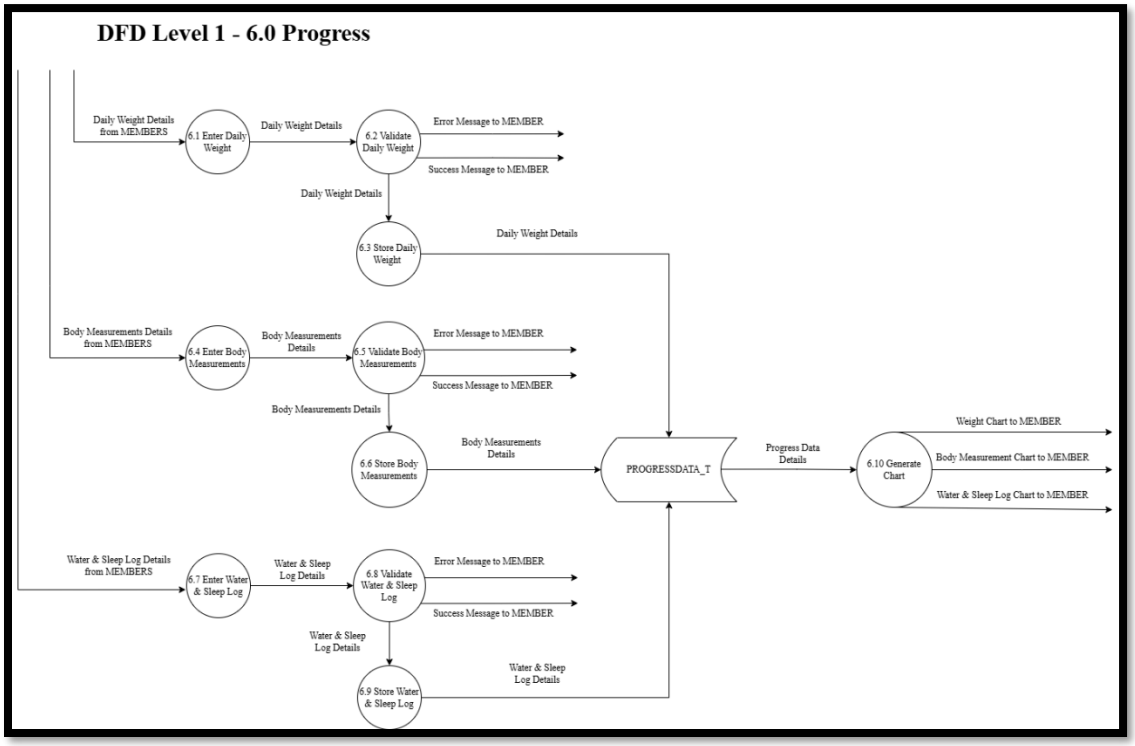


Figure 11: DFD Level 1 (Progress)

7.4.7 DFD Level 1 – Report (YONG XUAN LYN – TP076797)

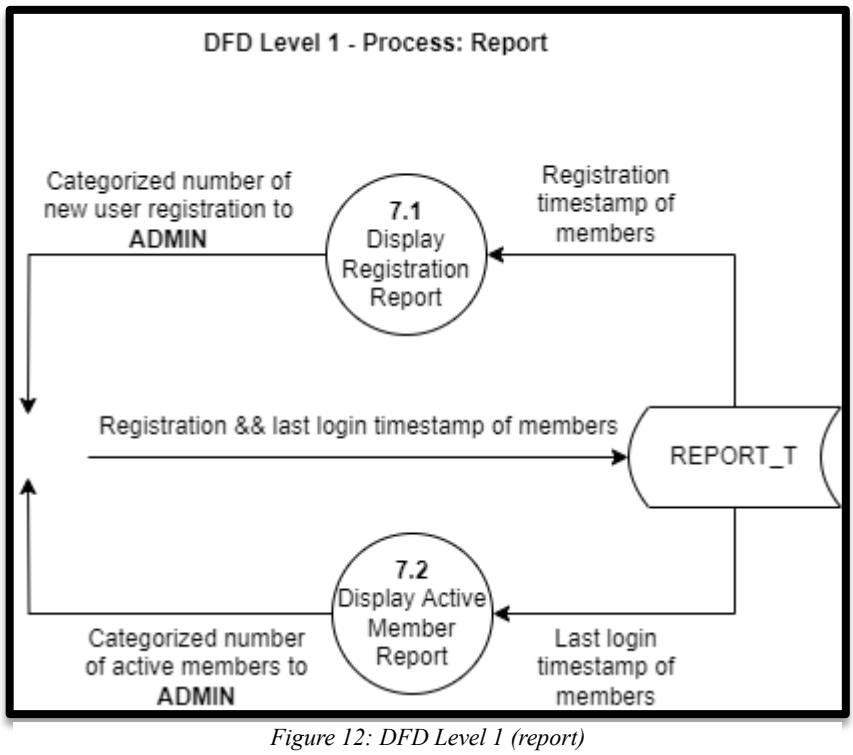


Figure 12: DFD Level 1 (report)

7.5 Program Flowcharts

7.5.1 Process 1 – Registration (YAP XIN LING – TP077224)

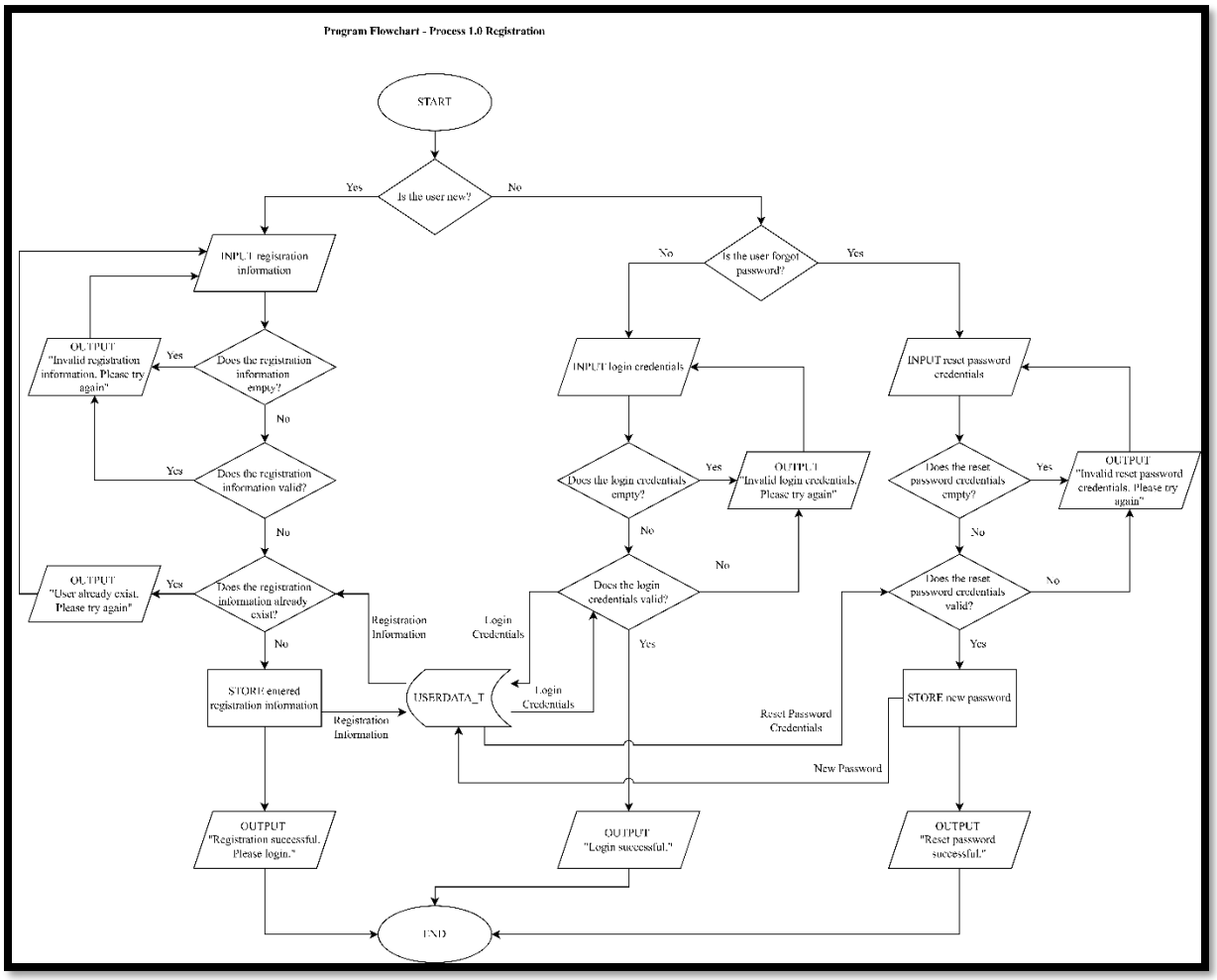


Figure 13: Process Flowchart (Registration Process)

7.5.2 Process 2 – Manage Profile (SERENE LOH ZI TING – TP075920)

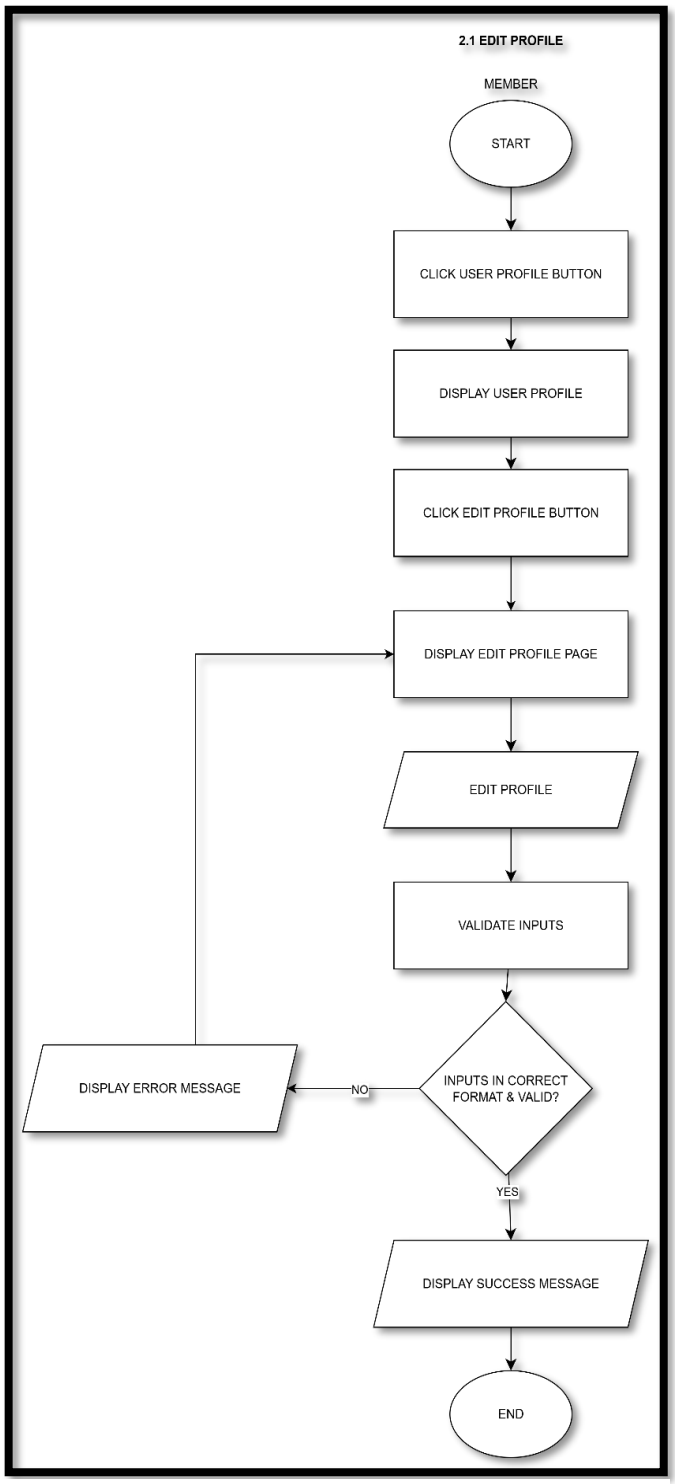


Figure 14: 2.1 Edit profile flowchart for member

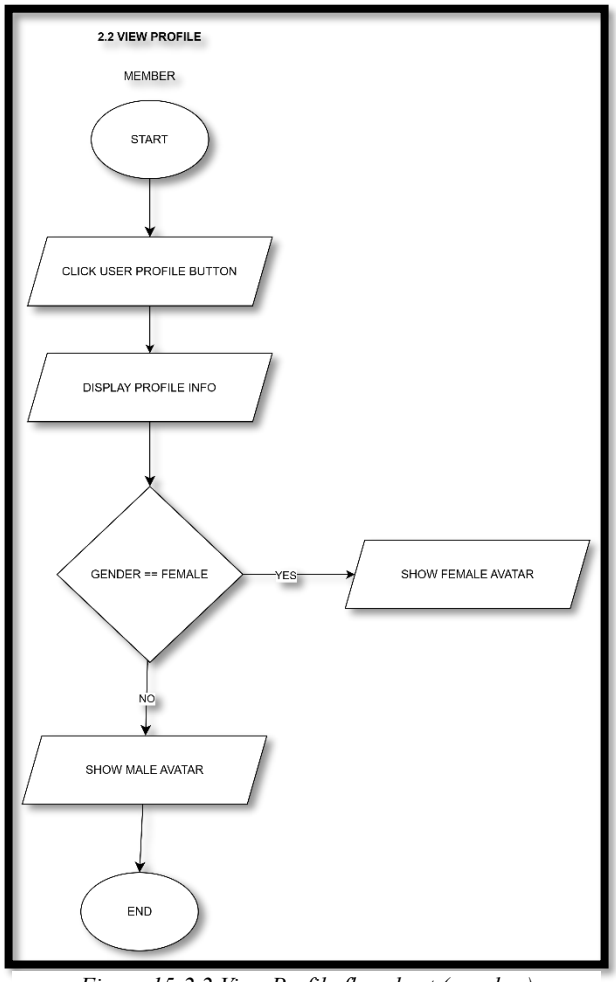


Figure 15:2.2 View Profile flowchart (member)

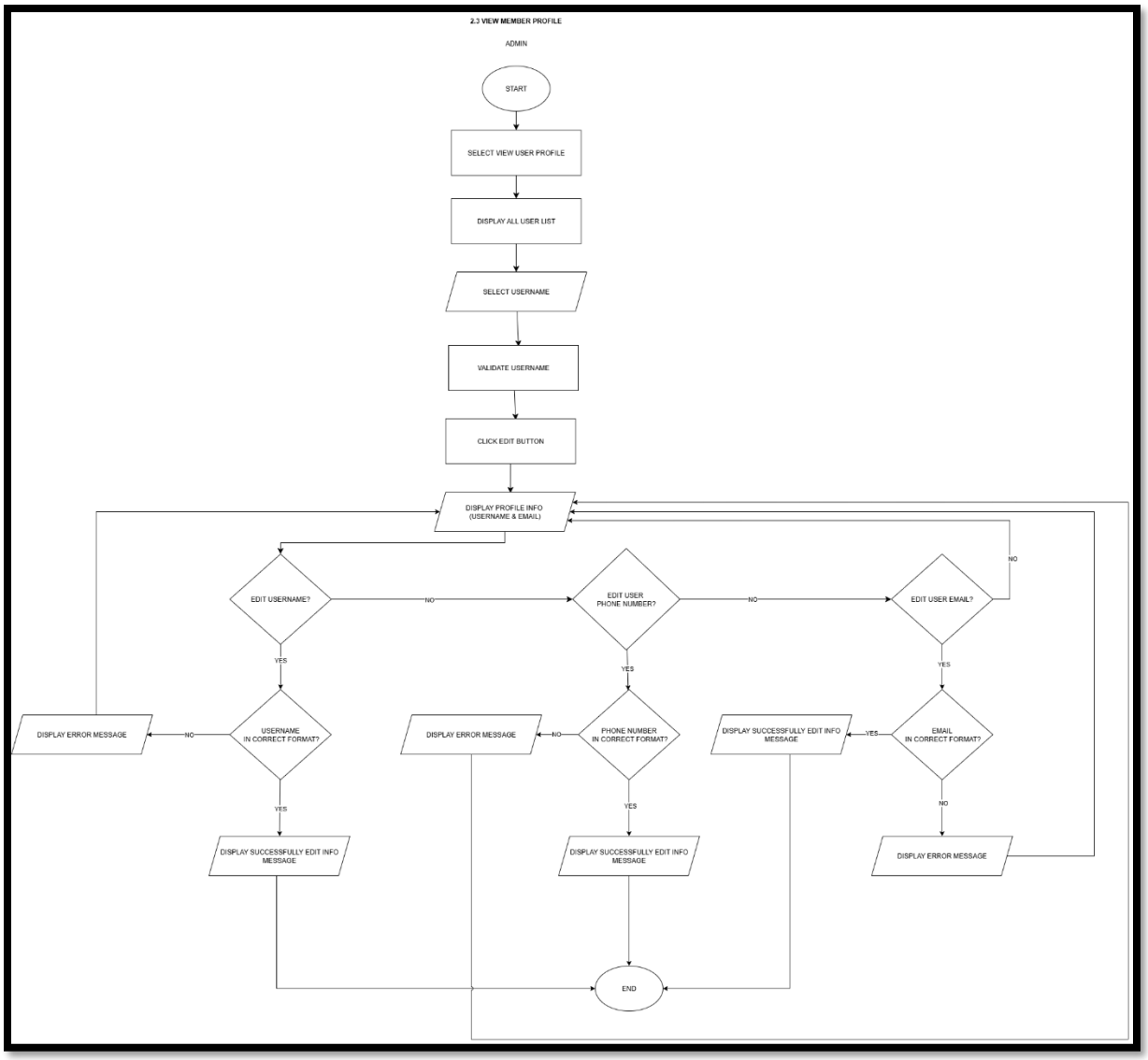


Figure 16: 2.3 View Member Profile Flowchart (Admin)

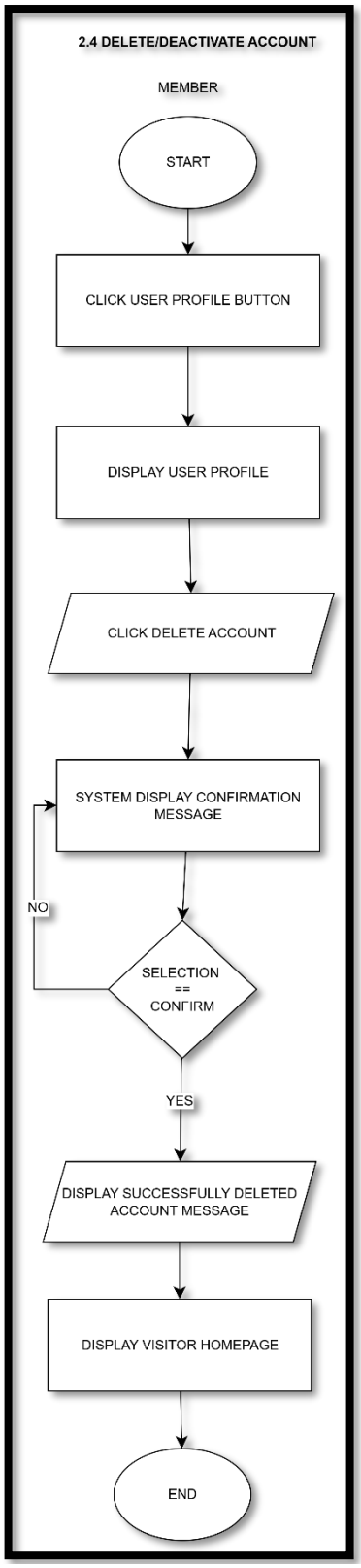


Figure 17: 2.4 Delete Account Flowchart for member

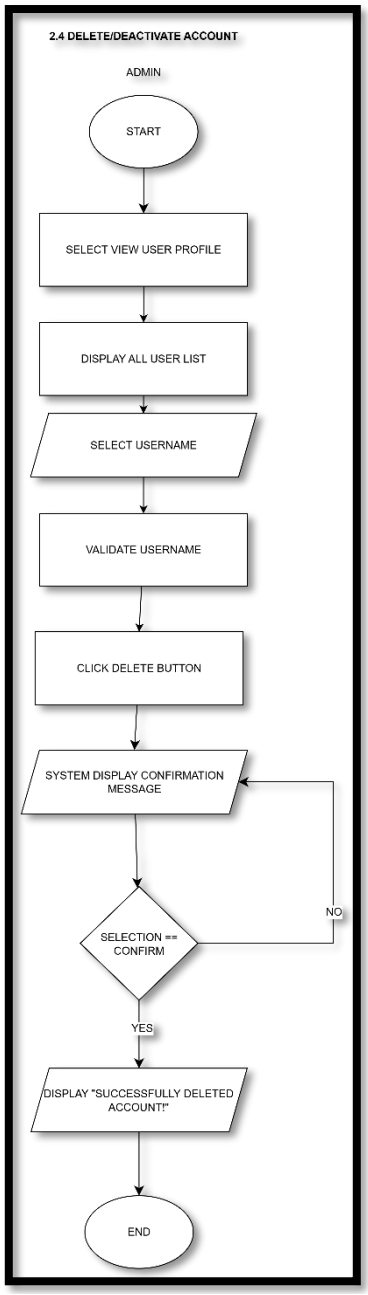


Figure 18: 2.4 Delete Member account flowchart for Admin

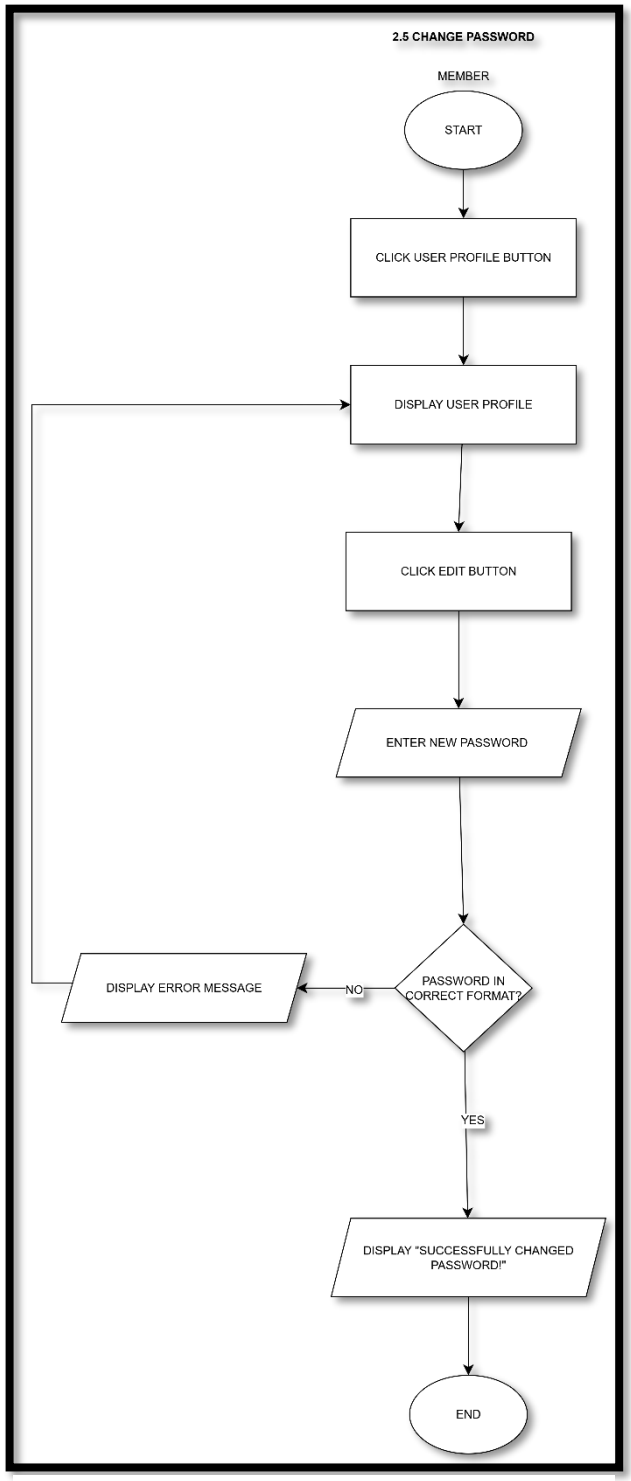
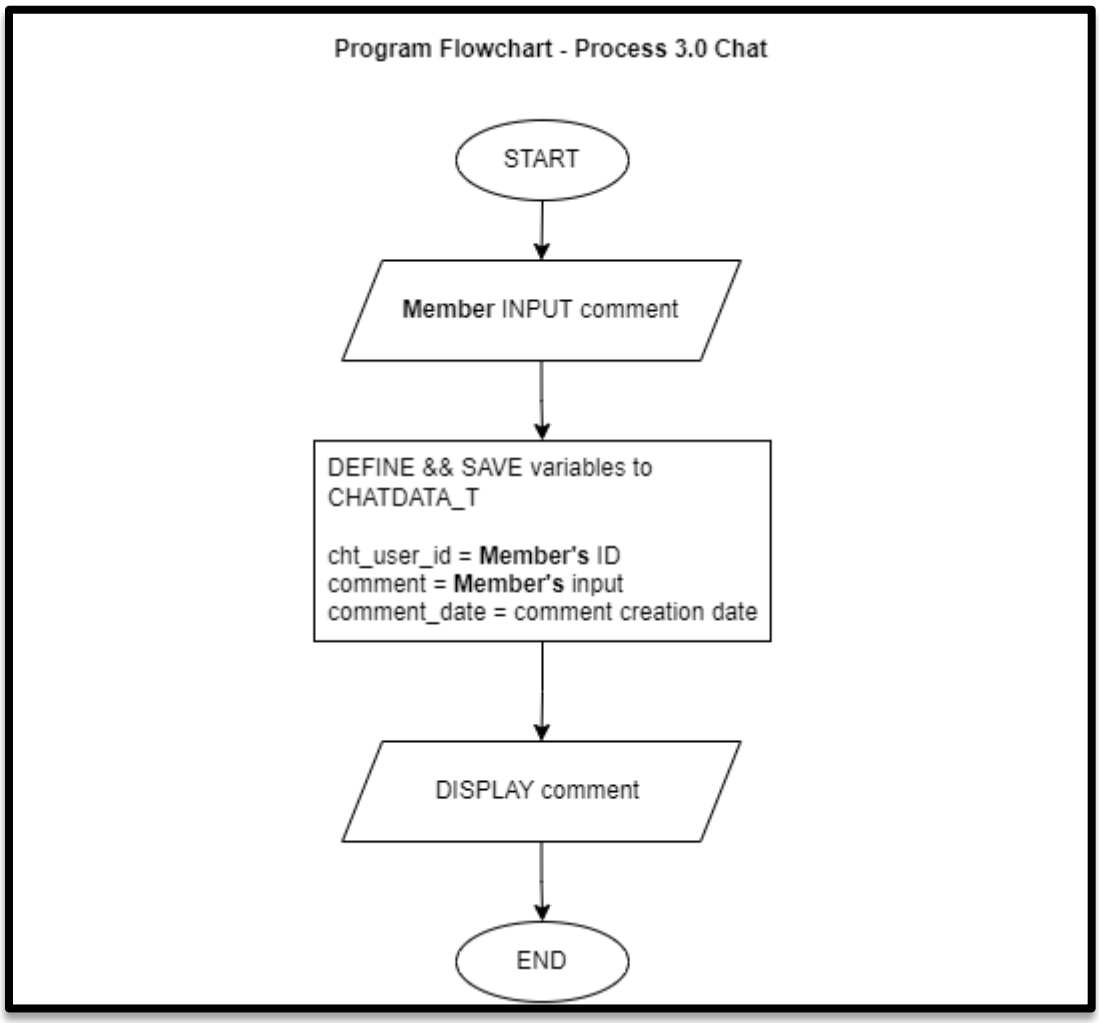


Figure 19: 2.5 Change password flowchart

7.5.3 Process 3 – Chat (YONG XUAN LYN – TP076797)



7.5.4 Process 4 – Diet Plan (SIEW ZHEN LYNN – TP076386)

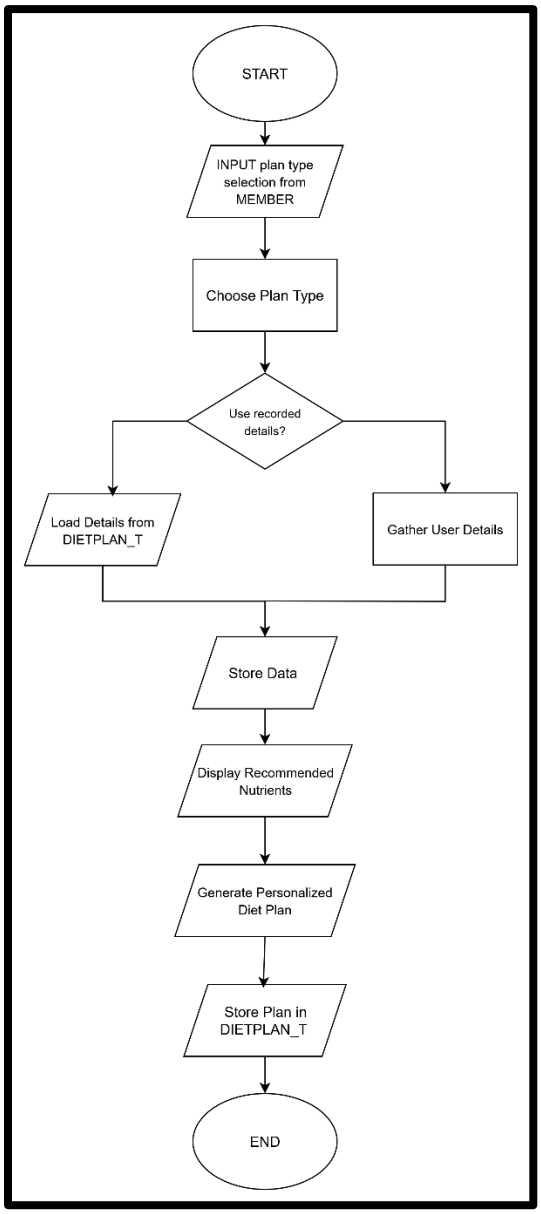


Figure 20: Personalized Diet Plan Flowchart

7.5.5 Process 5 – Workout (YEO PEI WEN - TP077057)

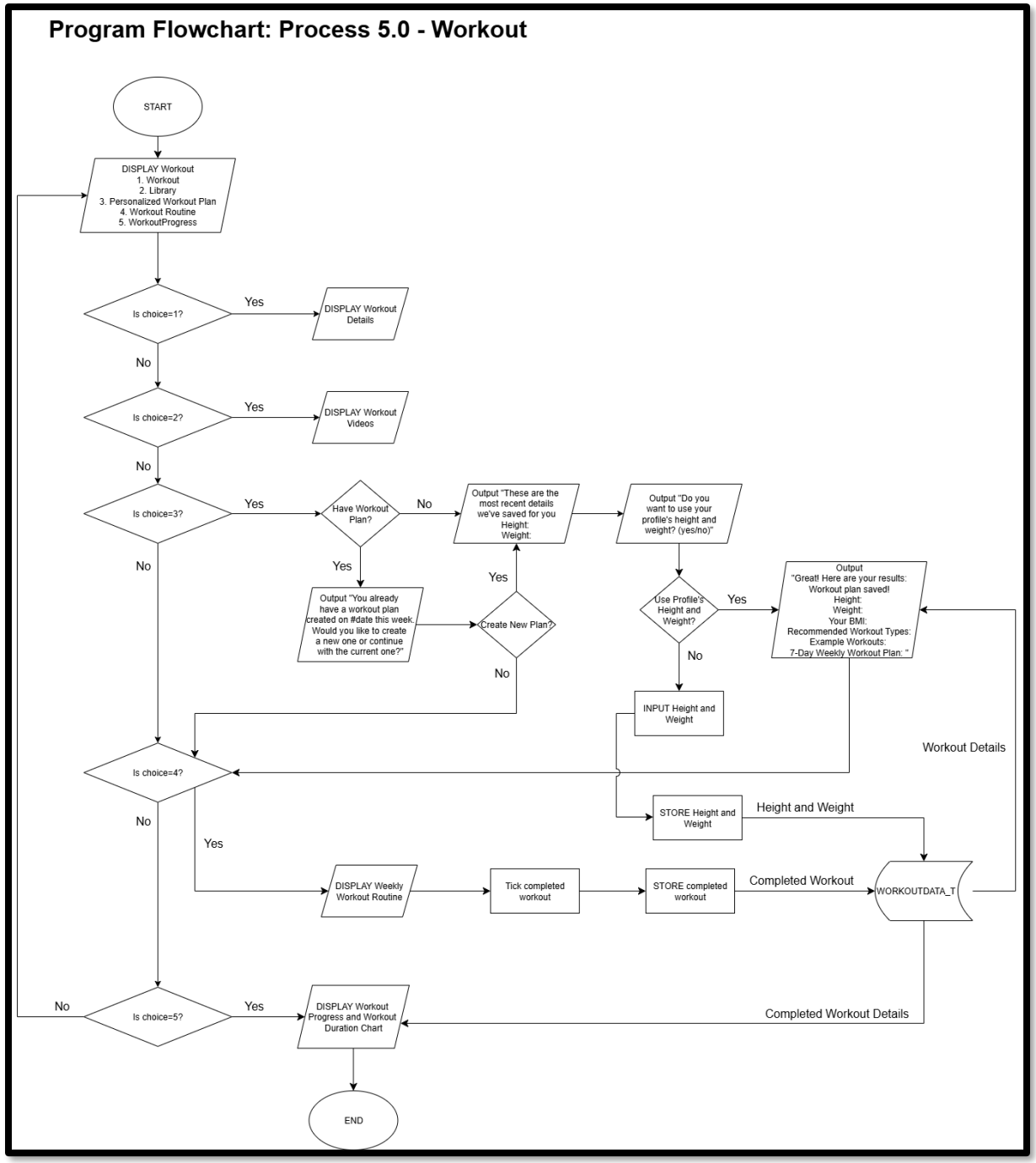


Figure 21: Program Flowchart: Process 5.0 (Workout)

7.5.6 Process 6 - Progress (SIM TIAN – TP077056)

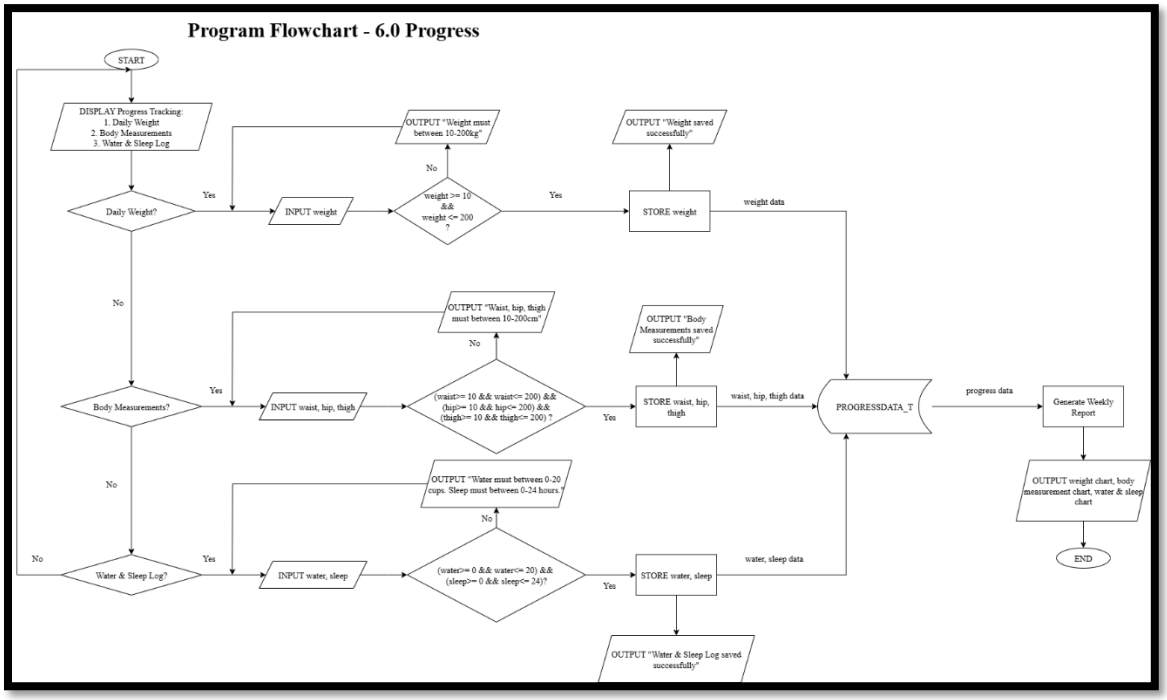
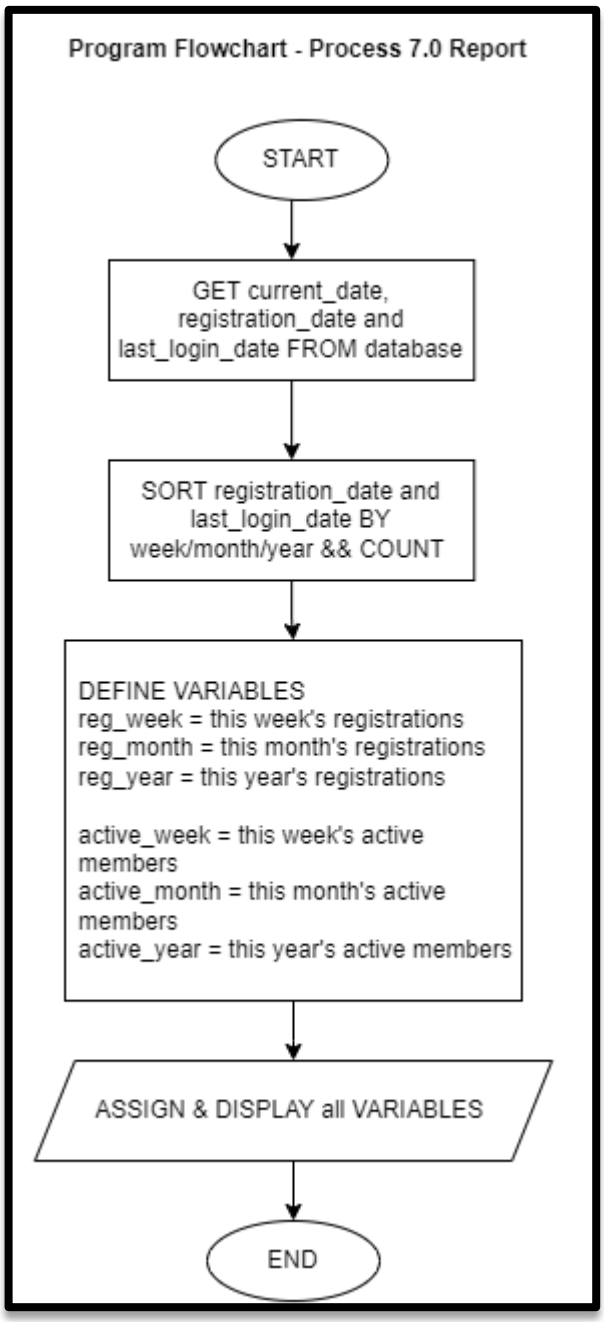


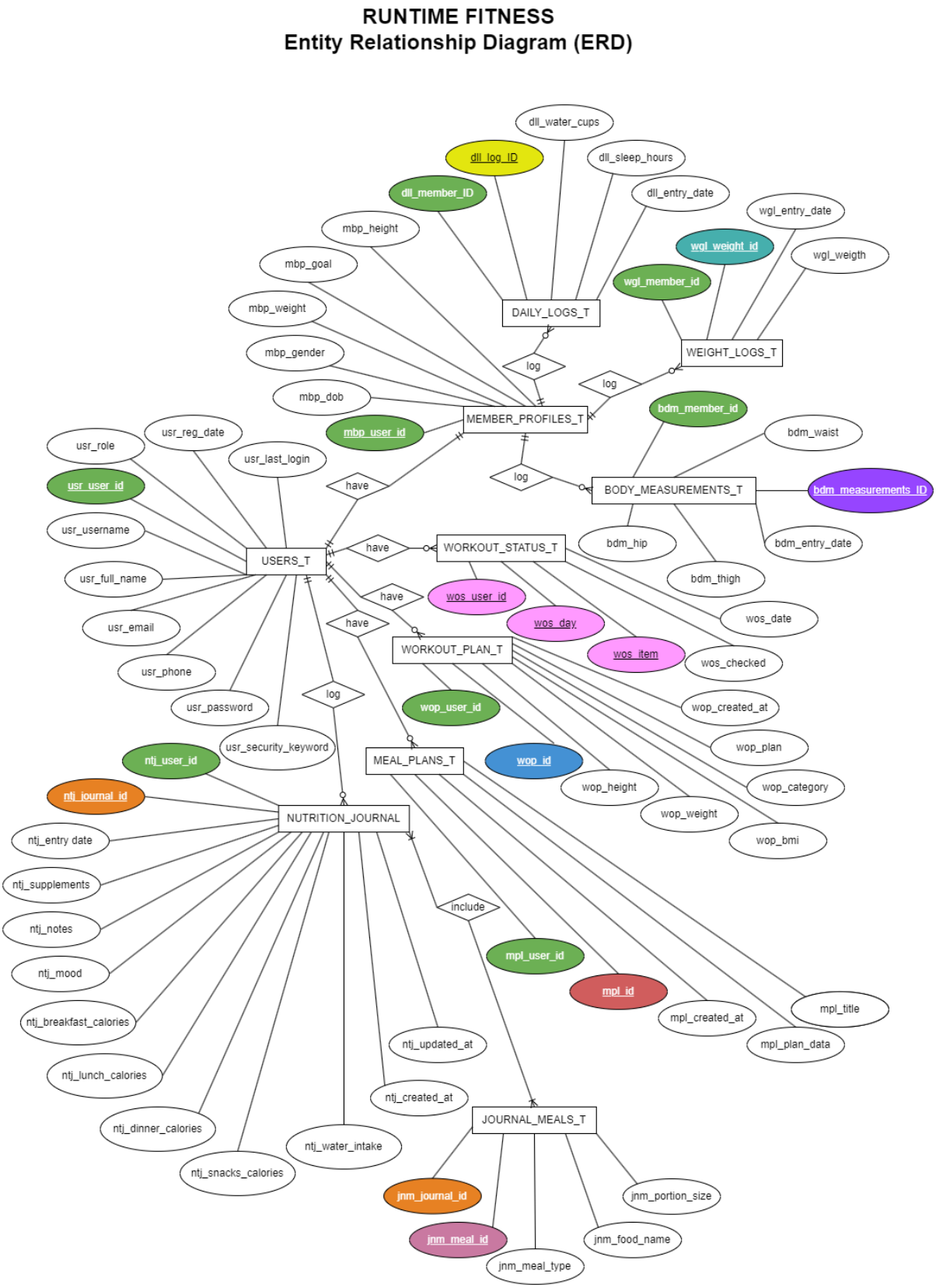
Figure 22: Program Flowchart (Progress Tracker)

7.5.7 Process 7 – Report (YONG XUAN LYN – TP076797)

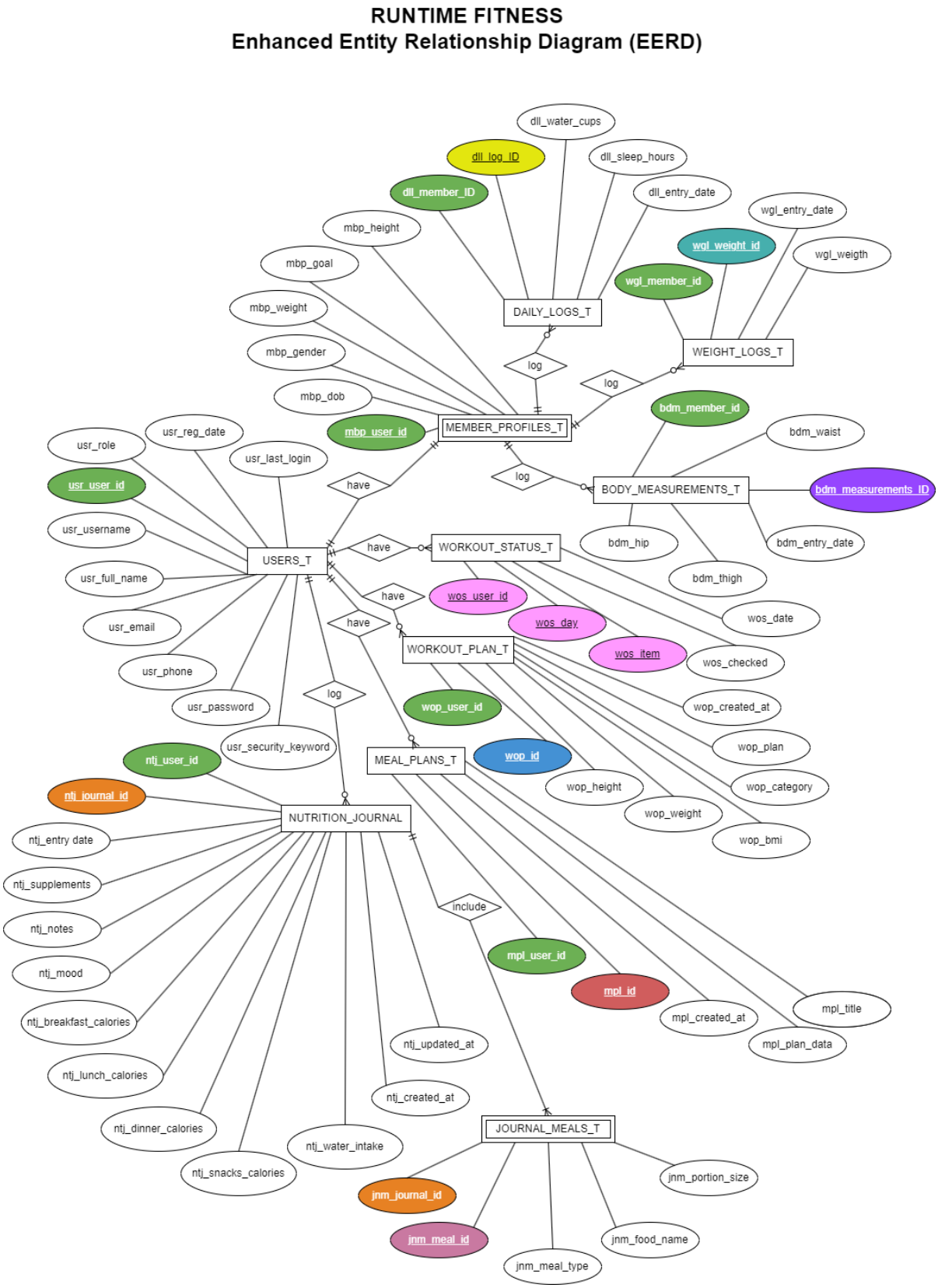


7.6 Database Design

7.6.1 Entity Relationship Diagram (ERD)



7.6.2 Enhanced Entity Relationship Diagram (EERD)



7.6.3 Database Table Structure

Name of Table	USERS_T			
Primary Key	usr_user_id			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
usr_user_id	Unique identification code for all users	int	11	PK, NOT NULL
usr_username	Unique username for users	varchar	50	DEFAULT NULL
usr_full_name	The full legal name of user	varchar	100	DEFAULT NULL
usr_email	User email	varchar	100	DEFAULT NULL
usr_phone	User phone number	varchar	15	DEFAULT NULL
usr_password	User password	varchar	255	DEFAULT NULL
usr_security_keyword	User security keyword for reset password	varchar	100	DEFAULT NULL
usr_role	User role type	enum('member','coach','nutritionist)		DEFAULT NULL
usr_reg_date	User registration date time	datetime		NOT NULL, DEFAULT current_timestamp()
usr_last_login	User last login date time	datetime		DEFAULT NULL
Name of Table	MEMBER_PROFILES_T			
Primary Key	mbp_user_ID			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
mbp_user_ID	Member ID, referencing usr_user_id	int	11	PK, FK, DEFAULT NULL
mbp_dob	Date of birth	date		DEFAULT NULL
mbp_gender	Gender	varchar	10	DEFAULT NULL
mbp_height	Height	float		DEFAULT NULL
mbp_weight	Weight	float		DEFAULT NULL
mbp_goal	Fitness Goal	varchar	255	DEFAULT NULL
Name of Table	NUTRITION_JOURNAL_T			
Primary Key	ntj_journal_id			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
ntj_journal_id	Unique identification code for all journal	int	11	PK, NOT NULL
ntj_user_id	Referencing usr_user_id	int	11	FK, NOT NULL
ntj_entry_date	The date where the record is dedicated to	date		NOT NULL
ntj_breakfast_calories	Total calories in breakfast	int	11	DEFAULT NULL
ntj_lunch_calories	Total calories in lunch	int	11	DEFAULT NULL
ntj_dinner_calories	Total calories in dinner	int	11	DEFAULT NULL
ntj_water_intake	Total water intake	varchar	20	DEFAULT NULL
ntj_supplements	Supplements taken	text		DEFAULT NULL
ntj_notes	Remarks of that particular journal	text		DEFAULT NULL
ntj_mood	Mood of the day	varchar	20	DEFAULT NULL
ntj_created_at	Date of creation of the record	datetime		DEFAULT current_timestamp()
ntj_updated_at	Date of update of the record	datetime		DEFAULT current_timestamp()
Name of Table	JOURNAL_MEALS_T			
Primary Key	jnm_meal_id			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
jnm_meal_id	Unique identification code for all meal	int	11	PK, NOT NULL
jnm_journal_id	Referencing ntj_journal_id	int	11	FK, NOT NULL
jnm_meal_type	Meal type	enum('breakfast','lunch','dinner','snack')		NOT NULL
jnm_food_name	Food name	varchar	100	NOT NULL
jnm_portion_size	Food portion	varchar	50	DEFAULT NULL
Name of Table	DAILY_LOGS_T			
Primary Key	dll_log_ID			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
dll_log_ID	Unique identification code for all daily logs	int	11	PK, NOT NULL
dll_member_ID	Referencing mbp_user_ID	int	11	FK, NOT NULL
dll_water_cups	Cups of water drank	int	11	DEFAULT NULL
dll_sleep_hours	Sleep hours	decimal	4, 2	DEFAULT NULL
dll_entry_date	Date of record entry	date		DEFAULT NULL
Name of Table	WEIGHT_LOGS_T			
Primary Key	wgl_weight_id			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
wgl_weight_id	Unique identification code for all weight logs	int	11	PK, NOT NULL
wgl_member_ID	Referencing mbp_user_ID	int	11	FK, NOT NULL
wgl_entry_date	Date of record entry	date		DEFAULT NULL
wgl_weight	Weight of user	decimal	5,2	DEFAULT NULL
Name of Table	BODY_MEASUREMENT_T			
Primary Key	bdm_measurement_ID			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
bdm_measurement_ID	Unique identification code for all measurement logs	int	11	PK, NOT NULL
bdm_member_ID	Referencing mbp_user_ID	int	11	FK, NOT NULL
bdm_waist	Waist measurement	decimal	5,2	DEFAULT NULL
bdm_hip	Hip measurement	decimal	5,2	DEFAULT NULL
bdm_thigh	Thigh measurement	decimal	5,2	DEFAULT NULL
bdm_entry_date	Date of record entry	date		DEFAULT NULL
Name of Table	MEAL_PLANS_T			
Primary Key	mpl_id			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
mpl_id	Unique identification code for all meal plans	int	11	PK, NOT NULL
mpl_user_id	Unique identification code for all users	int	11	NOT NULL
mpl_title	Title of meal plan	varchar	255	DEFAULT NULL
mpl_plan_data	Plan details	longtext		DEFAULT NULL
mpl_created_at	Date of creation of the plan	datetime		DEFAULT current_timestamp()
Name of Table	WORKOUT_STATUS_T			
Primary Key	wos_user_id, wos_day, wos_item			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
wos_user_id	Unique identification code for all users	varchar	50	PK, NOT NULL
wos_day	Day of the week	varchar	20	PK, NOT NULL
wos_item	Workout done	varchar	255	PK, NOT NULL
wos_checked	Workout status (1=done,0=undone)	tinyint	1	DEFAULT NULL
wos_date	Date when workout marcked as done	date		DEFAULT current_date()
Name of Table	WORKOUT_PLAN_T			
Primary Key	wop_id			
Attribute Name	**Description**	**Data Type**	**Size**	**PK / FK / NULL / NOT NULL**
wop_id	Unique identification code for all plans	int	11	PK, NOT NULL
wop_user_id	Unique identification code for all users	varchar	50	DEFAULT NULL
wop_height	User height	decimal	5,2	DEFAULT NULL
wop_weight	User weight	decimal	5,3	DEFAULT NULL
wop_bmi	User BMI	decimal	5,4	DEFAULT NULL
wop_category	Weight category	varchar	20	DEFAULT NULL
wop_plan	Workout plan	mediumtext		DEFAULT NULL
wop_created_at	Creation date of workout plan	datetime		DEFAULT current_timestamp()

7.7 Interface Design

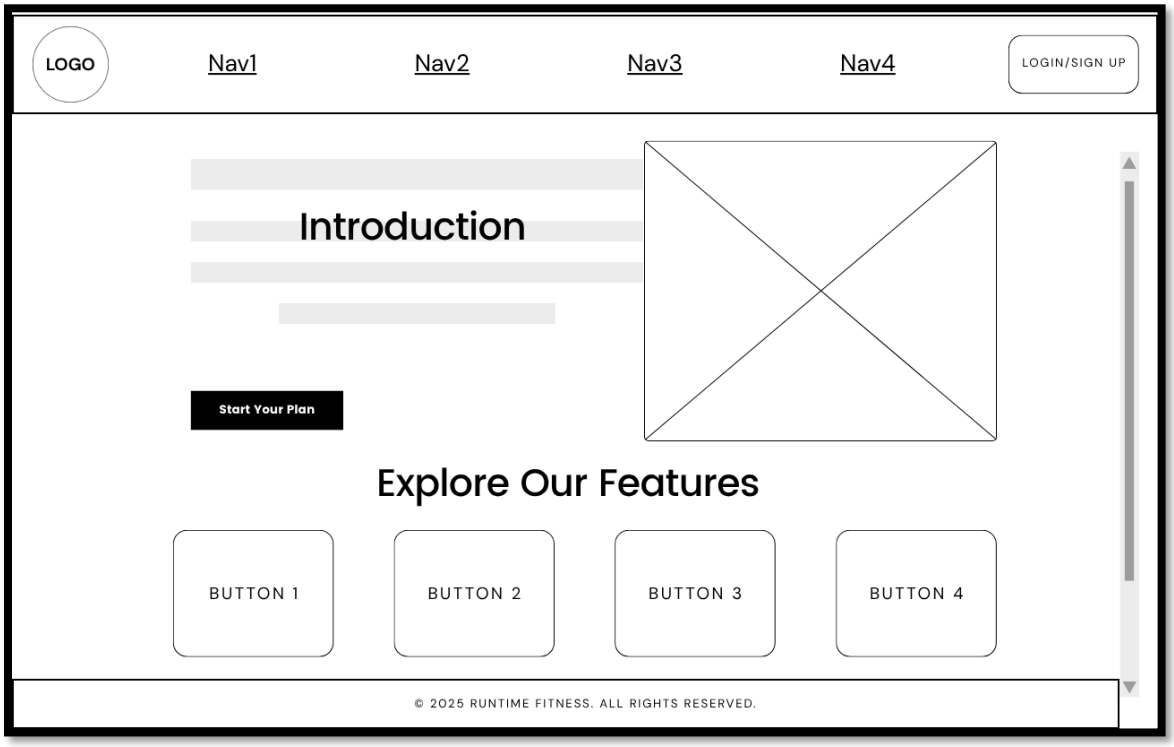


Figure 23: homepage

This is the homepage users see upon visiting the app. It features a clean navigation bar with a logo, four main sections (Nav1-Nav4), and a login/signup button. The introduction area includes a title, a brief description, and a call-to-action button labelled “Start Your Plan”. On the lower section, four neatly arranged buttons showcase key features of the app. The layout balances visuals and text to guide users smoothly into exploring the platform.

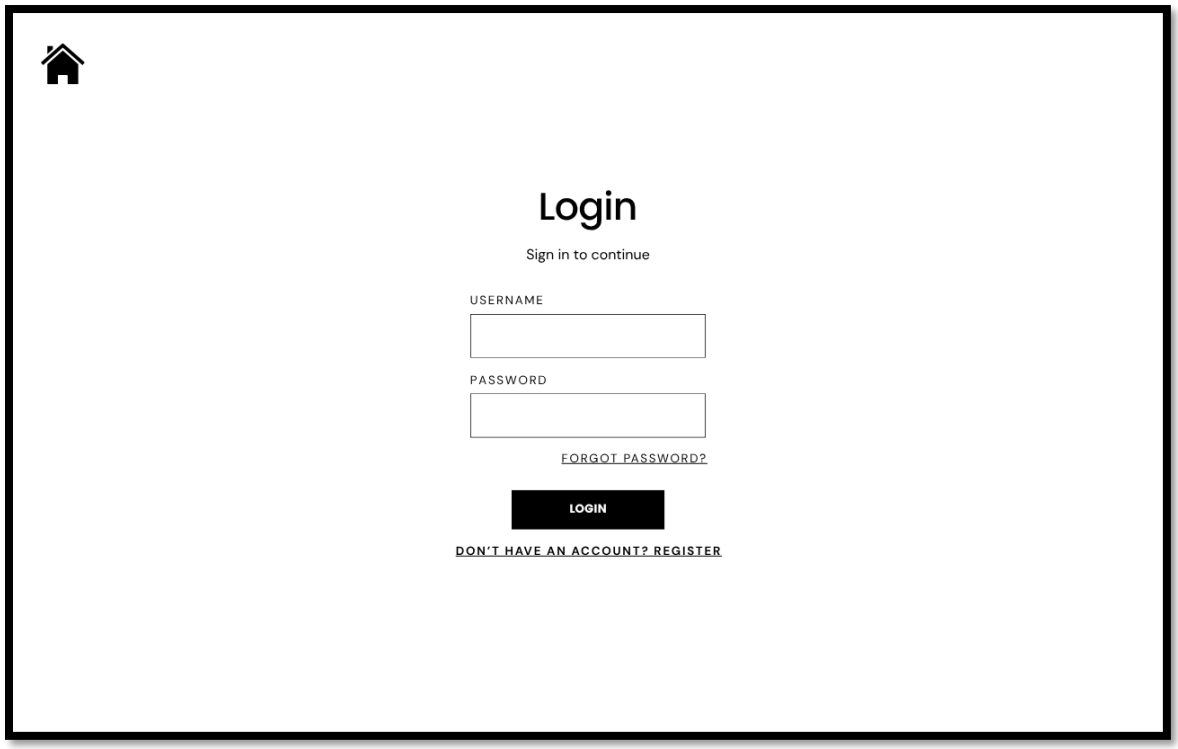


Figure 24: Log In Page

This is the login page that users are directed to after they click the "LogIn/Signup" button on the homepage. It features a home button on the top left of the screen that allows users to redirect to the homepage. The "Login" title is centred on the screen to clearly indicate the purpose of the page. To access the system, users must enter their username and password to log into the system. A "Forgot Password" link is provided to allow user to reset their password. Below the login form, a "Don't have an account? Register" link enables new users to manually register themselves.



Figure 25: Registration Page

This is the login page that users are directed to after they click the “Don’t have an account? Register” button on the Log In page. It features a home button on the top left of the screen that allows users to redirect to the homepage. The “Registration” title is centred on the screen to clearly indicate the purpose of the page. To create an account, users must enter all the fields, including username, full name, email, phone, password, confirm password, security keyword, date of birth, gender, height, weight and fitness goal. Once all fields are filled, users can click the “Done” button to submit the registration form. An “Already have an account? Login” link is provided to allow users to go back to the Log In page if they wish to.

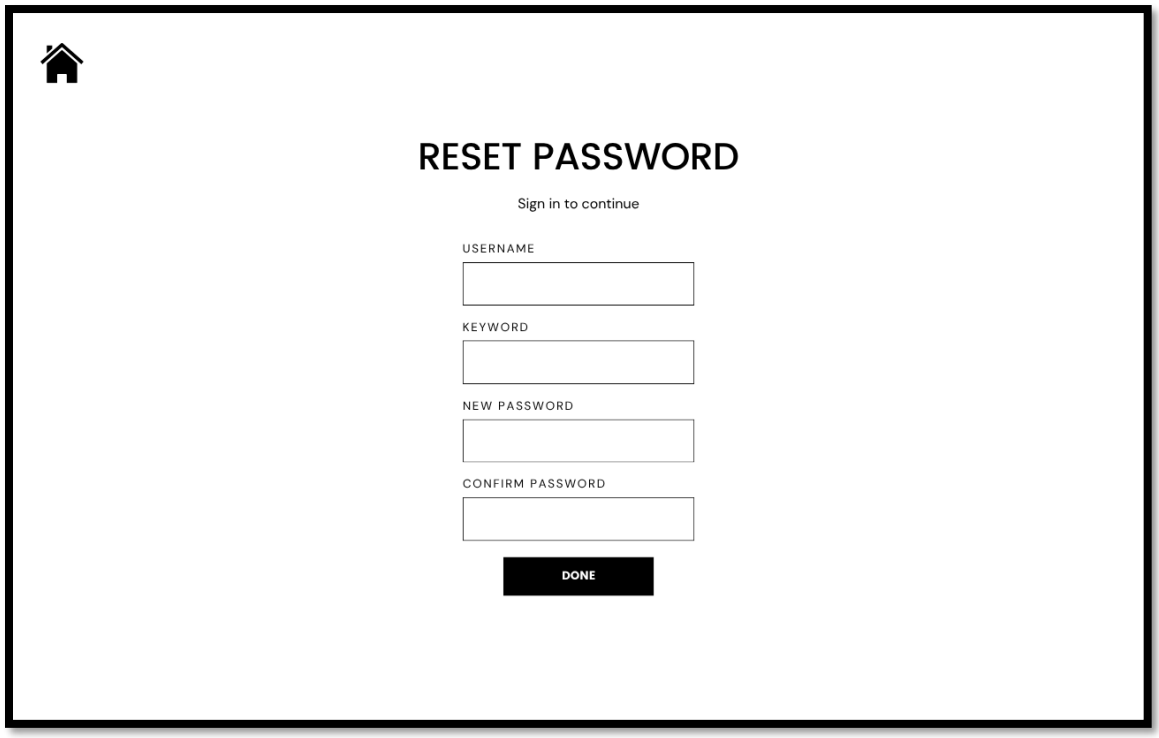


Figure 26: Reset Password Page

This is the reset password page that users are directed to after they click the "Forgot Password" Link on the Log In page. It features a home button on the top left of the screen that allows users to redirect to the homepage. "Reset Password" title is centred on the screen to clearly indicate the purpose of the page. To reset the password, users must enter all the fields including username, keyword, new password and confirm password. A "Done" button is provided below the reset password form, which allows user to submit their form after they fill in all the fields.

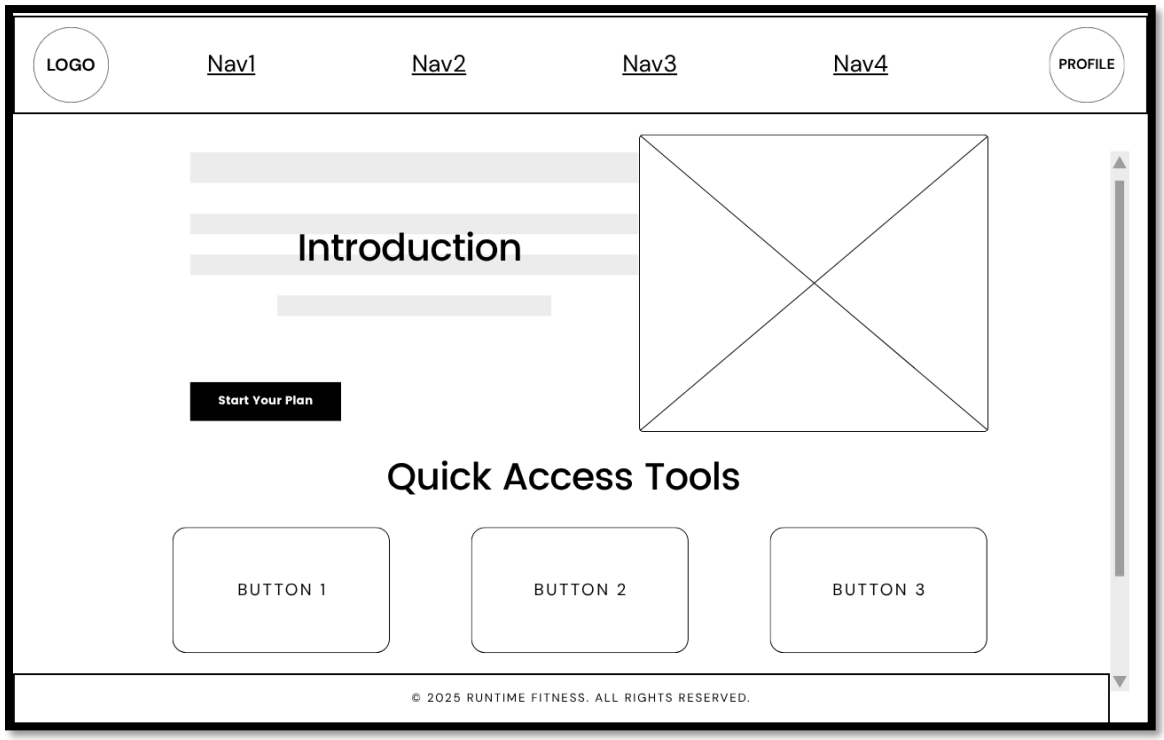


Figure 27: Member Homepage

This is the Member Homepage that users are directed to after they are successfully logged into the system. At the top, it features a navigation bar that has a logo, navigation links and a profile button. The centre highlights an “Introduction” section with text an image place holder, and a “Start Your Plan” call-to-action button. Below that, a “Quick Access Tools” section displays three buttons that toggle contents, providing brief overviews of the diet plan, work out plan, and key system features. The page ends with a copyright notice in the footer.

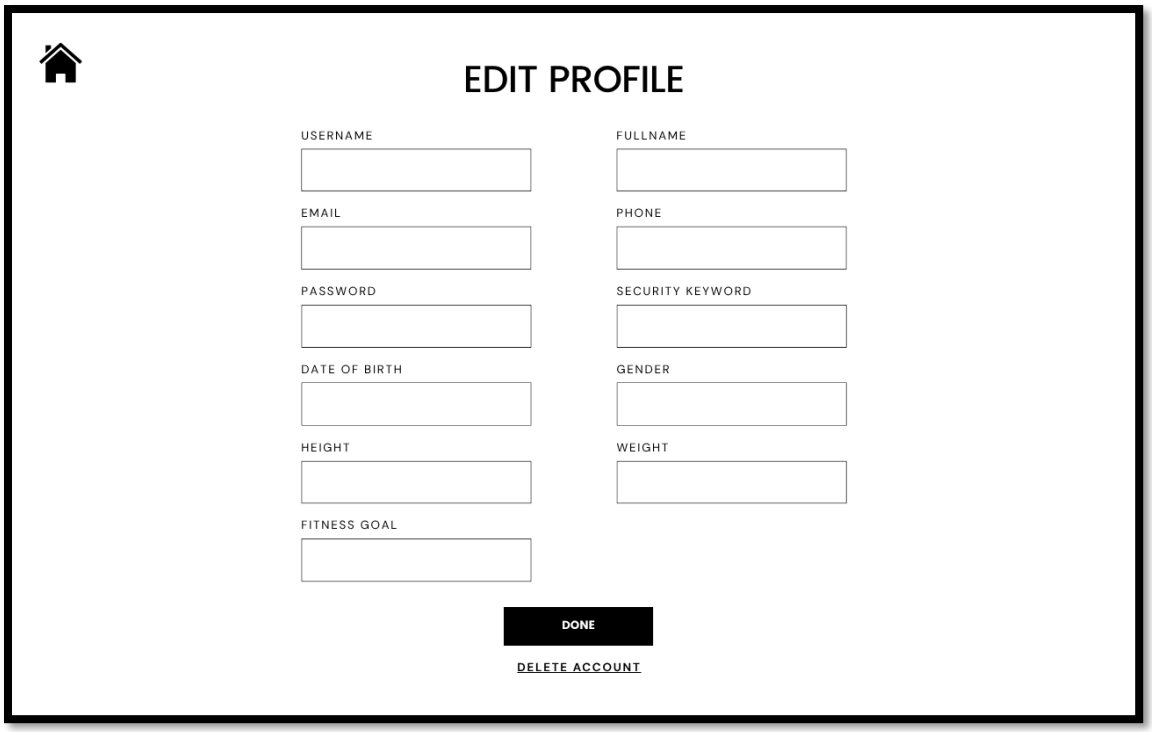


Figure 28: Edit Profile Page

This is the Edit Profile page that users are directed to after they clicked the Profile icon on Member Homepage. It features a house button on the top left of the screen that allows user redirecting to Member Homepage. “Edit Profile” title is centred on the screen to clearly indicate the purpose of the page. All existing user information is displayed first, and users may choose to edit fields which they wish to edit. A “Done” button is provided below the edit profile form which allows user to submit their form after they fill in all the fields.

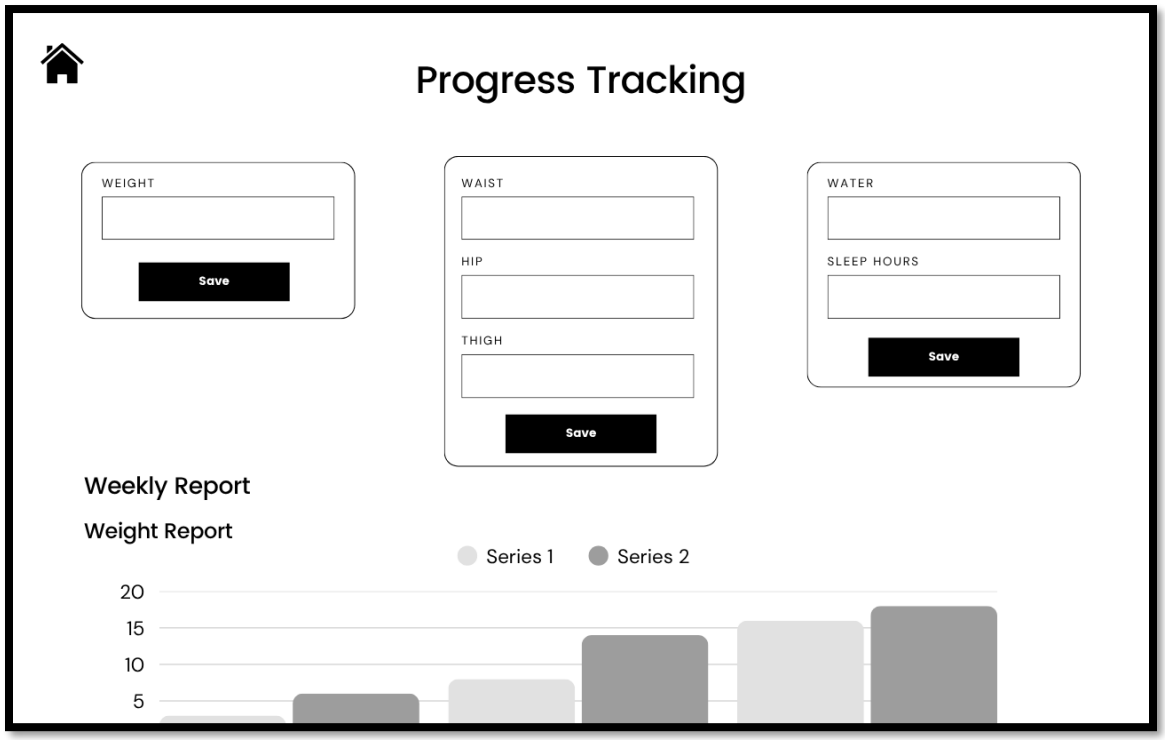


Figure 29: Progress Tracking - Part 1

This is the Progress Tracking page that users are directed to after they clicked the Progress Tracking link on the navigation bar of the Member Homepage. It features a house button on the top left of the screen that allows user redirecting to Member Homepage. “Progress Tracking” title is displayed on the top middle of the page to show the purpose of the page. Below the title, there are three input boxes, each for different types of data: weight, body measurements (waist, hip, thigh) and water & sleep log. At the bottom of the page, users can view a weekly report presented in chart format, displaying trends for their weight, body measurements, and hydration & sleep hours.



Figure 30: Progress Tracking - Part 2

This is the Progress Tracking page that users are directed to after they clicked the Progress Tracking link on the navigation bar of the Member Homepage. This figure is showing the chart when users continue to scroll down from the previous figure. At the bottom of the page, users can view a weekly report presented in chart format, displaying trends for their weight, body measurements, and hydration & sleep hours.

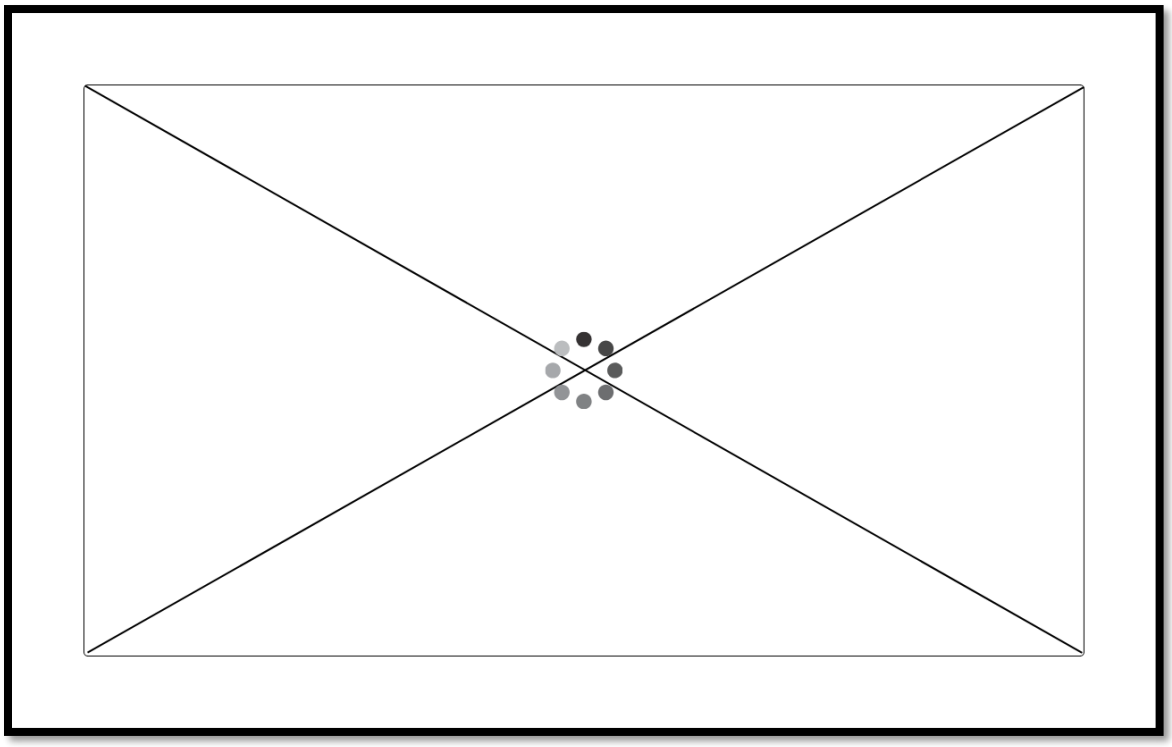


Figure 31: DP.Loading

This is the DP.Loading page that users are directed to after clicking the Diet Plan navigation link on the Member Homepage. A loading icon is displayed alongside a short video to engage the user while the system prepares to redirect them to the Diet Plan page. This loading page serves to make the redirection process more interactive and visually engaging, helping to reduce user impatience and avoid a static or dull waiting experience.

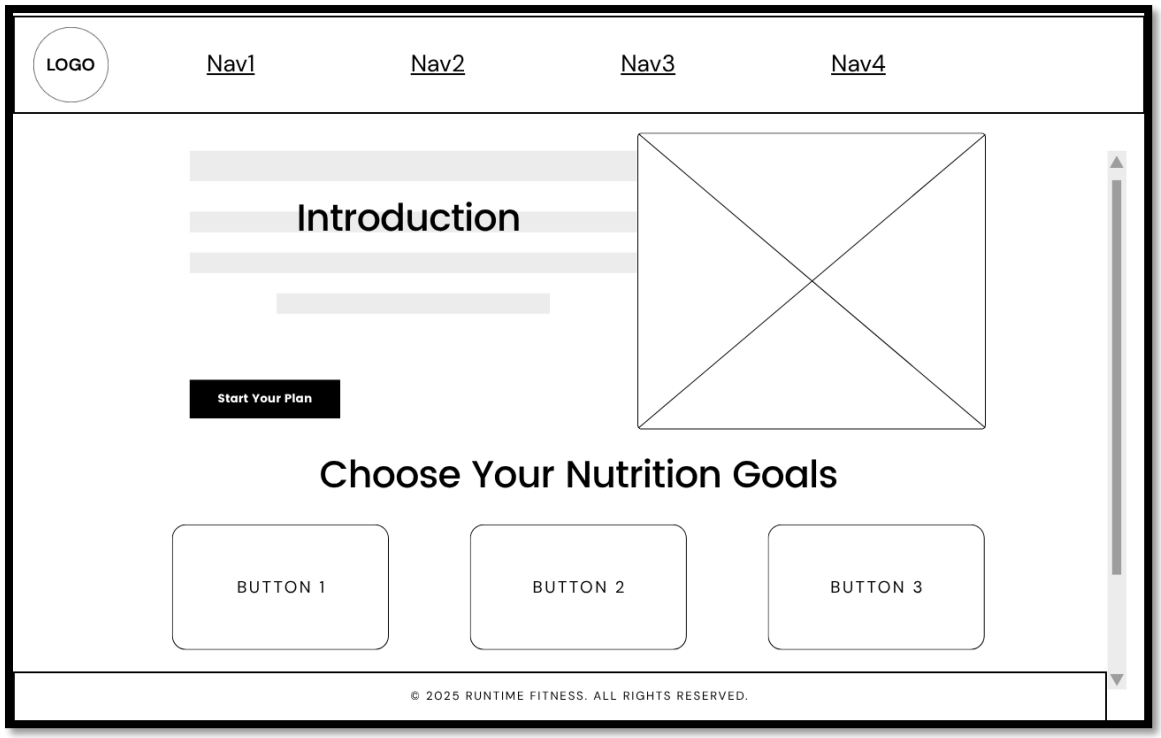


Figure 32: DP.Main

This is the DP.Main page, which acts as the homepage of the Diet Plan section. At the top, it features a navigation bar that has a logo, navigation links and a profile button. The centre highlights an “Introduction” section with text and an image placeholder, and a “Start Your Plan” call-to-action button. Below the introduction, a “Quick Access Tools” section displays three buttons that toggle contents, providing brief overviews of the diet plan and highlighting key features available in this module. This page has a copyright notice in the footer.

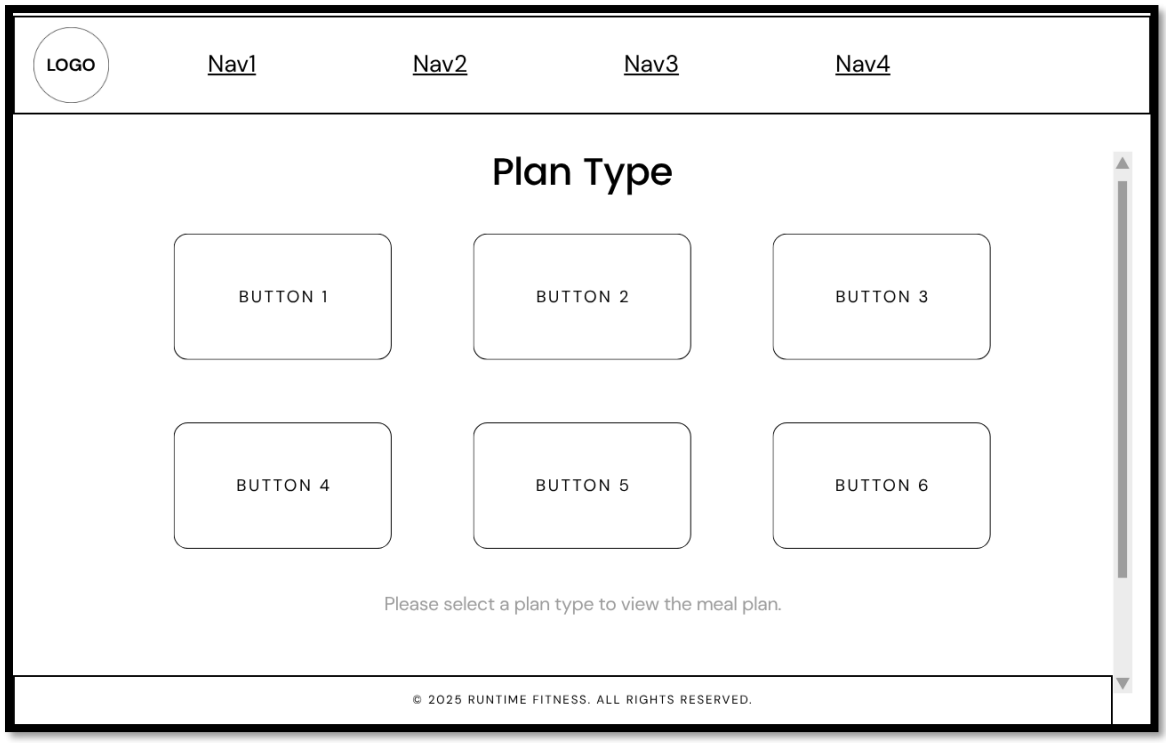


Figure 33: DP.DF-Part 1

This is the DP.DF page, which shows some default meal plans to the user. At the top, it features a navigation bar that has a logo, navigation links and a profile button. The screen has six buttons that represent different types of meal plans. Users may click on any button to view the corresponding meal plan in detail. Below the buttons, a short instruction “Please select a plan type to view the meal plan” is provided to guide users on how to interact with the page. This page has a copyright notice in the footer.

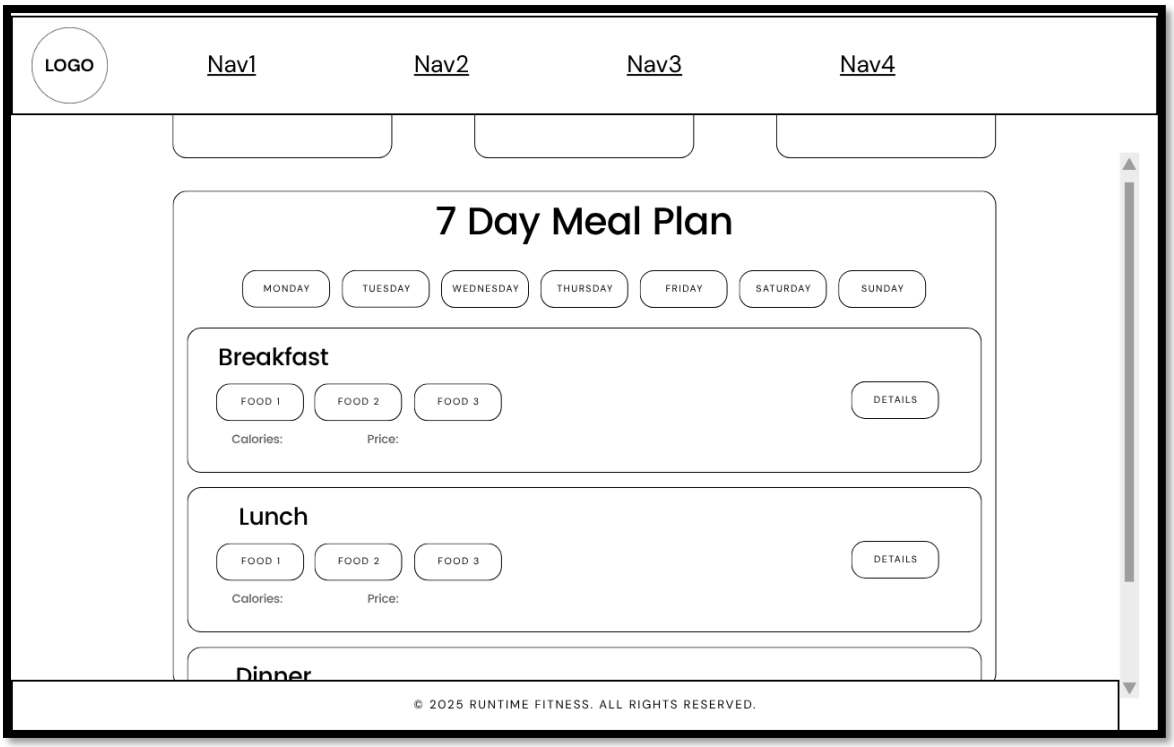


Figure 34: DP.DF-Part 2

This is the DP.DF page which shows some default meal plans to the user when they click the button to select a plan type. At the top, it features a navigation bar that has a logo, navigation links and a profile button. Below the section containing six plan type buttons, the selected meal plan is displayed. Each plan is structured as a seven-day schedule that includes details for breakfast, lunch, dinner, and snacks, along with information such as included foods, meal details, calorie count, price per meal, total daily calories, and estimated daily cost. This page has a copyright notice in the footer.

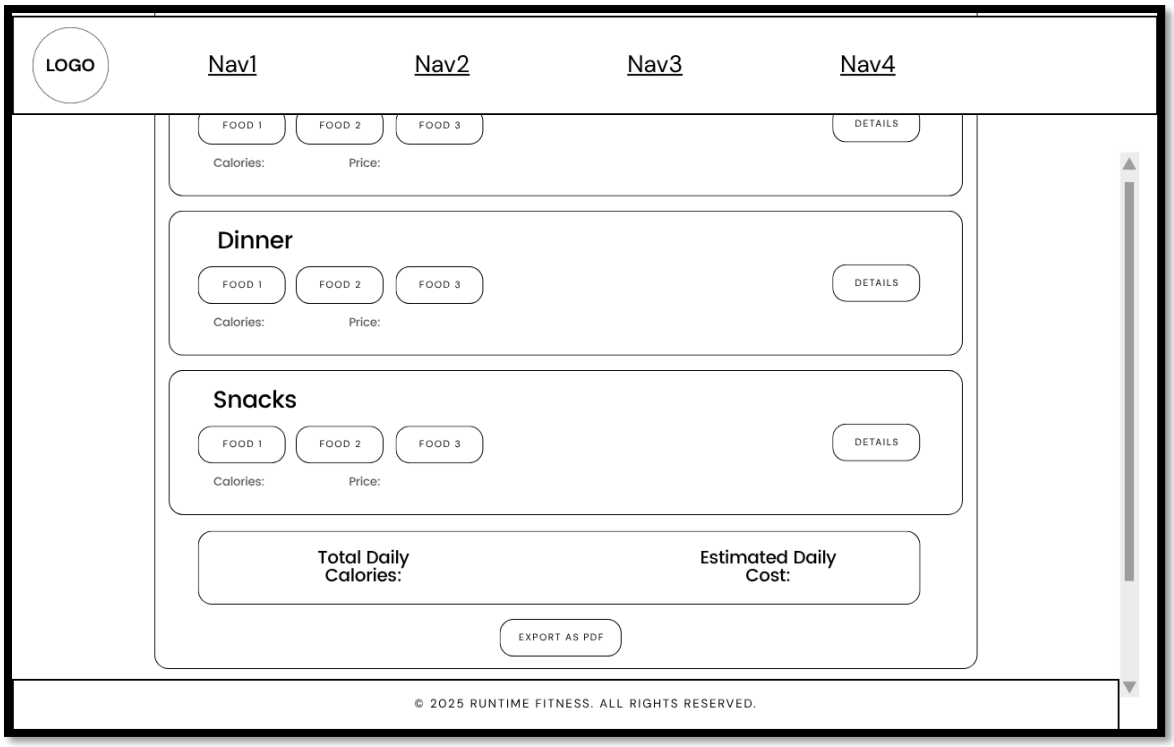


Figure 35: DP.DF-Part 3

This is the DP.DF page which shows some default meal plans to the user when they click the button to select a plan type. It is continued from the previous figure. At the top, it features a navigation bar that has a logo and navigation links. Below the section containing six plan type buttons, the selected meal plan is displayed. Each plan is structured as a seven-day schedule that includes details for breakfast, lunch, dinner, and snacks, along with information such as included foods, meal details, calorie count, price per meal, total daily calories, and estimated daily cost. This page has a copyright notice in the footer.

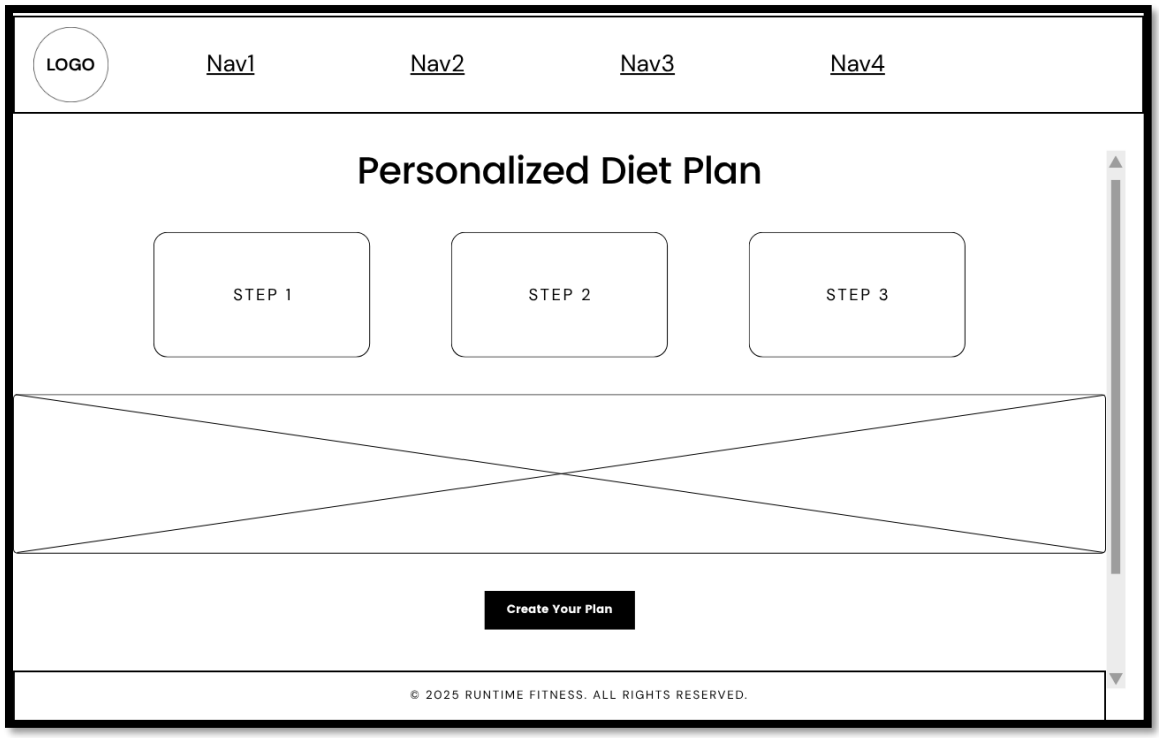


Figure 36: DP.PS.Personalized Diet Plan

This is the DP.PS.Personalized Diet Plan page, which guides users through creating a personalised meal plan. At the top, it features a navigation bar that has a logo and navigation links. A “Personalised Diet Plan” title is displayed to show the purpose of the page. Below the title, there are three boxes that toggle contents, showing the three steps to create a personalised diet plan. Below the boxes, it will display an embedded video that makes the experience more interactive and engaging. A “Create Your Plan” button will redirect the user to begin creating their personalised diet plan. This page has a copyright notice in the footer.

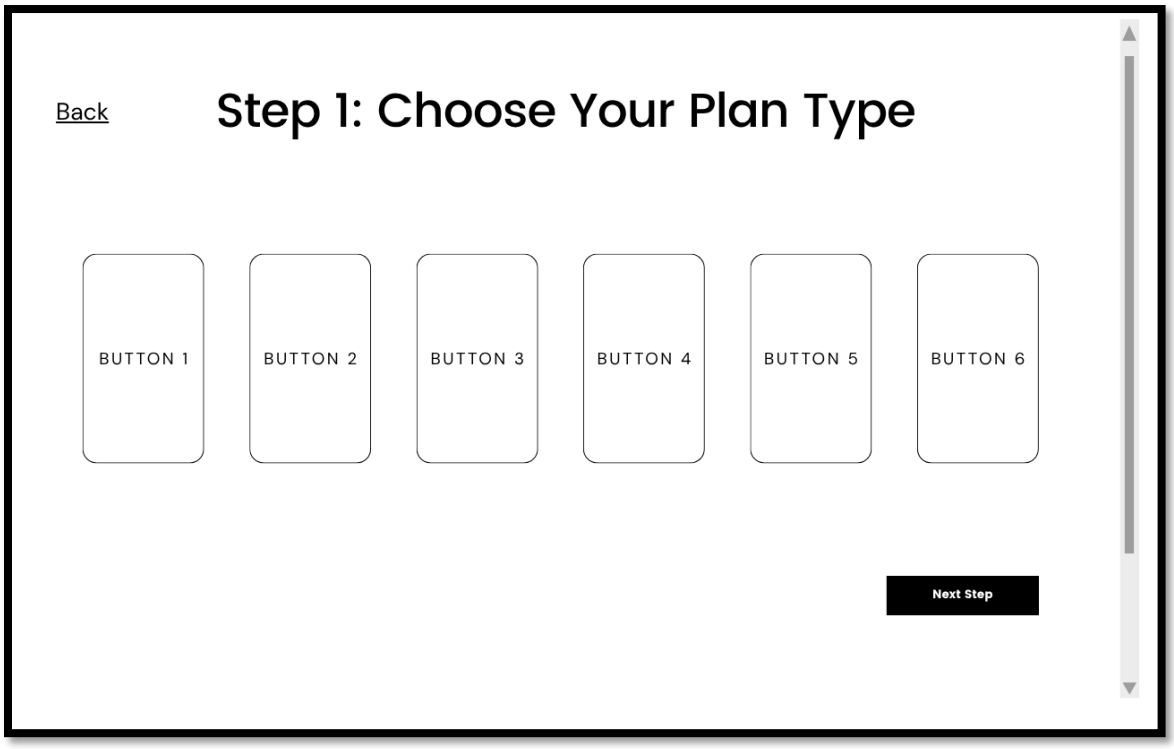


Figure 37: DP.PS.Step1

This is the DP.PS.Step1 page where users begin the process of creating their personalised diet plan. At the top left, it features a back button that allows users to return to the previous page. “Step 1: Choose Your Plan Type” is displayed to show the purpose of the page. At the centre of the page, there are six buttons that represent different meal types. Users may select their preferred meal type and click the “Next Step” button to proceed to the next stage of the plan creation process.

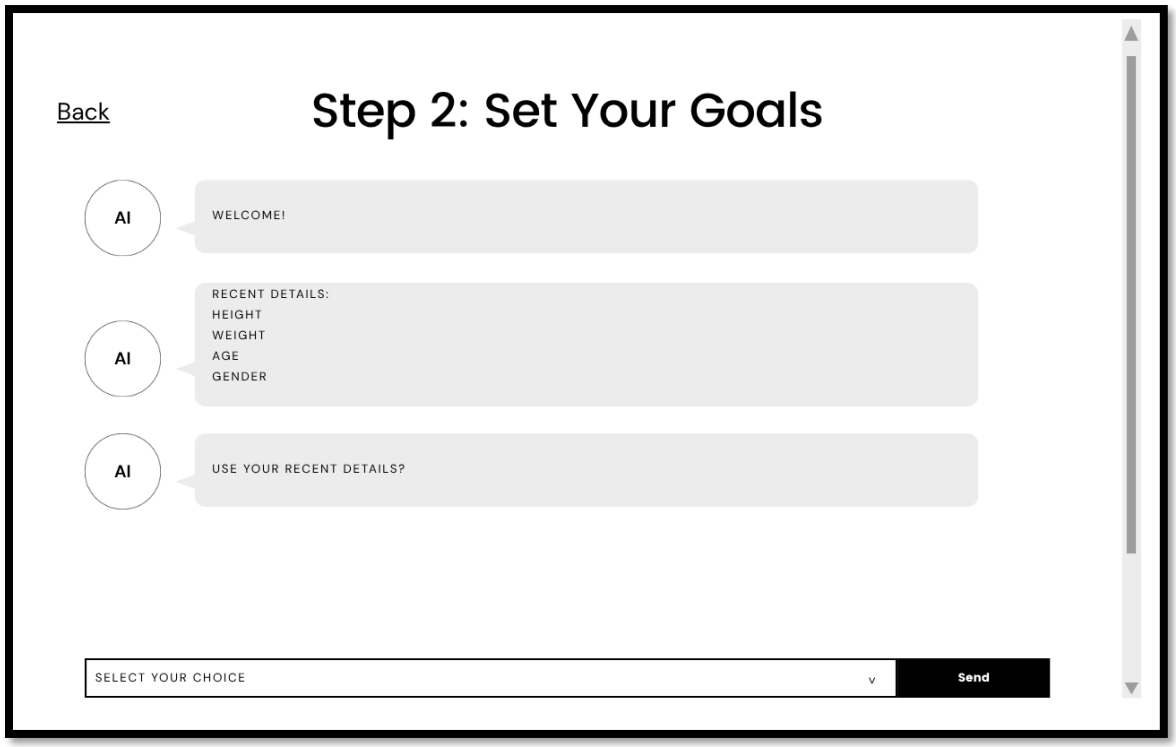


Figure 38: DP.PS.Step2 – Part 1

This is the DP.PS.Step2 page, where users provide personal details and information about their exercise habits to help tailor their diet plan. At the top left, it features a back button that allows users to return to the previous page. “Step 2: Set Your Goals” is displayed to show the purpose of the page. This step is uniquely designed as an interactive chatbot interface, where the system engages users through a conversational flow, asking questions based on their selections to gather relevant data for personalising the plan.

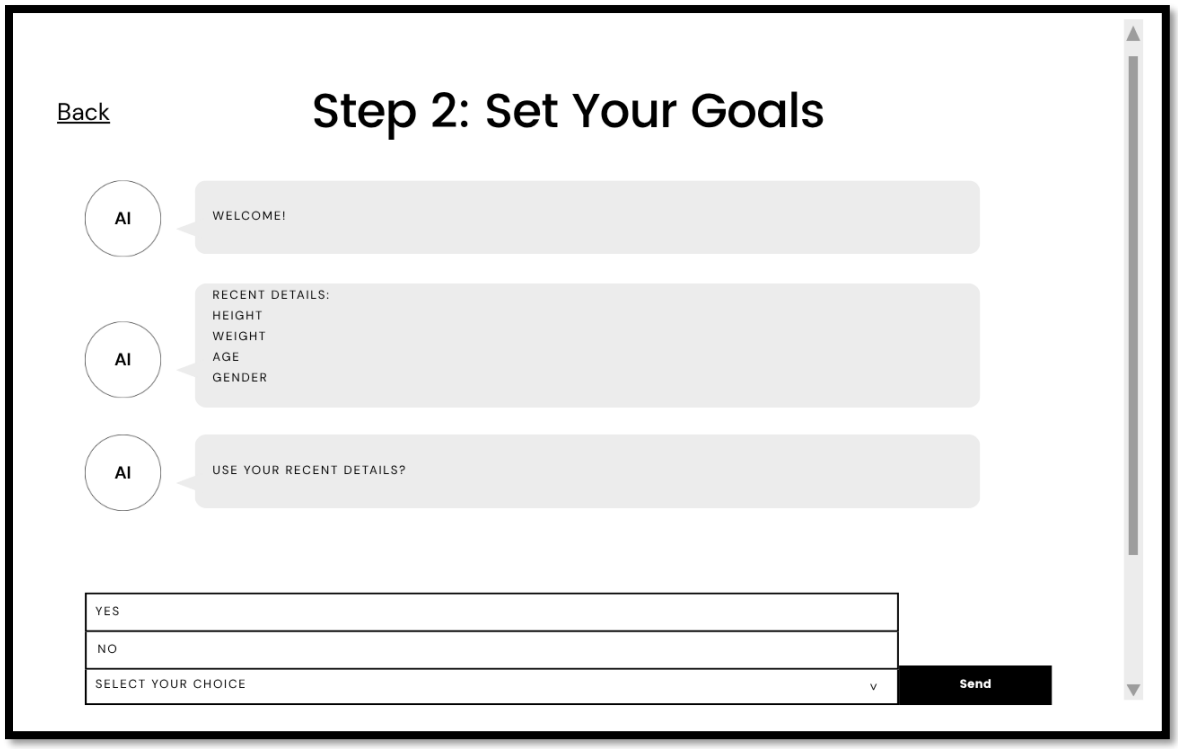


Figure 39: DP.PS.Step2 - Part 2

This is the DP.PS.Step2 page, where users provide personal details and information about their exercise habits to help tailor their diet plan. At the top left, it features a back button that allows users to return to the previous page. “Step 2: Set Your Goals” is displayed to show the purpose of the page. This step is uniquely designed as an interactive chatbot interface, where the system engages users through a conversational flow, asking questions based on their selections to gather relevant data for personalising the plan. Users may respond by selecting predefined options, such as “Yes” or “No”, and sending their answers directly to the chatbot.

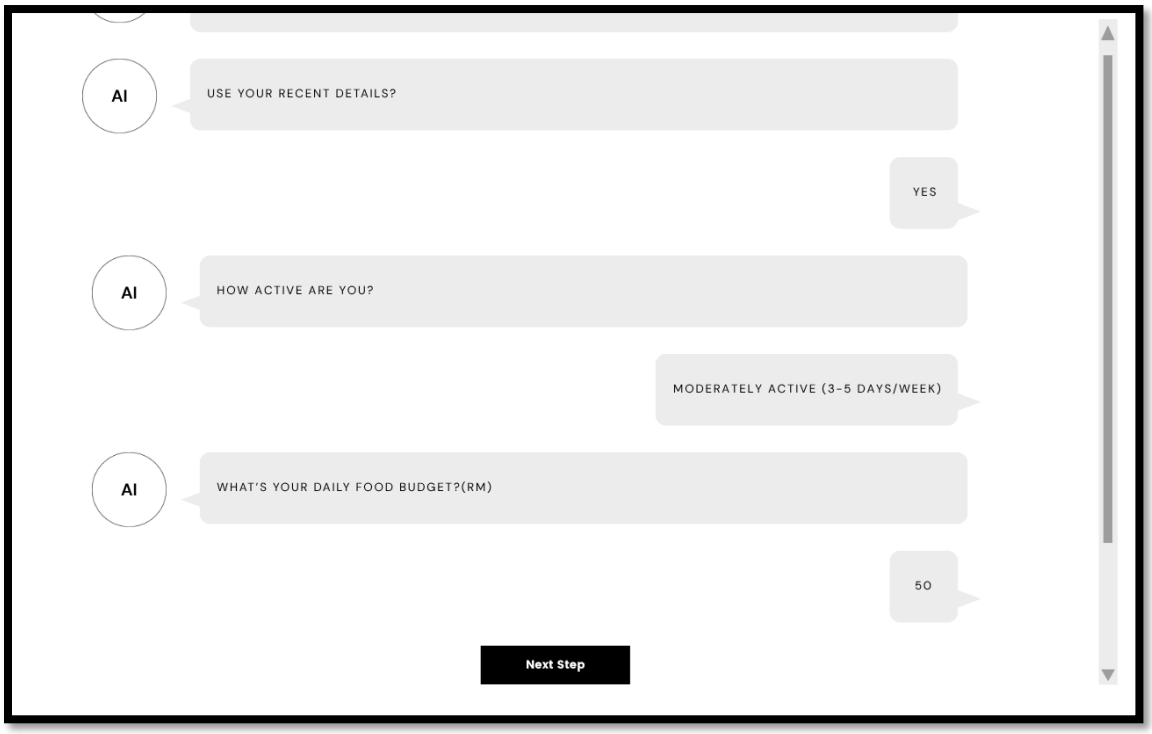


Figure 40: DP.PS.Step2 - Part 3

This is the DP.PS.Step2 page, where users provide personal details and information about their exercise habits to help tailor their diet plan. At the top left, it features a back button that allows users to return to the previous page. “Step 2: Set Your Goals” is displayed to show the purpose of the page. This step is uniquely designed as an interactive chatbot interface, where the system engages users through a conversational flow, asking questions based on their selections to gather relevant data for personalising the plan. This figure shows the flow of the chatbot interface. When users answer all the questions, a “Next Step” button will be shown and will redirect users to the next page.

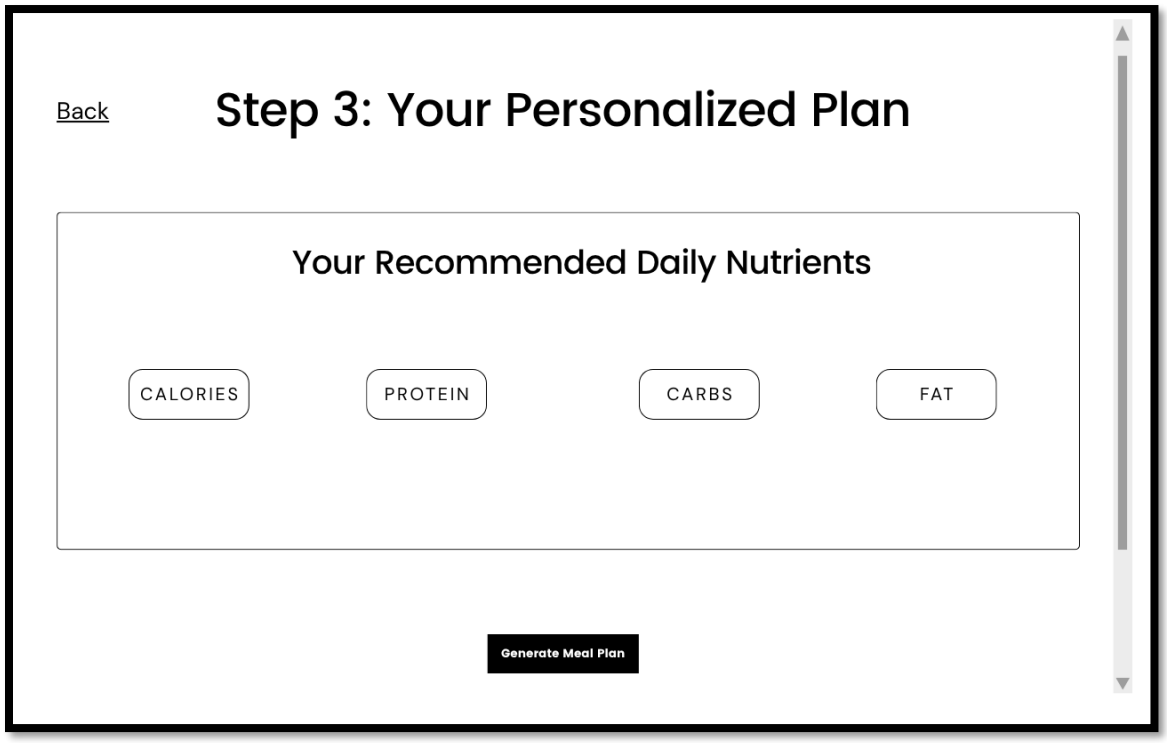


Figure 41: DP.PS.Step3

This is the DP.PS.Step3 page, which will display their personalised nutritional breakdown, including calculated values for calories, protein, carbs and fat. At the top left, it features a back button that allows users to return to the previous page. “Step 3: Your Personalised Plan” is displayed to show the purpose of the page. At the bottom, a “Generated” button is provided, enabling users to generate a personalised meal plan based on the calculated nutritional data.

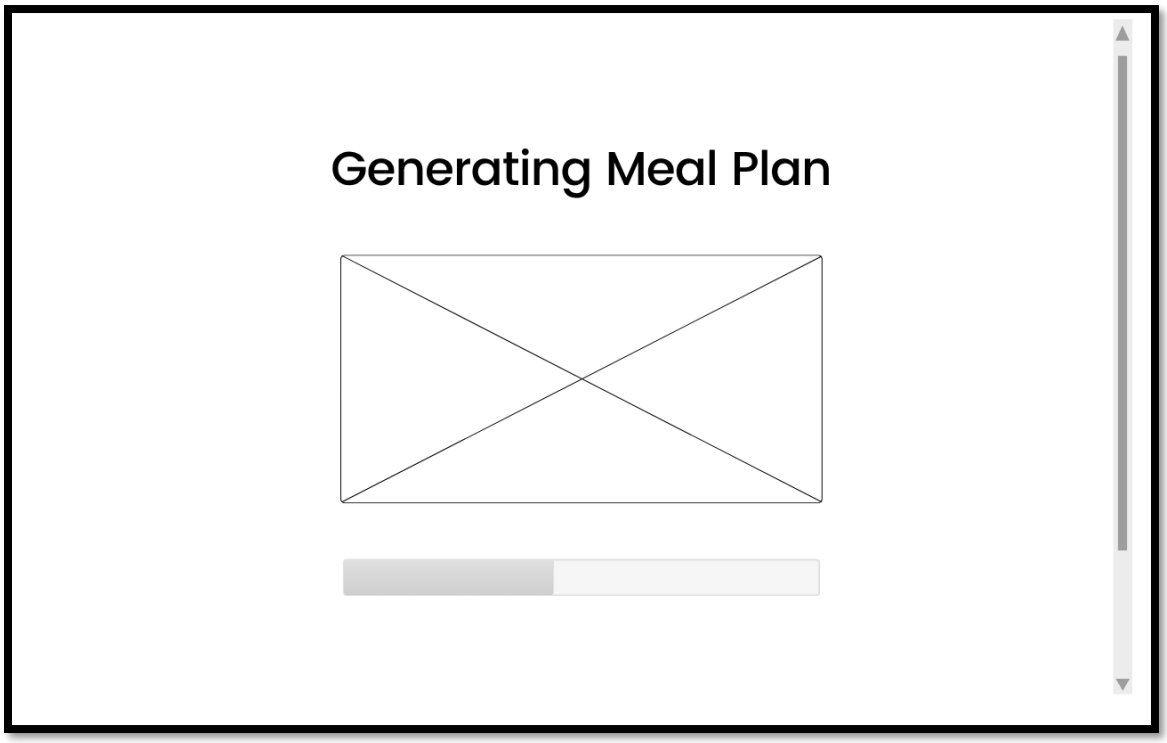


Figure 42: Generating Meal Plan

This is the Generating Meal Plan page, which informs users that the system is currently creating their personalised meal plan. During this process, a video is displayed along with a loading bar, providing visual feedback and making the waiting experience more interactive and engaging. This design helps reduce user impatience and enhances the overall user experience while the system processes the data.

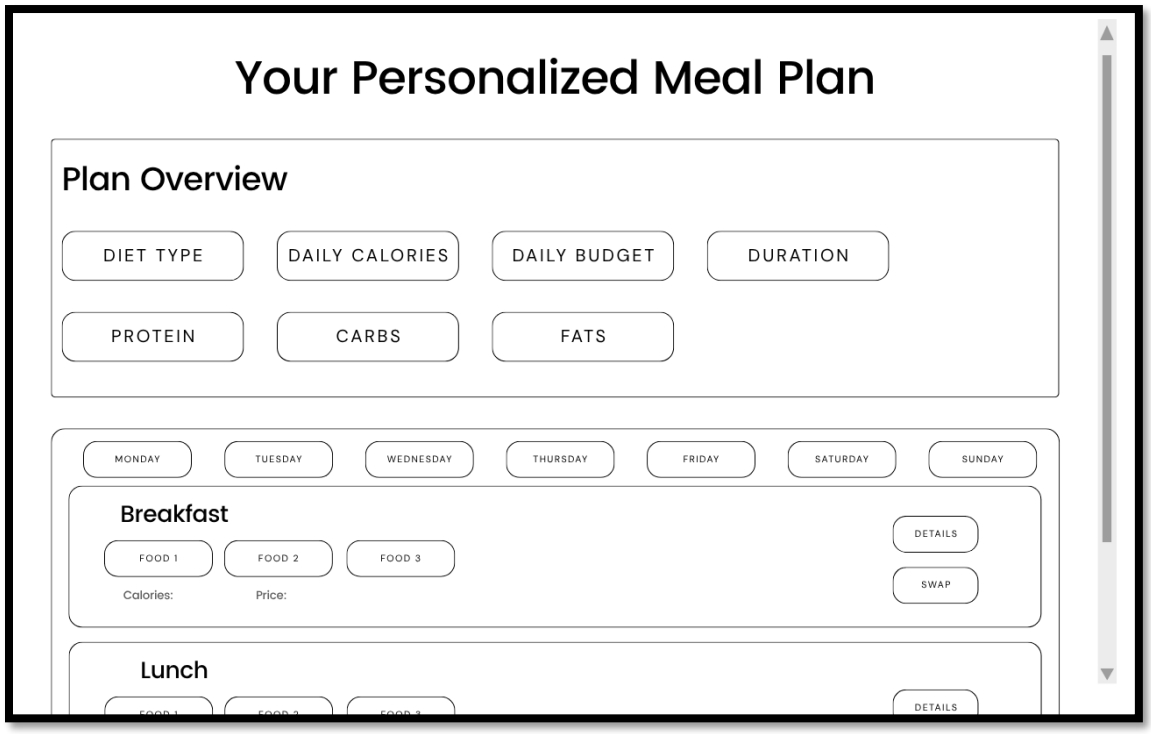


Figure 43: DP.PS.GeneratedPlan - Part 1

This is the DP.PS.GeneratedPlan page shows the generated plan based on user selection. A "Your Personalised Meal Plan" title is displayed at the top to show the purpose of the page. A plan overview section is shown to display the summarised user inputs from the previous steps, including diet type, daily calories, daily budget, duration, protein, carbs and fats. Below the plan overview, the personalised diet plan is shown and structured in a seven-day meal plan. Each day will include breakfast, lunch, dinner, snacks, details, included foods, calories, and price. Users are also given the option to swap meals, offering flexibility to customise their plan further.

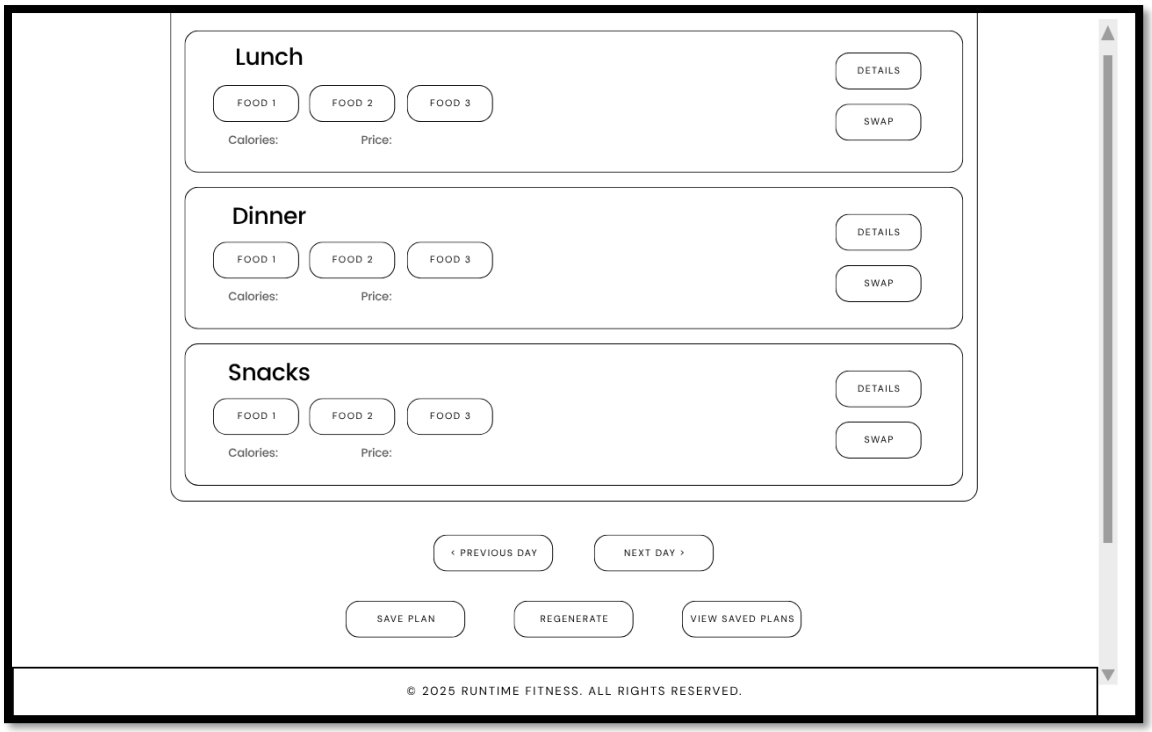


Figure 44: DP.PS.GeneratedPlan - Part 2

This is the DP.PS.GeneratedPlan page shows the generated plan based on user selection. When the users scroll down, they can navigate through the plan using the “Previous Day” or “Next Day” buttons to view the corresponding day’s meal plan. Additionally, users may choose to click the “Save Plan”, “Regenerate”, and “View Saved Plan” buttons, allowing them to save the current plan, generate a new one, or view previously saved plans. This page has a copyright notice in the footer.

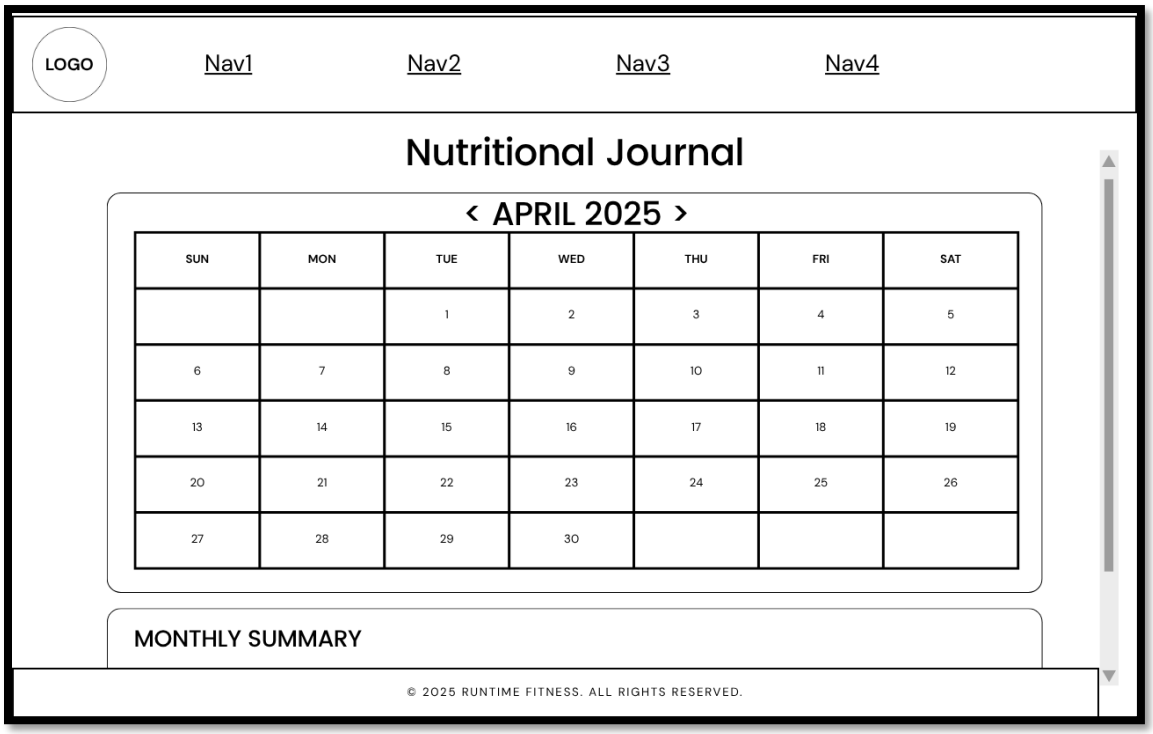


Figure 45: DP.NT.Main – Part 1

This is the DP.NT.Main page shows a nutritional journal. At the top, it features a navigation bar that has a logo and navigation links. A “Nutritional Journal” title is displayed to show the purpose of the page. The journal is presented in a calendar format, making it more intuitive and user-friendly for users to record and track their daily nutrition entries.

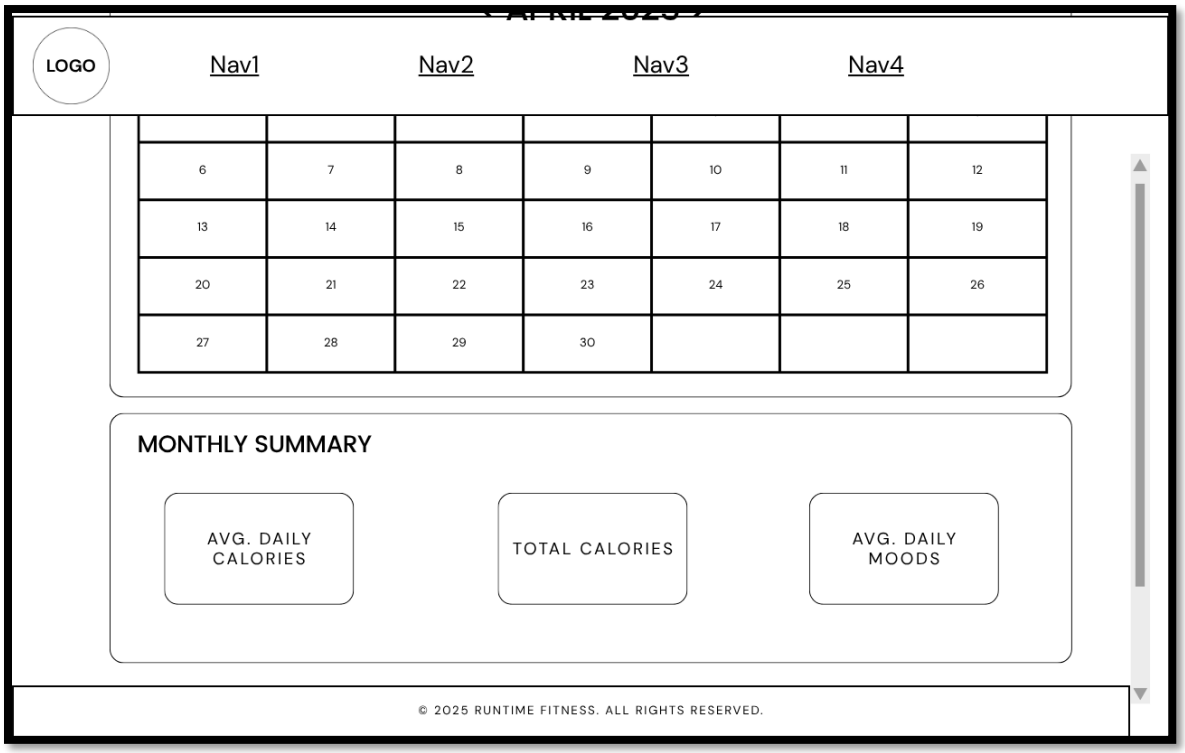


Figure 46: DP.NT.Main - Part 2

This is the DP.NT.Main page shows a nutritional journal. At the top, it features a navigation bar that has a logo and navigation links. A “Nutritional Journal” title is displayed to show the purpose of the page. When the users scroll down, they will see a monthly summary that will display the average daily calories, total calories and average daily moods based on their journal.

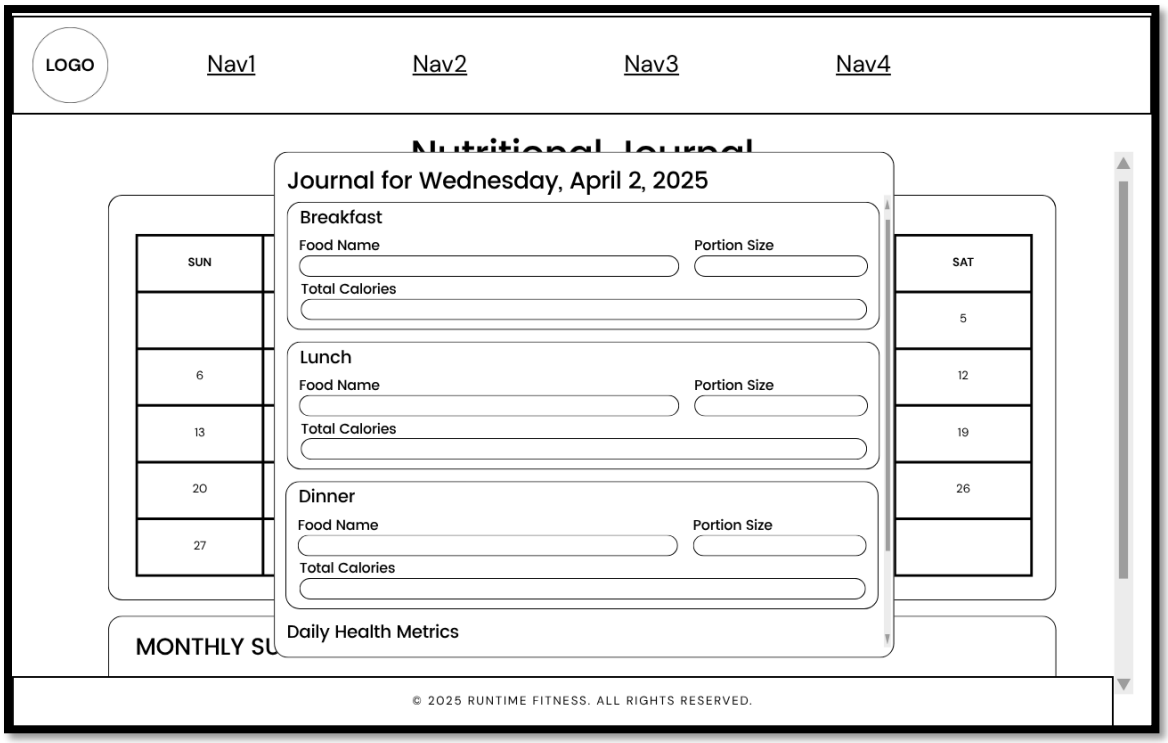


Figure 47: DP.NT.Main-Write Journal - Part 1

This is the DP.NT.Main page that allows user to write and manage their nutritional journal entries. The page features a calendar interface, enabling users to click on a specific date and begin journaling for that day. For each selected date, users can fill in text fields for breakfast, lunch, dinner and snack, where they can record the food name, portion size and total calories consumed. This structured format helps users easily track their daily nutritional intake and maintain consistent records.

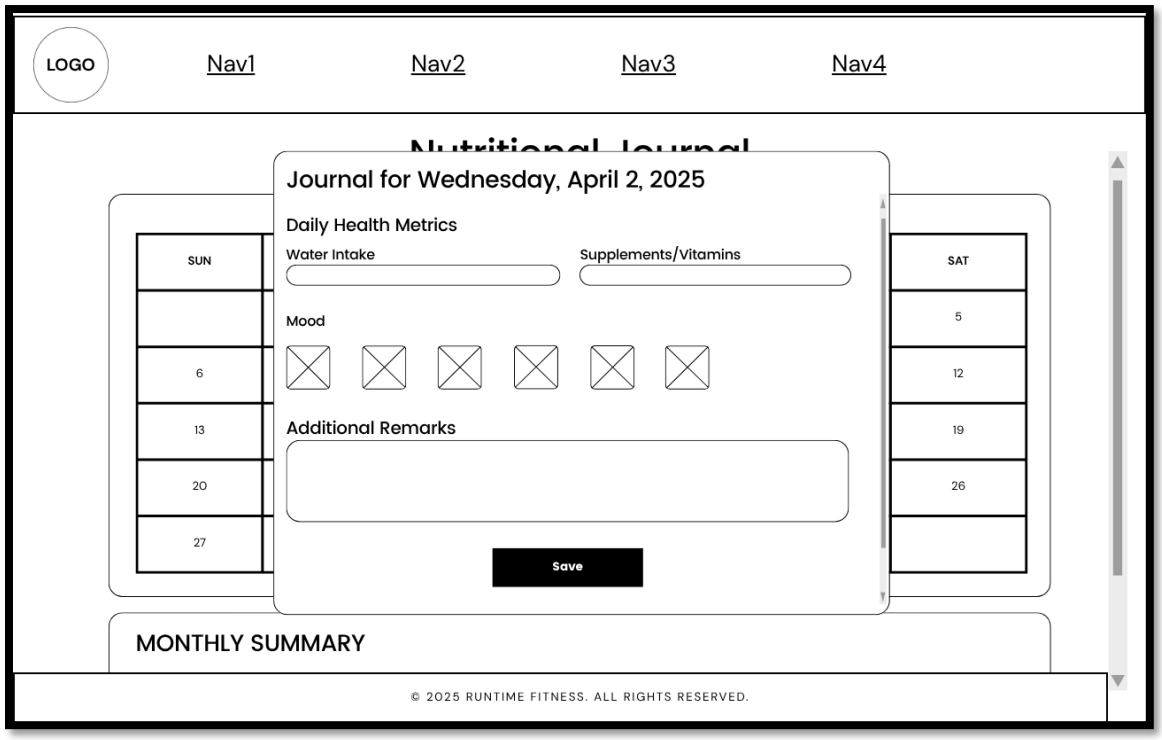


Figure 48: DP.NT.Main-Write Journal - Part 2

This is the DP.NT.Main page that allows user to write and manage their nutritional journal. In addition to logging meals, users can scroll down to record daily health metrics, including water intake, supplements or vitamins taken, and other wellness-related data. Users can also select their mood for the day using emoji icons, providing a quick and visual way to track emotional well-being. An “Additional Remarks” text box is available for users to write any extra notes or reflections. Once completed, users can click the “Save” button to store their journal entries in the database.



Figure 49：View Saved Plans

This is the View Saved Plans page, which allows users to view their saved personalised diet plan. At the top, it features a navigation bar that has a logo and navigation links. A “My Saved Plans” title is displayed to show the purpose of the page. Each saved plan is presented within a neatly styled box that includes the creation date and time of the meal plan. A “View Details” button is provided within each box, allowing users to click and view the full details of their selected meal plan.

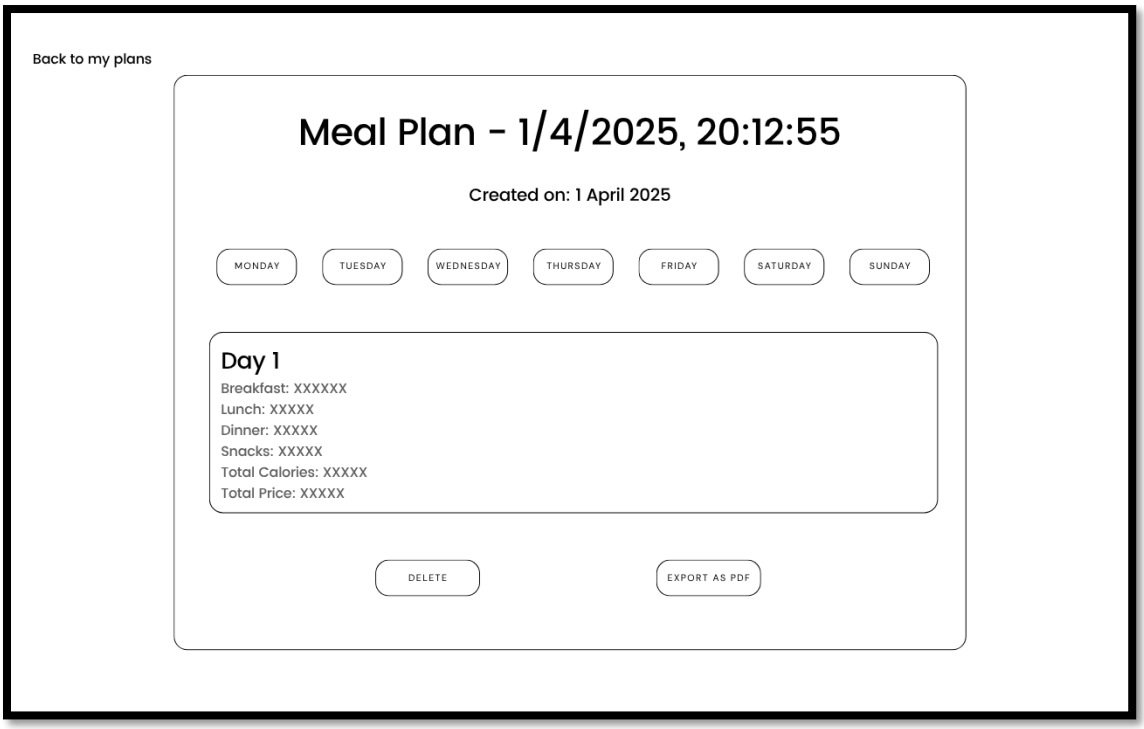


Figure 50: View Saved Plan – Details

This figure shows the meal plan details page, which displays a specific saved meal plan. At the top, it shows the plan's title with the creation date and time. Below that, day buttons (Monday to Sunday) allow users to switch between different days of the meal plan. The meal details of the selected day – including breakfast, lunch, dinner, snacks, total calories, and price, are shown in a summary box. Users can choose to delete the plan or export it as a PDF using the buttons at the bottom. A "Back to my plans" link is provided in the top-left for easy navigation.

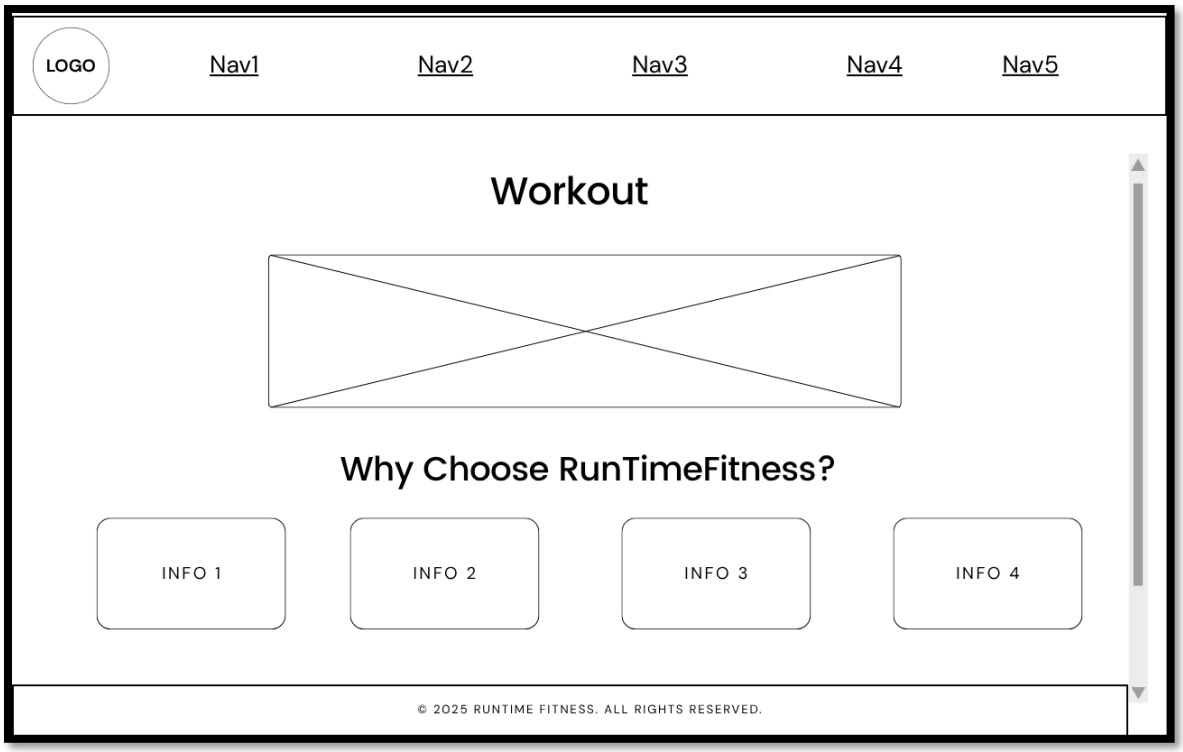


Figure 51: W.Main

This is the W.Main page, which acts as a homepage of the Workout section. At the top, it features a navigation bar that has a logo and navigation links. The centre of the page shows an interactive video to engage users. Below the video, a “Quick Access Tools” section presents four toggleable buttons that provide brief overviews of the diet plan and highlight key features of the workout module. A copyright notice is included in the footer.

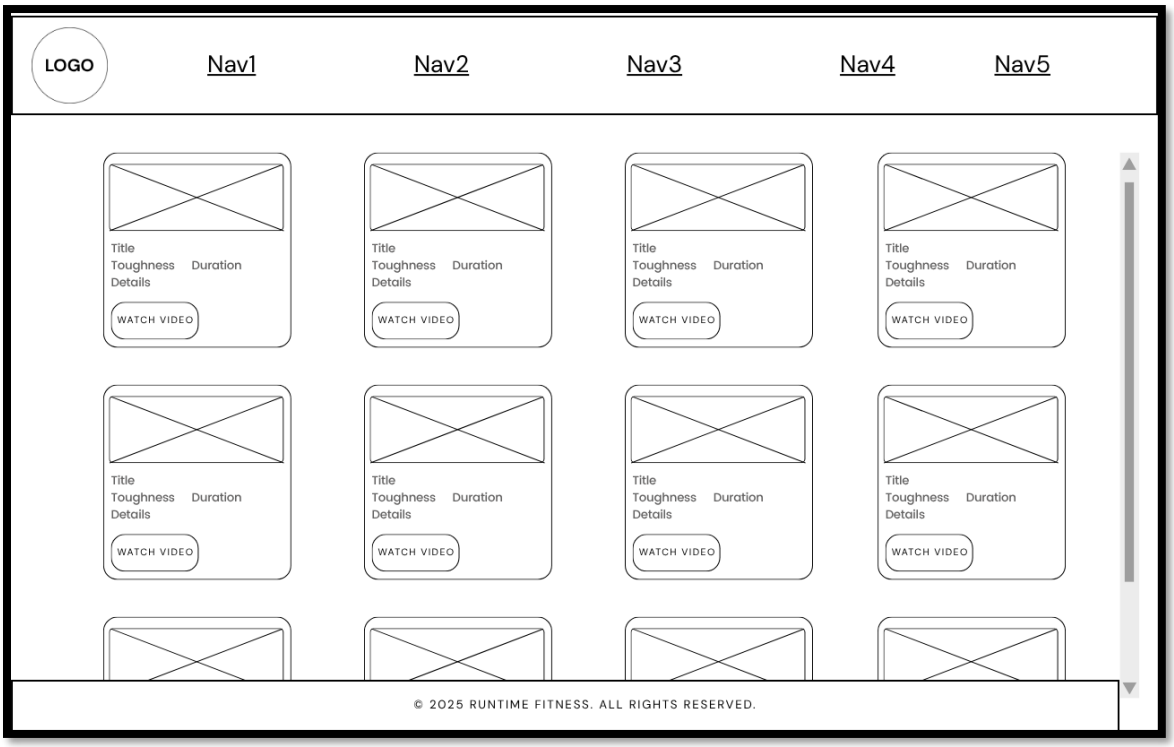


Figure 52: W.Library

This is the default workout page, which has a list of workout videos for the users to choose. At the top, it features a navigation bar that has a logo and navigation links. Each workout video is presented as a thumbnail along with its title, toughness level, duration, and a short description. A “Watch Video” button is provided for each workout, which redirects users to the corresponding video on YouTube. A copyright notice is included in the footer.

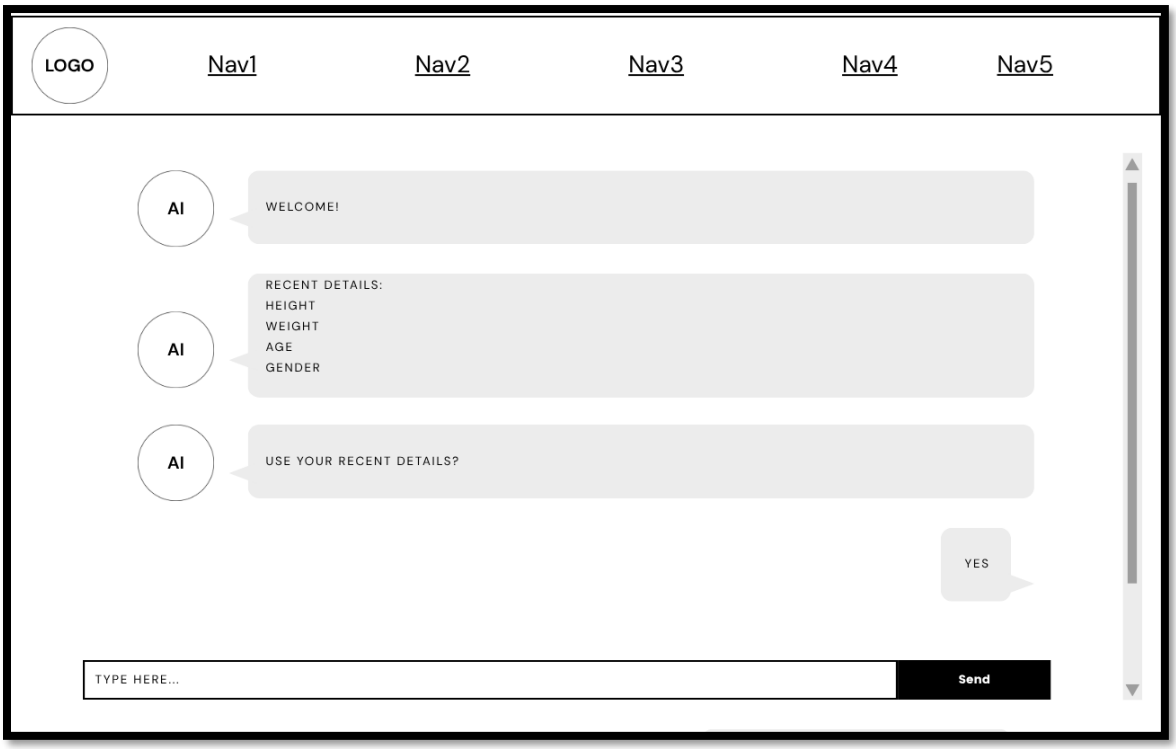


Figure 53: W.P.plan - Part 1

This is the W.P.plan page where users provide personal details and information about their exercise habits to help tailor their workout plan. At the top, it features a navigation bar that has a logo and navigation links. This step is uniquely designed as an interactive chatbot interface, where the system engages users through a conversational flow, asking questions based on their selections to gather relevant data for personalising the plan.

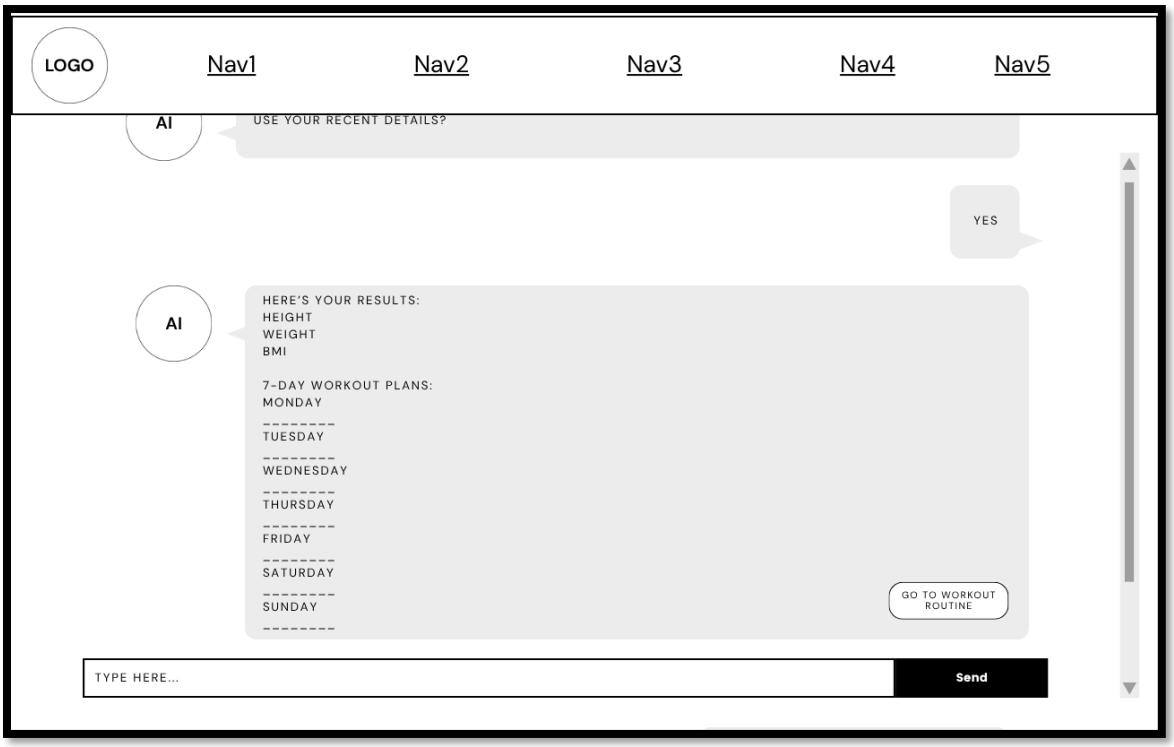


Figure 54: W.P.plan - Part 2

This is the W.P.plan page where users provide personal details and information about their exercise habits to help tailor their workout plan. At the top, it features a navigation bar that has a logo and navigation links. When the chatbot receives the information from the users, it will compile and display all the provided information. Based on the input, a personalised workout plan is generated and shown to the user. The user can then navigate to the Workout Routine page to view their customised workout plan in detail.

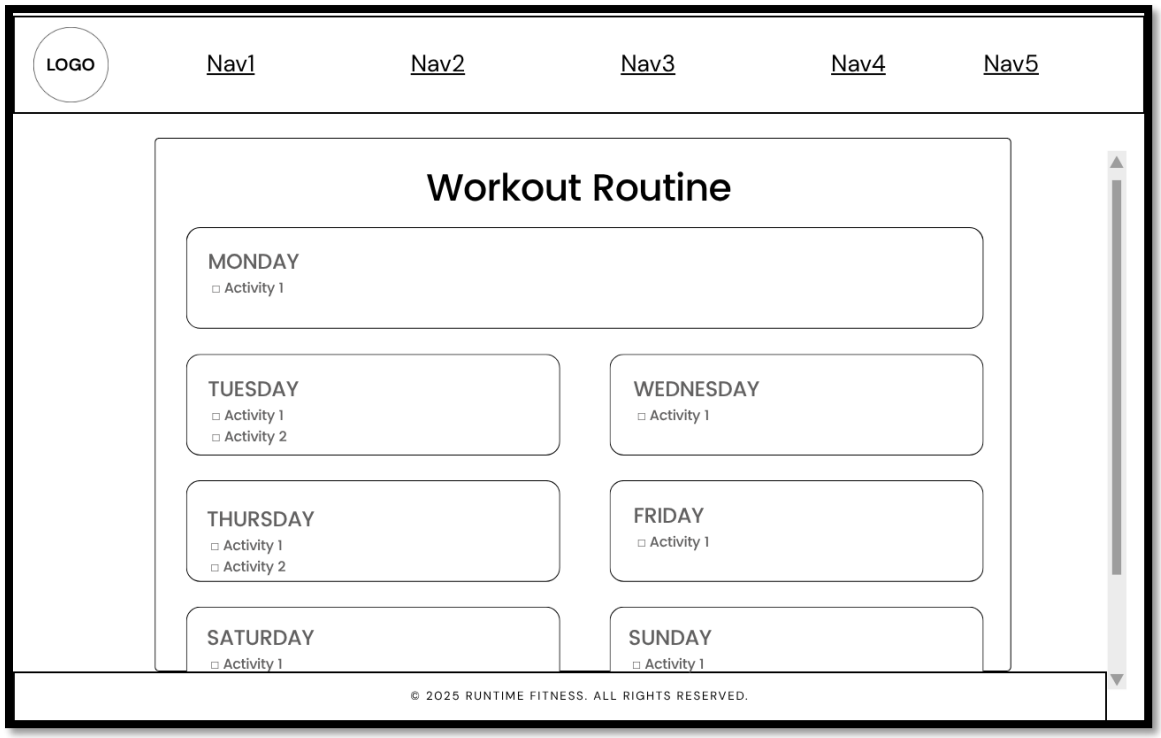


Figure 55: W.Routine

This is the W.P.plan page where users provide personal details and information about their exercise habits to help tailor their workout plan. At the top, it features a navigation bar that has a logo and navigation links. The workout routine is organised by days of the week – Monday through Sunday. The workout tasks for the current day are highlighted and shown at the top of the page for easy access. Only the tasks scheduled for today can be ticked off upon completion, helping users stay focused and track daily progress. A copyright notice is included in the footer.

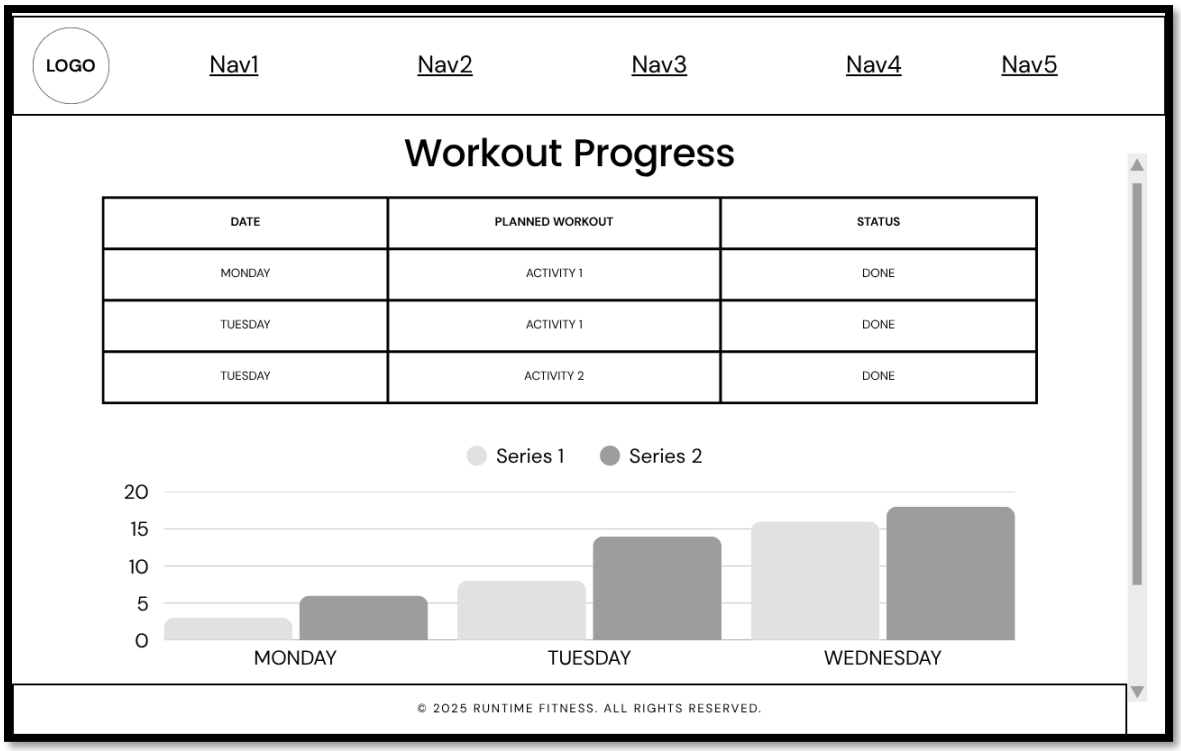


Figure 56: W.Progress

This is the W.Progress page that records the workout done by the users. At the top, it features a navigation bar that has a logo and navigation links. The page features a table summarising the user's progress, showing the date, planned workout, and status of each task from the Workout Routine page. Below the table, a chart visually represents the activities completed by the user, indicating the type of workout, day, and duration.

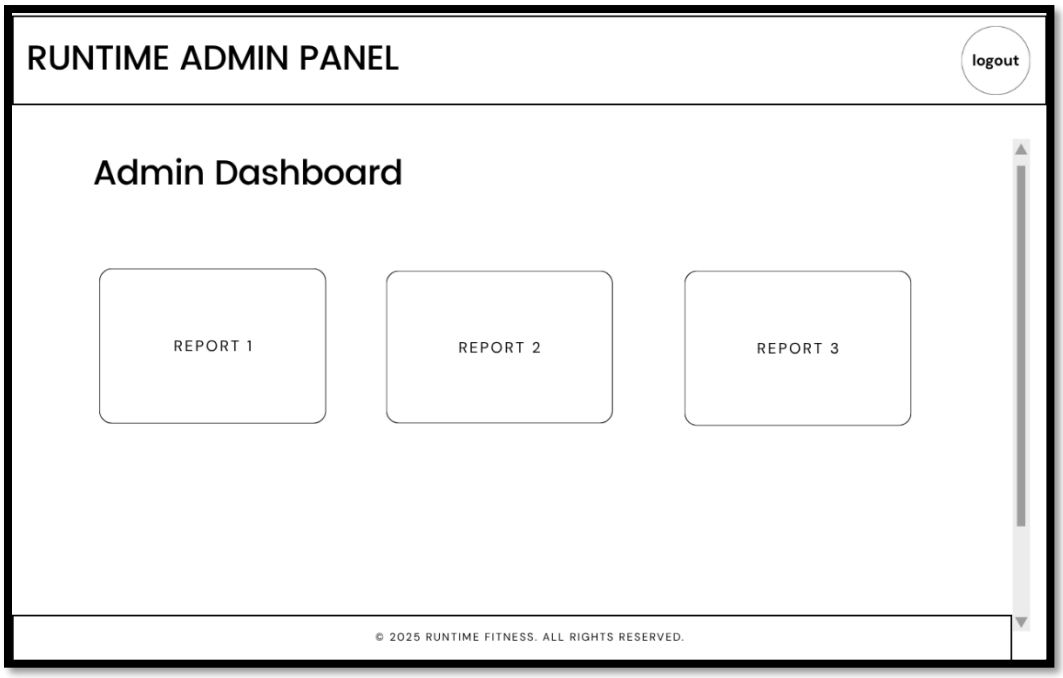


Figure 57: Admin Dashboard

This is the Admin Dashboard of the Runtime Fitness Admin Panel. It presents a clean and minimal interface that provides quick access to three key reports: Report 1, Report 2, and Report 3. Each report is displayed as a clickable card for easy navigation. The layout emphasises simplicity and user-friendliness, with a prominent logout button for secure session control. The design ensures admins can monitor and manage system reports efficiently. A copyright notice is included in the footer.

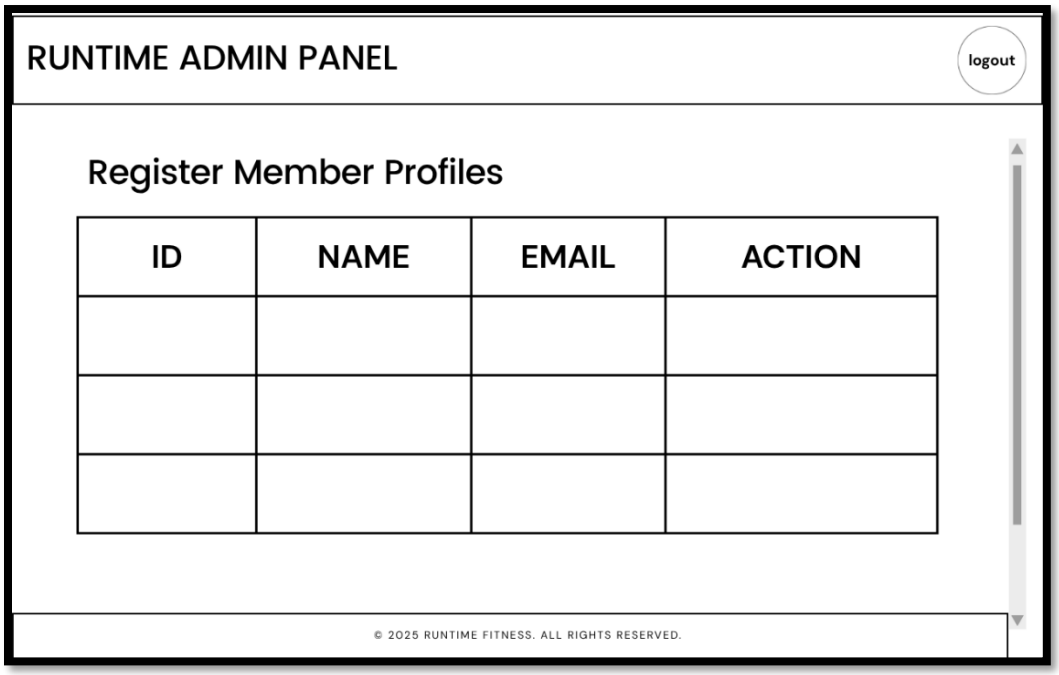


Figure 58: Reports-Register Member Profiles

This figure allows the admin to view registered member profiles. It displays members in a structured table format with columns for ID, Name, Email and Action. The “Action” column likely provides options such as edit or delete. The clean and minimal layout ensures that the admin can easily track and control member records. A copyright notice is included in the footer.

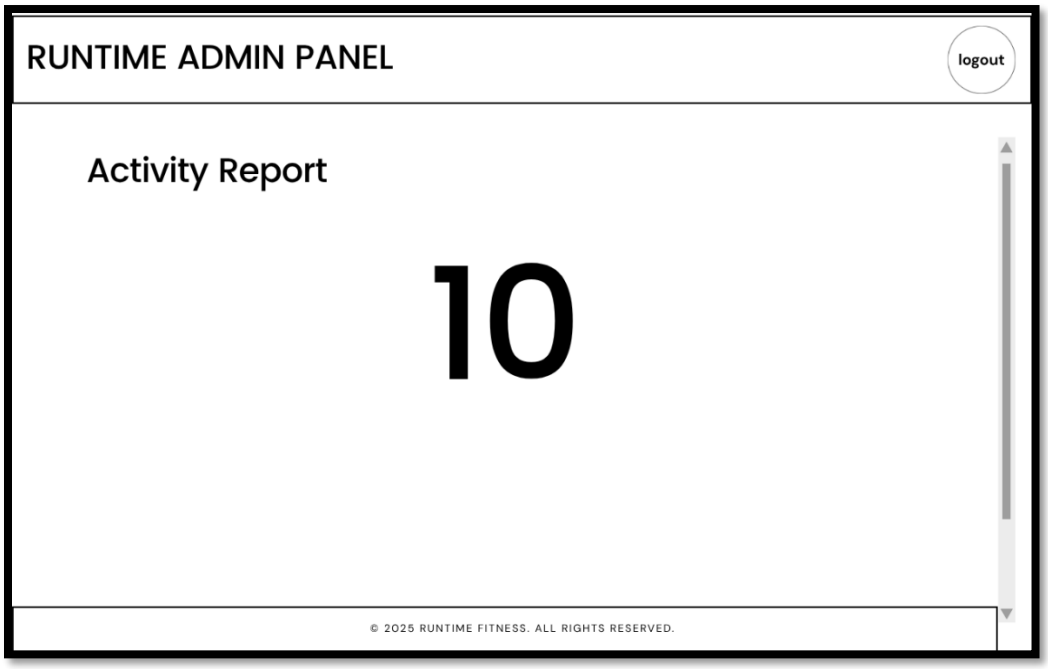


Figure 59: Activity Report

This figure shows the activity report, providing a concise summary of system activity. It will display the number “10” at the centre, likely indicating recent activity or usage metrics. A clearly labelled logout button is positioned in the top-right corner, supporting secure session management. A copyright notice is included in the footer.

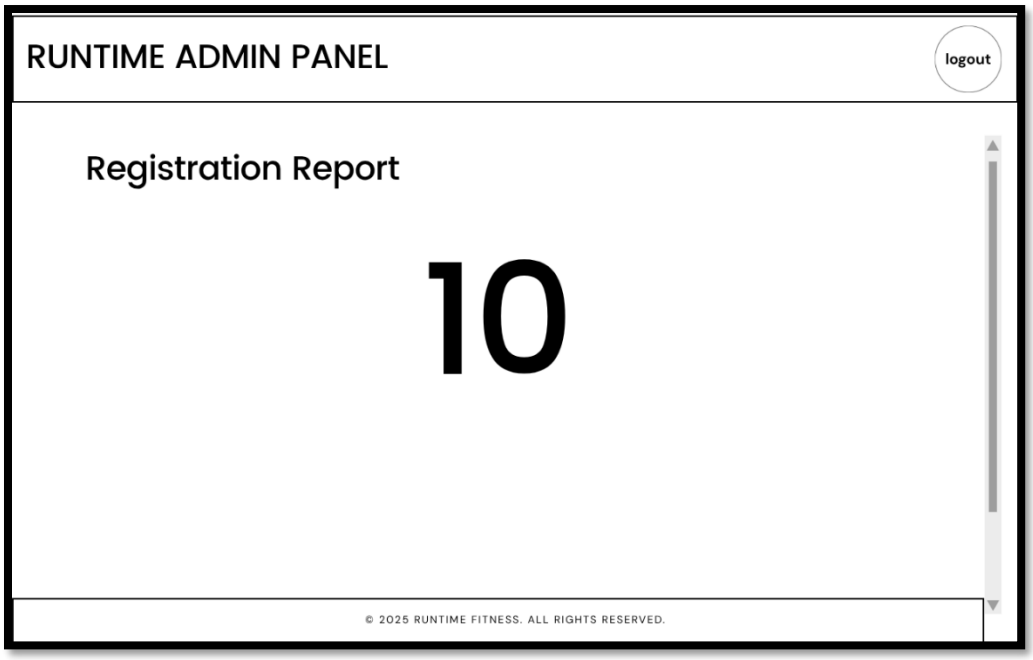


Figure 60: Registration Report

This figure shows the registration report, providing a concise summary of registration. It will display the number “10” at the centre, likely indicating recent activity or usage metrics. A clearly labelled logout button is positioned in the top-right corner, supporting secure session management. A copyright notice is included in the footer.

7.8 Report Design

There are two (2) main types of report: Member Registration Report and Member Activity

7.8.1 Member Registration Report (Week)



Description: This report displays the number of new registrations by week.

7.8.2 Member Registration Report (Month)



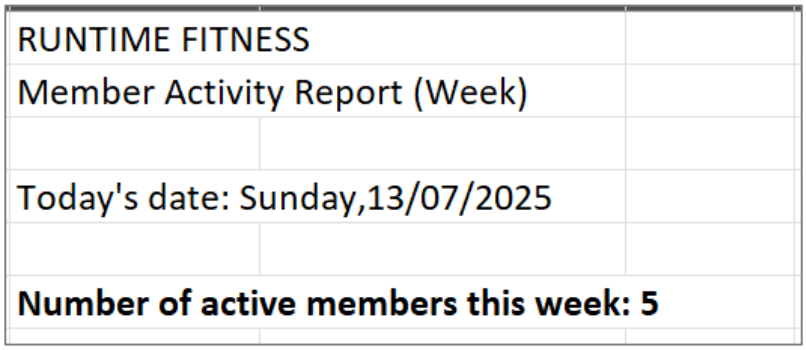
Description: This report displays the number of new registrations by month.

7.8.3 Member Registration Report (Year)



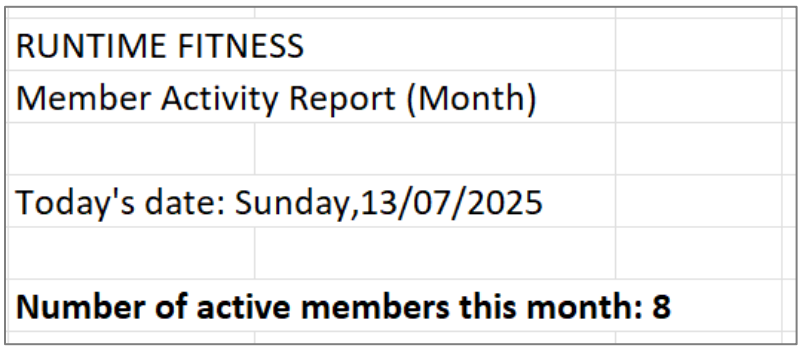
Description: This report displays the number of new registrations by year.

7.8.4 Member Activity Report (Week)



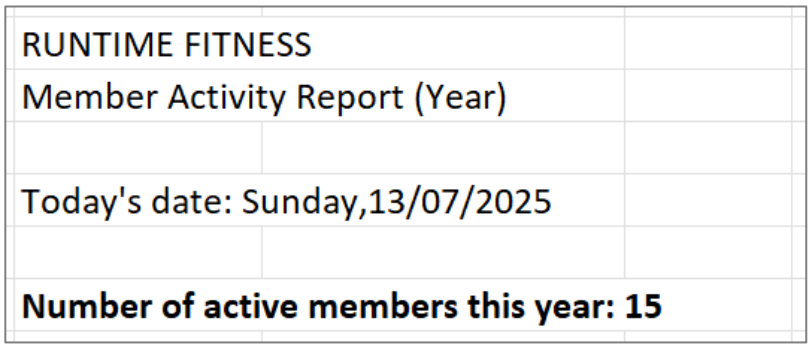
Description: This report displays the number of active members by week.

7.8.5 Member Activity Report (Month)



Description: This report displays the number of active members by month.

7.8.6 Member Activity Report (Year)



Description: This report displays the number of active members by year.

7.9 Test Cases Preparation

7.9.1 Unit Test Plan

Name of the program: login.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Password	Expected Message	Actual Message	Result
P-LT001	blank	blank	Please filled in the empty field.		
P-LT002	blank	cl@12345	Please filled in the empty field.		
P-LT003	CL	blank	Please filled in the empty field.		
P-LT004	ABC	cl@12345	User don't have an account. Directing to registration page…		
P-LT005	CL	888	Incorrect password format! Please try again.		
P-LT006	CL	Cl@13901	Incorrect password. Please try again!		
P-LT007	CL	cl@12345	**The user sees a Welcome Page**		
P-LT008	admin	i_am_admin_123	**The user sees a Admin Page**		

Name of the program: register.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Full Name	Email	Phone	Password	Confirm Password	Security keyword	Date of Birth	Gender	Height (cm)	Weight (kg)	Fitness Goal	Expected Message	Actual Message
P-RT001	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT002	abc12345	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT003	abc12345	ABC	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT004	abc12345	ABC	abc@gmail.com	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT005	abc12345	ABC	abc@gmail.com	012-3456789	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT006	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT007	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT008	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT009	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT010	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.	
P-RT011	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	*blank*	*blank*	Please filled in the empty field.	
P-RT012	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	*blank*	Please select your fitness goal.	
P-RT013	abc	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for username!	
P-RT014	abc12345	ABC-DEF	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for FULL NAME!	
P-RT015	abc12345	ABC	abc.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for EMAIL!	
P-RT016	abc12345	ABC	abc@gmail.com	123456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for Phone number!	
P-RT017	abc12345	ABC	abc@gmail.com	012 3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for Phone number!	
P-RT018	abc12345	ABC	abc@gmail.com	012-3456789	abc123	abc123	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for password!	
P-RT019	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@45678	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct password to confirm your password!	
P-RT020	abc12345	ABC	abc@gmail.com	012-3456789	abc123	abc@45678	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for password!	
P-RT021	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc123	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct password to confirm your password!	
P-RT022	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABC*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for Security Keyword!	
P-RT023	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	0.01	45	Gain weight	Invalid height value!	
P-RT024	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	0.5	Gain weight	Invalid weight value!	
P-RT025	*blank*	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your username.	
P-RT026	abc12345	*blank*	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your full name.	
P-RT027	abc12345	ABC	*blank*	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your email.	
P-RT028	abc12345	ABC	abc@gmail.com	*blank*	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your phone number.	
P-RT029	abc12345	ABC	abc@gmail.com	012-3456789	*blank*	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your password.	
P-RT030	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	*blank*	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please confirm your password.	
P-RT031	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*blank*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your security keyword.	
P-RT032	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	*blank*	FEMALE	160	45	Gain weight	Please input your B.O.D.	
P-RT033	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	*blank*	160	45	Gain weight	Please select your gender.	
P-RT034	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	*blank*	45	Gain weight	Please filled in your height	
P-RT035	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	*blank*	Gain weight	Please filled in your weight.	
P-RT036	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	**The user sees a login Page**	

Name of the program: login.php (forgot password)

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Keyword	New Password	Confirm Password	Expected Message	Actual Message	Result
P-RST001	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-RST002	abc12345	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-RST003	abc12345	ABC123	*blank*	*blank*	Please filled in the empty field.		
P-RST004	abc12345	ABC123	ABC@45678	*blank*	Please filled in the empty field.		
P-RST005	abc12345	ABC123	ABC@45678	ABC@45678	Please filled in the empty field.		
P-RST007	abc12345	ABC123	ABC@45678	ABC	Password does not match		
P-RST008	abc12345	ABC	ABC@45678	ABC@45678	Keyword does not found!		
P-RST009	abc12345	ABC123	abc@12345	ABC@45678	New password cannot same as old password! + Password does not match		
P-RST010	abc12345	*ABC123*	abc@12345	abc@12345	Your password has been successfully reset.		
P-RST011	abc	ABC123	ABC@45678	ABC@45678	Please fill in your username in the correct format.		
P-RST012	abc12345	ABC123	abc	abc	Please fill in your new password in the correct format.		
P-RST013	abc12345	ABC123	abc	efg	Password does not match		
P-RST014	abc12345	ABC123	ABC@45678	ABC@45678	Your password has been reset successfully.		

Name of the program: tracking.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Today's Weight (kg)	Waist (cm)	Hip (cm)	Thigh (cm)	Water Intake (cups)	Sleep (hours)	Expected Message	Actual Message	Result
P-PT001	blank	blank	blank	blank	blank	blank	Please filled in the empty field.		
P-PT002	50	blank	blank	blank	blank	blank	Successfullt saved weight!		
P-PT003	5	blank	blank	blank	blank	blank	Weight must be greater than 10.		
P-PT004	blank	f	blank	blank	blank	blank	Invalid input.		
P-PT005	blank	blank	f	blank	blank	blank	Invalid input.		
P-PT006	blank	blank	blank	f	blank	blank	Invalid input.		
P-PT007	blank	f	f	f	blank	blank	Invalid input.		
P-PT008	blank	5	blank	blank	blank	blank	Invalid input.		
P-PT009	blank	blank	5	blank	blank	blank	Invalid input.		
P-PT010	blank	blank	blank	5	blank	blank	Invalid input.		
P-PT011	blank	5	5	5	blank	blank	Invalid input.		
P-PT012	blank	58	88	25	blank	blank	Sucessfully saved body measurement!		
P-PT013	blank	blank	blank	blank	1	blank	Display drink and sleep more reminder and successfully saved daily logs message.		
P-PT014	blank	blank	blank	blank	blank	1	Display drink and sleep more reminder and successfully saved daily logs message.		
P-PT015	blank	blank	blank	blank	1	8	Display drink more water reminder and successfully saved daily logs message.		

Name of the program: DP.PS.Main.php

Name of the tester: Ms. YAP XIN LING

TC-ID	Use Recorded Details	Height (cm)	Weight (kg)	Age	Gender	Activity Level	Goal	Budget (RM)	Expected Message	Actual Message	Result
P-DT001	No	blank	blank	blank	blank	blank	blank	blank	Please fill in the empty field		
P-DT002	No	-170	blank	blank	blank	blank	blank	blank	Value must be greater than or equal to 80		
P-DT003	No	170	blank	blank	blank	blank	blank	blank	Please fill in the empty field		
P-DT004	No	170	-60	blank	blank	blank	blank	blank	Value must be greater than or equal to 20		
P-DT005	No	170	60	blank	blank	blank	blank	blank	Please fill in the empty field		
P-DT006	No	170	60	-25	blank	blank	blank	blank	Value must be greater than or equal to 5		
P-DT007	No	170	60	130	blank	blank	blank	blank	Value must be less than or equal to 120		
P-DT008	No	170	60	25	blank	blank	blank	blank	Please select an item in the list		
P-DT009	No	170	60	25	Female	blank	blank	blank	Please select an item in the list		
P-DT010	No	170	60	25	Male	Moderately active (3-5 days/week)	blank	blank	Please select an item in the list		
P-DT011	No	170	60	25	Male	Moderately active (3-5 days/week)	Lose Weight	blank	Please fill in the empty field		
P-DT012	No	170	60	25	Male	Moderately active (3-5 days/week)	Lose Weight	-10	Value must be greater than or equal to 1		
P-DT013	No	170	60	25	Male	Moderately active (3-5 days/week)	Lose Weight	30	Personalized diet plan is ready		
P-DT014	Yes	-	-	-	-	blank	blank	blank	Please select an item in the list		
P-DT015	Yes	-	-	-	-	Moderately active (3-5 days/week)	blank	blank	Please select an item in the list		
P-DT016	Yes	-	-	-	-	Moderately active (3-5 days/week)	Lose Weight	-10	Value must be greater than or equal to 1		
P-DT017	Yes	-	-	-	-	Moderately active (3-5 days/week)	Lose Weight	50	Personalized diet plan is ready		
P-DT018	blank	-	-	-	-	blank	blank	blank	Please select an item in the list		

Name of the program: DP.NT.Main.php

Name of the tester: Ms. YAP XIN LING

TC-ID	Selected Date	Breakfast Food Name	Breakfast Portion Size	Total Calories for Breakfast	Lunch Food Name	Lunch Portion Size	Total Calories for Lunch	Dinner Food Name	Dinner Portion Size	Total Calories for Dinner	Snacks Food Name	Snacks Portion Size	Total Calories for Snacks	Water Intake	Supplements /Vitamins	Mood	Additional Remarks	Expected Message	Actual Message	Result
P-NT001	July 1, 2025	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	Please enter at least one meal to save your journal		
P-NT002	July 1, 2025	Omelette	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	Vitamin C	Happy	Felt good	Please enter a portion size for breakfast		
P-NT003	July 1, 2025	blank	1 plate	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	6 cups	blank	Energetic	blank	Please enter a food name for breakfast.		
P-NT004	July 1, 2025	Toast	2 slices	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	Calm	blank	Please enter calories for breakfast.		
P-NT005	July 1, 2025	Toast	2 slices	300	blank	blank	blank	blank	blank	blank	blank	blank	blank	8 cups	Omage 3	Calm	Slept well	Saved successfully		
P-NT006	July 1, 2025	Toast	2 slices	-100	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	Multivitamin	Calm	blank	Value must be greater than or equal to 0		
P-NT007	July 1, 2025	Bread	2 slices	250	Rice	1 bowl	600	Chicken	1 thigh	350	Apple	1	80	10 cups	Vitamin B	Happy	Productive day	Saved successfully		
P-NT008	July 1, 2025	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	blank	Calm	blank	Please enter at least one meal to save your journal		
P-NT009	July 1, 2025	Cereal	1 bowl	200	blank	blank	blank	blank	blank	blank	blank	blank	blank	4 cups	blank	Calm	blank	Saved successfully		
P-NT010	July 1, 2025	Oatmeal	1 bowl	300	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	blank	blank	blank	Saved successfully		
P-NT011	July 1, 2026	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	You can only log meals for today or earlier		

Name of the program: P.editProfile.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Full Name	Email	Phone	Security keyword	Date of Birth	Gender	Height (cm)	Weight (kg)	Fitness Goal	Expected Message	Actual Message	Result
P-ET001	*blank*	*blank*	*blank*	*blank*	*blank*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET002	abc12345	*blank*	*blank*	*blank*	*blank*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET003	abc12345	ABC	*blank*	*blank*	*blank*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET004	abc12345	ABC	abc@gmail.com	*blank*	*blank*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET005	abc12345	ABC	abc@gmail.com	012-3456789	*blank*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET006	abc12345	ABC	abc@gmail.com	012-3456789	*blank*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET007	abc12345	ABC	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET008	abc12345	ABC	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET009	abc12345	ABC	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.		
P-ET010	abc12345	ABC	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	160	*blank*	*blank*	Please filled in the empty field.		
P-ET011	abc12345	ABC	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	160	50	*blank*	Please filled in the empty field.		
P-ET012	abc	ABCDEF	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	160	50	Gain Weight	Incorrect username format		
P-ET013	abc12345	A	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	160	50	Gain Weight	Successfully updated profile!		
P-ET014	abc12345	ABCDEF	abc.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	160	50	Gain Weight	Invalid email format		
P-ET015	abc12345	ABC	abc@gmail.com	123456789	*ABCDEF*	1/1/2005	FEMALE	160	50	Gain Weight	Incorrect phone number format		

P-ET016	abc12345	ABCDEF	abc@gmail.com	012-3456789	*ABCD*	1/1/2005	FEMALE	160	50	Gain Weight	Incorrect security keyword format		
P-ET017	abc12345	ABCDEF	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	1	50	Gain Weight	Incorrect height format		
P-ET018	abc12345	ABCDEF	abc@gmail.com	012-3456789	*ABCDEF*	1/1/2005	FEMALE	160	3	Gain Weight	Incorrect weight format		
P-ET019	abc12345	ABCDEF	abc@gmail.com	012 3456789	*ABCDEF*	1/1/2005	MALE	160	50	Lose Weight	Successfully updated profile!		

7.9.2 User Acceptance Testing Plan

Name of the user: Mr/Ms – Full Name:

Job Position:

Date:

Time From:

Location:

Category	0 Mark(Very Poor)	1 Mark	2 Mark	3 Mark	4 Mark	5 Mark(Excellent)
User-friendly						
Free-from errors						
Secure						
Meeting its objectives						
Performance						
Feedback From the user	Date: Signature					
Response from Project Development Team	Name: Signature Date:					

Name of the user: Mr/Ms – Full Name:

Job Position:

Date:

Time From:

Location:

Category	0 Mark(Very Poor)	1 Mark	2 Mark	3 Mark	4 Mark	5 Mark(Excellent)
User-friendly						
Free-from errors						
Secure						
Meeting its objectives						
Performance						
Feedback From the user	Date: Signature					
Response from Project Development Team	Name: Signature Date:					

Name of the user: Mr/Ms – Full Name:

Job Position:

Date:

Time From: Time To:

Location: (where the testing is conducted)

Category	0 Mark(Very Poor)	1 Mark	2 Mark	3 Mark	4 Mark	5 Mark(Excellent)
User-friendly						
Free-from errors						
Secure						
Meeting its objectives						
Performance						
Feedback From the user	Date: Signature					
Response from Project Development Team	Name: Signature Date:					

8.0 Execution / Implementation

8.1 Screenshots

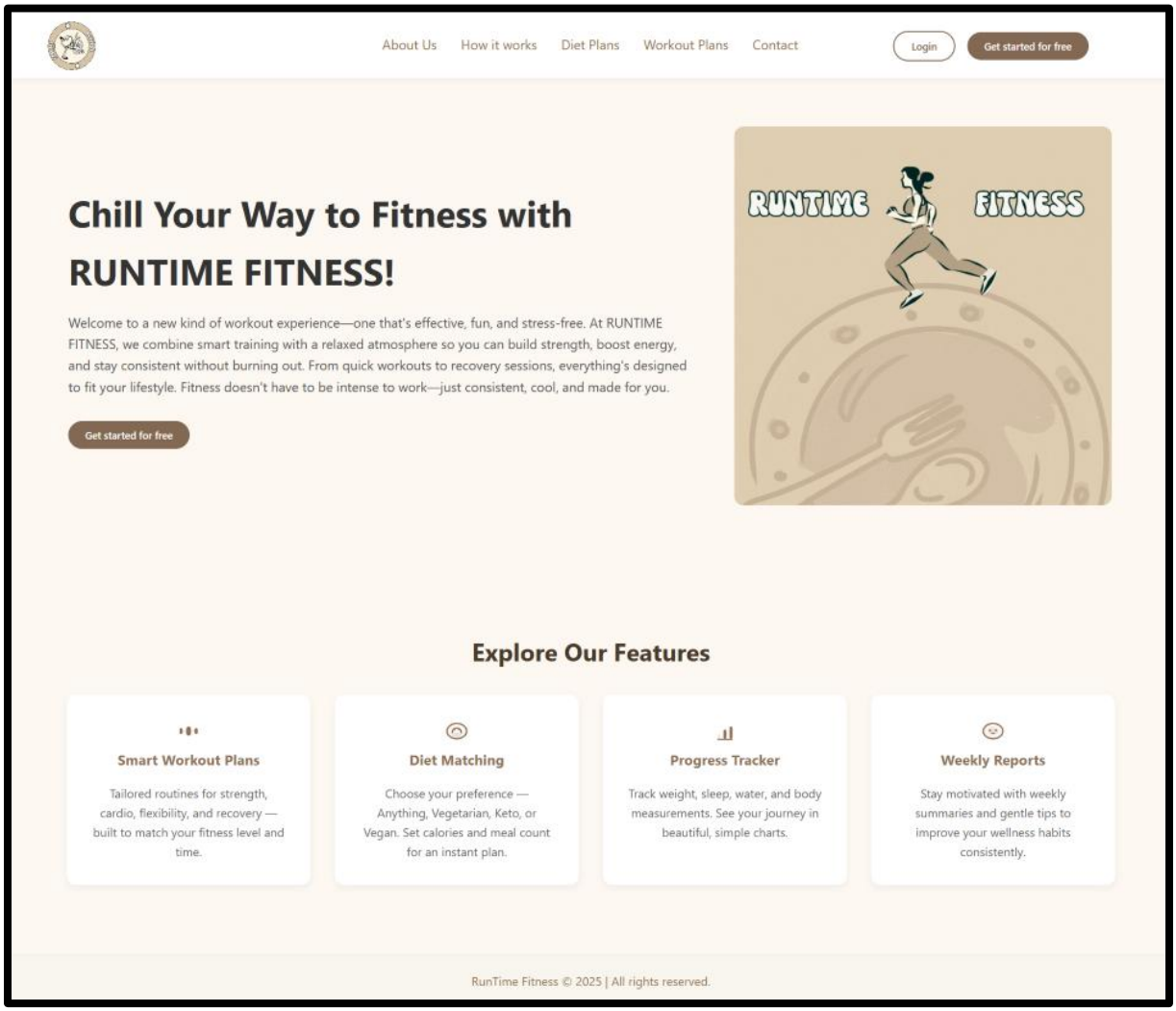


Figure 61: Home Page

This is the main landing page displayed before user login. It introduces new users to RunTime Fitness and features a header navigation with links to About Us, How It Works, Diet Plans, Workout Plans, and Contact pages. Users can click the Login button or the "Get Started for Free" button.

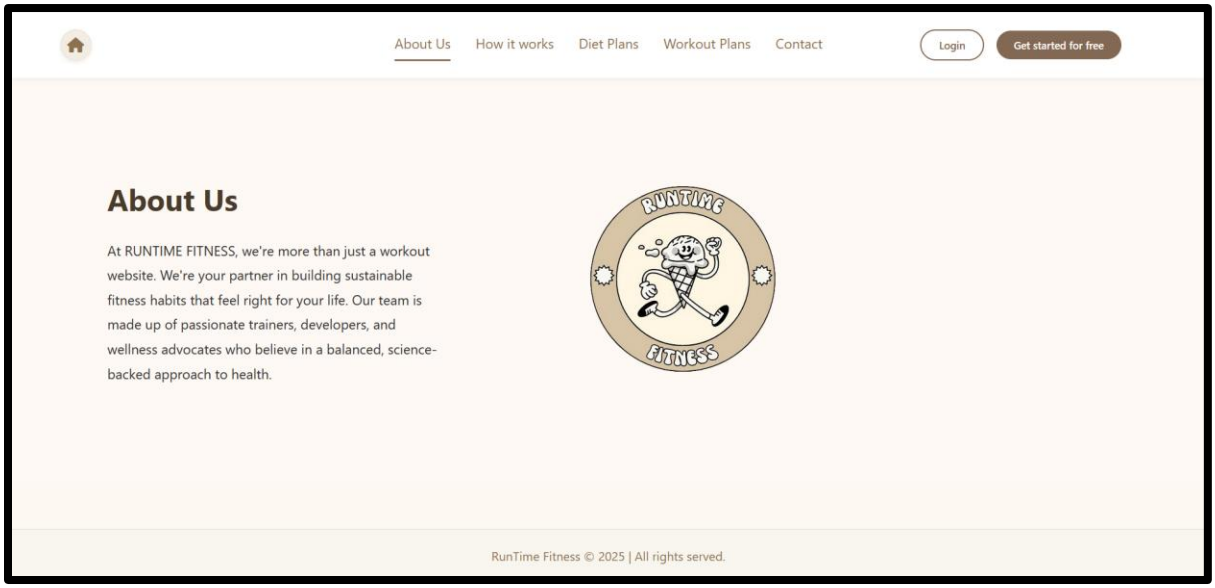


Figure 62: About Us

The About Us page shares RunTime Fitness's mission and team values, focusing on sustainable, science-backed fitness and a passion for healthy living.

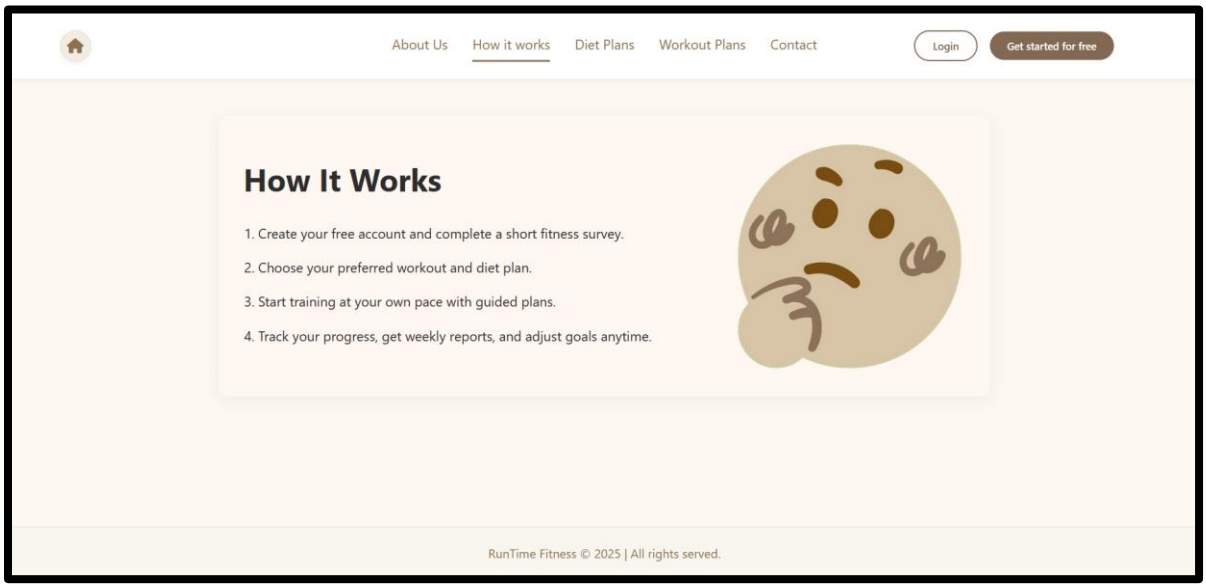


Figure 63: How It Works

This page outlines simple steps for new users—sign up, take a fitness survey, choose plans, start training, and track progress with weekly updates.

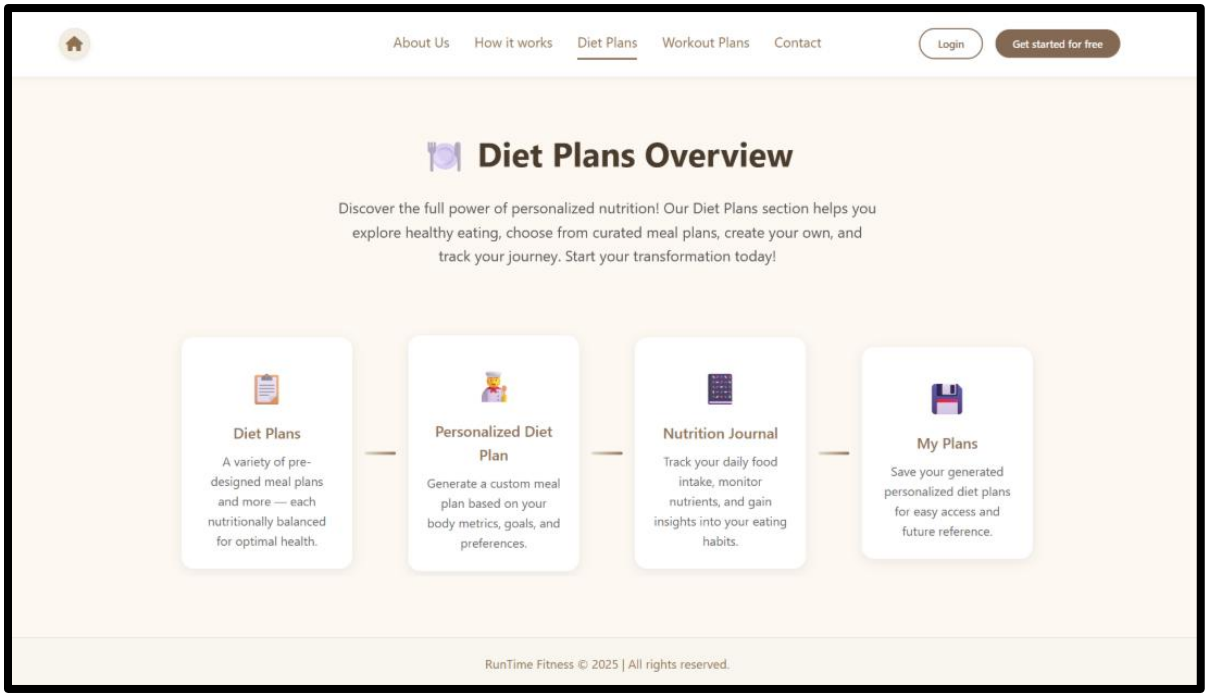


Figure 64: Diet Plans Overview

Diet Plans Overview highlights the benefits of personalized nutrition, offering access to pre-designed meal plans, custom diet generation based on user data, a nutrition journal to track food intake, and a "My Plans" area to save and manage personalized diets.

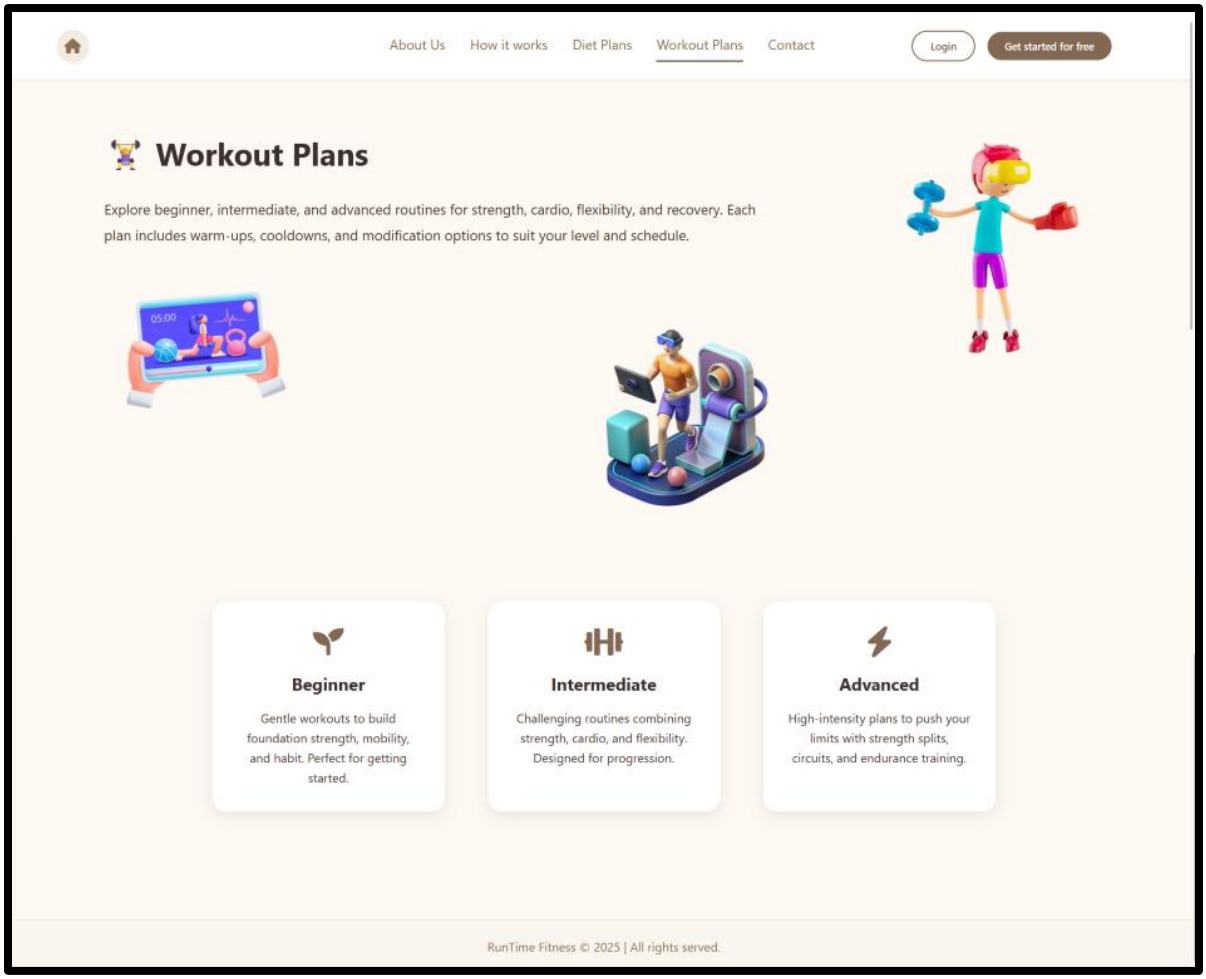


Figure 65: Workout Plans Overview

The Workout Plans page introduces users to a variety of exercise routines tailored for different fitness levels—beginner, intermediate, and advanced—helping users find the right plan to match their goals and experience.

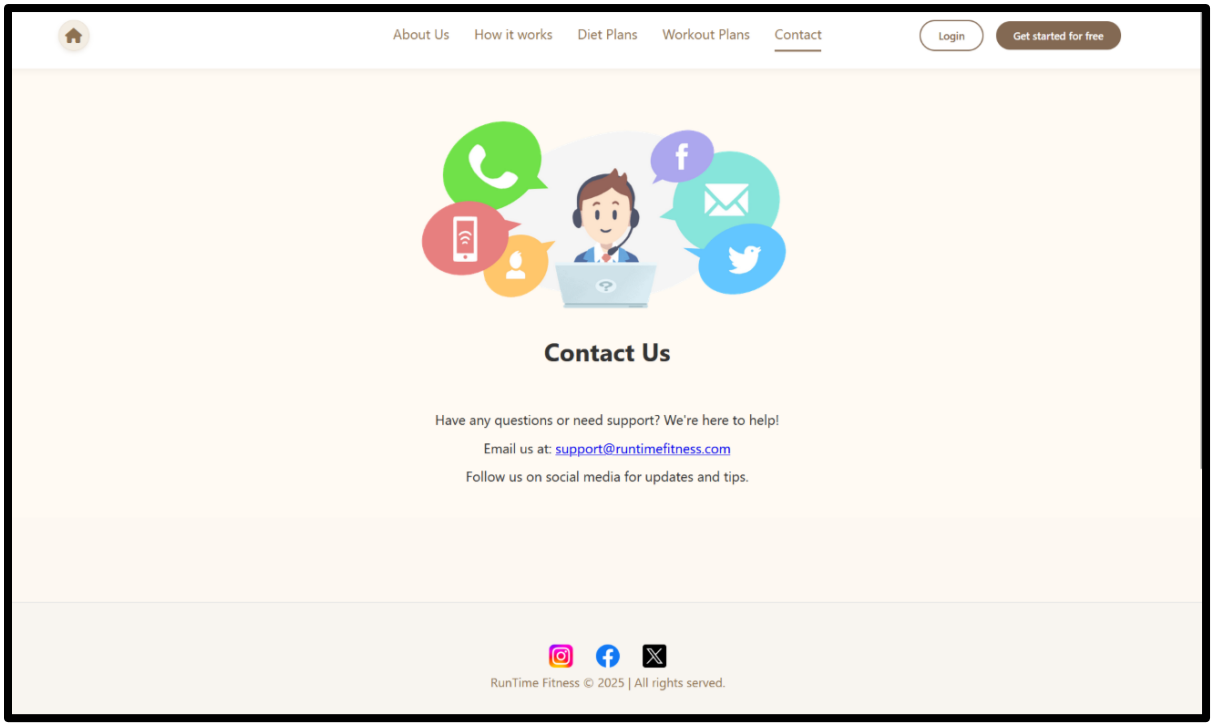


Figure 66: Contact Us

The Contact Us page provides users with ways to reach out for support or inquiries, and introduces the available communication channels, including email and social media.

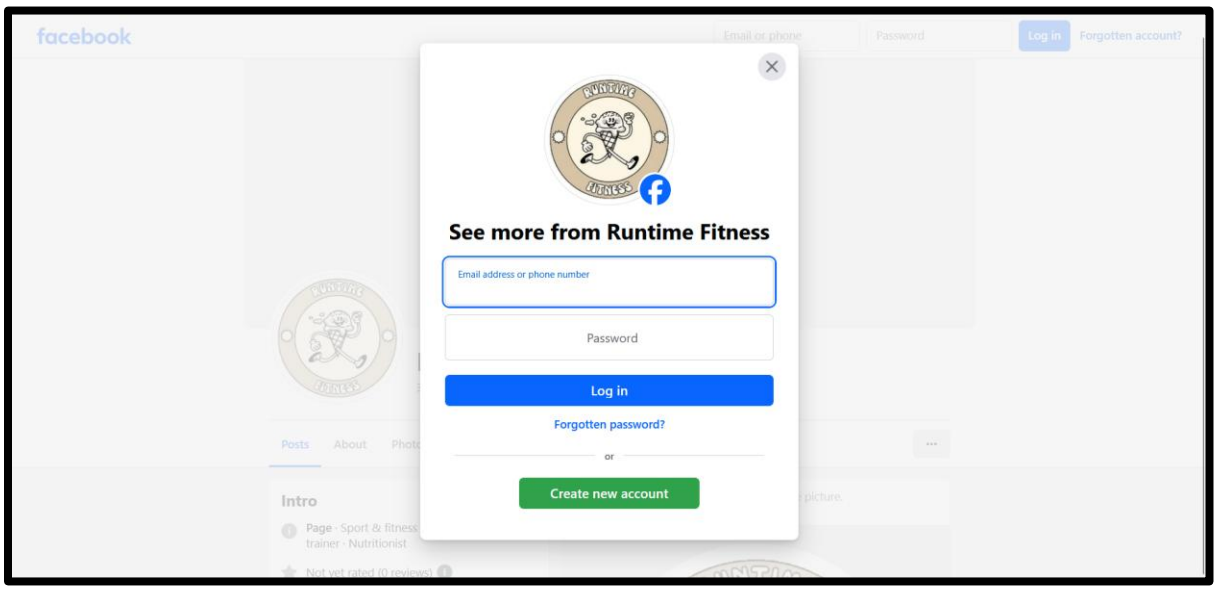


Figure 67: Runtime Fitness Facebook Page I

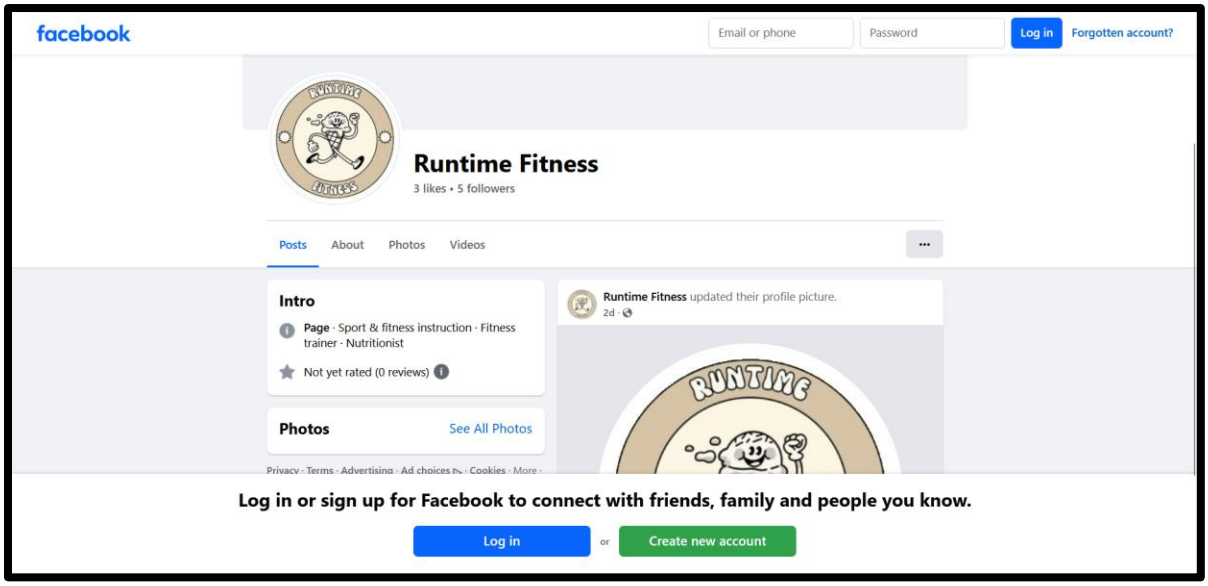


Figure 68: Runtime Fitness Facebook Page II

After clicking the Facebook icon, users are directed to the official Runtime Fitness Facebook page. Here, users can view more content, follow updates, and connect with the Runtime Fitness community on social media.

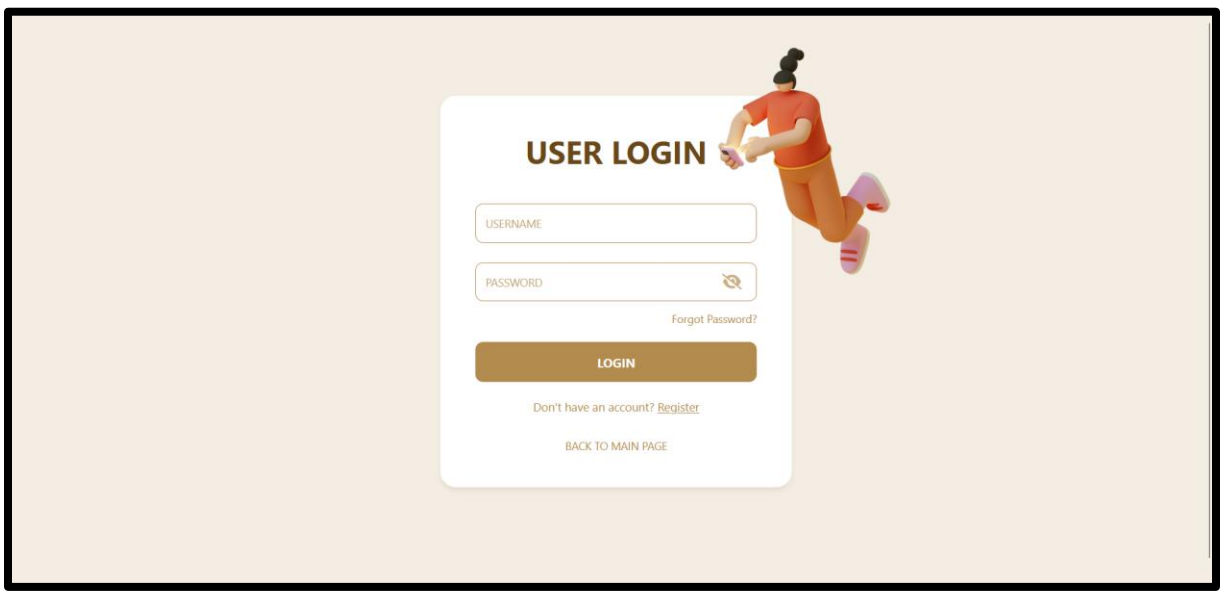


Figure 69: User Login

The User Login page allows users to sign in by entering their username and password. They can also click "Register" to create a new account or "Forgot password?" to reset a forgotten password.

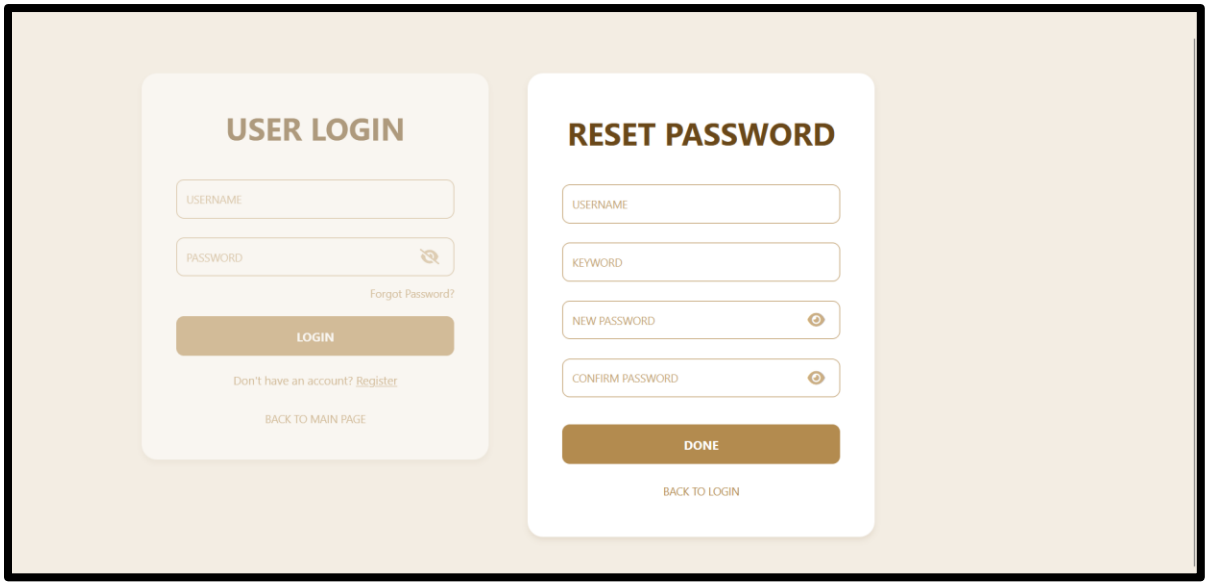


Figure 70: Reset Password

The Reset Password page enables users to recover access to their account by entering their username, a security keyword, and setting a new password.

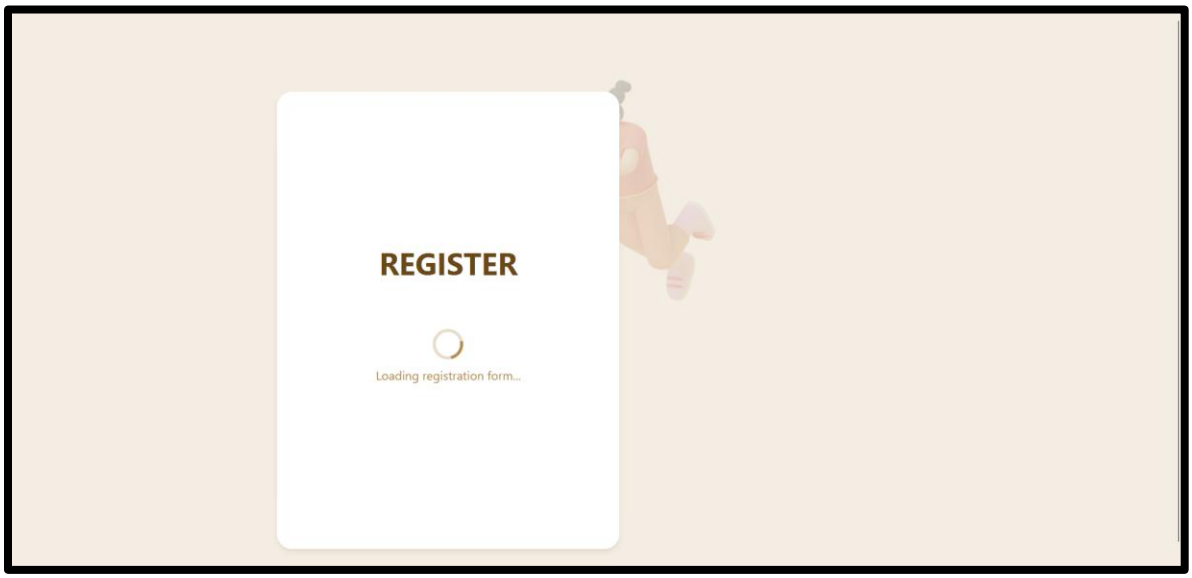


Figure 71: Registration Form

This page is the registration loading screen. It appears while the system is loading the registration form for new users.

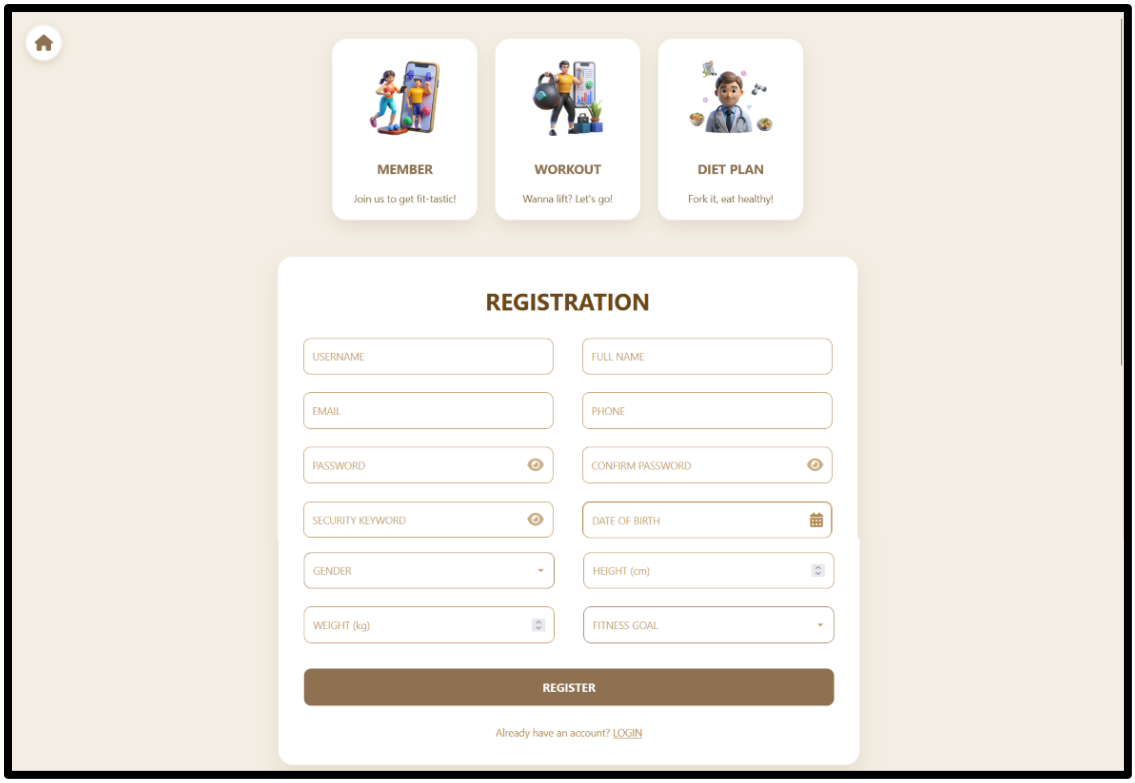


Figure 72: Registration

The Registration page allows new users to create an account by providing personal details such as username, email, password, and fitness goals. This information helps personalize the user's experience on the platform.



Figure 73: Loading Page

After a user log in, a loading page is displayed to provides a smooth transition while the system prepares the user's personalized dashboard.

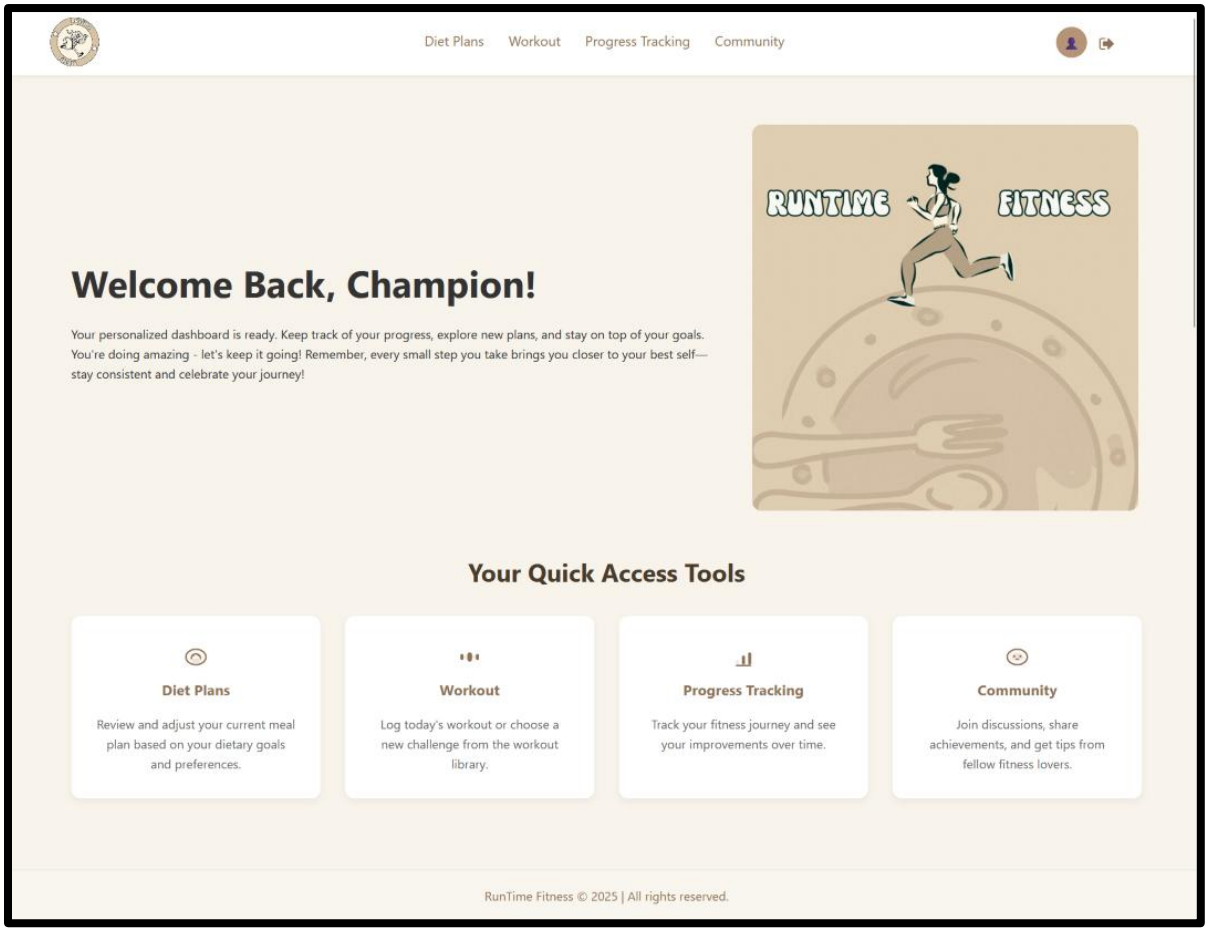


Figure 74: Home Page

The user home page welcomes the user and serves as a central dashboard. Users can quickly access Diet Plans, Workout routines, Progress Tracking, and the Community section either through the navigation bar or the "Your Quick Access Tools" area.



Figure 75: Splash Page

A splash page appears before the user is directed to the diet plan section.

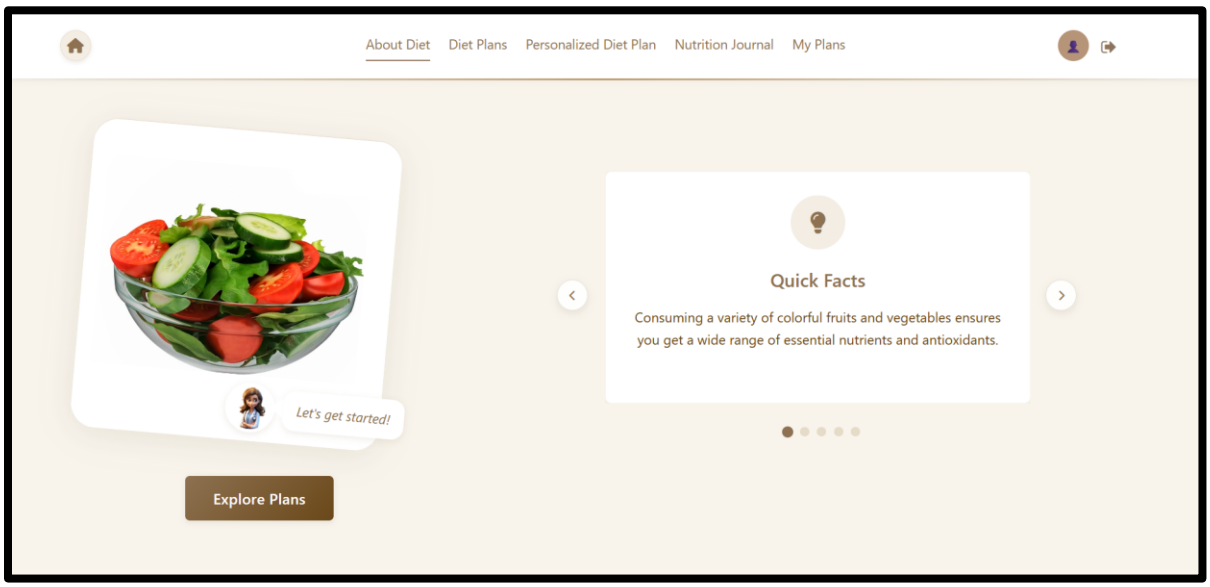


Figure 76: About Diet I

About Diet offer quick facts about healthy eating and guiding them to choose their nutrition goals.

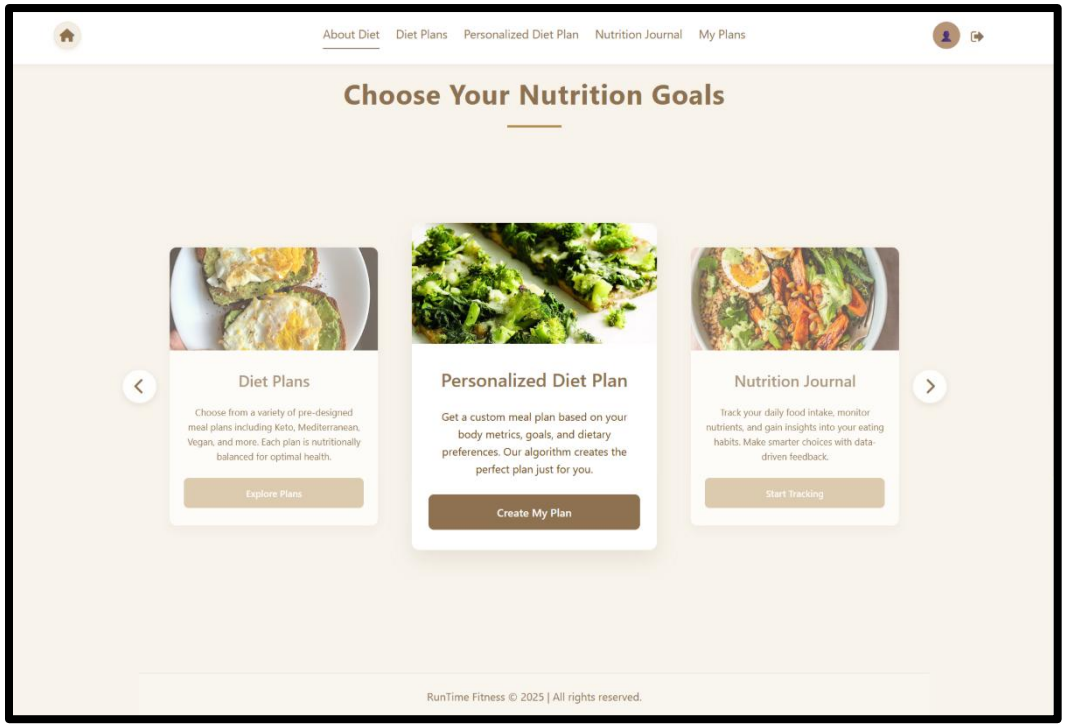


Figure 77: About Diet II

Users can select from three main options—Diet Plans (pre-designed meal plans), Personalized Diet Plan (customized based on user data), and Nutrition Journal (track daily food intake).



Figure 78: Diet Plans

This page displays six diet plan types: Regular, Keto, Vegan, Vegetarian, Dairy-Free, and Gluten-Free. Each type is briefly described, helping users select a plan that matches their dietary preferences or restrictions.

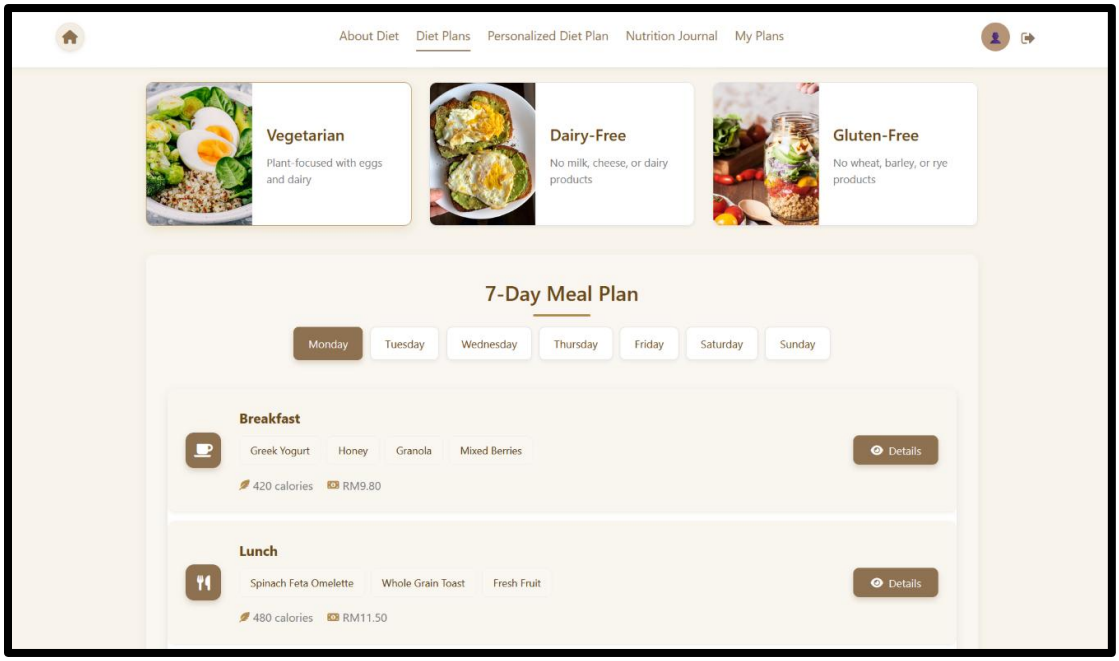


Figure 79: 7-Day Meal Plan I

For each selected plan type, a 7-day meal plan is displayed, with daily breakdowns for breakfast, lunch, dinner, and snacks. Each meal includes calorie counts, estimated prices, and a "Details" button for more information.

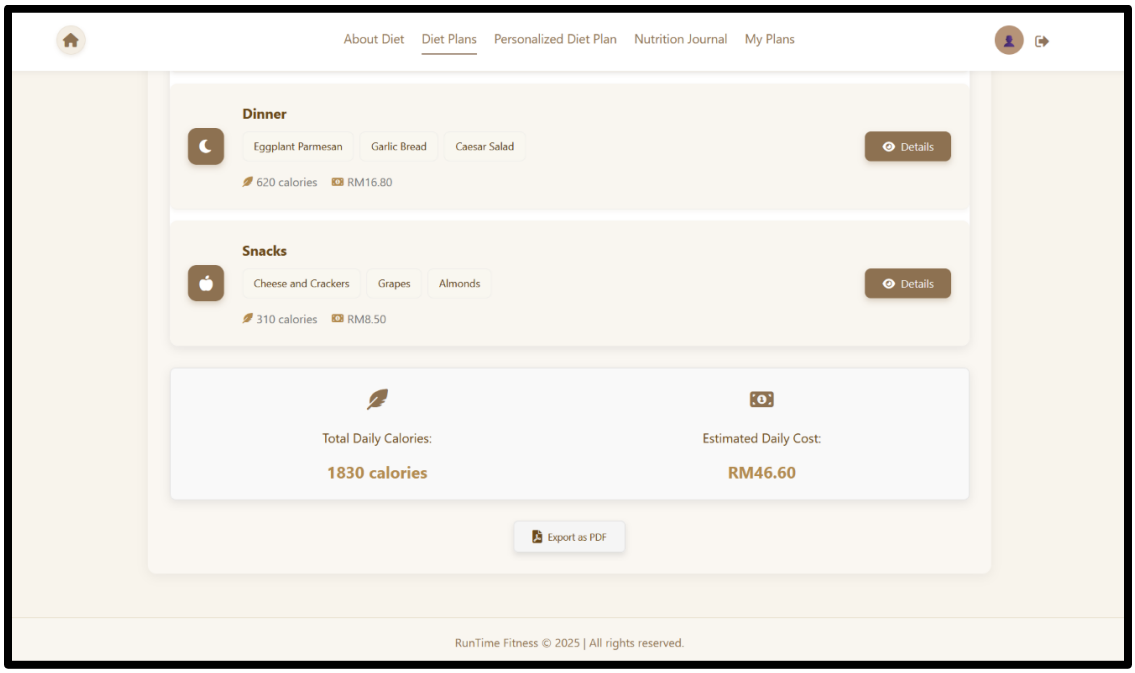


Figure 80: 7-Day Meal Plan II

Each daily meal plan includes total daily calories and estimated daily costs for all meals combined. An "Export as PDF" button, allowing users to save their meal plan.

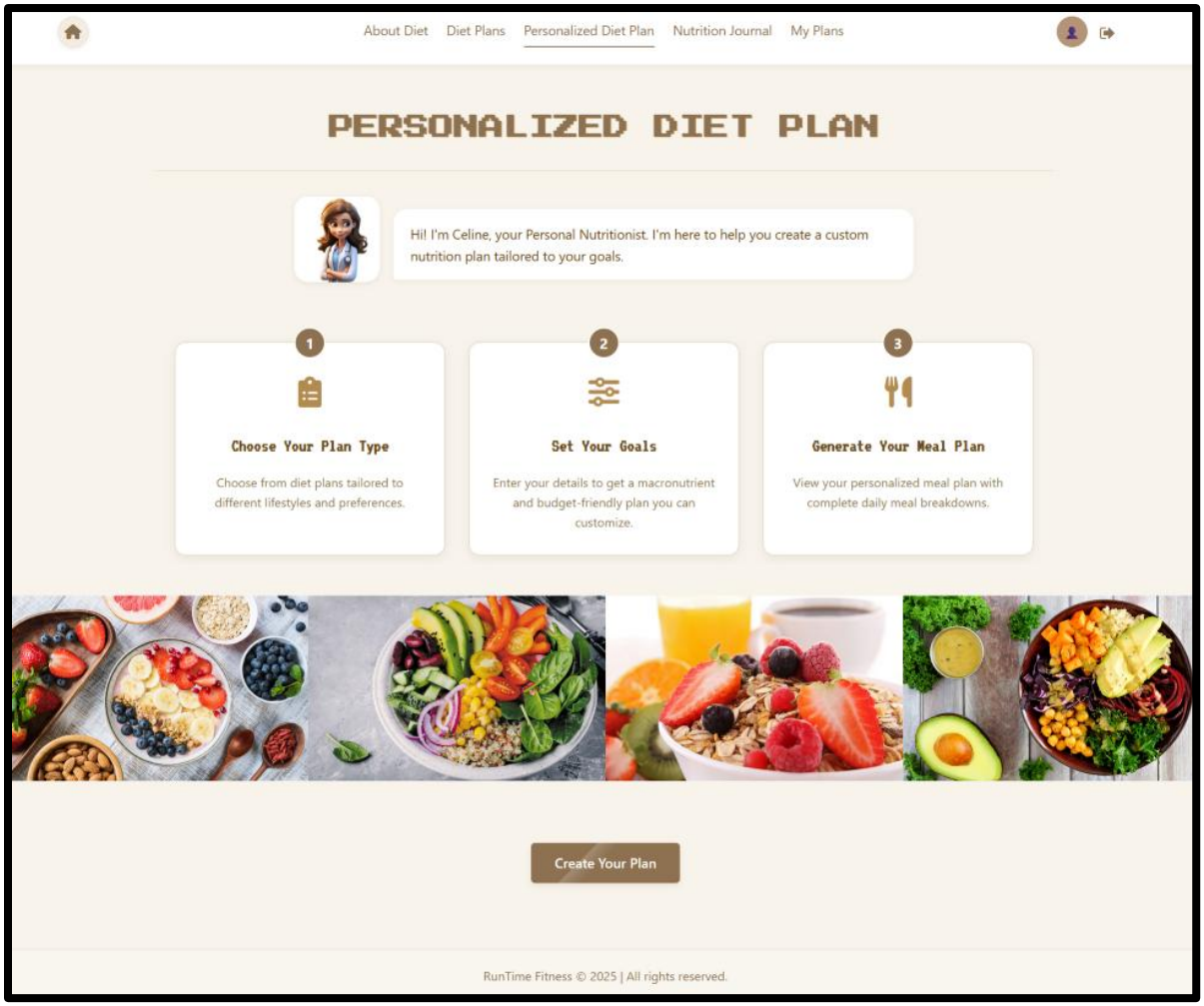


Figure 81: Personalized Diet Plan

The Personalized Diet Plan page introduces a 3-step process for creating a custom nutrition plan: (1) Choose Your Plan Type, (2) Set Your Goals, and (3) Generate Your Meal Plan. A button is provided to start the process.



Figure 82: Step 1 of Personalized Diet Plan

In the first step, users select their preferred diet type (e.g., Regular, Keto, Vegan, Vegetarian, Dairy-Free, Gluten-Free) to tailor the plan to their lifestyle and preferences.

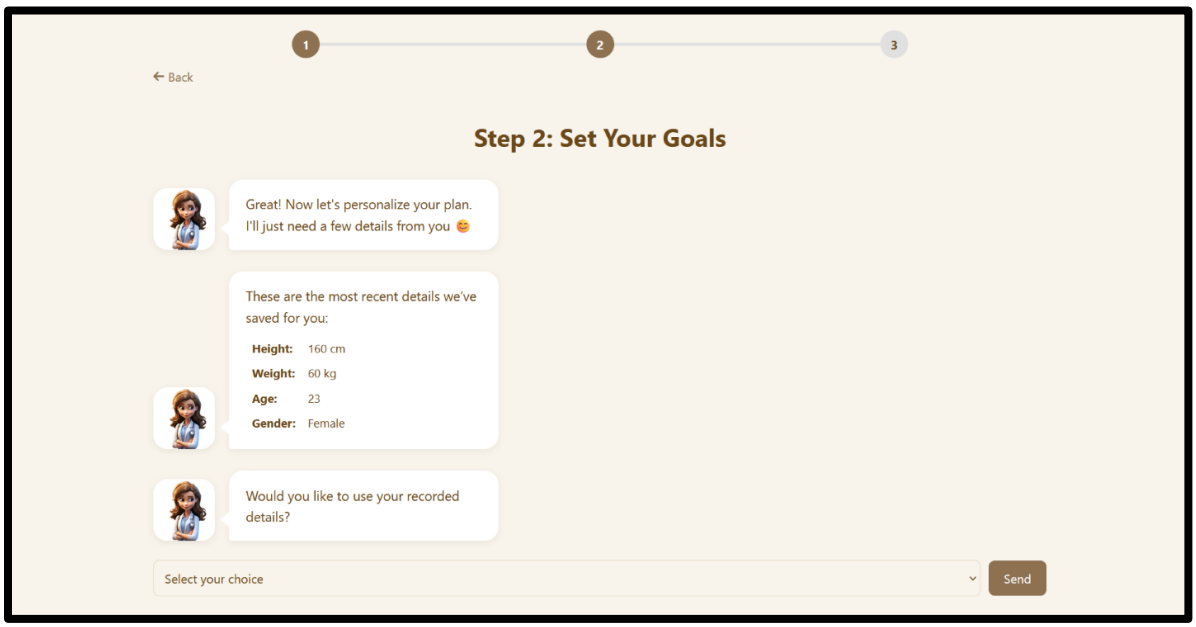


Figure 83: Step 2 of Personalized Diet Plan I

The system, via a chat interface with the virtual nutritionist, first asks if the user wants to use their previously recorded details (from registration). If the user agrees, those details are used; otherwise, the system will prompt for new information.

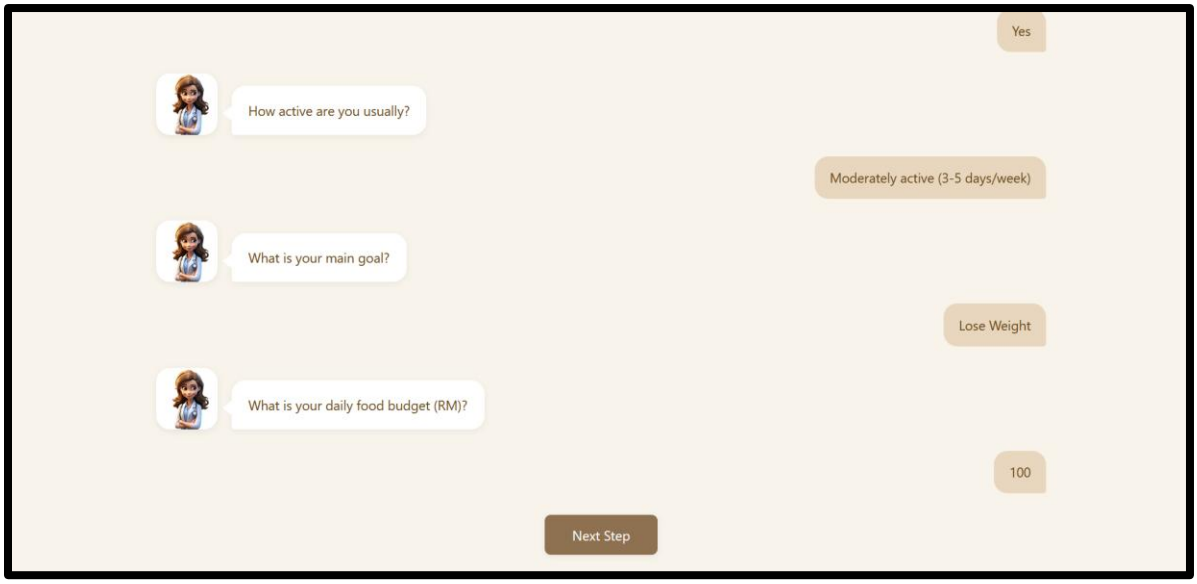


Figure 84: Step 2 of Personalized Diet Plan II

The nutritionist chat continues by asking about the user's activity level, main goal (e.g., lose weight), and daily food budget, gathering all necessary information to personalize the meal plan.

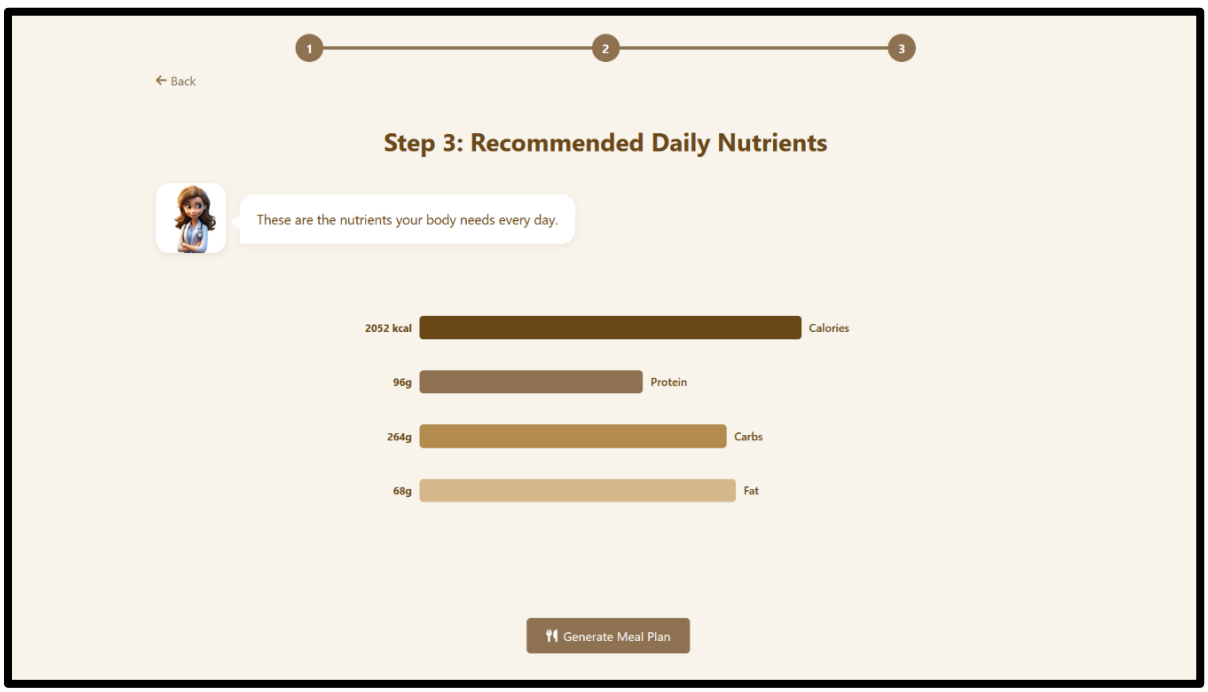


Figure 85: Step 3 of Personalized Diet Plan

Based on the user's information and goals, the system displays the recommended daily intake for calories, protein, carbs, and fats. A "Generate Meal Plan" button allows the user to proceed and create a personalized meal plan.



Figure 86: Loading Page of Personalized Diet Plan

A loading page appears while the system generates the user's personalized meal plan.

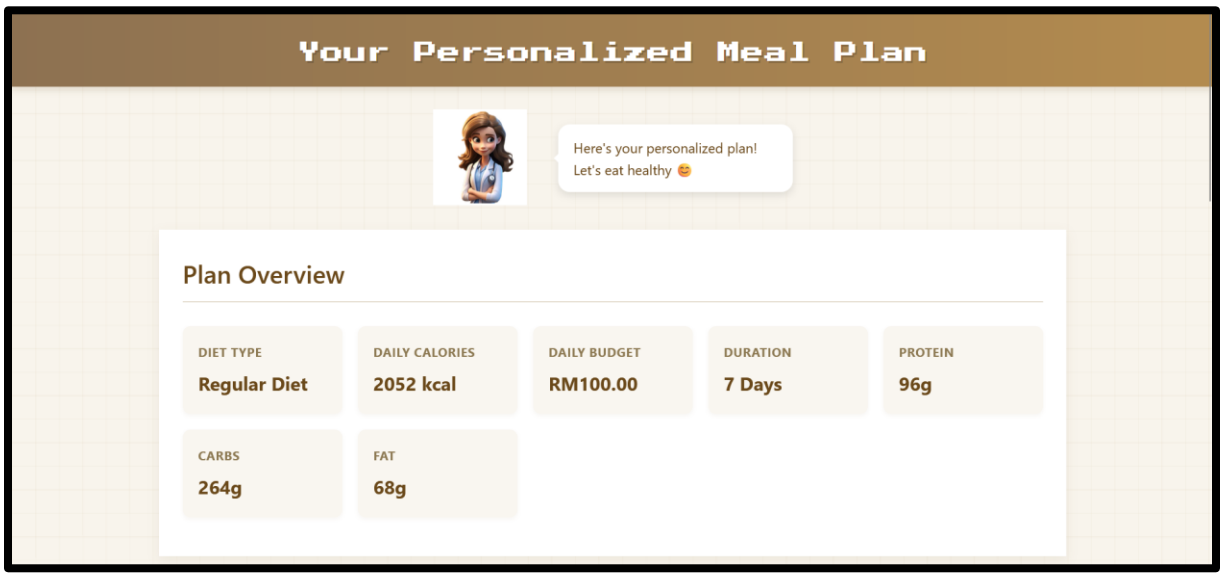


Figure 87: Generated Personalized Diet Plan I

The generated meal plan begins with an overview, summarizing the selected diet type, daily calories, budget, duration, and macronutrient targets (protein, carbs, fat).

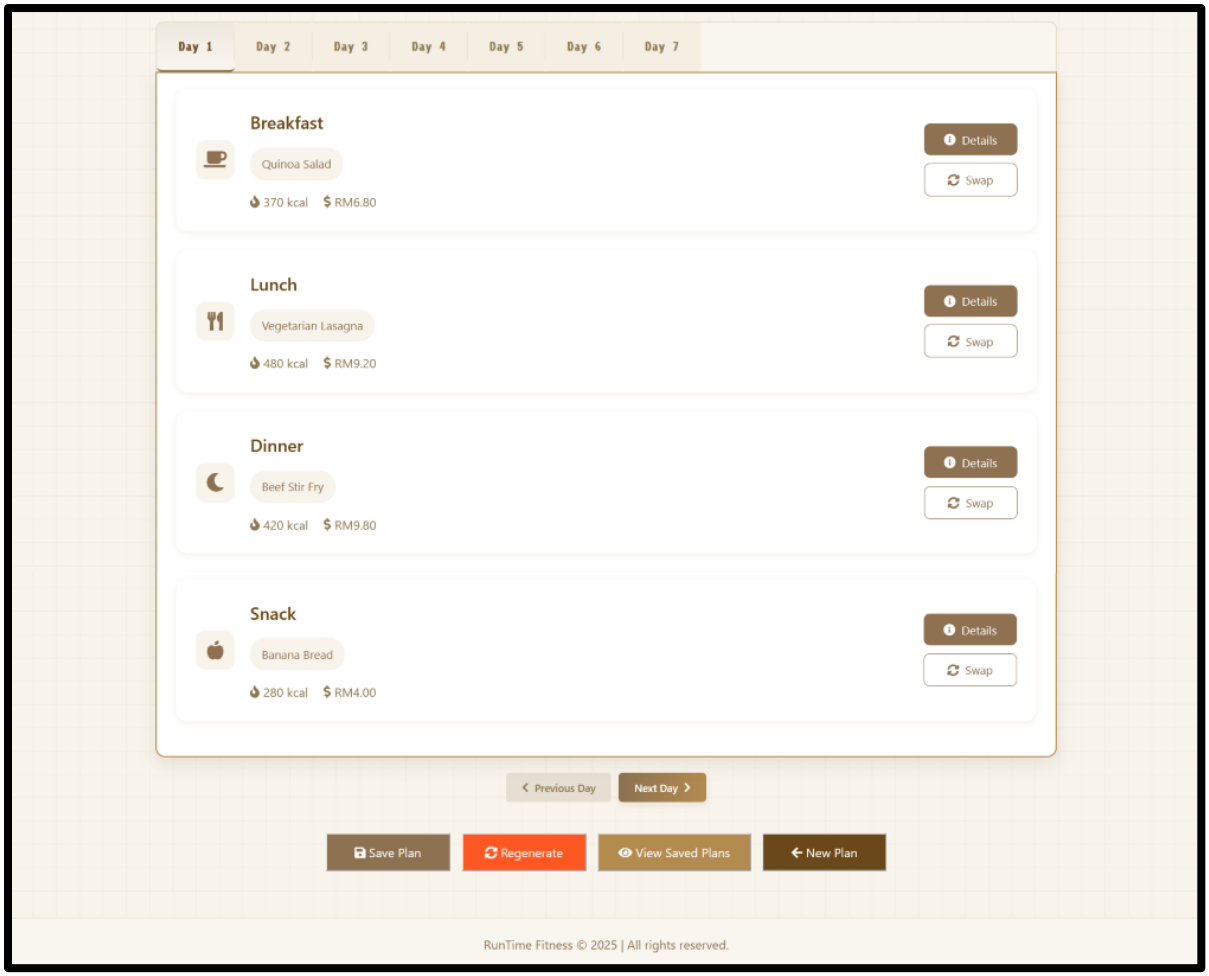


Figure 88: Generated Personalized Diet Plan II

The user receives a detailed 7-day meal plan, with each day broken down into meals (breakfast, lunch, dinner, snack), showing calories, estimated price, and options to view details or swap meals. Users can save the plan to "My Plans," regenerate a new plan, view all saved plans, or start a new plan from scratch.



Figure 89: Meal Details I

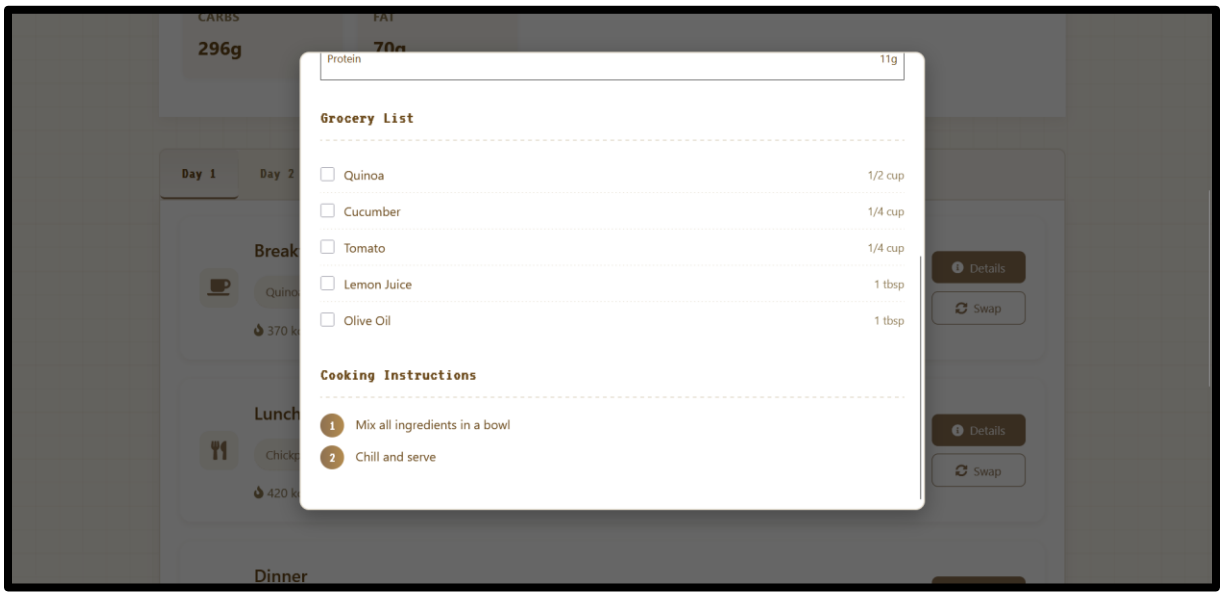


Figure 90: Meal Details II

When the user clicks the "Details" button for a meal, a popup appears with comprehensive information, including nutritional values (calories, fat, carbs, protein per serving), a grocery list with ingredient quantities, and step-by-step cooking instructions.

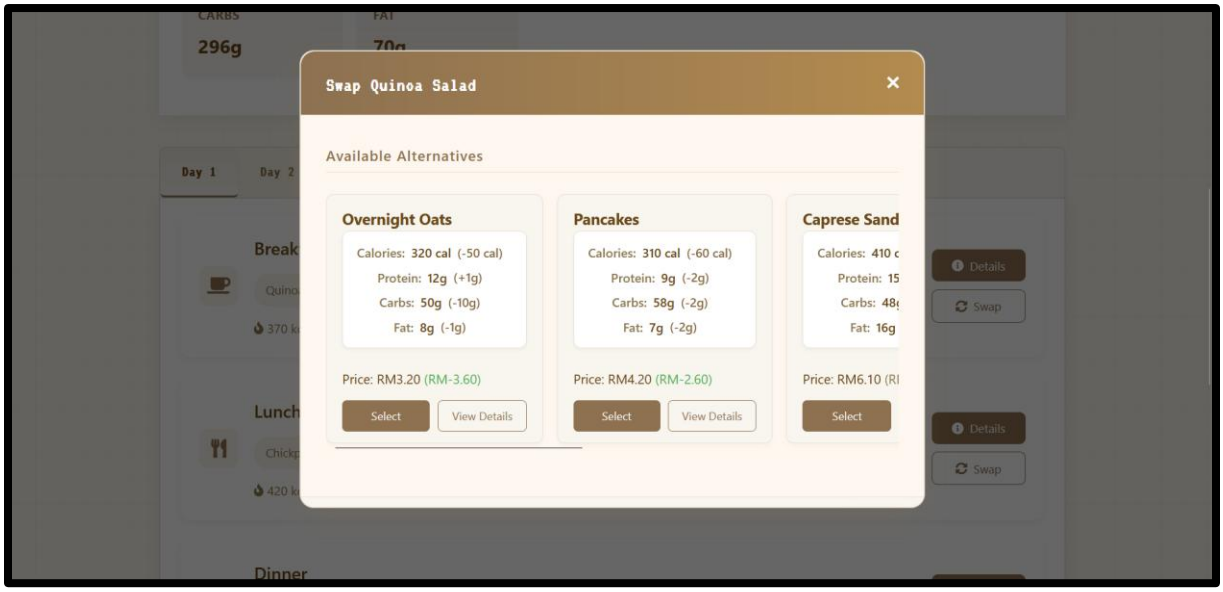


Figure 91: Swap Meal Popup I

Clicking the "Swap" button opens a list of available alternative meals, each displaying calories, macronutrients, price, and options to "Select" or "View Details" for more information.

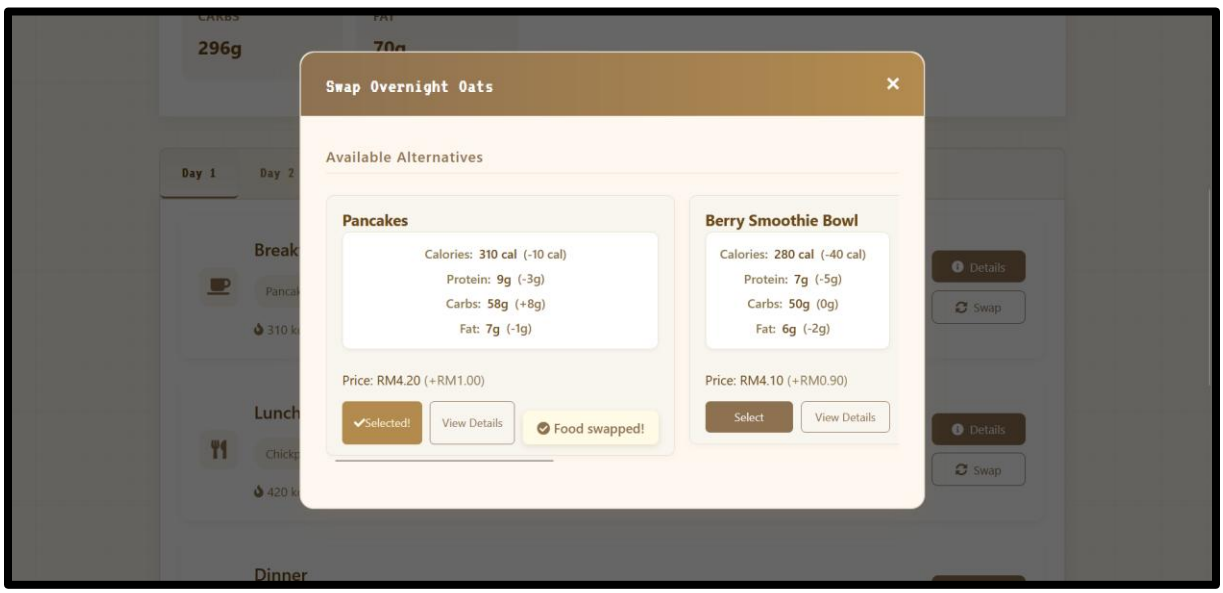


Figure 92; Swap Meal Popup II

After clicking "Select" on an alternative meal, the original meal is replaced with the chosen alternative in the meal plan, updating the plan accordingly.



Figure 93: Nutrition Journal I

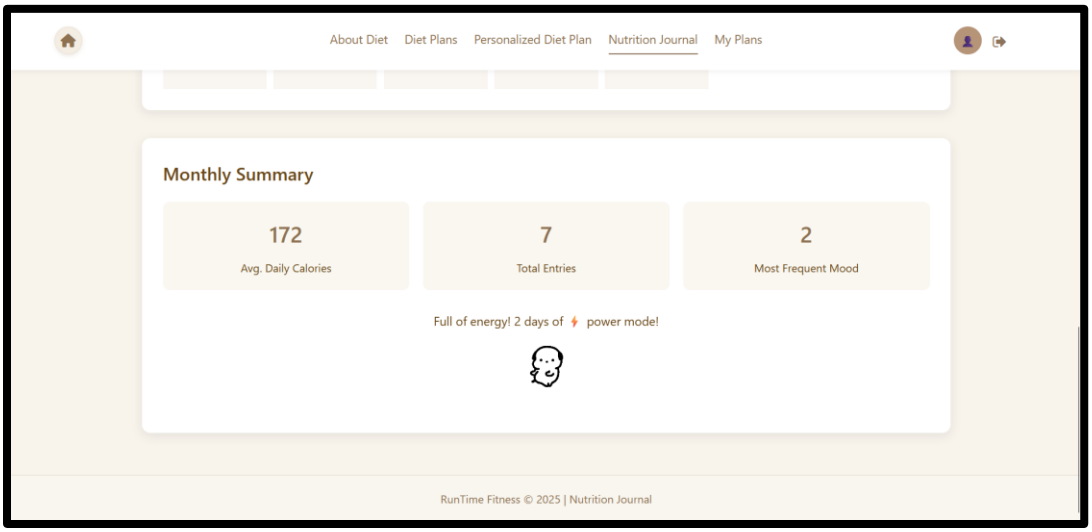


Figure 94: Nutrition Journal II

The Nutrition Journal page displays a calendar view where users can track their daily meals, calories, and moods. At the bottom, a Monthly Summary provides insights such as average daily calories, total entries, and the most frequent mood, helping users monitor their nutrition habits over time.

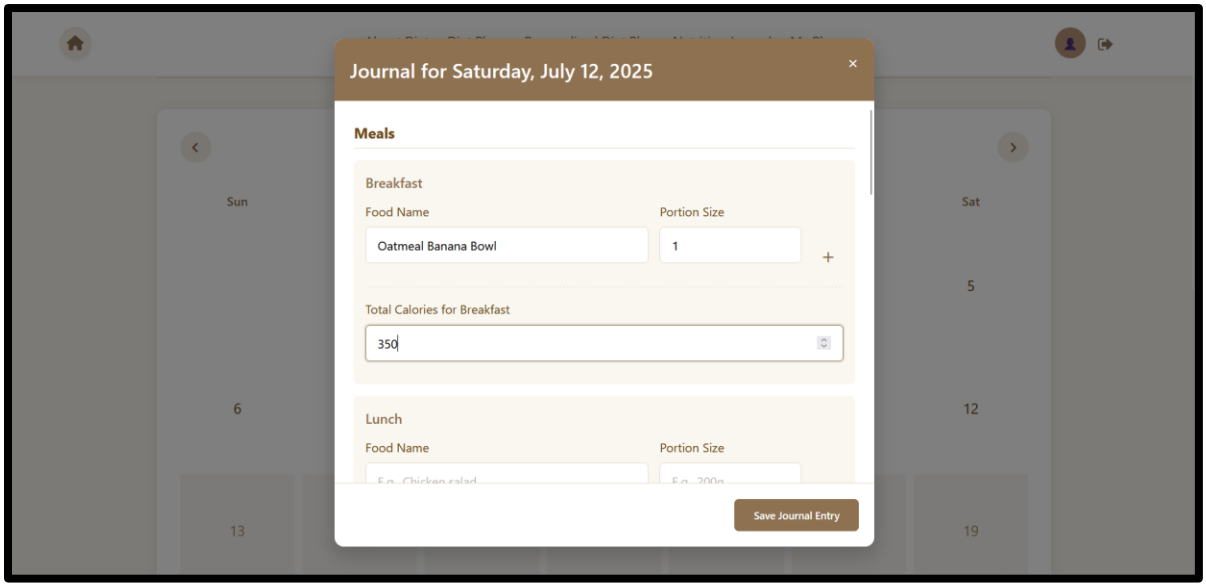


Figure 95: Journal Entry I

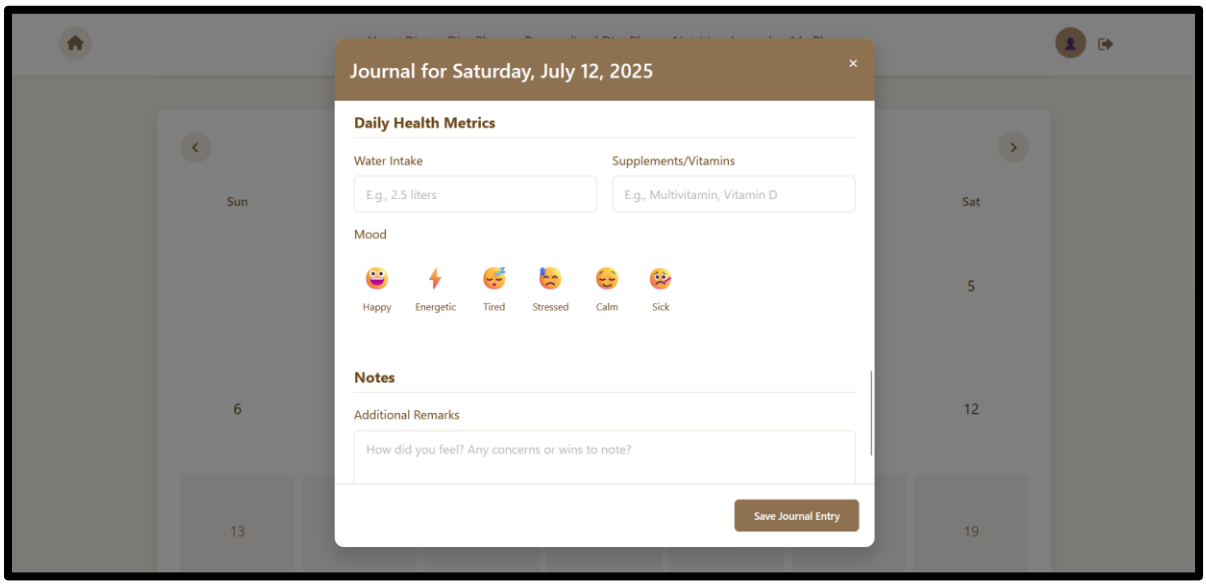


Figure 96: Journal Entry II

When adding or editing a journal entry for a specific day, users can log meal details (food name, portion size, calories), daily health metrics (water intake, supplements/vitamins), mood, and additional notes. This comprehensive entry helps users keep detailed records of their nutrition and well-being.

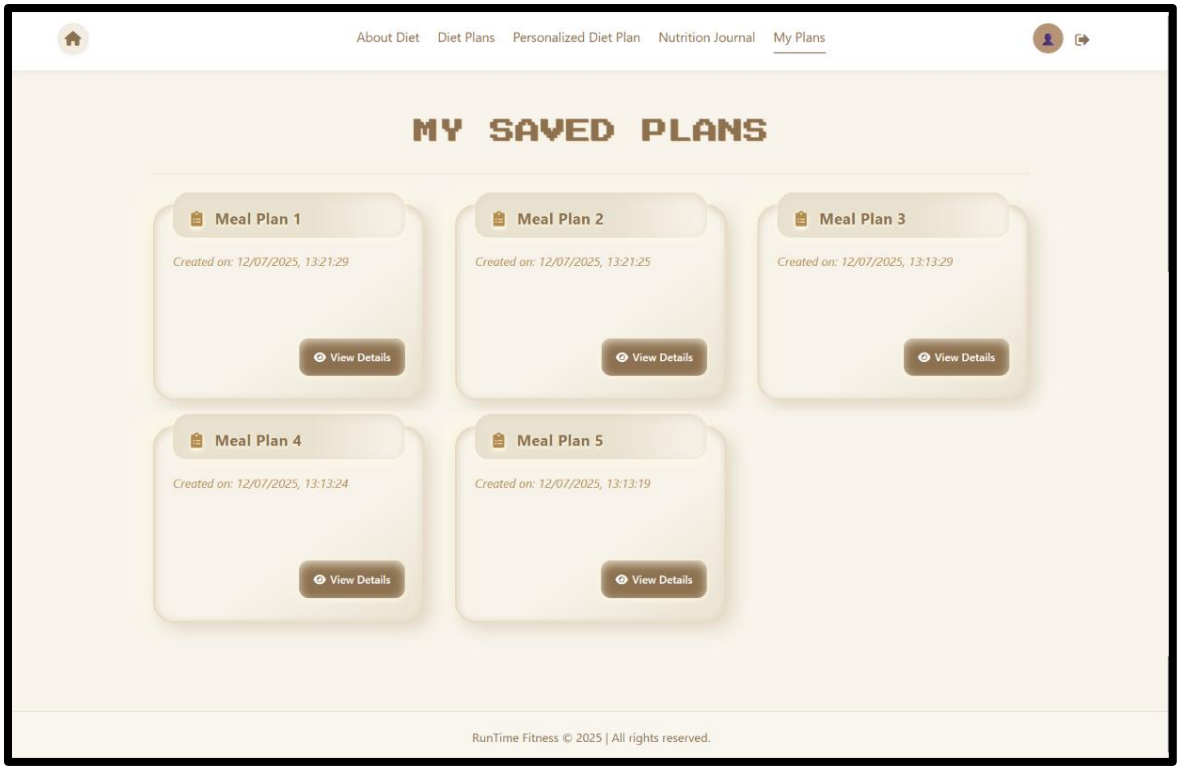


Figure 97: My Plans

The My Plans page lists all meal plans that the user has saved. Each plan shows its creation date and provides a "View Details" button, allowing users to revisit and manage their personalized or fixed meal plans at any time.



Figure 98: Saved Plan Details

This page displays the details of a saved meal plan, allowing users to review each day's meals, total calories, and price. Users can delete the plan or export it as a PDF for easy reference.

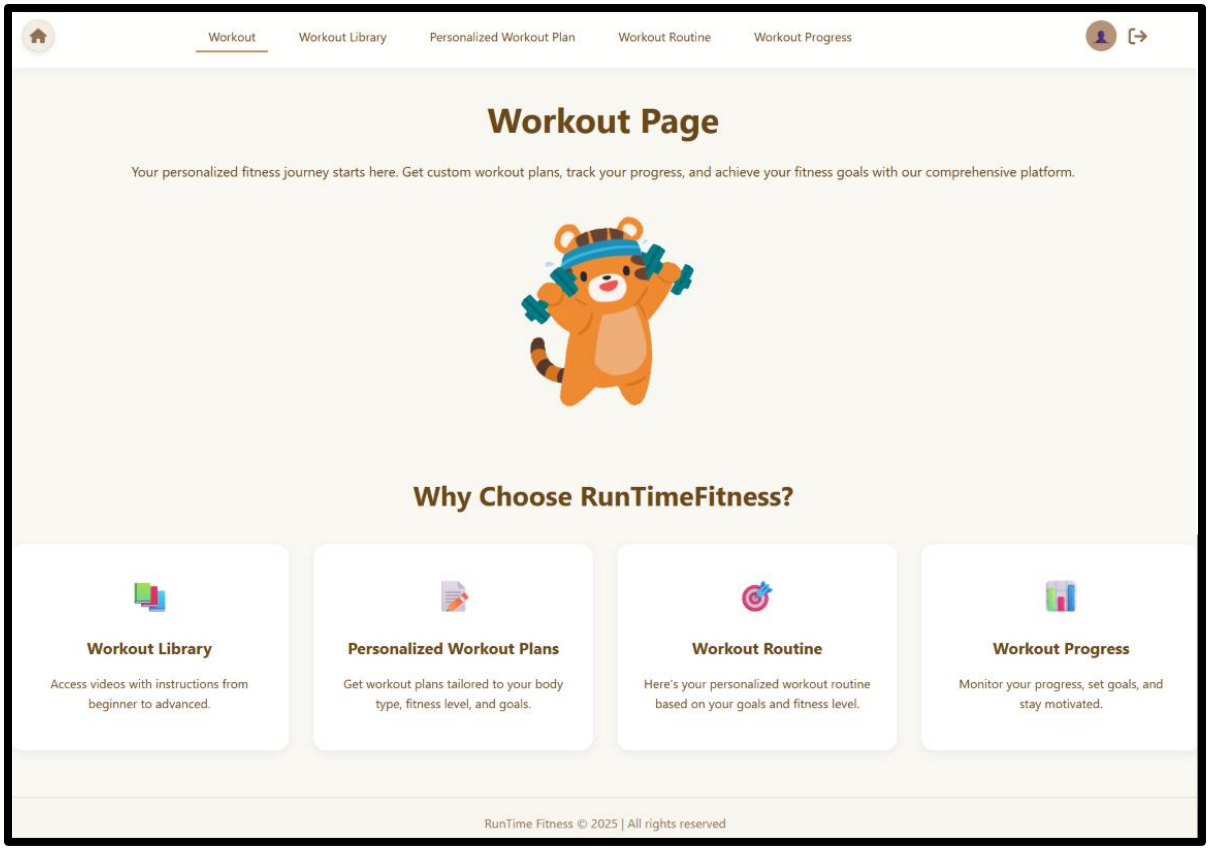


Figure 99: About Workout

About Workout welcomes users to the fitness section, highlighting features like the workout library, personalized workout plans, routines, and progress tracking to help users achieve their fitness goals.

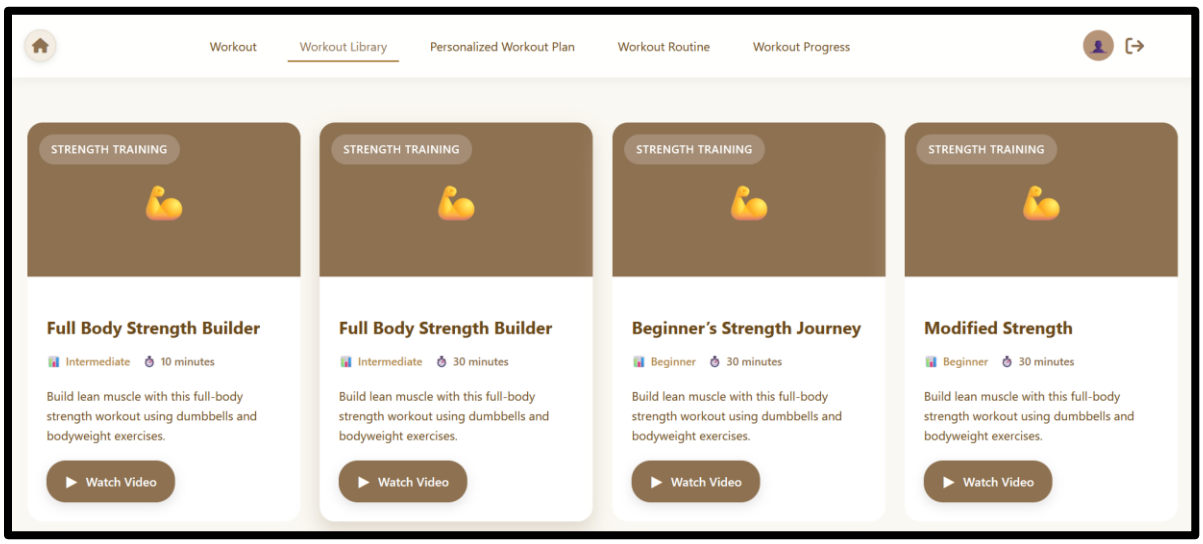


Figure 100: Workout Library I

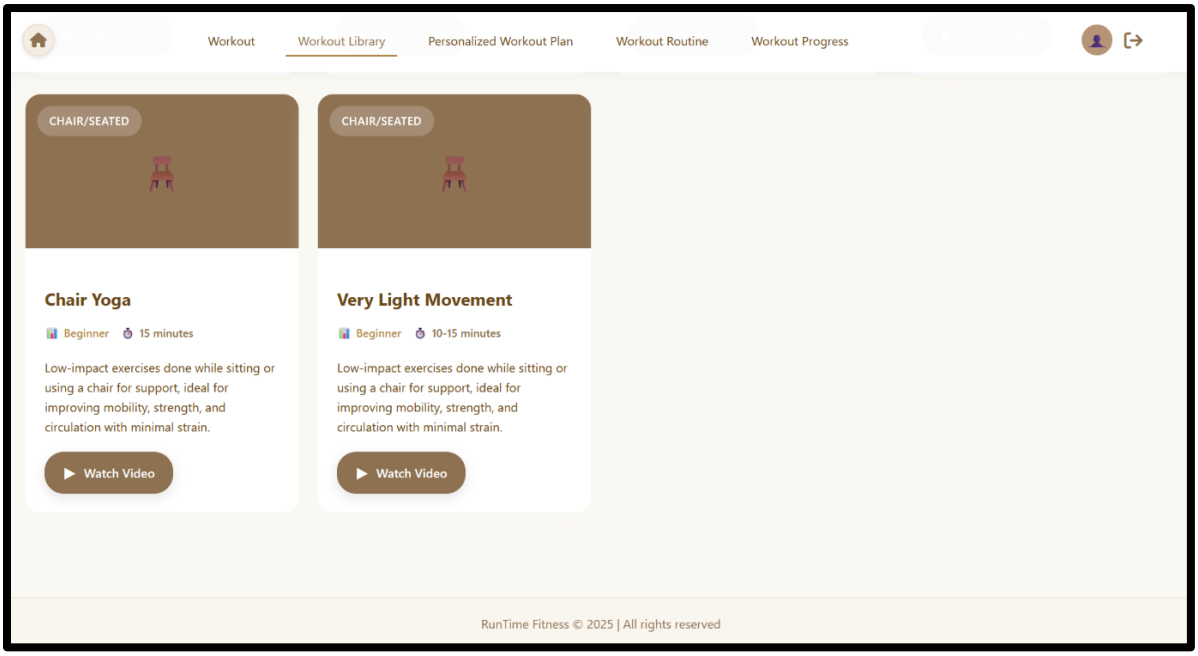


Figure 101: Workout Library II

The Workout Library provides access to a variety of workout videos and routines, categorized by type and difficulty, so users can find and follow exercises that suit their fitness level.

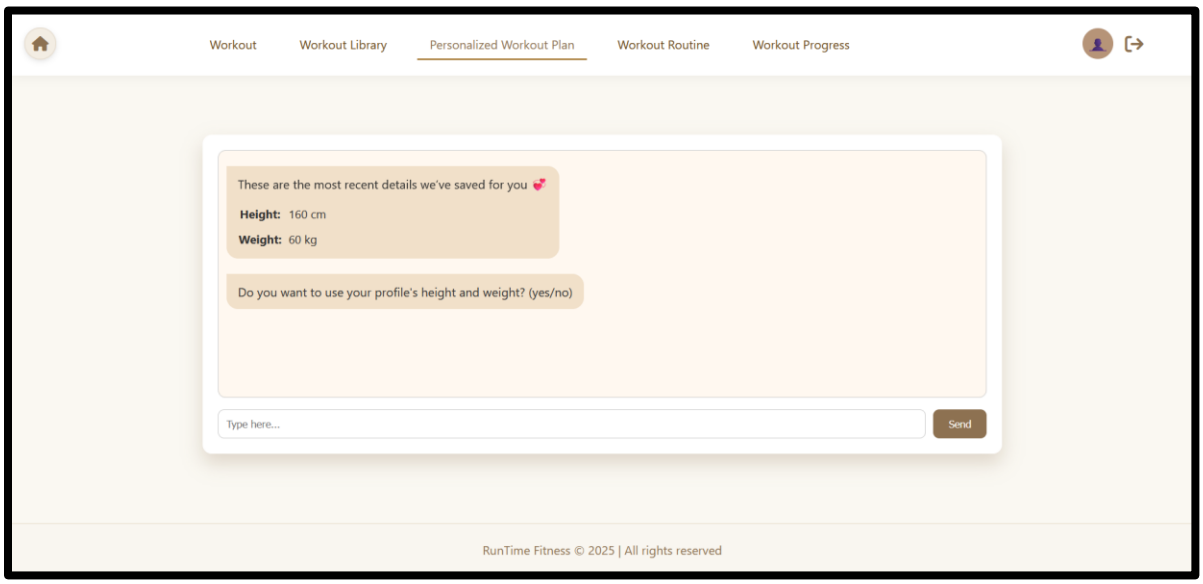


Figure 102: Personalized Workout Plan

The Personalized Workout Plan uses a chat interface to gather user details (such as height and weight) and then calculates the user's BMI. Based on the BMI and user profile, it recommends suitable workout types and generates a 7-day workout plan. The plan is displayed in a structured format and saved for the user to access later in the "Workout Routine" section.

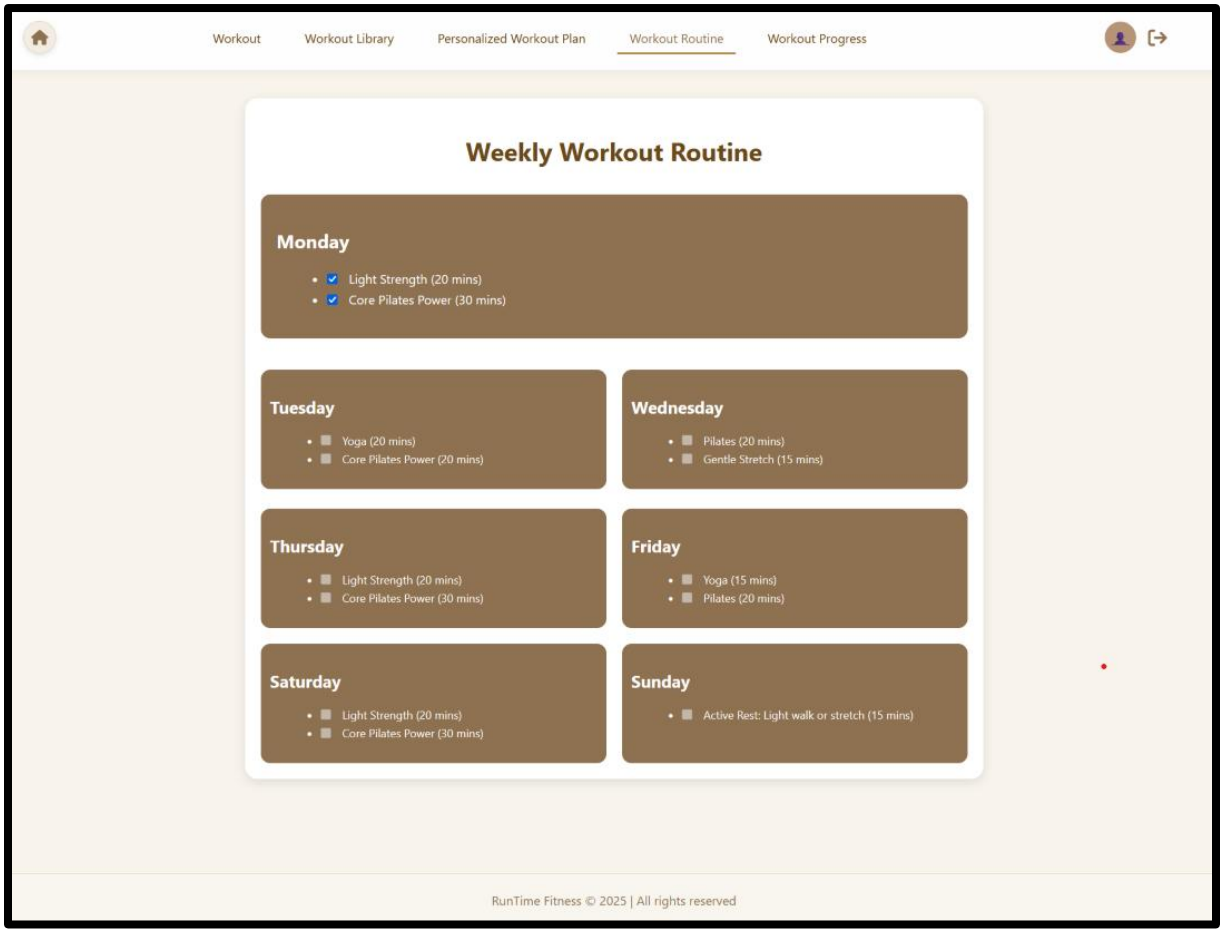


Figure 103: Workout Routine

This page displays the user's weekly workout routine, listing exercises for each day. Users can tick off (mark as completed) the workouts for the current day only, helping them track their daily progress and stay accountable, just like checking off a real-life workout calendar.

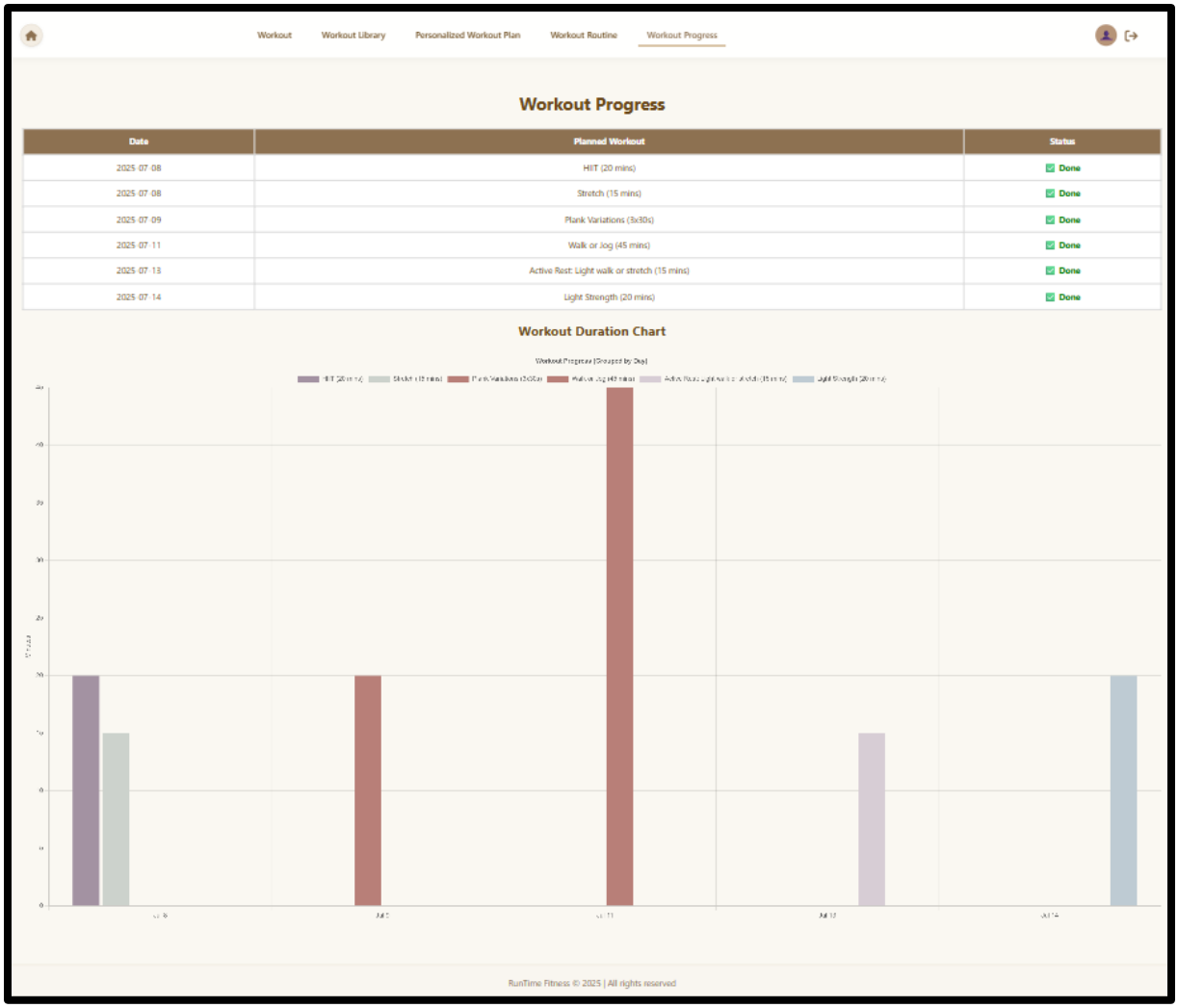


Figure 104: Workout Progress

The Workout Progress page provides a detailed report of the user's workout history. It includes a table listing each workout by date, planned activity, and completion status ("Done"), as well as a bar chart visualizing workout durations and types over time, making it easy to monitor consistency and achievements.

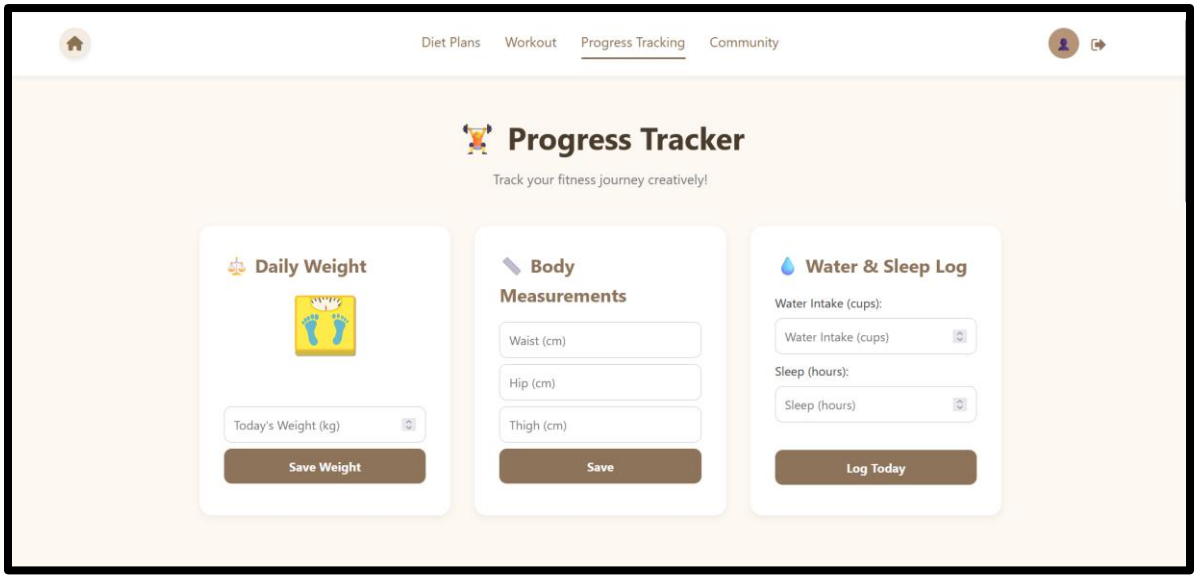


Figure 105: Progress Tracker I

The Progress Tracker page allows users to log their daily weight, body measurements (waist, hip, thigh), and water/sleep intake. This helps users keep a detailed record of their fitness journey and lifestyle habits.

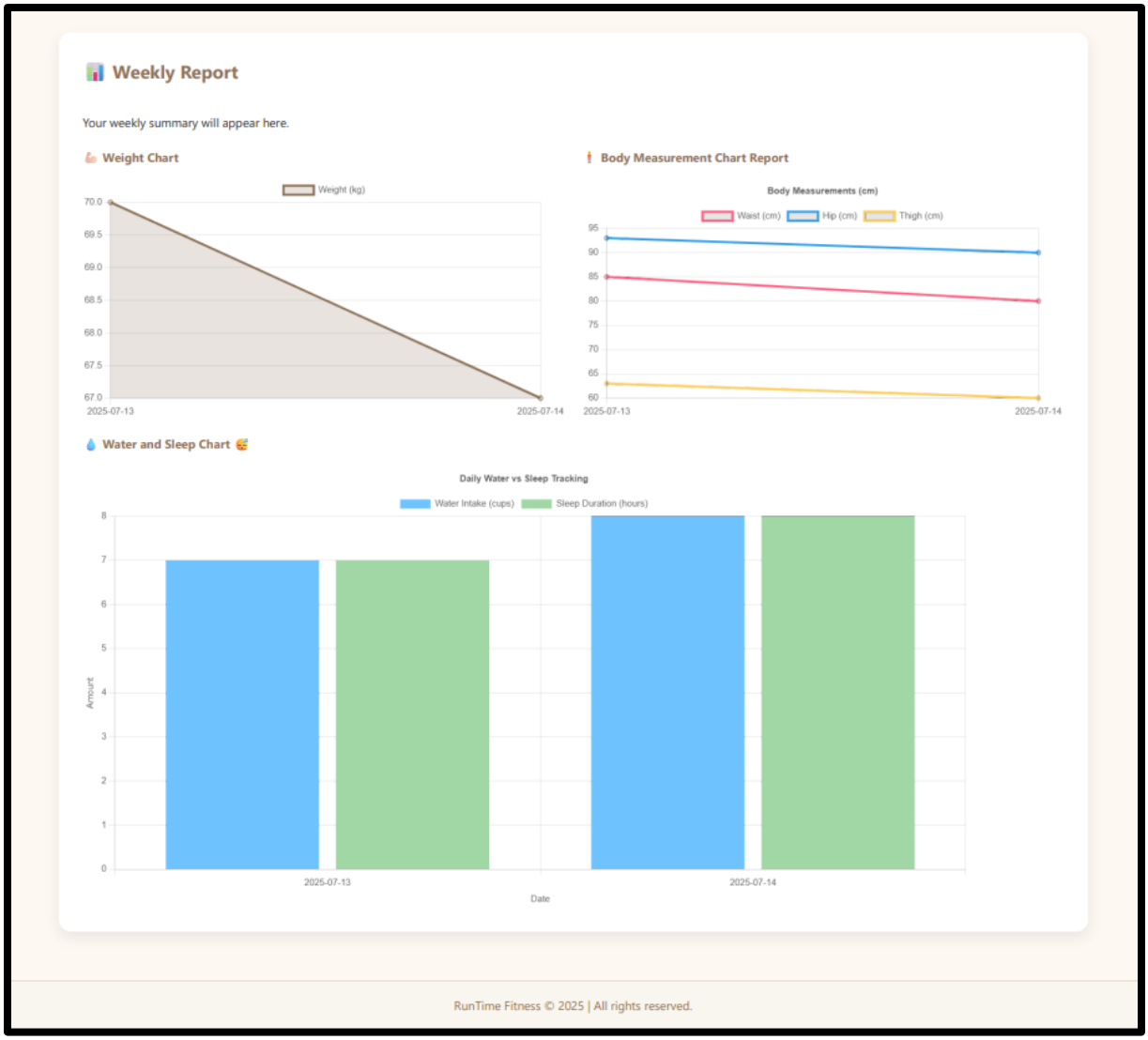


Figure 106: Progress Tracker II

The Weekly Report offers users a visual summary of their fitness progress over the week, featuring a Weight Chart that displays daily weight changes using a line graph, a Body Measurement Chart Report that tracks waist, hip, and thigh measurements through a multi-line graph, and a Water and Sleep Chart that uses a bar chart to compare daily water intake (in cups) and sleep duration (in hours).

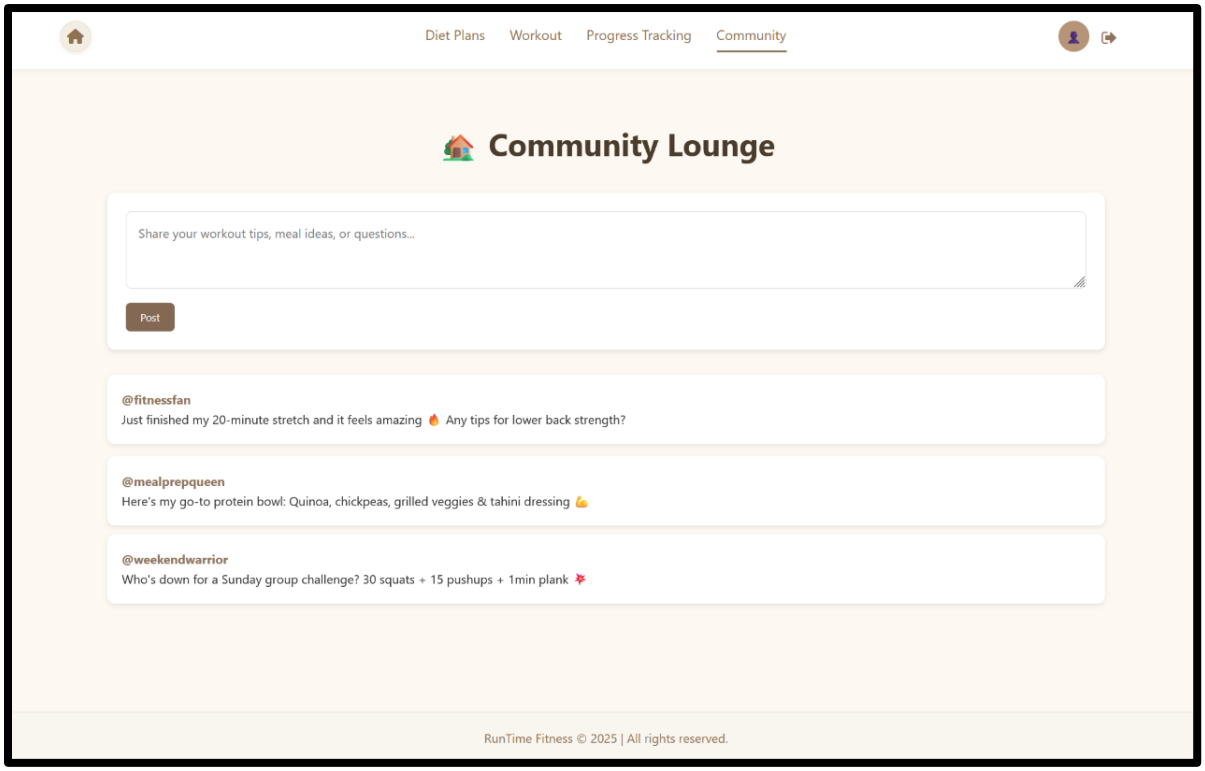


Figure 107: Community

The Community Lounge is a forum where users can share workout tips, meal ideas, achievements, and questions with others. Users can post new messages using a simple form, and all posts appear in a feed for the community to see. This feature fosters motivation, support, and knowledge sharing among members of RunTime Fitness.

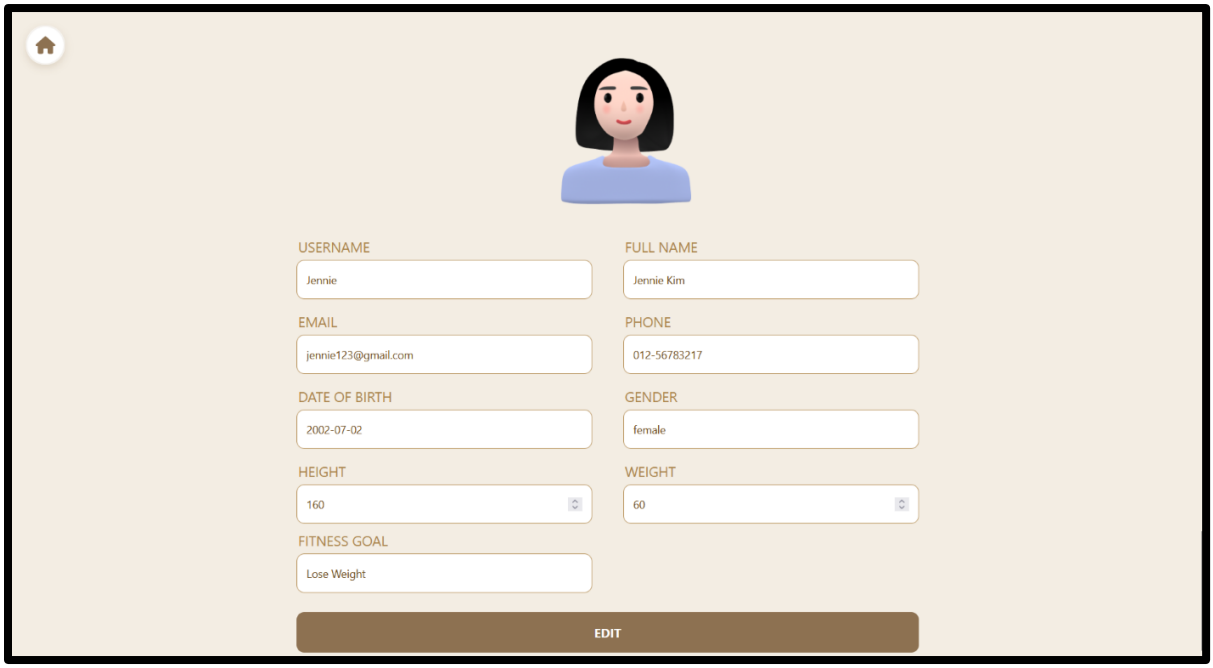


Figure 108: User Profile

This page displays the user's personal profile information, including username, full name, email, phone, date of birth, gender, height, weight, and fitness goal. The user can review and edit their details by clicking the "Edit" button.

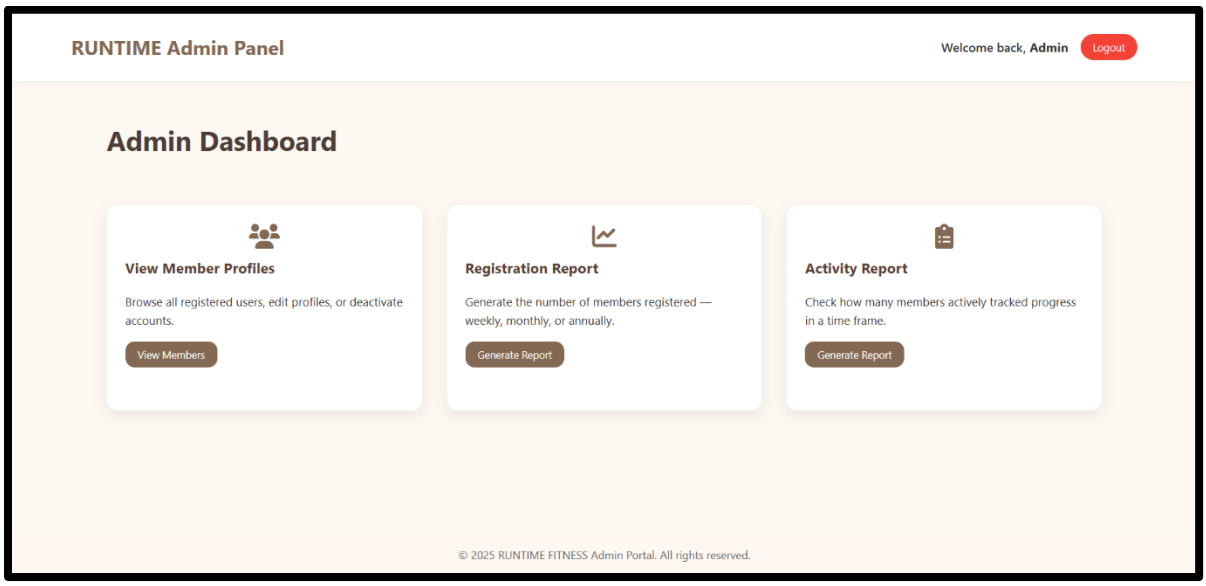


Figure 109: Admin Dashboard

The admin dashboard including viewing member profiles, generating registration reports, and generating activity reports to monitor user engagement and registrations.

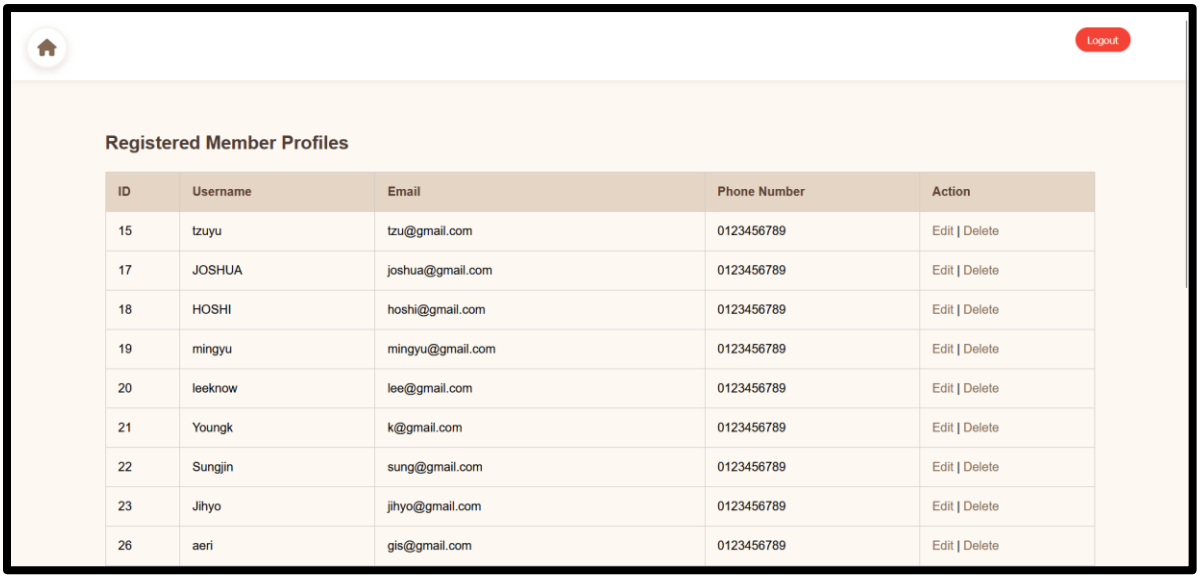


Figure 110: Registered Member Profiles

This page lists all registered members in a table format, showing their ID, username, email, and phone number. Admins can edit or delete each member's profile using the corresponding action buttons.

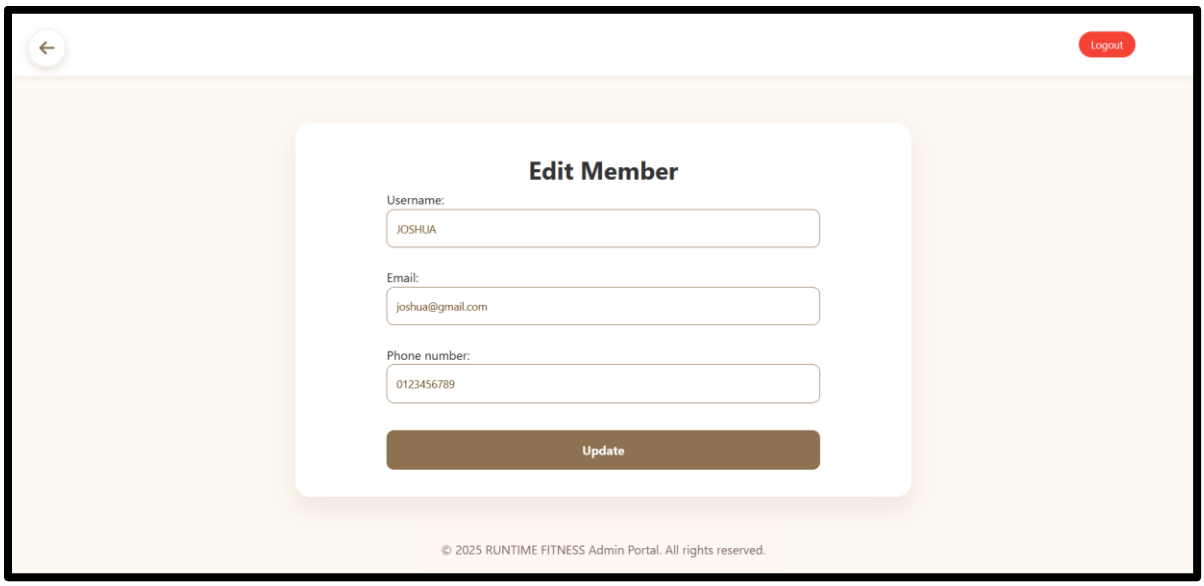


Figure 111: Edit Member Account Details

This page allows the admin to update a member's username, email, or phone number. After making changes, the admin can click "Update" to save the edits.

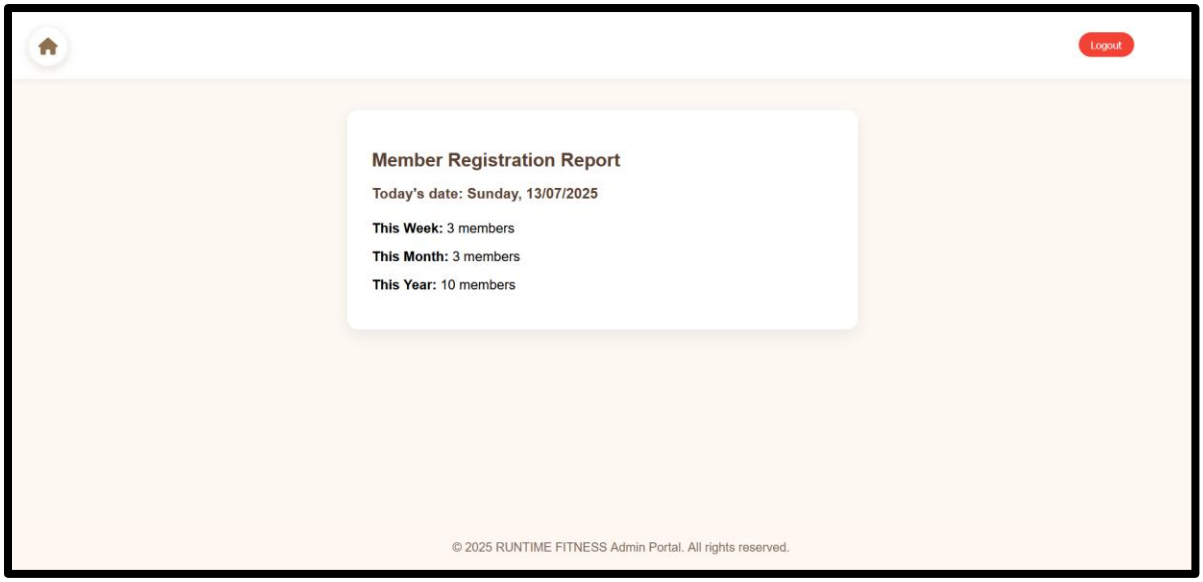


Figure 112: Member Registration Report

This page displays a summary of new member registrations, showing how many users registered this week, this month, and this year. The data is dynamically generated using the report_registration.php file, which queries the database for registration counts based on the current date, week, month, and year.

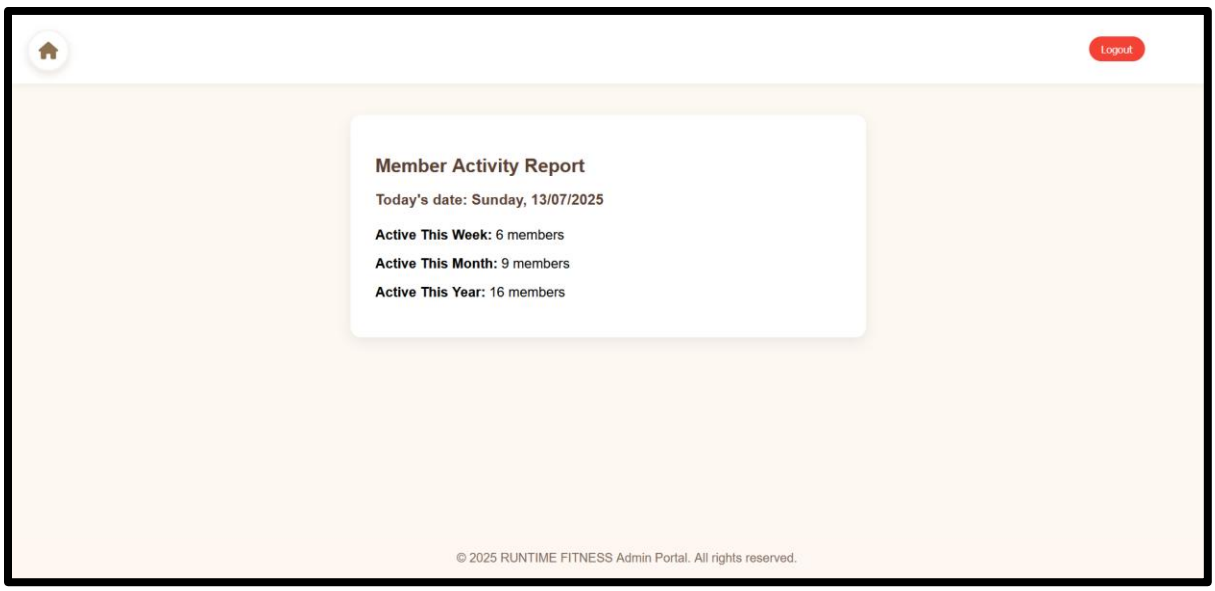


Figure 113: Member Activity Report

This page shows how many members have been active during the current week, month, and year. The statistics are generated by the report_activity.php file, which counts users based on their last login date for each time period.

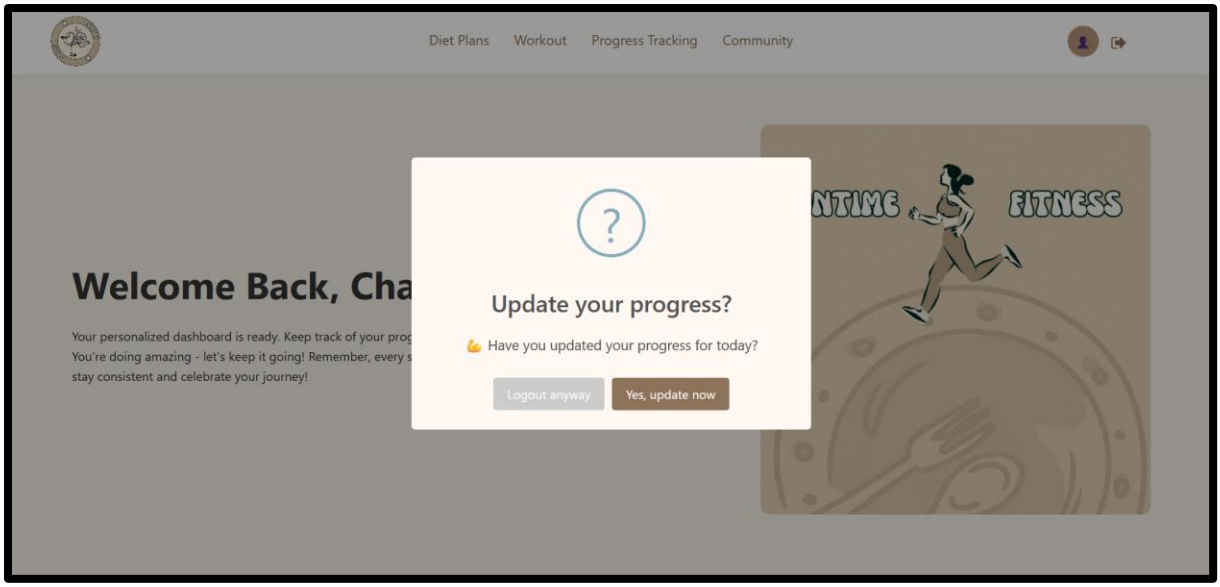


Figure 114: Logout I

When a user or admin logs out, a confirmation prompt may appear asking if they have updated their progress for the day.

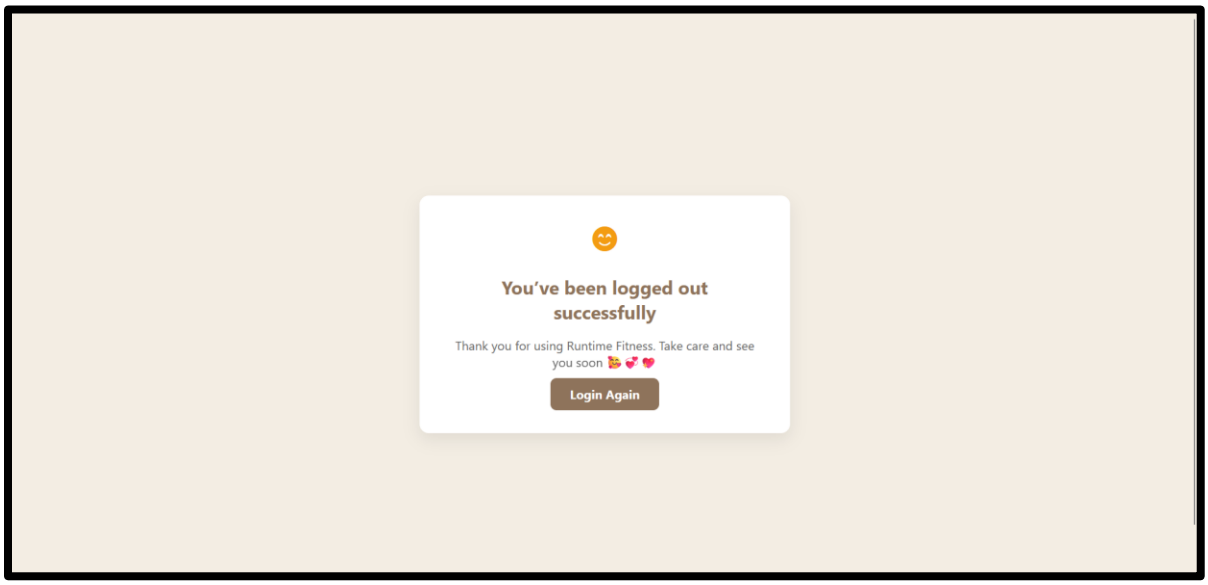


Figure 115: Logout II

After confirming logout, a message is displayed indicating that the user has been logged out successfully, with an option to log in again.

8.2 Significant Source Codes

Personalized Diet Plan —— DP.PS.GeneratedPlan.js

```
// Load user preferences from localStorage
function loadUserPreferences() {
    try {
        const planType = localStorage.getItem('ps_planType');
        const height = localStorage.getItem('ps_height');
        const weight = localStorage.getItem('ps_weight');
        const age = localStorage.getItem('ps_age');
        const gender = localStorage.getItem('ps_gender');
        const activity = localStorage.getItem('ps_activity');
        const goal = localStorage.getItem('ps_goal');
        const duration = localStorage.getItem('ps_duration');
        const budget = localStorage.getItem('ps_budget');

        const nutritionData = JSON.parse(localStorage.getItem('ps_nutrition') || '{}');

        // Store all preferences in the current state
        currentUserPreferences = {
            planType: planType || 'regular',
            height: parseFloat(height) || 170,
            weight: parseFloat(weight) || 70,
            age: parseInt(age) || 30,
            gender: gender || 'male',
            activity: activity || 'moderate',
            goal: goal || 'maintain',
            duration: duration || '7',
            budget: parseFloat(budget) || 50,
            calories: nutritionData.calories || 2000,
            protein: nutritionData.protein || 125,
            carbs: nutritionData.carbs || 250,
            fat: nutritionData.fat || 67
        };
        // Check if there's a saved meal plan
        const savedMealPlan = localStorage.getItem('generatedMealPlan');
        if (savedMealPlan) {
            currentMealPlan = JSON.parse(savedMealPlan);
        }

    } catch (error) {
        console.error('Error loading user preferences:', error);

        // Default values if loading fails
        currentUserPreferences = {
            planType: 'regular',
            height: 170,
            weight: 70,
            age: 30,
            gender: 'male',
            activity: 'moderate',
            goal: 'maintain',
            duration: '7',
            budget: 50,
            calories: 2000,
            protein: 125,
            carbs: 250,
            fat: 67
        };
    }
}
```

Figure 116: Load User Preferences

```
// Update plan summary information
function updateSummaryInfo() {
    const dietTypeNames = {
        'regular': 'Regular Diet',
        'keto': 'Keto Diet',
        'vegan': 'Vegan Diet',
        'vegetarian': 'Vegetarian Diet',
        'pescatarian': 'Pescatarian Diet',
        'dairy-free': 'Dairy-Free Diet',
        'gluten-free': 'Gluten-Free Diet',
        'low-carb': 'Low Carb Diet',
        'home-cook': 'Simple Home Cooking'
    };

    const durationText = {
        '1': '1 Day',
        '3': '3 Days',
        '7': '7 Days'
    };

    // Update DOM elements with user preferences
    dietTypeEl.textContent = dietTypeNames[currentUserPreferences.planType] || currentUserPreferences.planType;
    dailyCaloriesEl.textContent = `${currentUserPreferences.calories} kcal`;
    dailyBudgetEl.textContent = `RM${currentUserPreferences.budget.toFixed(2)}`;
    planDurationEl.textContent = durationText[currentUserPreferences.duration] || `${currentUserPreferences.duration} Days`;
    dailyProteinEl.textContent = `${currentUserPreferences.protein}g`;
    dailyCarbsEl.textContent = `${currentUserPreferences.carbs}g`;
    dailyFatEl.textContent = `${currentUserPreferences.fat}g`;
}
```

Figure 117: Update Summary Information

```
// Generate meal plan
function generatePlan() {
    loadingIndicator.style.display = 'block';

    daysTabsContainer.innerHTML = '';
    daysContentContainer.innerHTML = '';

    setTimeout(() => {
        try {
            const duration = 7;
            currentMealPlan = [];

            const usedMealsAcrossPlan = new Set();

            for (let i = 0; i < duration; i++) {
                const dailyPlan = window.PlanGenerator.generateDailyPlan(
                    currentUserPreferences.planType,
                    currentUserPreferences.calories,
                    currentUserPreferences.protein,
                    currentUserPreferences.carbs,
                    currentUserPreferences.fat,
                    currentUserPreferences.budget,
                    usedMealsAcrossPlan
                );
                currentMealPlan.push(dailyPlan);
            }
            localStorage.setItem('generatedMealPlan', JSON.stringify(currentMealPlan));
            renderTabbedMealPlan();
            loadingIndicator.style.display = 'none';
        } catch (error) {
            loadingIndicator.style.display = 'none';
            alert('Error generating meal plan. Please try again.');
        }
    }, 500);
}
```

Figure 118: Generate Meal Plan

```
function renderTabbedMealPlan() {
const numberOfDays = currentMealPlan.length;

daysTabsContainer.innerHTML = '';
daysContentContainer.innerHTML = '';

for (let i = 0; i < numberOfDays; i++) {
    const dayNumber = i + 1;
    // Create tab
    const tab = document.createElement('div');
    tab.className = `day-tab ${dayNumber === currentActiveDay ? 'active' : ''}`;
    tab.setAttribute('data-day', i);
    tab.textContent = `Day ${dayNumber}`;
    tab.addEventListener('click', () => selectDay(i, numberOfDays));
    daysTabsContainer.appendChild(tab);

    // Create content for this day
    const dayCard = document.createElement('div');
    dayCard.className = `day-card ${dayNumber === currentActiveDay ? 'active' : ''}`;
    dayCard.id = `day-${i}`;
    const mealContainer = document.createElement('div');
    mealContainer.className = 'meal-container';

    // Create sections for each meal type
    const mealTypes = ['breakfast', 'lunch', 'dinner', 'snack'];
    mealTypes.forEach(mealType => {
        const mealFoods = currentMealPlan[i][mealType];
        renderMealSection(mealType, Array.isArray(mealFoods) ? mealFoods : [mealFoods], mealContainer, i);
    });

    dayCard.appendChild(mealContainer);
    daysContentContainer.appendChild(dayCard);
}
```

Figure 119: Render Meal Plan

```
// Show meal details in modal
function showMealDetails(meal, title) {
    document.getElementById('modal-meal-title').textContent = title;
    document.getElementById('modal-calories').textContent = meal.calories + ' kcal';
    document.getElementById('modal-fat').textContent = meal.fat + 'g';
    document.getElementById('modal-carbs').textContent = meal.carbs + 'g';
    document.getElementById('modal-protein').textContent = meal.protein + 'g';

    const groceryContainer = document.getElementById('modal-grocery');
    groceryContainer.innerHTML = '';

    if (meal.groceryList && meal.groceryList.length > 0) {
        meal.groceryList.forEach(item => {
            const groceryItem = document.createElement('li');
            groceryItem.className = 'grocery-item';
            groceryItem.innerHTML = `
                <input type="checkbox" class="grocery-checkbox">
                <span class="grocery-name">${item.item || item.name}</span>
                <span class="grocery-quantity">${item.quantity}</span>
            `;
            groceryContainer.appendChild(groceryItem);
        });
    } else {
        groceryContainer.innerHTML = '<p>No grocery information available.</p>';
    }

    // Populate cooking instructions
    const instructionsContainer = document.getElementById('modal-instructions');
    instructionsContainer.innerHTML = '';

    if (meal.cookingInstructions && meal.cookingInstructions.length > 0) {
        meal.cookingInstructions.forEach(instruction => {
            const instructionItem = document.createElement('li');
            instructionItem.className = 'cooking-step';
            instructionItem.textContent = instruction;
            instructionsContainer.appendChild(instructionItem);
        });
    } else {
        instructionsContainer.innerHTML = '<p>No cooking instructions available.</p>';
    }

    // Show the modal
    const modal = document.getElementById('details-modal');
    if (modal) {
        modal.style.display = 'block';
    }
}
```

Figure 120: Show Meal Plan Details

Personalized Diet Plan —— DP.PS.Generator.js

```
const PlanGenerator = {
    // Generate meal plan for multiple days
    generateMealPlan: function (
        dietType,
        duration,
        targetCalories,
        targetProtein,
        targetCarbs,
        targetFat,
        budget
    ) {
        const fullPlan = [];

        for (let day = 0; day < duration; day++) {
            const dailyPlan = this.generateDailyPlan(
                dietType,
                targetCalories,
                targetProtein,
                targetCarbs,
                targetFat,
                budget
            );

            fullPlan.push(dailyPlan);
        }

        return fullPlan;
    },
```

Figure 121: Generate Multi-Day Meal Plan

```
// Generate meal plan for one day
generateDailyPlan: function (dietType, targetCalories, targetProtein, targetCarbs, targetFat, budgetPerDay) {
    const mealDistribution = {
        breakfast: 0.25,
        lunch: 0.30,
        dinner: 0.30,
        snack: 0.15
    };

    const mealTargetCalories = {
        breakfast: targetCalories * mealDistribution.breakfast,
        lunch: targetCalories * mealDistribution.lunch,
        dinner: targetCalories * mealDistribution.dinner,
        snack: targetCalories * mealDistribution.snack
    };

    const mealTargetProtein = {
        breakfast: targetProtein * mealDistribution.breakfast,
        lunch: targetProtein * mealDistribution.lunch,
        dinner: targetProtein * mealDistribution.dinner,
        snack: targetProtein * mealDistribution.snack
    };

    const mealTargetCarbs = {
        breakfast: targetCarbs * mealDistribution.breakfast,
        lunch: targetCarbs * mealDistribution.lunch,
        dinner: targetCarbs * mealDistribution.dinner,
        snack: targetCarbs * mealDistribution.snack
    };

    const mealTargetFat = {
        breakfast: targetFat * mealDistribution.breakfast,
        lunch: targetFat * mealDistribution.lunch,
        dinner: targetFat * mealDistribution.dinner,
        snack: targetFat * mealDistribution.snack
    };
const mealBudget = {
    breakfast: budgetPerDay * mealDistribution.breakfast,
    lunch: budgetPerDay * mealDistribution.lunch,
    dinner: budgetPerDay * mealDistribution.dinner,
    snack: budgetPerDay * mealDistribution.snack
};

const usedToday = {
    breakfast: new Set(),
    lunch: new Set(),
    dinner: new Set(),
    snack: new Set()
};

const dailyPlan = {
    breakfast: this.selectBestMeal('breakfast', dietType, mealTargetCalories.breakfast, mealTargetProtein.breal
    lunch: this.selectBestMeal('lunch', dietType, mealTargetCalories.lunch, mealTargetProtein.lunch, mealTarge
    dinner: this.selectBestMeal('dinner', dietType, mealTargetCalories.dinner, mealTargetProtein.dinner, mealT
    snack: this.selectBestMeal('snack', dietType, mealTargetCalories.snack, mealTargetProtein.snack, mealTarge
};

dailyPlan.totalCalories =
    dailyPlan.breakfast.calories +
    dailyPlan.lunch.calories +
    dailyPlan.dinner.calories +
    dailyPlan.snack.calories;

dailyPlan.totalPrice =
    dailyPlan.breakfast.price +
    dailyPlan.lunch.price +
    dailyPlan.dinner.price +
    dailyPlan.snack.price;

return dailyPlan;
```

Figure 122: Generate Single-Day Meal Plan

```
selectBestMeal: function (mealType, dietType, targetCalories, targetProtein, targetCarbs, targetFat, budget, usedMeals) {
    const availableFoods = FoodDB.getFilteredFoods(mealType, dietType);

    const budgetedAndUnusedFoods = availableFoods.filter(food => {
        return food.price <= budget && !usedMeals[mealType].has(food.name);
    });

    if (budgetedAndUnusedFoods.length === 0) {
        return this.defaultEmptyFood("No suitable food");
    }

    const scoredFoods = budgetedAndUnusedFoods.map(food => {
        const calorieDeviation = Math.abs(food.calories - targetCalories) / targetCalories;
        const proteinDeviation = Math.abs(food.protein - targetProtein) / (targetProtein || 1);
        const carbsDeviation = Math.abs(food.carbs - targetCarbs) / (targetCarbs || 1);
        const fatDeviation = Math.abs(food.fat - targetFat) / (targetFat || 1);

        // Score based on nutritional deviations
        const score =
            calorieDeviation * 0.4 +
            proteinDeviation * 0.2 +
            carbsDeviation * 0.2 +
            fatDeviation * 0.2;

        return { food, score };
    });

    scoredFoods.sort((a, b) => a.score - b.score);
    // Select from the top candidates
    const topCandidates = scoredFoods.slice(0, 5);

    if (topCandidates.length > 0) {
        const chosen = topCandidates[Math.floor(Math.random() * topCandidates.length)].food;
        usedMeals[mealType].add(chosen.name);
        return {
            name: chosen.name,
            calories: chosen.calories,
            protein: chosen.protein,
            carbs: chosen.carbs,
            fat: chosen.fat,
            price: chosen.price,
            groceryList: chosen.groceryList,
            cookingInstructions: chosen.cookingInstructions
        };
    }
    return this.defaultEmptyFood("No suitable food");
},
```

Figure 123: Select the Best Meal

Personalized Workout Plan —— W.P.plan.js

```
// Step 1: Fetch height & weight from DB first
function startBMIChat() {
  // Fetch height and weight
  fetch("W.P.getMemberData.php")
    .then(res => res.text())
    .then(text => {
      if (text.startsWith("<!DOCTYPE")) {
        throw new Error("Received HTML instead of JSON. Possibly logged out.");
      }
      const data = JSON.parse(text);
      if (data.mbp_height && data.mbp_weight) {
        member_profiles.mbp_height = parseFloat(data.mbp_height);
        member_profiles.mbp_weight = parseFloat(data.mbp_weight);
        addMessage(`
          <div>These are the most recent details we've saved for you 💕</div>
          <table class="chat-table">
            <tr><th>Height:</th><td>${data.mbp_height} cm</td></tr>
            <tr><th>Weight:</th><td>${data.mbp_weight} kg</td></tr>
          </table>
        `);
        addMessage("Do you want to use your profile's height and weight? (yes/no)");
        stage = 0;
      } else {
        addMessage("No height and weight found in your profile. Please enter your height in cm:");
        stage = 1;
      }
    })
    .catch(err => {
      console.error("Failed to fetch profile data:", err);
      addMessage("⚠️ Error retrieving profile data. Please make sure you're logged in.");
    });
}
```

Figure 124: Fetch Height and Weight

```
// Main handler
function handleUserInput() {
  const input = document.getElementById("userInput");
  const msg = input.value.trim();
  if (!msg) return;

  addMessage(msg, "user");
  input.value = "";

  if (stage === 0) {
    if (msg.toLowerCase() === "yes") {
      userHeight = member_profiles.mbp_height;
      userWeight = member_profiles.mbp_weight;
      proceedToGeneratePlan();
      stage = -1;
    } else {
      addMessage("Please enter your height in cm:");
      stage = 1;
    }
  } else if (stage === 1) {
    tempHeight = parseFloat(msg);
    if (isNaN(tempHeight) || tempHeight <= 0) {
      addMessage("Please enter a valid number for height.");
      return;
    }
    addMessage("Now enter your weight in kg:");
    stage = 2;
  } else if (stage === 2) {
    tempWeight = parseFloat(msg);
    if (isNaN(tempWeight) || tempWeight <= 0) {
      addMessage("Please enter a valid number for weight.");
      return;
    }
    userHeight = tempHeight;
    userWeight = tempWeight;
    proceedToGeneratePlan();
    stage = -1;
  }
}
```

Figure 125: Handle User Input

```
fetch("W.P.checkExistingPlan.php")
  .then(res => res.json())
  .then(data => {
    if (!data.success) {
      addMessage("⚠️ You must be logged in to continue.");
      return;
    }

    if (data.exists) {
      const createdDate = new Date(data.created_at).toLocaleDateString('en-GB');
            // Show chatbot message + buttons together
      const messageDiv = document.createElement("div");
      messageDiv.className = "bot";

      messageDiv.innerHTML = `
        💪 You already have a workout plan created on <b>${createdDate}</b> this week.<br>
       Would you like to create a new one or continue with the current one?
        <div class="chat-button-group">
          <button class="chat-option-btn" id="createBtn">🆕 Create New Plan</button>
          <button class="chat-option-btn" id="continueBtn">🔄 Continue Existing Plan</button>
        </div>
      `;

      const chatBox = document.getElementById("chatBox");
      chatBox.appendChild(messageDiv);
      chatBox.scrollTop = chatBox.scrollHeight;
    document.getElementById("createBtn").onclick = () => {
      // Reuse saved profile height/weight
      if (member_profiles.mbp_height && member_profiles.mbp_weight) {
        userHeight = member_profiles.mbp_height;
        userWeight = member_profiles.mbp_weight;
        proceedToGeneratePlan(); // ✅ Directly proceed to new plan
      } else {
        startBMIChat(); // fallback if no profile found
      }
    };


    document.getElementById("continueBtn").onclick = () => {
      messageDiv.remove();
      window.location.href = "W.routine.php"; // ✅ Redirect to workout routine
    };
  } else {
    startBMIChat();
  }
})
.catch(err => {
  console.error("Error:", err);
  addMessage("❌ Failed to check existing workout plan.");
});
```

Figure 126: Existing Plan Check and User Choice

```
// Save to database
function saveWorkoutPlan(planResult, weight, height) {
  const bmi = planResult.BMI;
  const category = planResult.Category;
  const plan = planResult.WeeklyPlan;

  console.log("📤 Sending to PHP:", {
    height,
    weight,
    bmi,
    category,
    plan
  });

  fetch("W.saveWorkoutPlan.php", {
    method: "POST",
    headers: { "Content-Type": "application/json" },
    body: JSON.stringify({
      height,
      weight,
      bmi,
      category,
      plan
    })
  })
  .then(res => res.json())
  .then(data => {
    if (data.success) {
      // Step 1: confirm saved
      addMessage("✅ Workout plan saved!");

      // Step 2: guide message after slight delay
      setTimeout(() => {
        addMessage("🎯 You can now visit the <b>Workout Routine</b> page to view your daily workouts.");
        // Step 3: show the button only after message
        const buttonDiv = document.createElement("div");
        buttonDiv.className = "bot";
        buttonDiv.innerHTML = `
          <div class="chat-button-group">
            <button class="chat-option-btn" onclick="window.location.href='W.routine.php'">
              🏋️ Go to Workout Routine
            </button>
          </div>
        `;
        document.getElementById("chatBox").appendChild(buttonDiv);
        document.getElementById("chatBox").scrollTop = document.getElementById("chatBox").scrollHeight;
      }, 4000);
    } else {
      addMessage("❌ Failed to save workout plan.");
      console.error("Server says:", data.message);
    }
  });
}
```

Figure 127: Save Workout Plan to Database

Progress Tracker —— tracking.php

```php
if ($_SERVER['REQUEST_METHOD'] === 'POST') {
    $date = date('Y-m-d');

    if (isset($_POST['save_weight'])) {
        $weight = $_POST['weight'];
        $stmt = $con->prepare("SELECT * FROM weight_logs WHERE wgl_member_ID = ? AND wgl_entry_date = ?");
        $stmt->bind_param("is", $memberID, $date);
        $stmt->execute();
        $result = $stmt->get_result();

        if ($result->num_rows > 0) {
            $stmt = $con->prepare("UPDATE weight_logs SET wgl_weight = ? WHERE wgl_member_ID = ? AND wgl_entry_date = ?");
            $stmt->bind_param("dis", $weight, $memberID, $date);
        } else {
            $stmt = $con->prepare("INSERT INTO weight_logs (wgl_member_ID, wgl_entry_date, wgl_weight) VALUES (?, ?, ?)");
            $stmt->bind_param("isd", $memberID, $date, $weight);
        }
        $stmt->execute();
        $stmt->close();

        echo "<script>alert('Weight saved successfully!'); window.location.replace('tracking.php');</script>";
    }

    if (isset($_POST['save_measurements'])) {
        $waist = $_POST['waist'];
        $hip = $_POST['hip'];
        $thigh = $_POST['thigh'];

        if (is_numeric($waist) && is_numeric($hip) && is_numeric($thigh)) {
            $stmt = $con->prepare("SELECT * FROM body_measurements WHERE bdm_member_ID = ? AND bdm_entry_date = ?");
            $stmt->bind_param("is", $memberID, $date);
            $stmt->execute();
            $result = $stmt->get_result();

            if ($result->num_rows > 0) {
                // Update existing record
                $stmt = $con->prepare("UPDATE body_measurements SET bdm_waist = ?, bdm_hip = ?, bdm_thigh = ? WHERE bdm_member_ID = ? AND bdm_entry_date = ?");
                $stmt->bind_param("ddd is", $waist, $hip, $thigh, $memberID, $date);
            } else {
                // Insert new record
                $stmt = $con->prepare("INSERT INTO body_measurements (bdm_member_ID, bdm_waist, bdm_hip, bdm_thigh, bdm_entry_date) VALUES (?, ?, ?, ?, ?)");
                $stmt->bind_param("iddds", $memberID, $waist, $hip, $thigh, $date);
            }

            $stmt->execute();
            $stmt->close();

            echo "<script>alert('Body measurements saved successfully!'); window.location.replace('tracking.php');</script>";
        } else {
            echo "<script>alert('Please enter a value between 10 cm and 200 cm for waist, hip, and thigh measurements.');</script>";
        }
    }
```

Figure 128: Handle and Save Daily Progress Updates I

```php
if (isset($_POST['save_logs'])) {
    $water = $_POST['water'];
    $sleep = $_POST['sleep'];

    $stmt = $con->prepare("SELECT * FROM daily_logs WHERE dll_member_ID = ? AND dll_entry_date = ?");
    $stmt->bind_param("is", $memberID, $date);
    $stmt->execute();
    $check = $stmt->get_result();

    if ($check->num_rows > 0) {
        $stmt = $con->prepare("UPDATE daily_logs SET dll_water_cups = ?, dll_sleep_hours = ? WHERE dll_member_ID = ? AND dll_entry_date = ?");
        $stmt->bind_param("diis", $water, $sleep, $memberID, $date);
    } else {
        $stmt = $con->prepare("INSERT INTO daily_logs (dll_member_ID, dll_water_cups, dll_sleep_hours, dll_entry_date) VALUES (?, ?, ?, ?)");
        $stmt->bind_param("iids", $memberID, $water, $sleep, $date);
    }

    $stmt->execute();
    $stmt->close();

    // Generate JS-friendly message
    $feedback = "";
    if ($water < 4) {
        $feedback .= "💧 You drank too little water today. Stay hydrated!\\n";
    } elseif ($water <= 10) {
        $feedback .= "👍 Great job! You're drinking a healthy amount of water.\\n";
    } else {
        $feedback .= "⚠️ Too much water isn't always good! Try to moderate.\\n";
    }
    if ($sleep < 5) {
        $feedback .= "😴 You didn't sleep enough. Try to rest more.\\n";
    } elseif ($sleep <= 9) {
        $feedback .= "🌟 Excellent sleep! Keep it up.\\n";
    } else {
        $feedback .= "💤 Too much sleep might make you feel sluggish. Stay balanced.\\n";
    }

    $feedback .= "✅ Daily logs saved successfully!";

    echo "<script>alert(\"$feedback\"); window.location.replace('tracking.php');</script>";
    exit;
}
```

Figure 129: Handle and Save Daily Progress Updates II

```php
// Fetch data for chart display
$weightDates = $weightValues = [];
$measurementDates = $measurementValues = [];
$waterDates = $waterValues = [];
$sleepDates = $sleepValues = [];

// Fetch weight logs
$result = $con->query("SELECT wgl_entry_date, wgl_weight FROM weight_logs WHERE wgl_member_ID = '$memberID' ORDER BY wgl_entry_date");
while ($row = $result->fetch_assoc()) {
    $weightDates[] = $row['wgl_entry_date'];
    $weightValues[] = $row['wgl_weight'];
}

// Fetch individual body measurements
$measurementDates = $waistValues = $hipValues = $thighValues = [];

$result = $con->query("SELECT bdm_entry_date, bdm_waist, bdm_hip, bdm_thigh FROM body_measurements WHERE bdm_member_ID = '$memberID' ORDER BY bdm_entry_date");
while ($row = $result->fetch_assoc()) {
    $measurementDates[] = $row['bdm_entry_date'];
    $waistValues[] = $row['bdm_waist'];
    $hipValues[] = $row['bdm_hip'];
    $thighValues[] = $row['bdm_thigh'];
}

// Fetch water/sleep logs
$result = $con->query("SELECT dll_entry_date, dll_water_cups, dll_sleep_hours FROM daily_logs WHERE dll_member_ID = '$memberID' ORDER BY dll_entry_date");
while ($row = $result->fetch_assoc()) {
    $waterDates[] = $row['dll_entry_date'];
    $waterValues[] = $row['dll_water_cups'];
    $sleepDates[] = $row['dll_entry_date'];
    $sleepValues[] = $row['dll_sleep_hours'];
}
```

Figure 130: Fetch Historical Data for Progress Charts

```
<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script>
  function validateWeightForm() {
    const weightInput = document.querySelector('input[name="weight"]').value;
    const weight = parseFloat(weightInput);
    if (!weightInput || isNaN(weight) || weight < 10 || weight > 200) {
      alert("Please enter a value between 10 to 200 for weight.");
      return false; // prevent submission
    }
    return true;
  }

  function validateMeasurementForm() {
    const waist = parseFloat(document.querySelector('input[name="waist"]').value);
    const hip = parseFloat(document.querySelector('input[name="hip"]').value);
    const thigh = parseFloat(document.querySelector('input[name="thigh"]').value);

    if (
      isNaN(waist) || waist < 10 || waist > 200 ||
      isNaN(hip) || hip < 10 || hip > 200 ||
      isNaN(thigh) || thigh < 10 || thigh > 200
    ) {
      alert("Please enter values between 10 to 200 for waist, hip, and thigh.");
      return false;
    }
    return true;
  }
```

Figure 131: Validate Weight and Measurement Forms

```
// Weight Chart
new Chart(document.getElementById('chartWeight'), {
  type: 'line',
  data: {
    labels: weightLabels,
    datasets: [{
      label: 'Weight (kg)',
      data: weightData,
      backgroundColor: 'rgba(142, 115, 91, 0.2)',
      borderColor: '#8E735B',
      fill: true,
      tension: 0.3
    }]
  }
});
const ctx = document.getElementById('measurementChart').getContext('2d');
const chart = new Chart(ctx, {
  type: 'line',
  data: {
    labels: <?= json_encode($measurementDates) ?>,
    datasets: [
      {
        label: 'Waist (cm)',
        data: <?= json_encode($waistValues) ?>,
        borderColor: '#FF6384',
        fill: false
      },
      {
        label: 'Hip (cm)',
        data: <?= json_encode($hipValues) ?>,
        borderColor: '#36A2EB',
        fill: false
        },
      {
        label: 'Thigh (cm)',
        data: <?= json_encode($thighValues) ?>,
        borderColor: '#FFCE56',
        fill: false
      }
    ]
  },
  options: {
    responsive: true,
    plugins: {
      title: {
        display: true,
        text: 'Body Measurements (cm)'
      }
    }
  }
});
```

Figure 132: Weight Chart Generation

```
// Water & Sleep Chart
const ctx2 = document.getElementById('waterandsleepChart').getContext('2d');
new Chart(ctx2, {
type: 'bar',
data: {
  labels: labels,
  datasets: [
    {
      label: 'Water Intake (cups)',
      data: waterData,
      backgroundColor: '#6EC6FF'
    },
    {
      label: 'Sleep Duration (hours)',
      data: sleepData,
      backgroundColor: '#A3D9A5'
    }
  ]
},
options: {
  responsive: true,
  plugins: {
    title: {
      display: true,
      text: 'Daily Water vs Sleep Tracking'
    }
  },

  scales: {
    y: {
      beginAtZero: true,
      title: {
        display: true,
        text: 'Amount'
      }
    },
    x: {
      title: {
        display: true,
        text: 'Date'
      }
    }
  }
}
});
```

Figure 133: Water and Sleep Chart Generation

Admin View Member Accounts —— view_members.php

```
<section class="dashboard-container">
<h2>Registered Member Profiles</h2>
<table>
    <tr><th>ID</th><th>Username</th><th>Email</th><th>Phone Number</th><th>Action</th></tr>
    <?php foreach ($users as $row): ?>
    <tr>
        <td><?= htmlspecialchars($row['usr_user_id']) ?></td>
        <td><?= htmlspecialchars($row['usr_username']) ?></td>
        <td><?= htmlspecialchars($row['usr_email']) ?></td>
        <td><?= htmlspecialchars($row['usr_phone']) ?></td>
        <td>
            <a href="edit_member.php?id=<?= $row['usr_user_id'] ?>">Edit</a> |
            <a href="deactivate_member.php?id=<?= $row['usr_user_id'] ?>" onclick="return confirm('Are you sure that you
        </td>
    </tr>
    <?php endforeach; ?>
</table>
</section>
```

Figure 134: Display Member Account Details

Member Registration Report —— report_registration.php

```
// Fetch registration counts

// Get current date
$currentDate = date('l, d/m/Y');
// Get current year, month, and week
$currentYear = date('Y');
$currentMonth = date('m');
$currentWeek = date('W');

// Annual registrations
$resultAnnual = mysqli_query($con, "SELECT COUNT(*) AS count FROM users WHERE YEAR(usr_reg_date) = '$currentYear'");
$annual = mysqli_fetch_assoc($resultAnnual)['count'];

// Monthly registrations
$resultMonth = mysqli_query($con, "SELECT COUNT(*) AS count FROM users WHERE YEAR(usr_reg_date) = '$currentYear' AND MONTH(usr_reg_date) = '$currentMonth'");
$month = mysqli_fetch_assoc($resultMonth)['count'];

// Weekly registrations
$resultWeek = mysqli_query($con, "SELECT COUNT(*) AS count FROM users WHERE YEAR(usr_reg_date) = '$currentYear' AND WEEK(usr_reg_date, 1) = '$currentWeek'");
$week = mysqli_fetch_assoc($resultWeek)['count'];
?>
```

Figure 135: Registration Count Retrieval

Member Activity Report —— report_activity.php

```
// Fetch registration counts

// Get current date
$currentDate = date('l, d/m/Y');
// Get current year, month, and week
$currentYear = date('Y');
$currentMonth = date('m');
$currentWeek = date('W');

// Annual registrations
$resultAnnual = mysqli_query($con, "SELECT COUNT(*) AS count FROM users WHERE YEAR(usr_last_login) = '$currentYear'");
$activeThisYear = mysqli_fetch_assoc($resultAnnual)['count'];

// Monthly registrations
$resultMonth = mysqli_query($con, "SELECT COUNT(*) AS count FROM users WHERE YEAR(usr_last_login) = '$currentYear' AND MONTH(usr_last_login) = '$currentMonth'");
$activeThisMonth = mysqli_fetch_assoc($resultMonth)['count'];

// Weekly registrations
$resultWeek = mysqli_query($con, "SELECT COUNT(*) AS count FROM users WHERE YEAR(usr_last_login) = '$currentYear' AND WEEK(usr_last_login, 1) = '$currentWeek'");
$activeThisWeek = mysqli_fetch_assoc($resultWeek)['count'];
?>
```

Figure 136: Active User Count Retrieval

9.0 Outcomes of Testing

9.1 Unit Testing

Name of the program: login.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Password	Expected Message	Actual Message	Result
LT001	blank	blank	Please filled in the empty field.	Please fill out this field. (username)	Display error message and stay in login page
LT002	blank	cl@12345	Please filled in the empty field.	Please fill out this field. (username)	Display error message and stay in login page
LT003	CL	blank	Please filled in the empty field.	Please fill out this field. (password)	Display error message and stay in login page
LT004	ABC	cl@12345	User don't have an account. Directing to registration page…	Localhost says User does not exist. Please register first!	Direct to registration page
LT005	CL	888	Incorrect password format!	Please match the requested format.	Display error message and stay in login page

			Please try again.	Password must be at least 8 characters, contain both letters and numbers, and only these symbols: -_.@	
LT006	CL	Cl@13901	Incorrect password. Please try again!	Localhost says Incorrect password. Please try again!	Display error message and stay in login page
LT007	CL	cl@12345	**The user sees a Welcome Page**	**The user sees a Welcome Page**	Direct to memberHomepage.php
LT008	admin	i_am_admin_123	**The user sees a Admin Page**	**The user sees a Admin Page**	Direct to adminHomepage.php

Name of the program: register.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Full Name	Email	Phone	Password	Confirm Password	Security keyword	Date of Birth	Gender	Height (cm)	Weight (kg)	Fitness Goal	Expected Message	Actual Message	Result
RT001	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for all field**	Passed
RT002	abc12345	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username field**	Passed
RT003	abc12345	ABC	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username & full name field**	Passed
RT004	abc12345	ABC	abc@gmail.com	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email & full name field**	Passed
RT005	abc12345	ABC	abc@gmail.com	012-3456789	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number & full name field**	Passed
RT006	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number, password & full name field**	Passed
RT007	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*blank*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number, password, confirm password& full name field**	Passed
RT008	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	*blank*	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number, password, confirm password, security keyword& full name field**	Passed
RT009	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number, password, confirm password, security keyword, date of birth& full name field**	Passed
RT010	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	*blank*	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number, password, confirm password, security keyword, date of birth, gender& full name field**	Passed
RT011	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	*blank*	*blank*	Please filled in the empty field.	**Display error message for field other than username, email, phone number, password, confirm password, security keyword, date of birth, gender, height& full name field**	Passed
RT012	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	*blank*	Please select your fitness goal.	Fitness goal is required	Passed
RT013	abc	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for username!	Username must be 5-15 characters, letters/numbers/underscores only	Passed
RT014	abc12345	ABC-DEF	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for FULL NAME!	Full name must contain only letters and spaces	Passed
RT015	abc12345	ABC	abc.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for EMAIL!	Invalid email format	Passed
RT016	abc12345	ABC	abc@gmail.com	123456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for Phone number!	Please match the format such as 012-3456789 or 011-23456789	Passed
RT017	abc12345	ABC	abc@gmail.com	012 3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for Phone number!	Please match the format such as 012-3456789 or 011-23456789	Passed
RT018	abc12345	ABC	abc@gmail.com	012-3456789	abc123	abc123	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for password!	Password must be at least 8 characters with uppercase, lowercase, number, and special character	Passed
RT019	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@45678	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct password to confirm your password!	Password do not match	Passed
RT020	abc12345	ABC	abc@gmail.com	012-3456789	abc123	abc@45678	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for password!	Password must be at least 8 characters with uppercase, lowercase, number, and special character	Passed
RT021	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc123	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct password to confirm your password!	Password do not match	Passed
RT022	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABC*	1/1/2005	FEMALE	160	45	Gain weight	Please enter the correct format for Security Keyword!	Security keyword must be at least 5 characters	Passed
RT023	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	0.01	45	Gain weight	Invalid height value!	Height cannot be negative and must be at least 50cm	Passed
RT024	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	0.5	Gain weight	Invalid weight value!	Weight cannot be negative and must be at least 10kg	Passed
RT025	blank	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your username.	Username must be 5-15 characters, letters/numbers/underscores only	Passed
RT026	abc12345	blank	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your full name.	Full name must contain only letters and spaces	Passed
RT027	abc12345	ABC	blank	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your email.	Invalid email format	Passed
RT028	abc12345	ABC	abc@gmail.com	blank	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your phone number.	Please match the format such as 012-3456789 or 011-23456789	Passed
RT029	abc12345	ABC	abc@gmail.com	012-3456789	blank	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please enter your password.	Password must be at least 8 characters with uppercase, lowercase, number, and special character	Passed
RT030	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	blank	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	Please confirm your password.	Password do not match	Passed
RT031	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	blank	1/1/2005	FEMALE	160	45	Gain weight	Please enter your security keyword.	Security keyword must be at least 5 characters	Passed
RT032	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	blank	FEMALE	160	45	Gain weight	Please input your B.O.D.	Date of birth is required	Passed
RT033	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	blank	160	45	Gain weight	Please select your gender.	Gender is required	Passed
RT034	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	blank	45	Gain weight	Please filled in your height	Height cannot be negative and must be at least 50cm	Passed
RT035	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	blank	Gain weight	Please filled in your weight.	Weight cannot be negative and must be at least 10kg	Passed
RT036	abc12345	ABC	abc@gmail.com	012-3456789	abc@12345	abc@12345	*ABCDEF*	1/1/2005	FEMALE	160	45	Gain weight	**The user sees a login Page**	**The user sees a login Page**	Passed

Name of the program: login.php (forgot password)

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Username	Keyword	New Password	Confirm Password	Expected Message	Actual Message	Result
RST001	*blank*	*blank*	*blank*	*blank*	Please filled in the empty field.	Please filled out this field.	Passed
RST002	abc12345	*blank*	*blank*	*blank*	Please filled in the empty field.	Please filled out this field.	Passed
RST003	abc12345	*ABC123*	*blank*	*blank*	Please filled in the empty field.	Please filled out this field.	Passed
RST004	abc12345	ABC123	ABC@45678	*blank*	Please filled in the empty field.	Please filled out this field.	Passed
RST005	abc12345	*ABC123*	ABC@45678	ABC@45678	Please filled in the empty field.	Please filled out this field.	Passed
RST007	abc12345	*ABC123*	ABC@45678	*ABC*	Password does not match	New Password and Confirm Password are not match. Please try again!	Passed
RST008	abc12345	*ABC*	ABC@45678	ABC@45678	Keyword does not found!	User does not exist or wrong keyword. Please try again!	Passed
RST009	abc12345	*ABC123*	abc@12345	ABC@45678	Password does not match	New Password and Confirm Password are not match. Please try again!	Passed
RST010	abc12345	*ABC123*	abc@12345	abc@12345	Your password has been successfully reset.	Your password has been successfully reset.	Passed
RST011	abc	*ABC123*	ABC@45678	ABC@45678	User not found!	User does not exist or wrong keyword. Please try again!	Passed
RST012	abc12345	*ABC123*	abc	abc	Please fill in your new password in the correct format.	Please match the requested format. Password must be at least 8 characters, contain both letters and numbers, and only these symbols: -_.@*	Passed
RST013	abc12345	*ABC123*	abc	*efg*	Password does not match	New Password and Confirm Password are not match. Please try again!	Passed
RST014	abc12345	*ABC123*	ABC@45678	ABC@45678	Your password has been reset successfully.	Your password has been successfully reset.	Passed

Name of the program: tracking.php

Name of the tester: Ms. SERENE LOH ZI TING

TC-ID	Today's Weight (kg)	Waist (cm)	Hip (cm)	Thigh (cm)	Water Intake (cups)	Sleep (hours)	Expected Message	Actual Message	Result
PT001	blank	blank	blank	blank	blank	blank	Please filled in the empty field.	Display error message for all field	Passed
PT002	50	blank	blank	blank	blank	blank	Successfullt saved weight!	Weight saved successfully!	Passed
PT003	5	blank	blank	blank	blank	blank	Weight must be greater than 10.	Value must greater than or equal to 10	Passed
PT004	blank	f	blank	blank	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT005	blank	blank	f	blank	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT006	blank	blank	blank	f	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT007	blank	f	f	f	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT008	blank	5	blank	blank	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT009	blank	blank	5	blank	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT010	blank	blank	blank	5	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT011	blank	5	5	5	blank	blank	Invalid input.	Please enter values between 10 to 200 for waist, hip, and thigh.	Passed
PT012	blank	58	88	25	blank	blank	Sucessfully saved body measurement!	Body measurements saved successfully!	Passed
PT013	blank	blank	blank	blank	1	blank	Display drink and sleep more reminder and successfully saved daily logs message.	You drank too little water today. Stay hydrated! You didn't sleep enough. Try to rest more. Daily logs saved successfully!	Passed
PT014	blank	blank	blank	blank	blank	1	Display drink and sleep more reminder and successfully saved daily logs message.	You drank too little water today. Stay hydrated! You didn't sleep enough. Try to rest more. Daily logs saved successfully!	Passed
PT015	blank	blank	blank	blank	1	8	Display drink more water reminder and successfully saved daily logs message.	You drank too little water today. Stay hydrated! Excellent sleep! Keep it up. Daily logs saved successfully!	Passed
PT016	blank	blank	blank	blank	5	2	Display sleep more alert and successfully saved daily logs message.	Great job! You're drinking a healthy amount of water. You didn't sleep enough. Try to rest more. Daily logs saved successfully!	Passed
PT017	blank	blank	blank	blank	5	8	Display successfully saved daily logs message.	Great job! You're drinking a healthy amount of water. Excellent sleep! Keep it up. Daily logs saved successfully!	Passed

Name of the program: P.editProfile.php

Name of the tester: Ms SERENE LOH ZI TING

TC-ID	Username	Full Name	Email	Phone	Security keyword	Date of Birth	Gender	Height (cm)	Weight (kg)	Fitness Goal	Expected Message	Actual Message	Result
ET001	blank	blank	blank	blank	blank	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET002	abc12345	blank	blank	blank	blank	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET003	abc12345	ABC	blank	blank	blank	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET004	abc12345	ABC	abc@gmail.com	blank	blank	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET005	abc12345	ABC	abc@gmail.com	012-3456789	blank	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET006	abc12345	ABC	abc@gmail.com	012-3456789	blank	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET007	abc12345	ABC	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET008	abc12345	ABC	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET009	abc12345	ABC	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	blank	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET010	abc12345	ABC	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	160	blank	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET011	abc12345	ABC	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	160	50	blank	Please filled in the empty field.	Please fill out this field.	Passed
ET012	abc	ABCDEF	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	160	50	Gain Weight	Incorrect username format	Username must be 5-15 characters, letters/numbers/underscores only	Passed
ET013	abc12345	A	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	160	50	Gain Weight	Successfully updated profile!	Profile updated successfully	Passed
ET014	abc12345	ABCDEF	abc.gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	160	50	Gain Weight	Invalid email format	Please include an '@' in the email address. 'abc.com' is missing an '@'.	Passed
ET015	abc12345	ABC	abc@gmail.com	123456789	ABCDEF	1/1/2005	FEMALE	160	50	Gain Weight	Incorrect phone number format	Please match the format such as 012-3456789 or 011-23456789	Passed
ET016	abc12345	ABCDEF	abc@gmail.com	012-3456789	ABCD	1/1/2005	FEMALE	160	50	Gain Weight	Incorrect security keyword format	Security keyword must be at least 5 characters	Passed
ET017	abc12345	ABCDEF	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	1	50	Gain Weight	Incorrect height format	Height cannot be negative and must be at least 50cm	Passed
ET018	abc12345	ABCDEF	abc@gmail.com	012-3456789	ABCDEF	1/1/2005	FEMALE	160	3	Gain Weight	Incorrect weight format	Weight cannot be negative and must be at least 10kg	Passed
ET019	abc12345	ABCDEF	abc@gmail.com	012 3456789	ABCDEF	1/1/2005	MALE	160	50	Lose Weight	Successfully updated profile!	Profile updated successfully	Passed

Name of the program: DP.PS.Main.php

Name of the tester: Ms. YAP XIN LING

TC-ID	Use Recorded Details	Height (cm)	Weight (kg)	Age	Gender	Activity Level	Goal	Budget (RM)	Expected Message	Actual Message	Result
P-DT001	No	blank	blank	blank	blank	blank	blank	blank	Please fill in the empty field	Please fill in the empty field	Passed
P-DT002	No	-170	blank	blank	blank	blank	blank	blank	Value must be greater than or equal to 80	Value must be greater than or equal to 80	Passed
P-DT003	No	170	blank	blank	blank	blank	blank	blank	Please fill in the empty field	Please fill in the empty field	Passed
P-DT004	No	170	-60	blank	blank	blank	blank	blank	Value must be greater than or equal to 20	Value must be greater than or equal to 20	Passed
P-DT005	No	170	60	blank	blank	blank	blank	blank	Please fill in the empty field	Please fill in the empty field	Passed
P-DT006	No	170	60	-25	blank	blank	blank	blank	Value must be greater than or equal to 5	Value must be greater than or equal to 5	Passed
P-DT007	No	170	60	130	blank	blank	blank	blank	Value must be less than or equal to 120	Value must be less than or equal to 120	Passed
P-DT008	No	170	60	25	blank	blank	blank	blank	Please select an item in the list	Please select an item in the list	Passed
P-DT009	No	170	60	25	Female	blank	blank	blank	Please select an item in the list	Please select an item in the list	Passed
P-DT010	No	170	60	25	Male	Moderately active (3-5 days/week)	blank	blank	Please select an item in the list	Please select an item in the list	Passed
P-DT011	No	170	60	25	Male	Moderately active (3-5 days/week)	Lose Weight	blank	Please fill in the empty field	Please fill in the empty field	Passed
P-DT012	No	170	60	25	Male	Moderately active (3-5 days/week)	Lose Weight	-10	Value must be greater than or equal to 1	Value must be greater than or equal to 1	Passed
P-DT013	No	170	60	25	Male	Moderately active (3-5 days/week)	Lose Weight	30	Personalized diet plan is ready	Personalized diet plan is ready	Passed
P-DT014	Yes	-	-	-	-	blank	blank	blank	Please select an item in the list	Please select an item in the list	Passed
P-DT015	Yes	-	-	-	-	Moderately active (3-5 days/week)	blank	blank	Please select an item in the list	Please select an item in the list	Passed
P-DT016	Yes	-	-	-	-	Moderately active (3-5 days/week)	Lose Weight	-10	Value must be greater than or equal to 1	Value must be greater than or equal to 1	Passed
P-DT017	Yes	-	-	-	-	Moderately active (3-5 days/week)	Lose Weight	50	Personalized diet plan is ready	Personalized diet plan is ready	Passed
P-DT018	blank	-	-	-	-	blank	blank	blank	Please select an item in the list	Please select an item in the list	Passed

Name of the program: DP.NT.Main.php

Name of the tester: Ms. YAP XIN LING

TC-ID	Selected Date	Breakfast Food Name	Breakfast Portion Size	Total Calories for Breakfast	Lunch Food Name	Lunch Portion Size	Total Calories for Lunch	Dinner Food Name	Dinner Portion Size	Total Calories for Dinner	Snacks Food Name	Snacks Portion Size	Total Calories for Snacks	Water Intake	Supplements /Vitamins	Mood	Additional Remarks	Expected Message	Actual Message	Result
P-NT001	July 1, 2025	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	Please enter at least one meal to save your journal	Please enter at least one meal to save your journal	Passed
P-NT002	July 1, 2025	Omelette	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	Vitamin C	Happy	Felt good	Please enter a portion size for breakfast	Please enter a portion size for breakfast	Passed
P-NT003	July 1, 2025	blank	1 plate	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	6 cups	blank	Energetic	blank	Please enter a food name for breakfast.	Please enter a food name for breakfast.	Passed
P-NT004	July 1, 2025	Toast	2 slices	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	Calm	blank	Please enter calories for breakfast.	Please enter calories for breakfast.	Passed
P-NT005	July 1, 2025	Toast	2 slices	300	blank	blank	blank	blank	blank	blank	blank	blank	blank	8 cups	Omage 3	Calm	Slept well	Saved successfully	Saved successfully	Passed
P-NT006	July 1, 2025	Toast	2 slices	-100	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	Multivitamin	Calm	blank	Value must be greater than or equal to 0	Value must be greater than or equal to 1	Passed
P-NT007	July 1, 2025	Bread	2 slices	250	Rice	1 bowl	600	Chicken	1 thigh	350	Apple	1	80	10 cups	Vitamin B	Happy	Productive day	Saved successfully	Saved successfully	Passed
P-NT008	July 1, 2025	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	blank	Calm	blank	Please enter at least one meal to save your journal	Please enter at least one meal to save your journal	Passed
P-NT009	July 1, 2025	Cereal	1 bowl	200	blank	blank	blank	blank	blank	blank	blank	blank	blank	4 cups	blank	Calm	blank	Saved successfully	Saved successfully	Passed
P-NT010	July 1, 2025	Oatmeal	1 bowl	300	blank	blank	blank	blank	blank	blank	blank	blank	blank	5 cups	blank	blank	blank	Saved successfully	Saved successfully	Passed
P-NT011	July 1, 2026	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	You can only log meals for today or earlier	You can only log meals for today or earlier	Passed

9.2 User Acceptance Testing Plan

Name of the user: Ms – Full Name: HO YUEN YEN

Job Position: Diploma in Psychology Student

Date: 1 July 2025

Time From: 2.30pm Time To: 2.55pm

Location: Restaurant

Category	0 Mark(Very Poor)	1 Mark	2 Mark	3 Mark	4 Mark	5 Mark(Excelent)
User-friendly						☑
Free-from errors					☑	
Secure					☑	
Meeting its objectives						☑
Performance						☑
Feedback From the user	I love the design, which is a bit like Muji style. The system can function well, and it also has a nutritionist journal which includes a daily emotional record that takes care of the mental health of the user too. But the navigation for each page in member menu might be a bit inconsistent. If it could be the same for every page, it would be good! Date: 1 July 2025 Signature:					
Response from Project Development Team	Thank you for highlighting the navigation inconsistency. We acknowledge that a consistent navigation layout enhances user experience and usability. We have reviewed the affected pages and identified the mismatches in menu placement, styling, and responsive behavior. The following actions will be taken: - Standardized the header and sidebar navigation structure across all pages - Ensures consistent font, icon style, and button alignment - Applied a common layout template					

	Name: Ms SERENE LOH ZI TING Signature: serene Date: 5 July 2025

Name of the user: Ms – Full Name: WONG SIN YEE

Job Position: Degree in IT Student

Date: 2 July 2025

Time From: 6.35pm Time To: 7.30pm

Location: Cafeteria APU

Category	0 Mark(Very Poor)	1 Mark	2 Mark	3 Mark	4 Mark	5 Mark(Excelent)
User-friendly					☑	
Free-from errors					☑	
Secure				☑		
Meeting its objectives						☑
Performance						☑
Feedback From the user	Can set password authentication to enhance overall password security. But the overall design and animations impressed me Date: 2 July 2025 Signature: sinyee					
Response from Project	Thank you for your valuable feedback. We acknowledge the importance of improving password security. In response, we will implement stronger					

Development Team	password validation rules such as minimum length, combination of characters, and are exploring options for two-factor authentication in future phases. Passwords are already encrypted using industry-standard hashing. Name: Ms SERENE LOH ZI TING Singature: serene Date: 7 July 2025

Name of the user: Ms – Full Name: WONG KID-ZEN

Job Position: Degree in business student

Date: 9 July 2025

Time From: 1.40pm Time To: 2.20pm

Location: Ms WONG KID-ZEN apartment

Category	0 Mark(Very Poor)	1 Mark	2 Mark	3 Mark	4 Mark	5 Mark(Excelent)
User-friendly						☑
Free-from errors					☑	
Secure						☑
Meeting its objectives						☑
Performance						☑
Feedback From the user	The overall system functionality is solid. I'm especially impressed by the dynamic avatar that changes based on gender and follows the mouse pointer, great touch! However, I noticed that some footers on the homepage are positioned a bit too high, and the height of footer sections linked through the navigation bar is inconsistent. Other than that, everything seems good. Date: 9 July 2025 Signature: Zen					
Response from Project	Thank you for your detailed feedback. We acknowledge the inconsistency in footer alignment and spacing. The design team has received this and will					

Development Team	adjust the footer heights and padding across all relevant pages to ensure a consistent and visually balanced layout. Name: Ms SERENE LOH ZI TING Signature: serene Date: 12 July 2025

10.0 User Manual

10.1 User

10.1.1. How Users Can Access RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

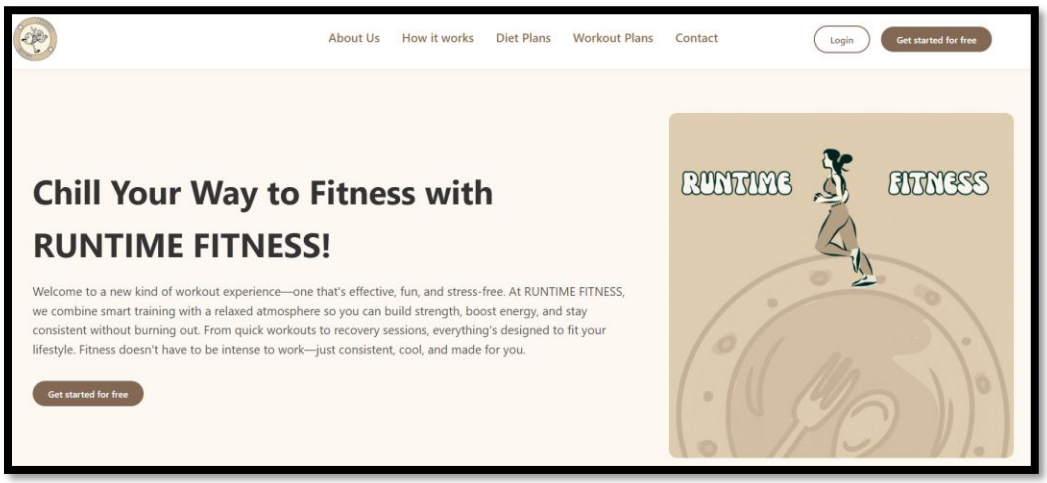


Figure 137: User Manual (homePage.php)

10.1.2 How Users Can Access About Us of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

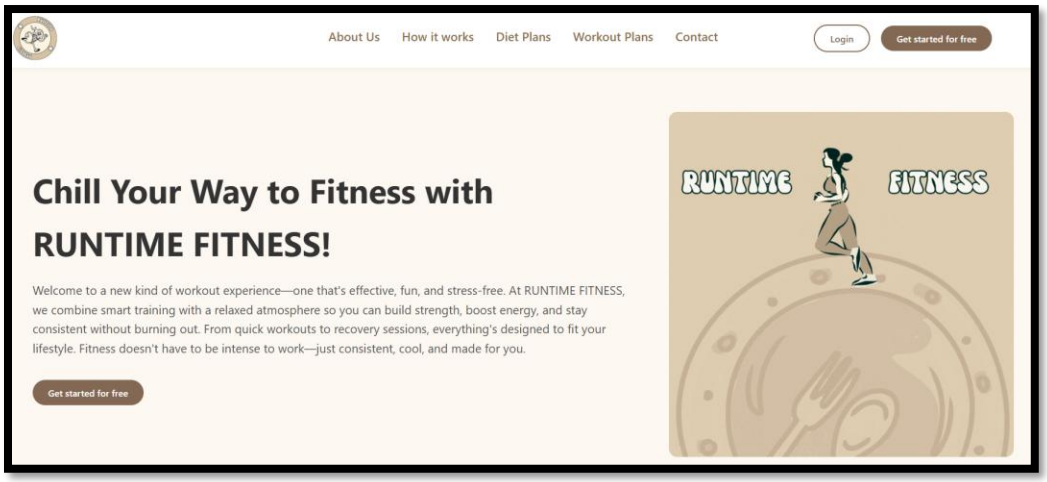


Figure 138: User Manual (homePage.php)

STEP 3:

Click About Us

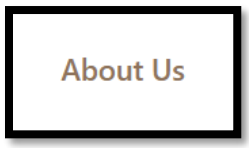


Figure 139: "About Us" link

STEP 4:

aboutUs.php is shown

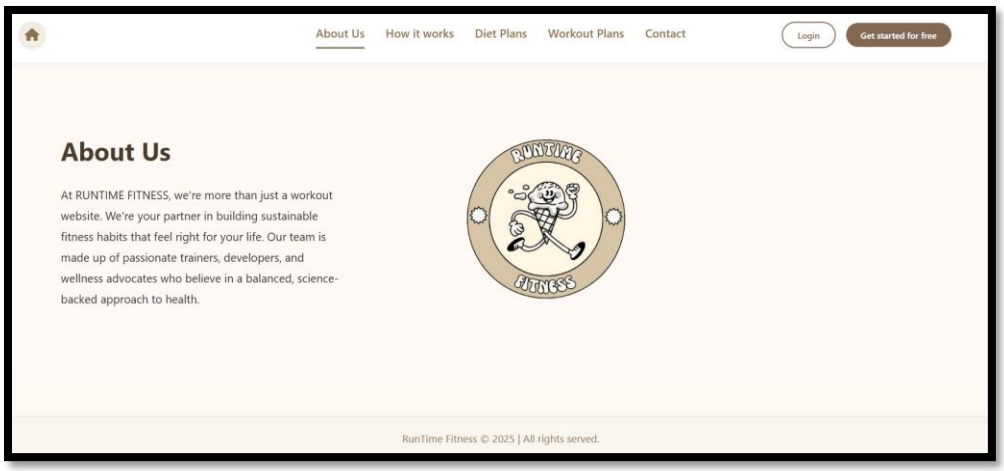


Figure 140: aboutUs.php

10.1.3 How Users Can Access How It Works of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

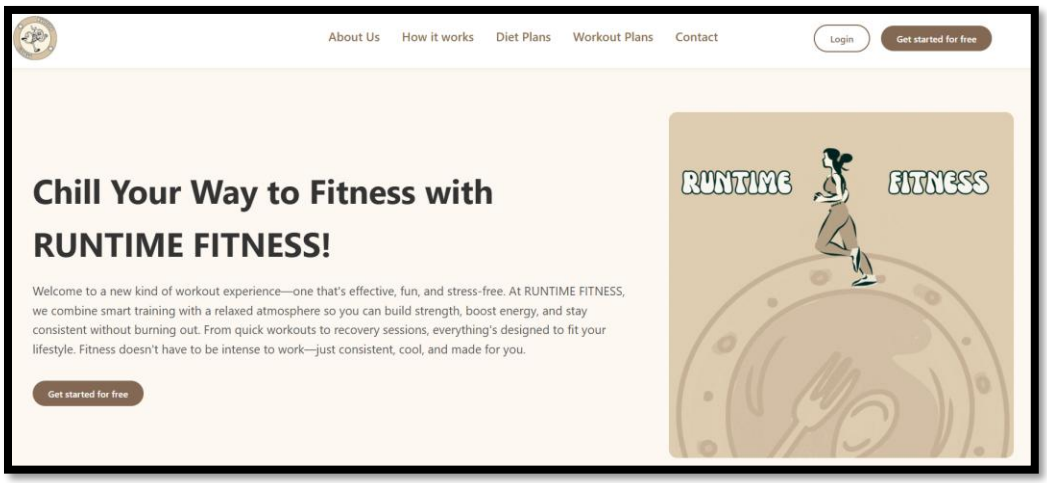


Figure 141: User Manual (homePage.php)

STEP 3:

Click How it works



Figure 142: "How it works" link

STEP 4:

howItWorks.html is shown

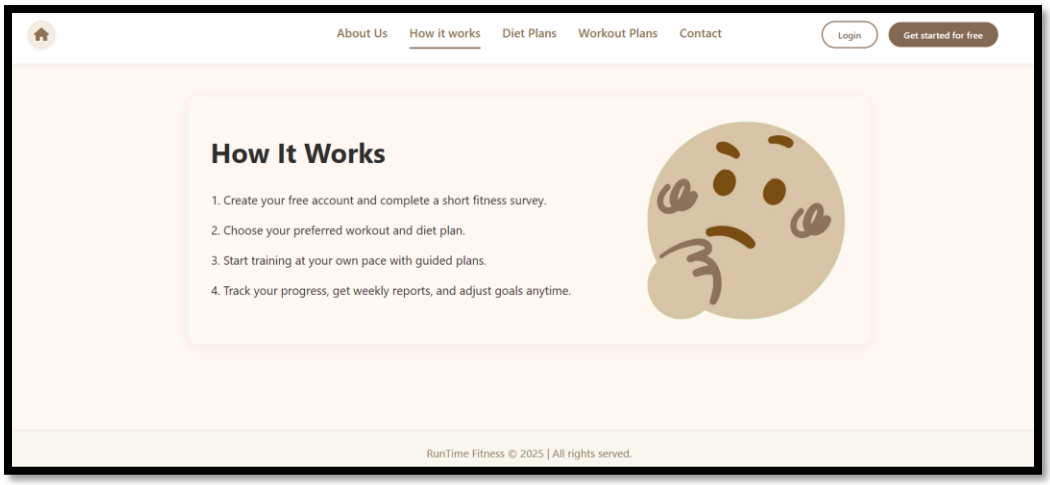


Figure 143: User Manual (howItWorks.html)

10.1.4 How Users Can Access Diet Plans of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

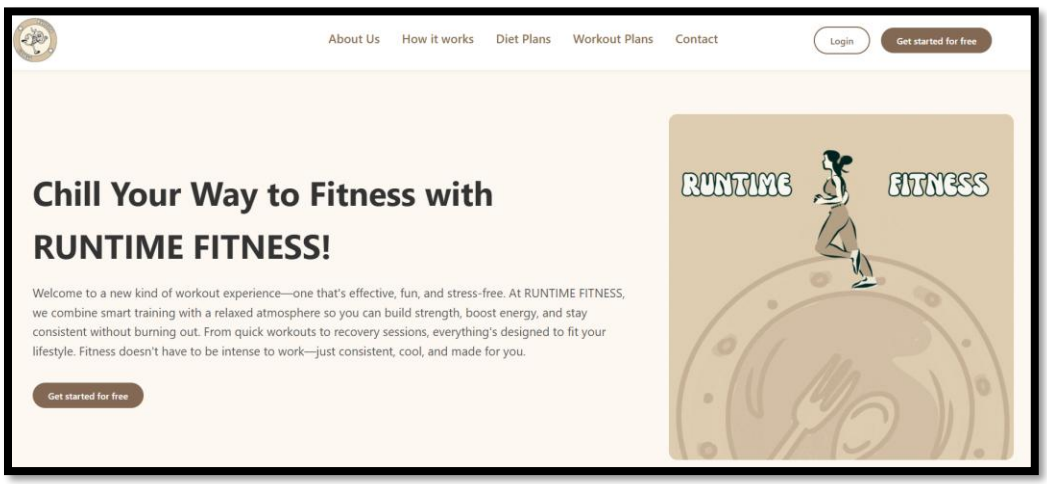


Figure 144: User Manual (homePage.php)

STEP 3:

Click Diet Plans



Figure 145: "Diet Plans" link

STEP 4:

dietPlan.html is shown

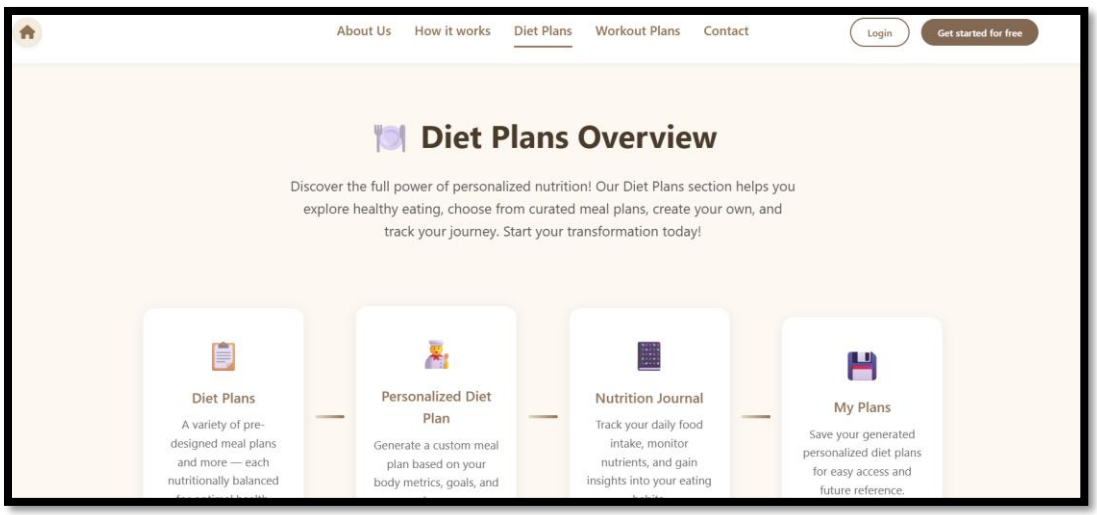


Figure 146: dietPlan.html

10.1.5 How Users Access Workout Plans of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

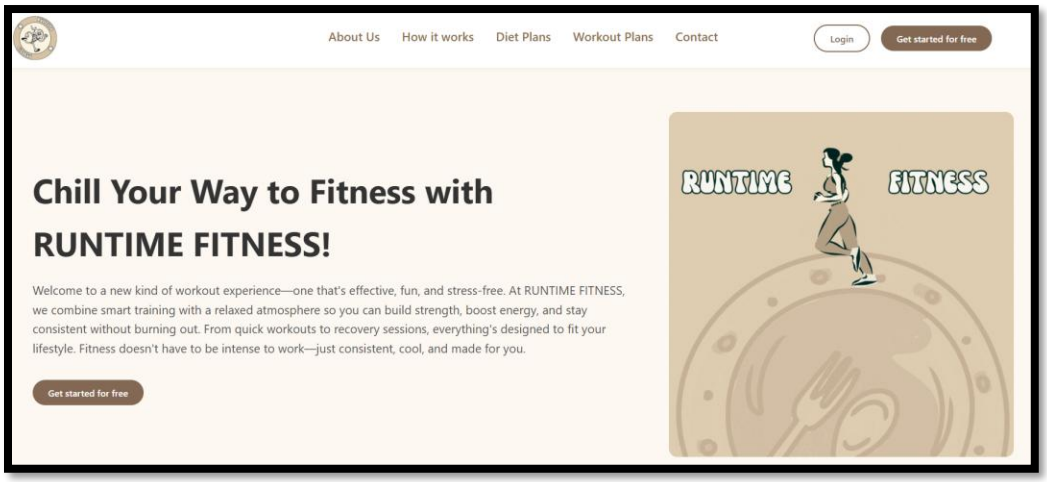


Figure 147: User Manual (homePage.php)

STEP 3:

Click Workout Plans

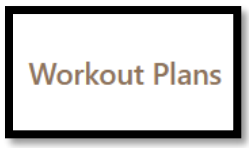


Figure 148: "Workout Plans" link

STEP 4:

workoutPlan.html is shown

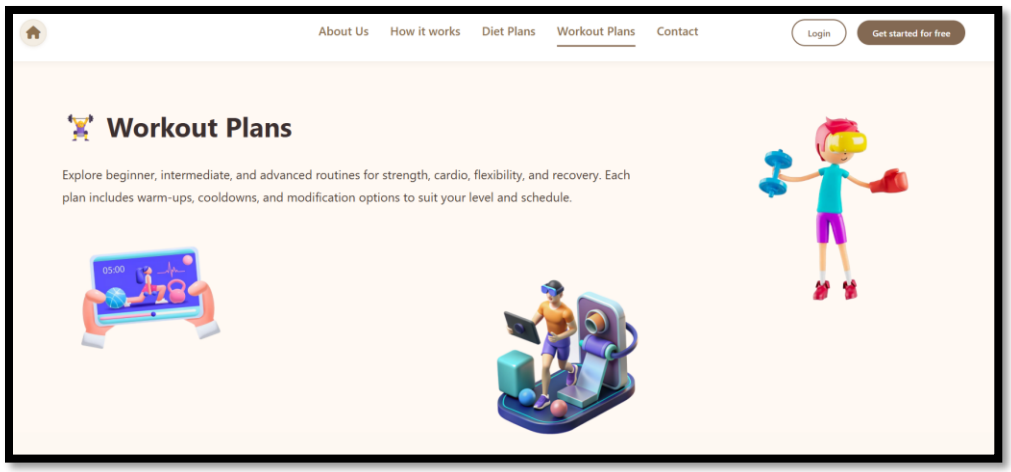


Figure 149: User Manual (workoutPlan.html)

10.1.6 How Users Can Access the Contact of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

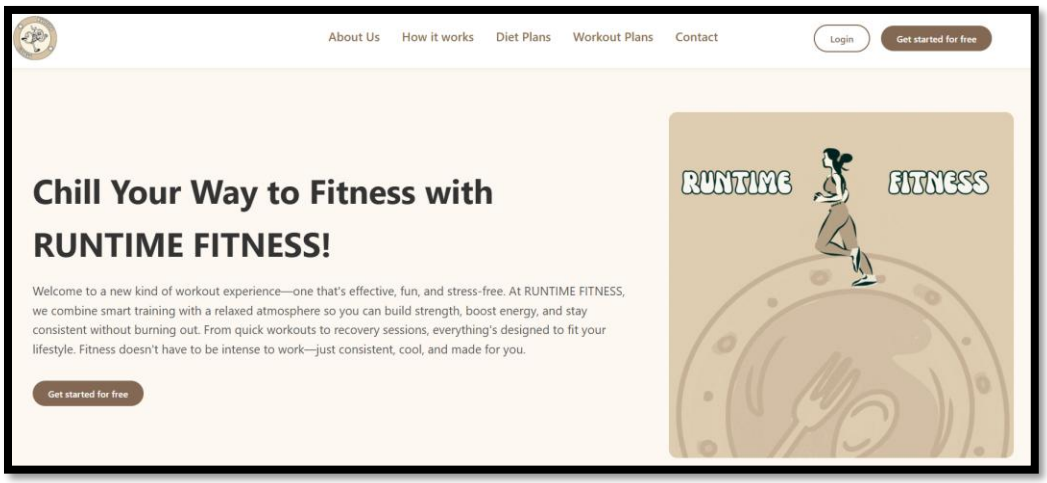


Figure 150: User Manual (homePage.php)

STEP 3:

Click Contact

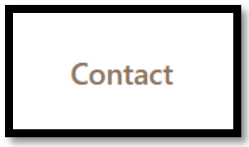


Figure 151: "Contact" link

STEP 4:

contact.php is shown

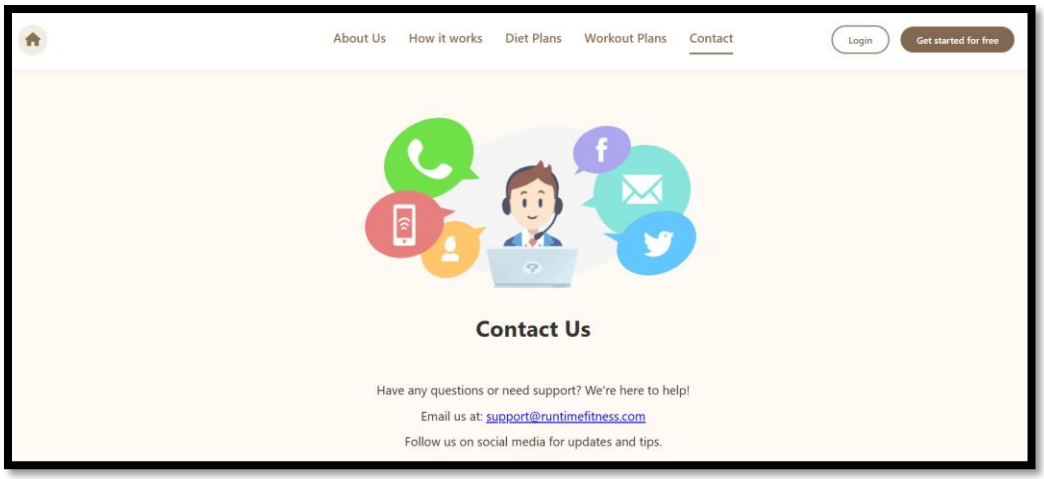


Figure 152: User Manual (contact.php)

10.1.7 How Users Can Register as a Member in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

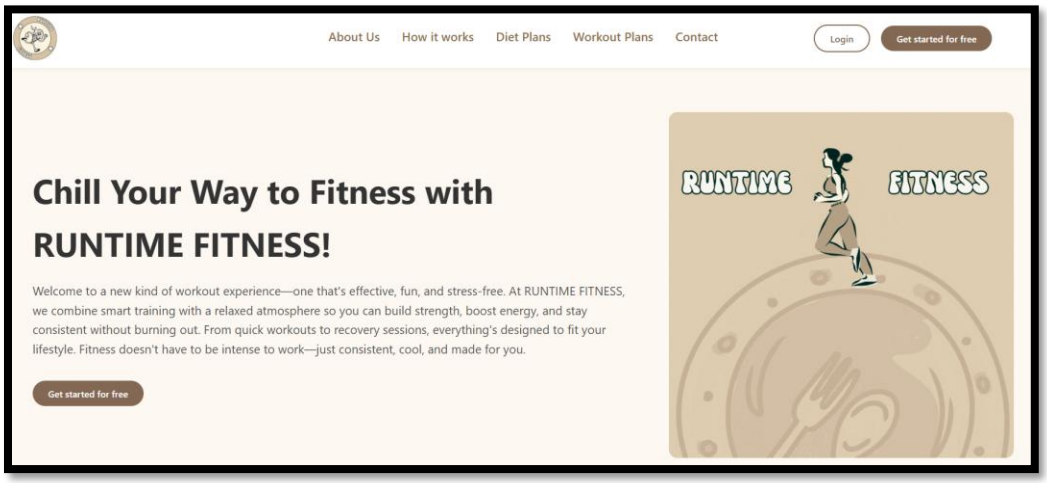


Figure 153: User Manual (homePage.php)

STEP 3:

Click the “Get started for free” button

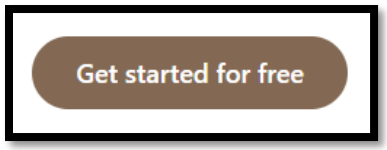


Figure 154: "Get started for free" button

STEP 4:

Enter username, full name, e-mail, phone, password, confirm password, security keyword, date of birth, gender, height, weight, fitness goal.

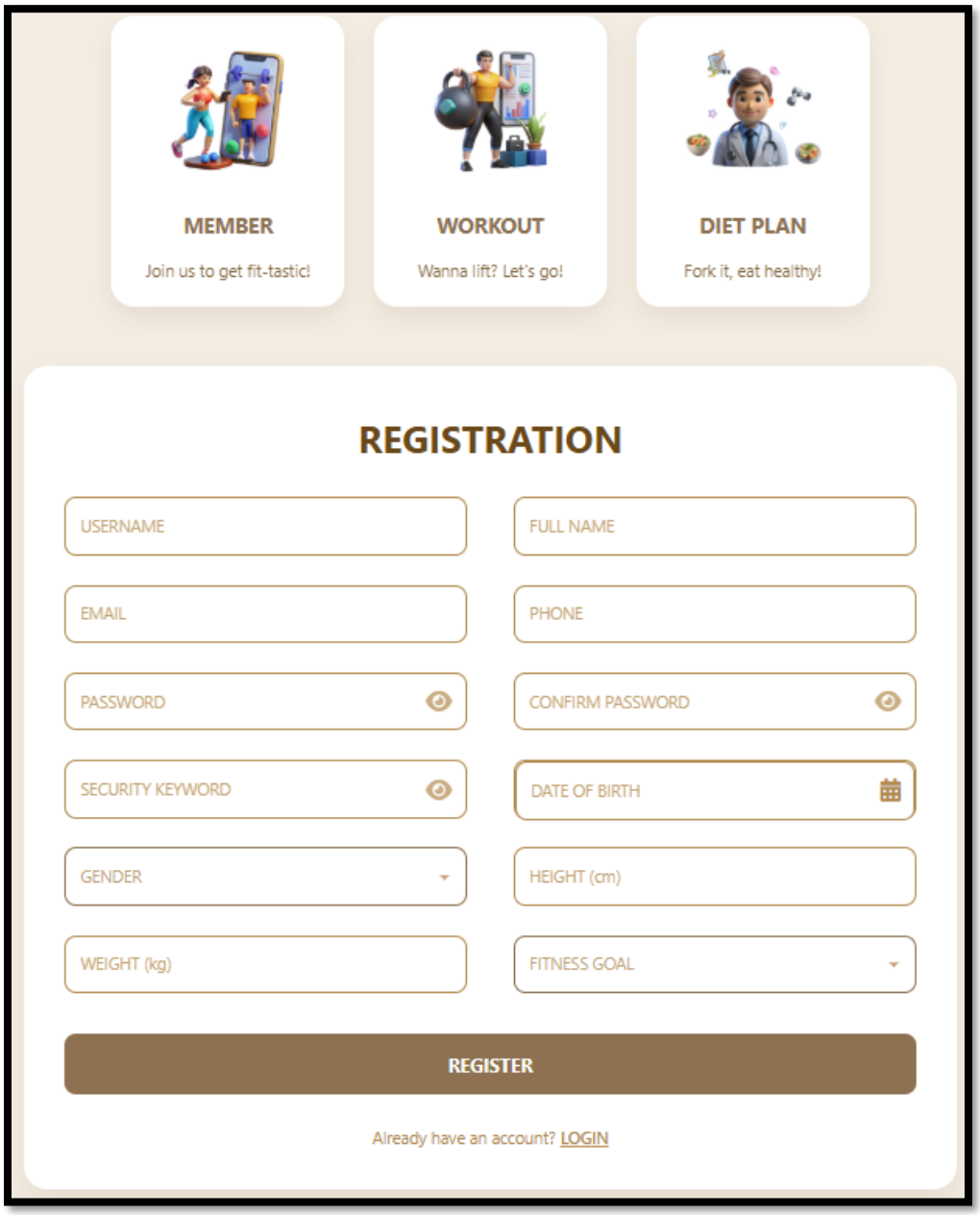


Figure 155: register.php

STEP 5:

Click the “REGISTER” button

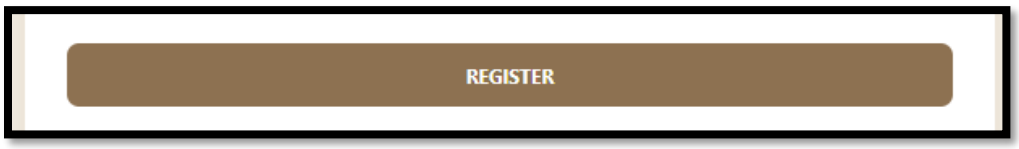


Figure 156: "REGISTER" button

10.2 Member

10.2.1 How Members Can Log into RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

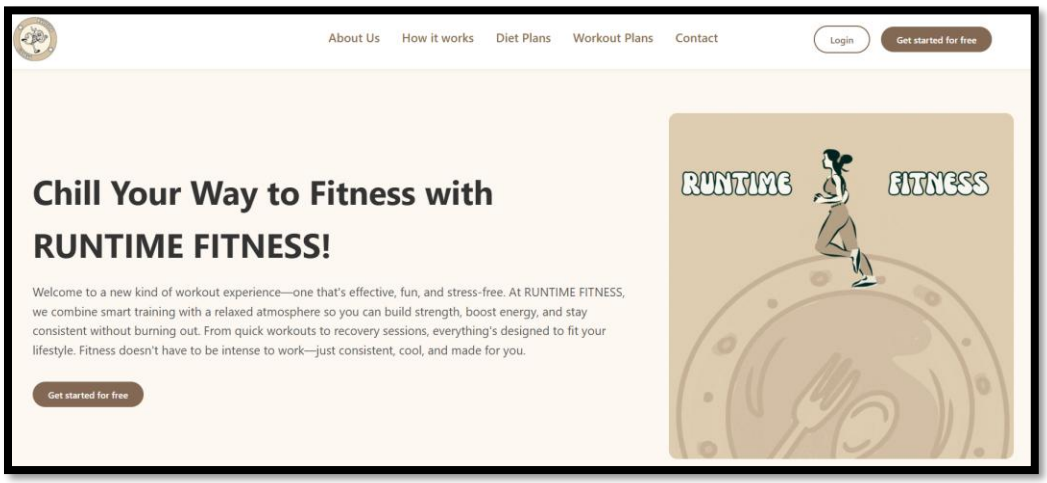


Figure 157: User Manual (homePage.php)

STEP 3:

Click the “Login” button



Figure 158: "Login" button

STEP 4:

Enter username and password

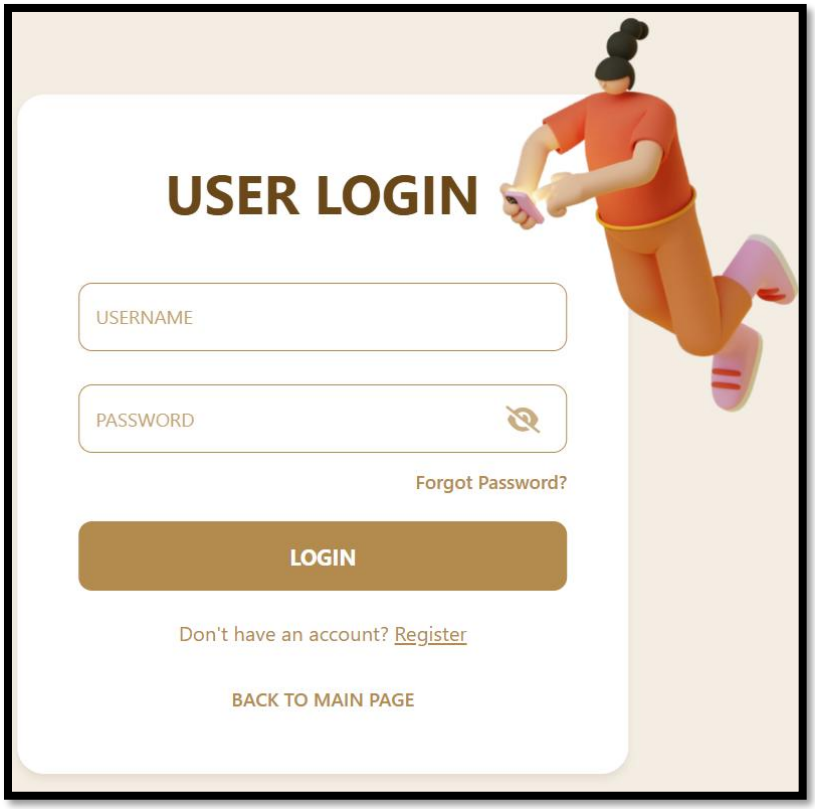


Figure 159: User Manual (login.html)

STEP 5:

Click the “LOGIN” button

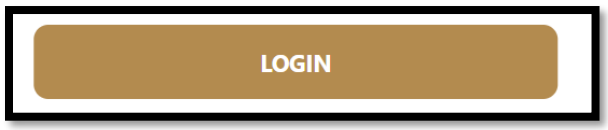


Figure 160: "LOGIN" button

STEP 6:

Once successfully logged in, it will be redirected to memberHomepage.php



Figure 161: indexMain.php

STEP 7:

memberHomepage.php is shown

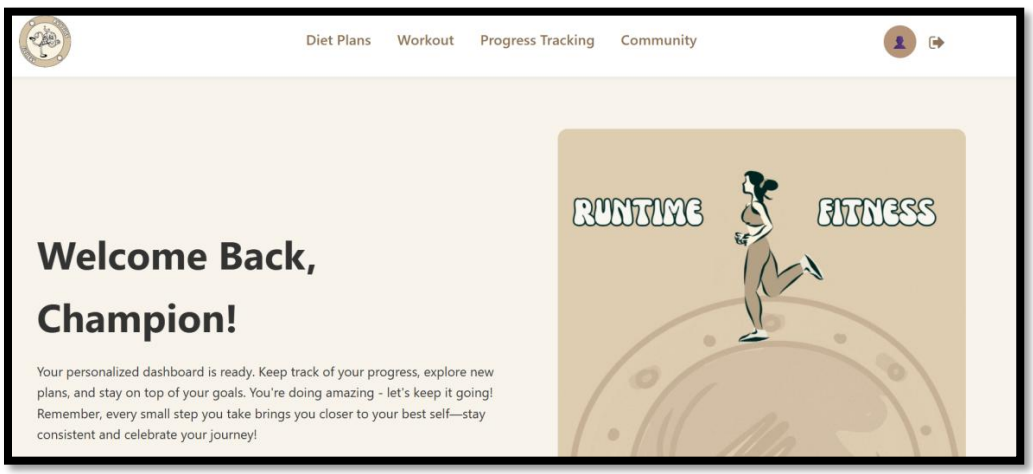


Figure 162: memberHomepage.php

10.2.2 How Members Can Reset Password in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

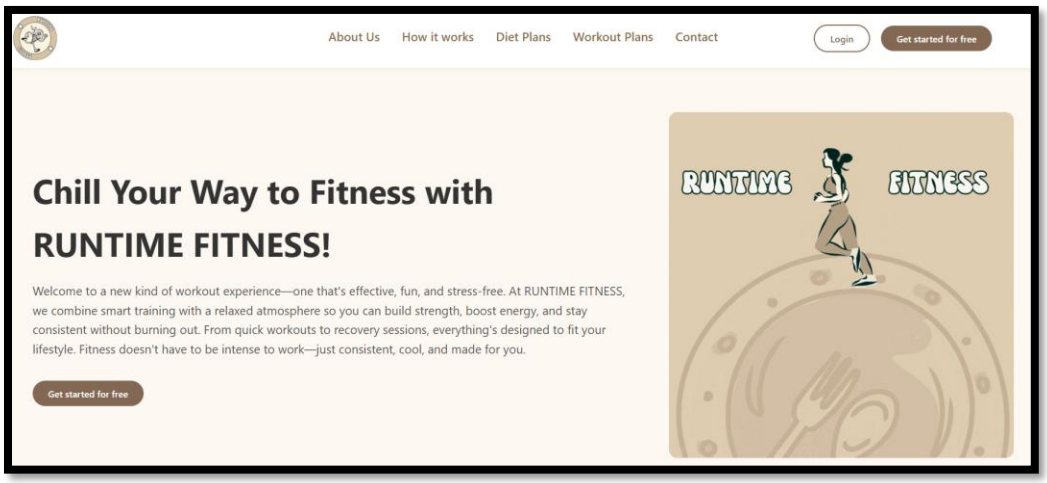


Figure 163: User Manual (homePage.php)

STEP 3:

Click the “Login” button



Figure 164: "Login" button

STEP 4:

login.html is shown.

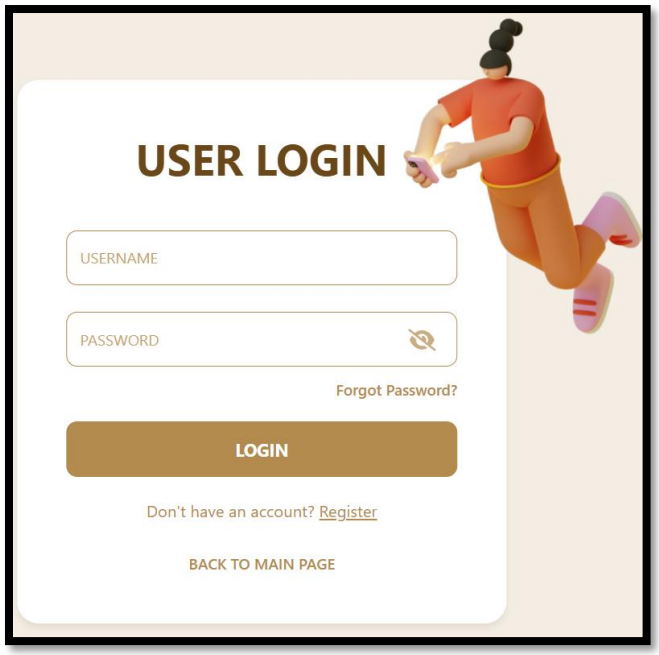


Figure 165: User Manual (login.html)

STEP 5:

Click “Forgot Password?”



Figure 166: "Forgot Password?" link

STEP 6:

Enter username, keyword, new password and confirm password

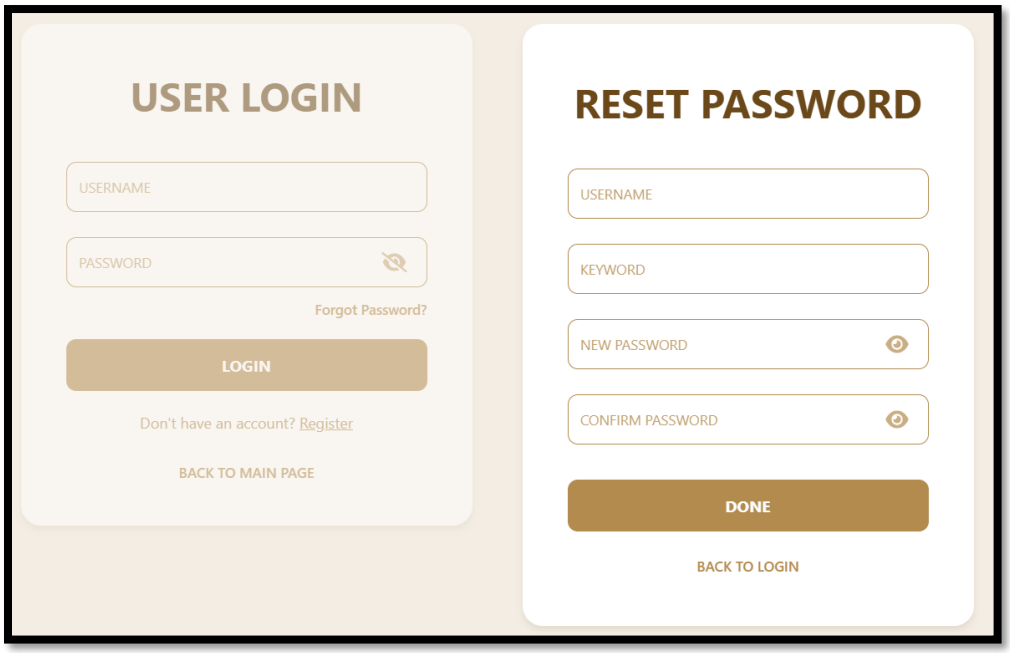


Figure 167: User Manual (Reset Password)

STEP 7:

Click the “DONE” button



Figure 168: "DONE" button

STEP 8:

Once the password is reset successfully, it will be redirect to login.html

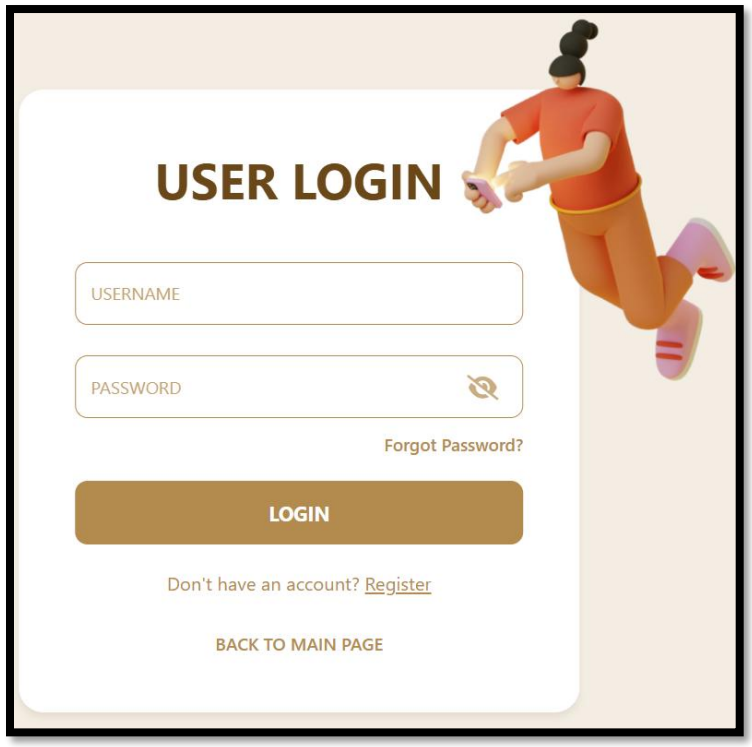


Figure 169: User Manual (login.html)

10.2.3 How Members Can Access the Diet Plans in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

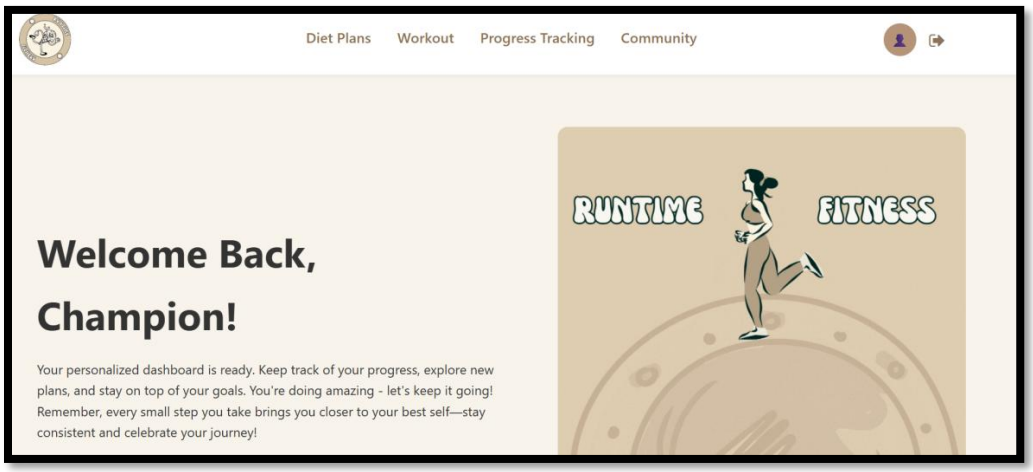


Figure 170: User Manual (memberHomepage.php)

STEP 2:

Click Diet Plans



Figure 171: "Diet Plans" link

STEP 3:

DP.MAIN.php is shown

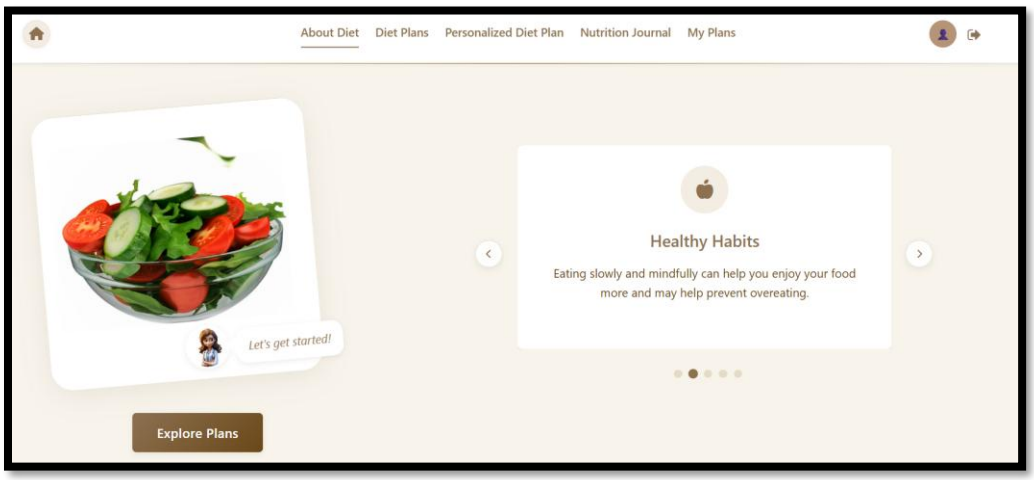


Figure 172: User Manual (DP.MAIN.php)

10.2.4 How Members Can View Default Diet Plans in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/DP.MAIN.php

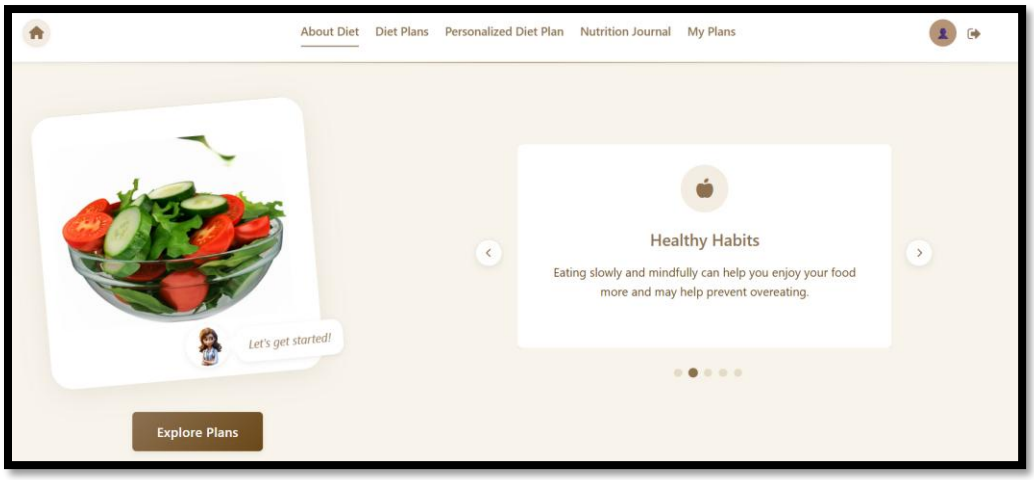


Figure 173: User Manual (DP.MAIN.php)

STEP 2:

Click Diet Plans



Figure 174: "Diet Plans" link

STEP 3:

Click one of the plan types



Figure 175: Sample Plan Type

STEP 4:

The 7-Day Meal Plan is shown below the plan type

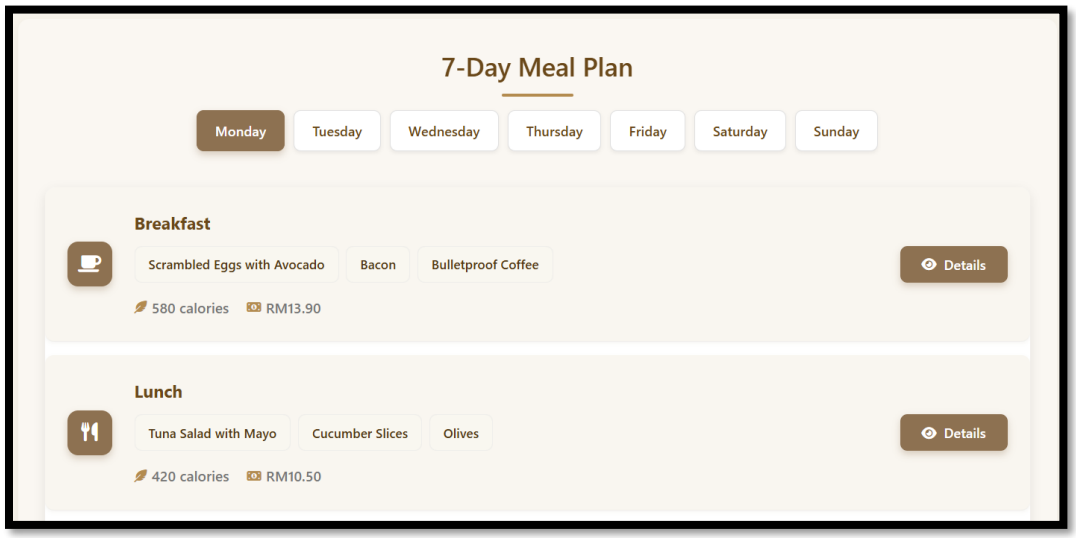


Figure 176: Meal Plan based on the Plan Type

STEP 5:

Click the “Details” button

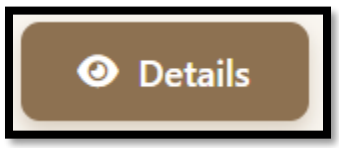


Figure 177: "Details" button

STEP 6:

Breakfast details are shown

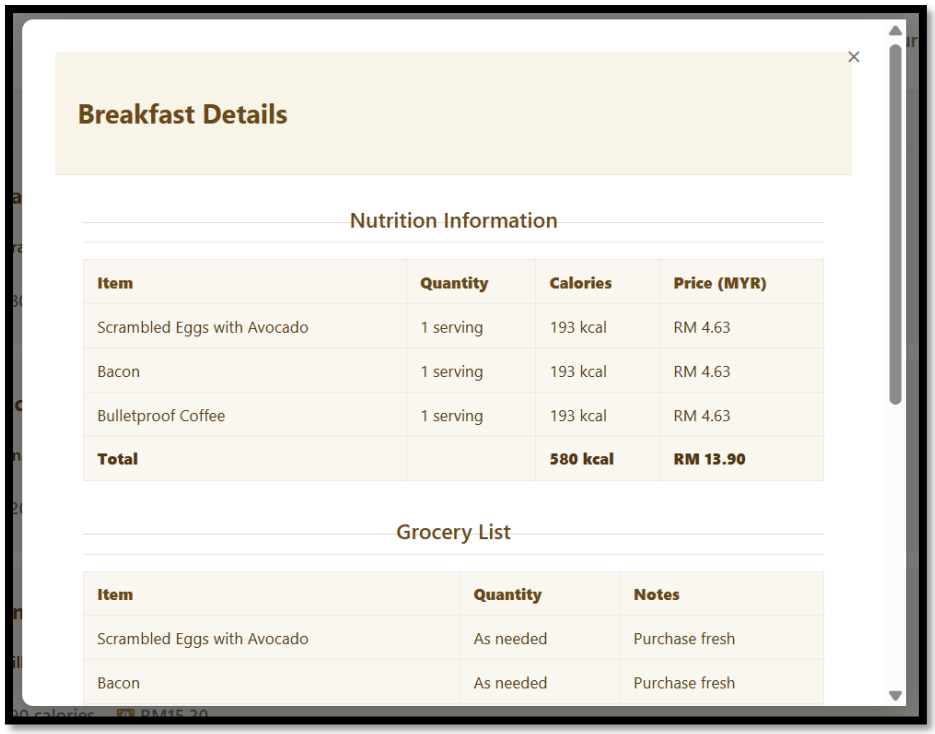


Figure 178: Breakfast Details from the Meal Plan

10.2.5 How Members Can Make Personalized Diet Plans in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/DP.MAIN.php

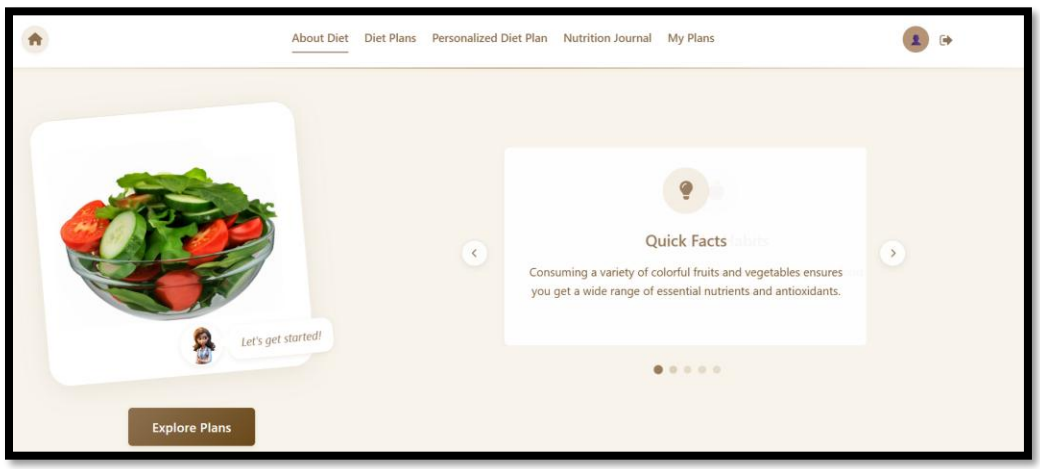


Figure 179: User Manual (DP.MAIN.php)

STEP 2:

Click Personalized Diet Plan



Figure 180: "Personalized Diet Plans" link

STEP 3:

DP.PS.Main.php is shown

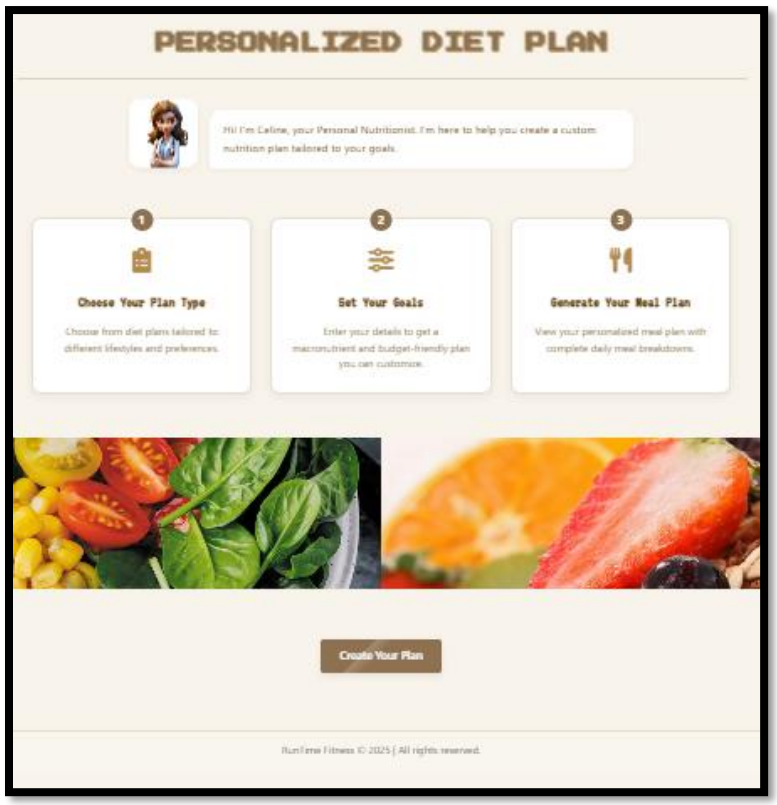


Figure 181: User Manual (DP.PS.Main.php)

STEP 4:

Click the “Create Your Plan” button



Figure 182: "Create Your Plan" button

STEP 5:

DP.PS.Step1.php is shown



Figure 183: User Manual (DP.PS.Step1.php)

STEP 6:

Select one of the Plan Types

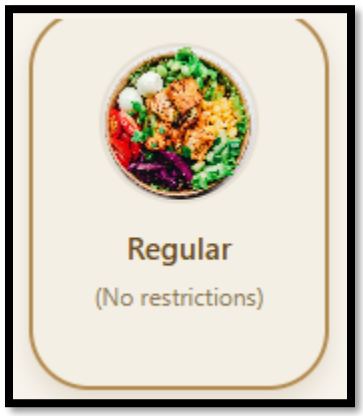


Figure 184: Sample Plan Type

STEP 7:

Click the “Next Step” button



Figure 185: "Next Step" button

STEP 7:

DP.PS.Step2.php is shown

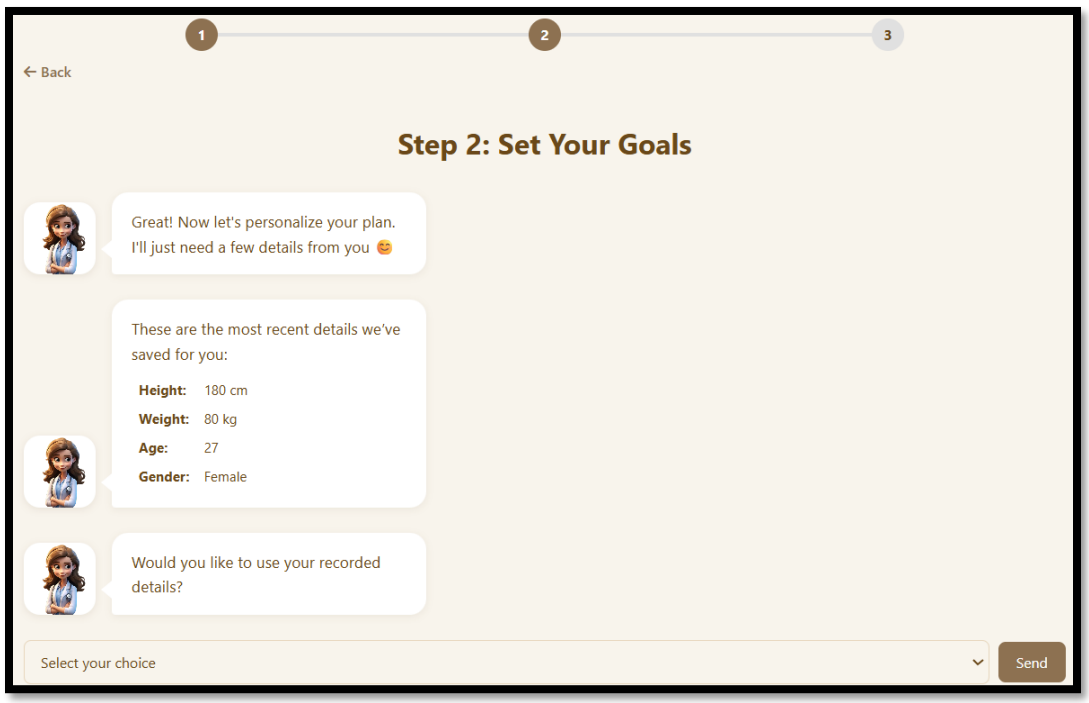


Figure 186: User Manual (DP.PS.Step2.php)

STEP 8:

Click “Select your choice” and choose “Yes” or “No”

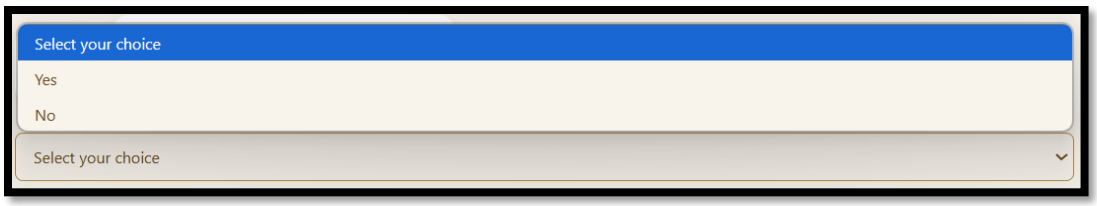


Figure 187: Select your choice

STEP 9:

Click the “Send” button



Figure 188: "Send" button

STEP 10:

Click “Select your choice” and choose one of the activity levels

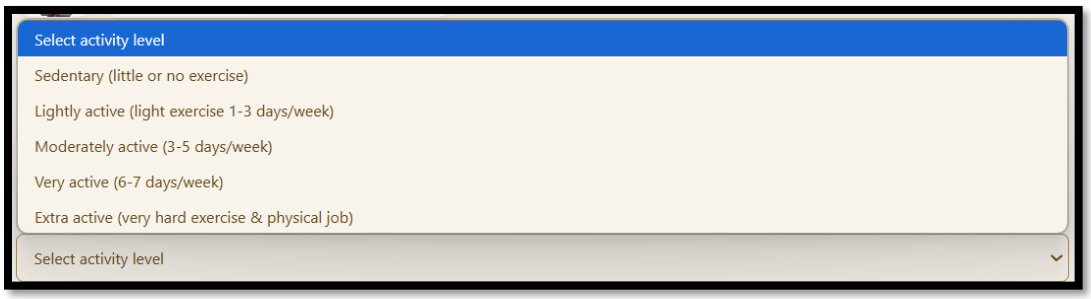


Figure 189: Select activity level

STEP 11:

Click the “Send” button



Figure 190: "Send" button

STEP 12:

Click “Select your goal” and choose one of the goal

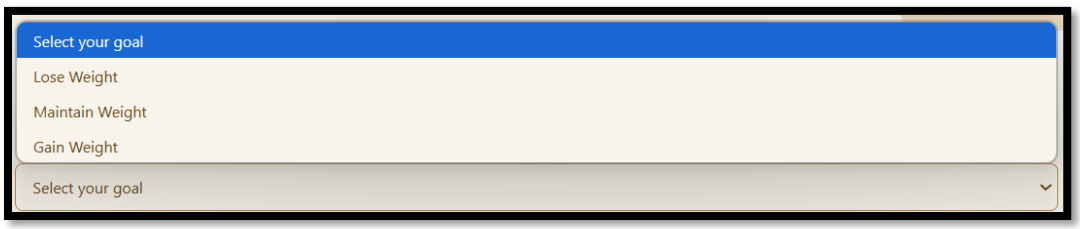


Figure 191: Select your goal

STEP 13:

Click the “Send” button



Figure 192: "Send" button

STEP 14:

Enter daily food budget

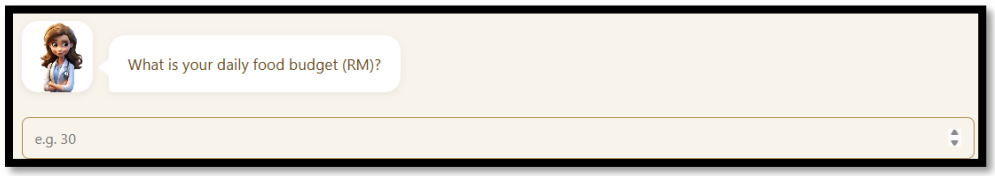


Figure 193: Enter daily food budget

STEP 15:

Click the “Send” button



Figure 194: "Send" button

STEP 16:

Click the “Next Step” button



Figure 195: "Next Step" button

STEP 17:

DP.PS.Step3.php is shown

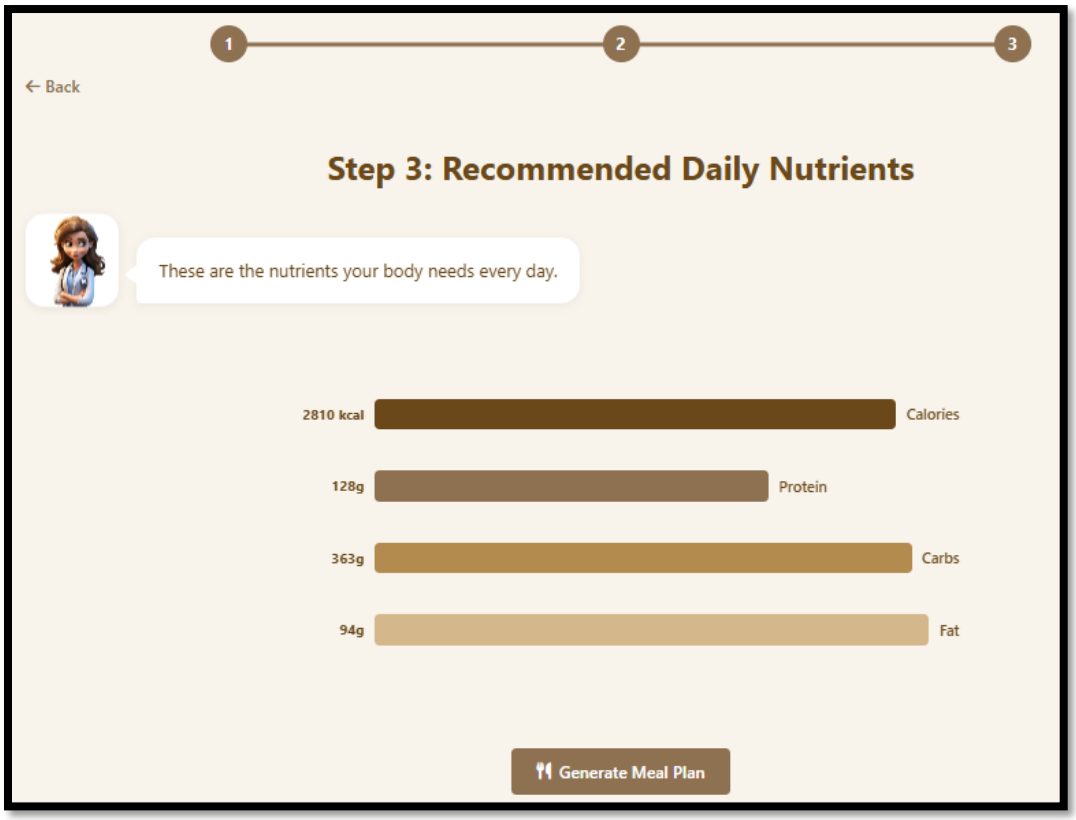


Figure 196: User Manual (DP.PS.Step3.php)

STEP 18:

Click the “Generate Meal Plan” button



Figure 197: "Generate Meal Plan" button

STEP 19:

DP.PS.Loading.php will be shown while the system is generating a meal plan

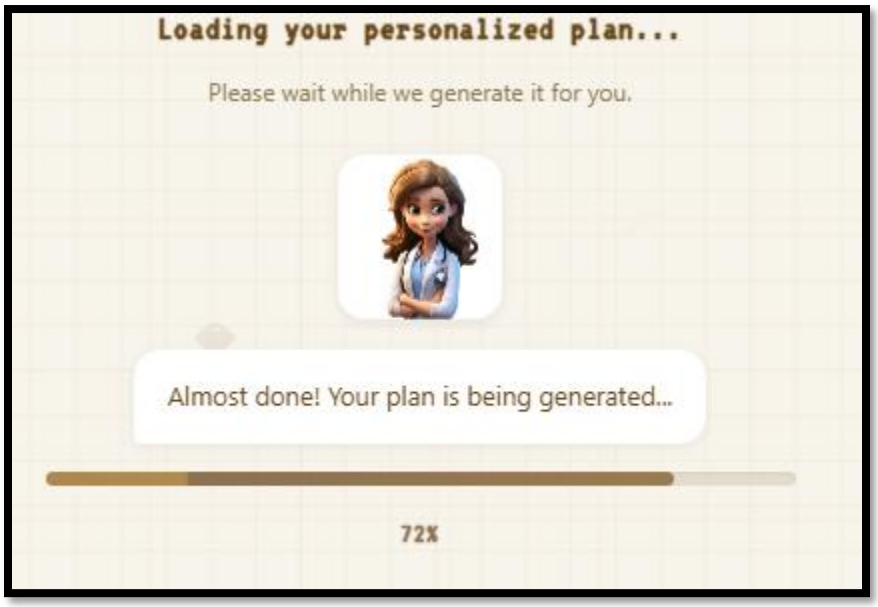


Figure 198: User Manual (DP.PS.Loading.php)

STEP 19:

DP.PS.GeneratedPlan.php will be shown while the system is generating a meal plan

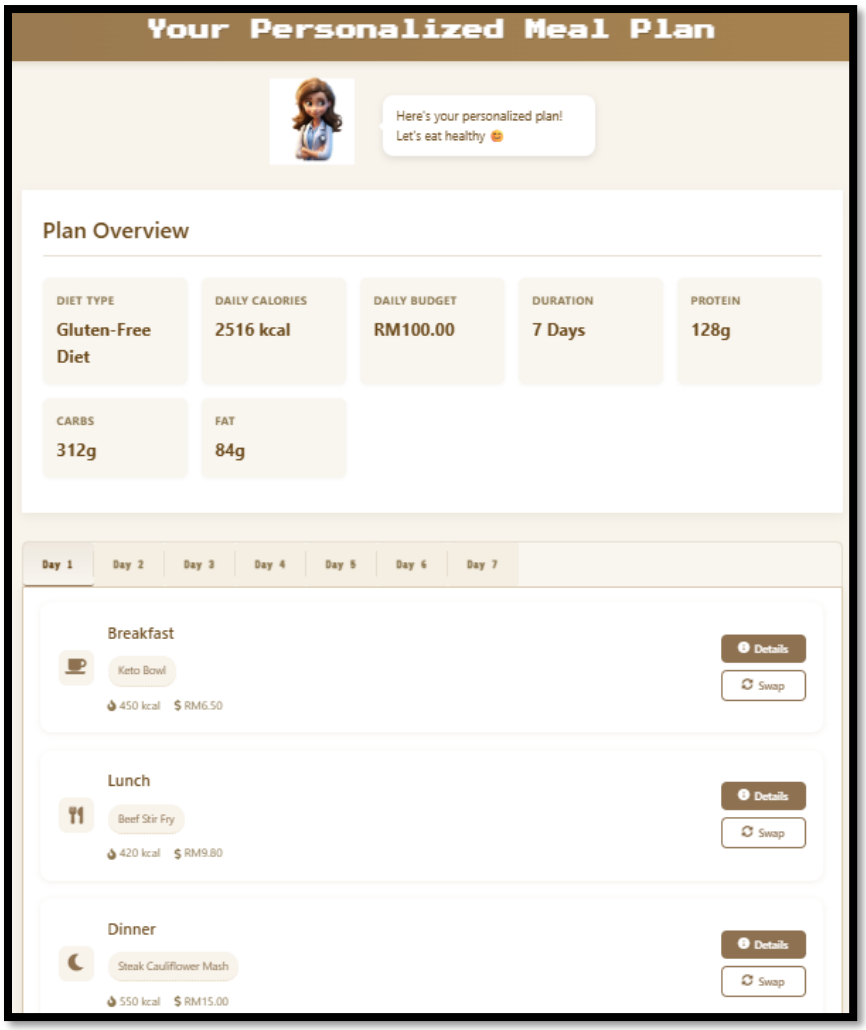


Figure 199: User Manual (DP.PS.GeneratedPlan.php)

STEP 20:

Click the “Details” button



Figure 200: "Details" button

STEP 20:

Meal Plan details will be shown

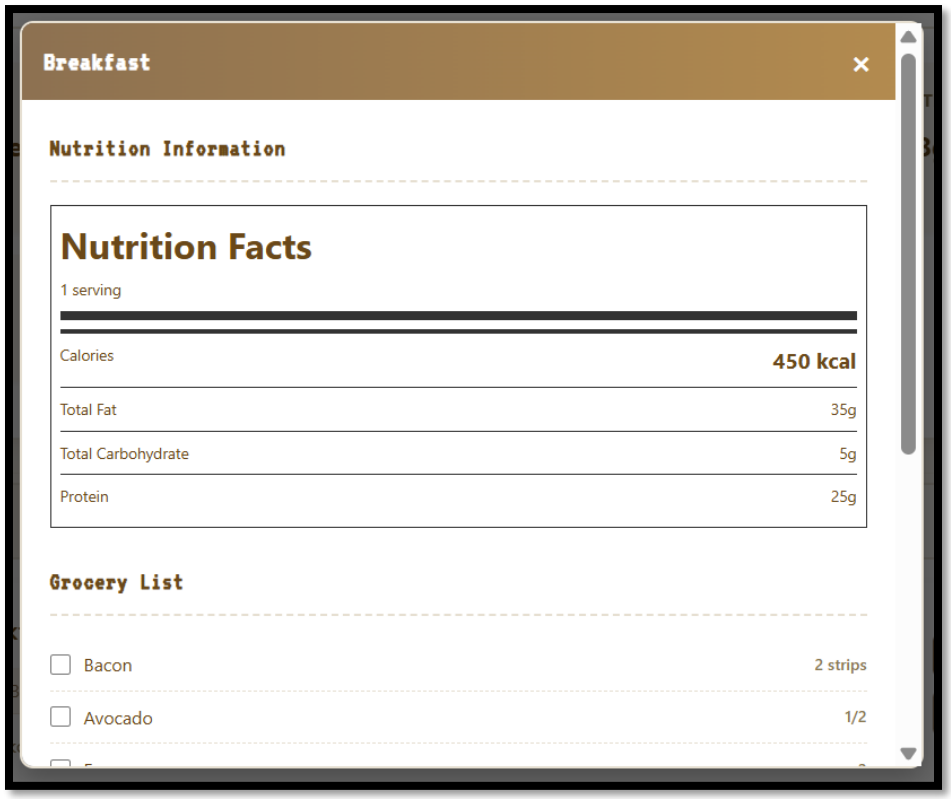


Figure 201: "Meal Plan details"

STEP 21:

Click the “Swap” button



Figure 202: "Swap" button

STEP 22:

Available meals will be shown for swap purposes

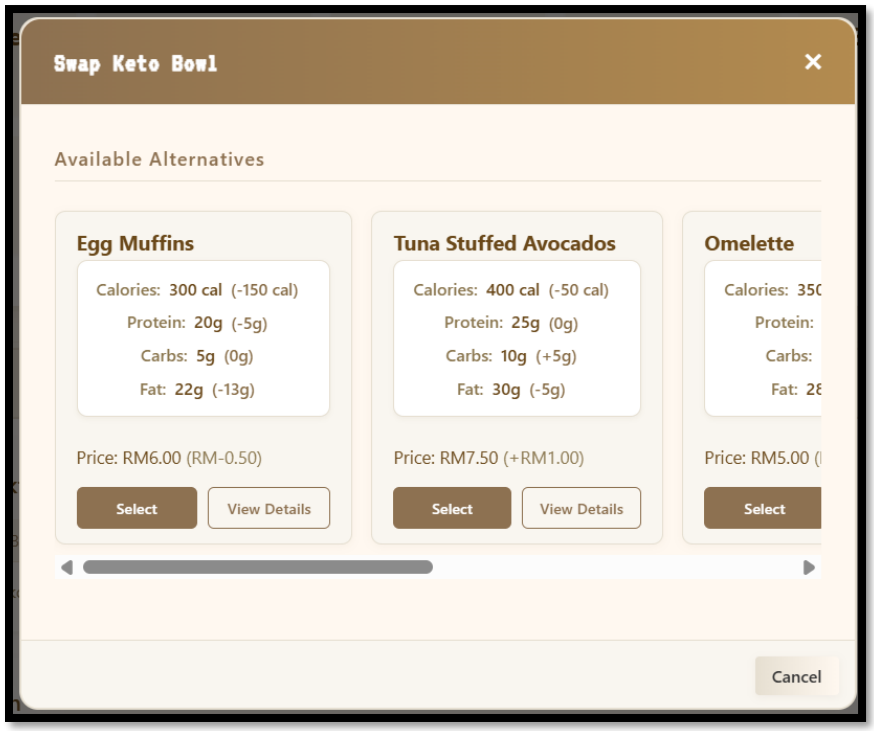


Figure 203: Available Meals

STEP 23:

Click the “Select” button



Figure 204: "Select" button

STEP 23:

Successfully to swapped the meal

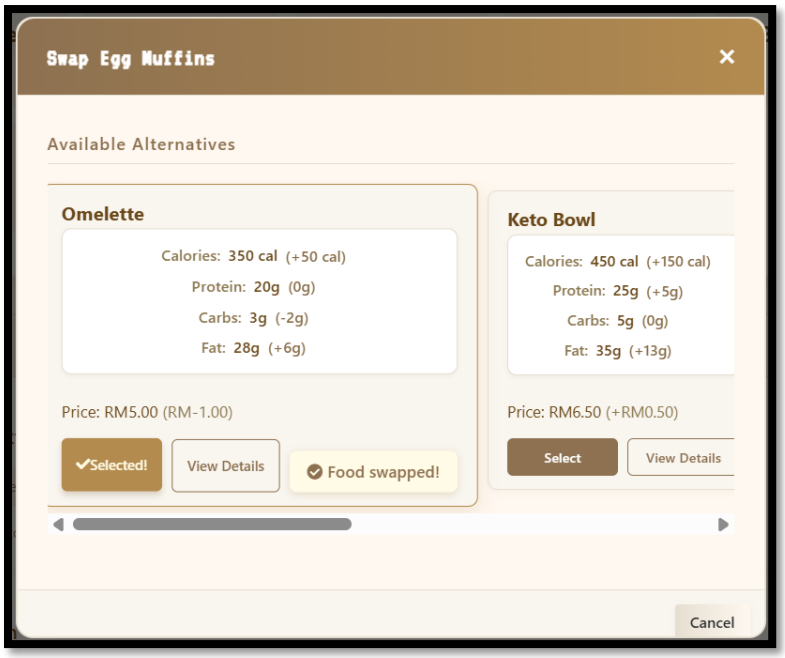


Figure 205: Successfully Swap Meals

STEP 24:

Click the “Save” Plan



Figure 206: "Save Plan" button

STEP 25:

Click “Save” Plan

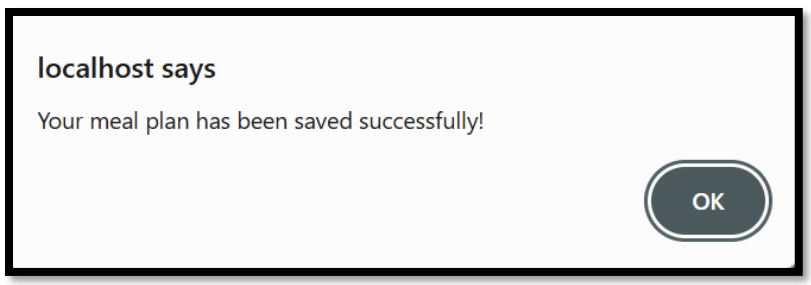


Figure 207: Meal plan saved successfully message

STEP 26:

Click the “Regenerate” Plan



Figure 208: "Regenerate" button

STEP 27:

Regenerated Meal Plan is shown

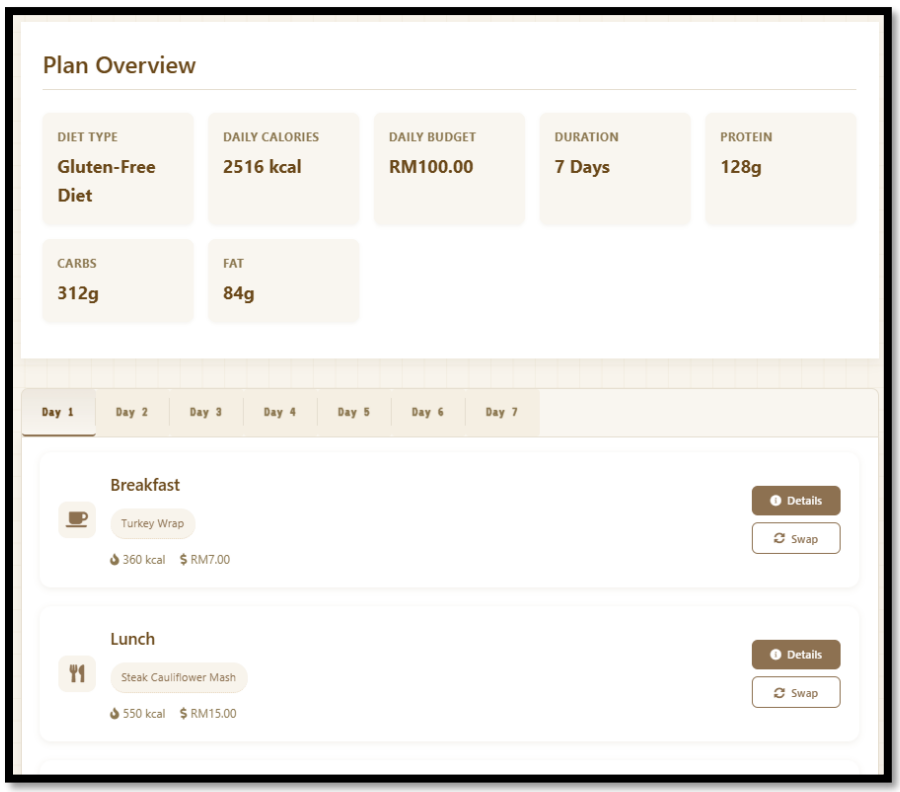


Figure 209: Regenerated Meal Plan

STEP 28:

Click the “View Saved Plans” Plan



Figure 210: "View Saved Plan" button

STEP 29:

DP.SP.PlansList.php is shown, displaying all saved plans

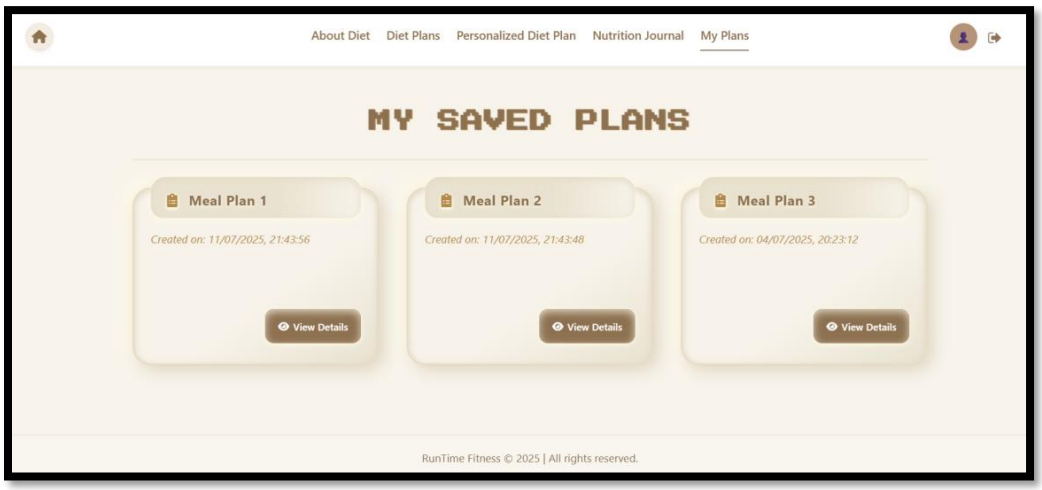


Figure 211: User Manual (DP.SP.PlansList.php)

STEP 30:

Click the “New Plan” Plan



Figure 212: "New Plan" button

STEP 31:

DP.PS.Main.php is shown

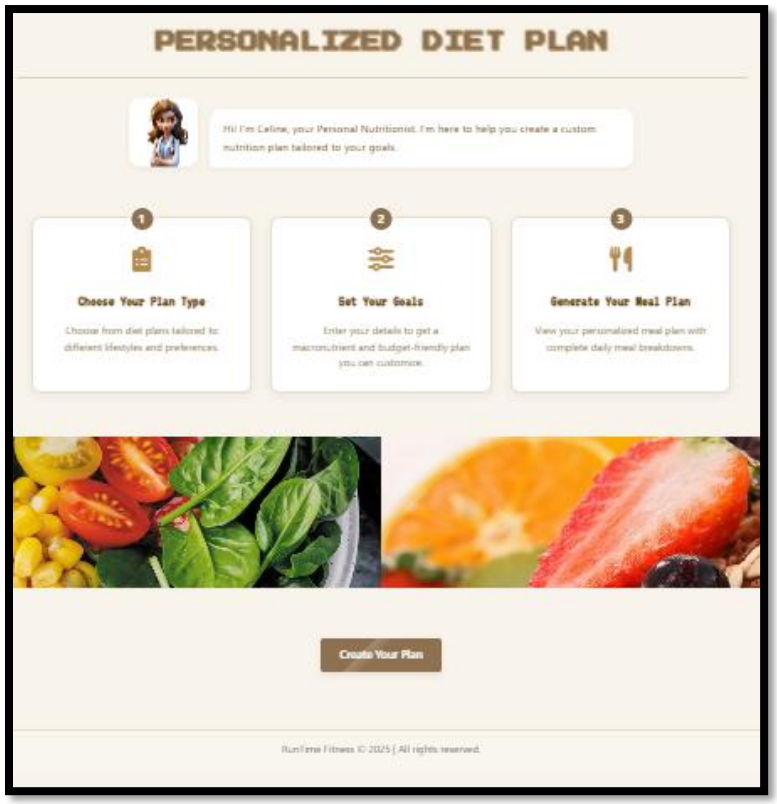


Figure 213: User Manual (DP.PS.Main.php)

10.2.6 How Members Can Write in Nutrition Journal in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/DP.MAIN.php

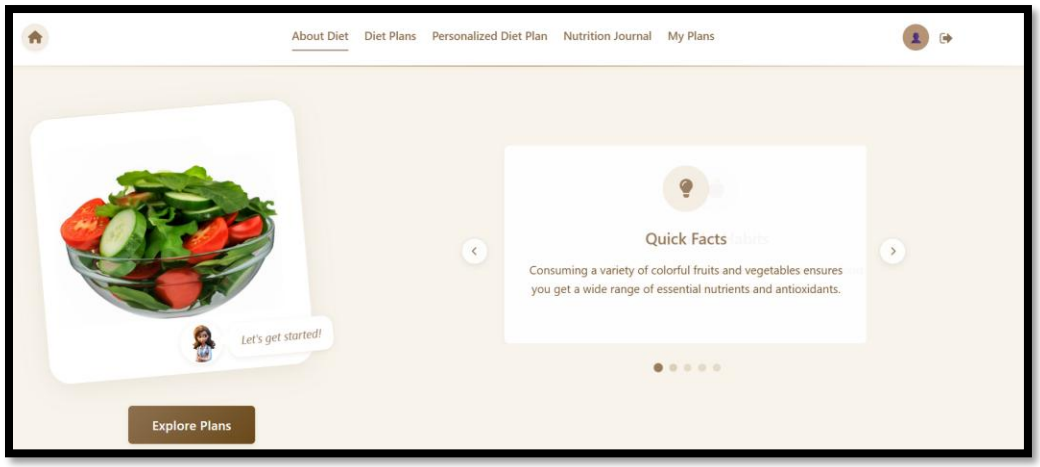


Figure 214: User Manual (DP.MAIN.php)

STEP 2:

Click Nutrition Journal



Figure 215: "Nutrition Journal" link

STEP 3:

DP.NT.Main.php is shown

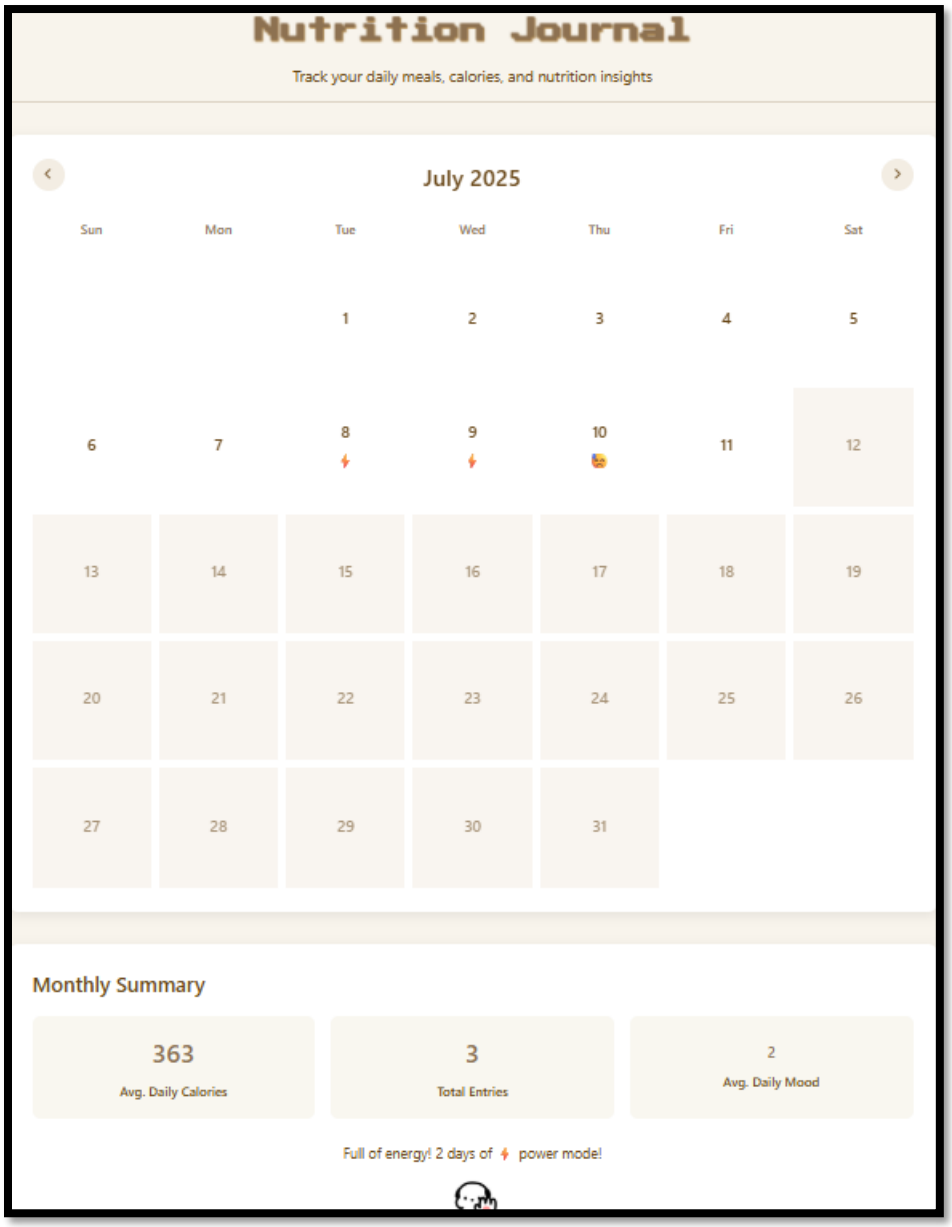


Figure 216: User Manual (DP.NT.Main.php)

STEP 3:

Click one of the dates from the calendar

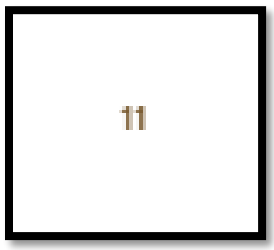


Figure 217: Selected Date

STEP 4:

Enter food name, portion size, total calories for breakfast

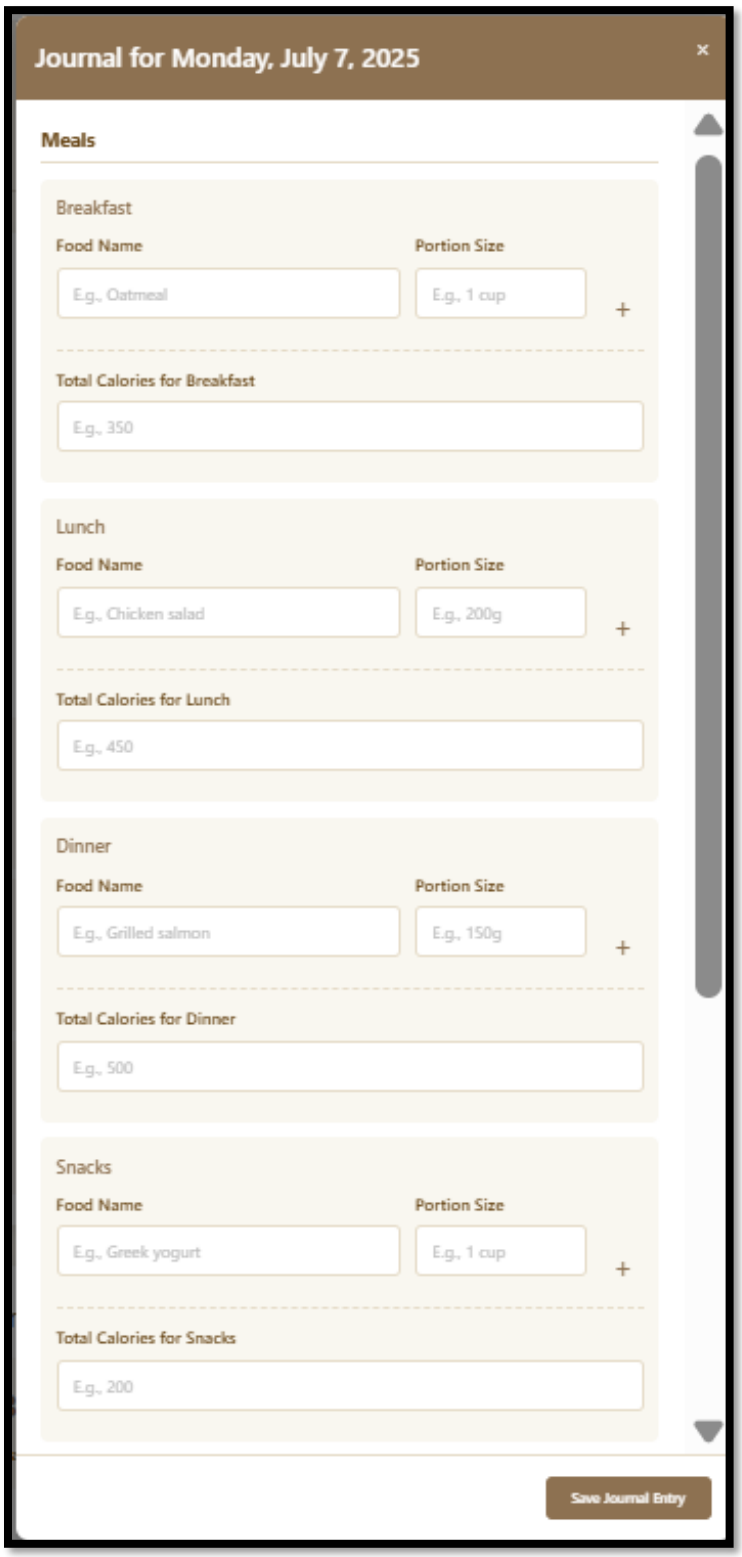


Figure 218: Journal Modal-Part 1

STEP 5:

Enter water intake, supplements/vitamins, additional remarks

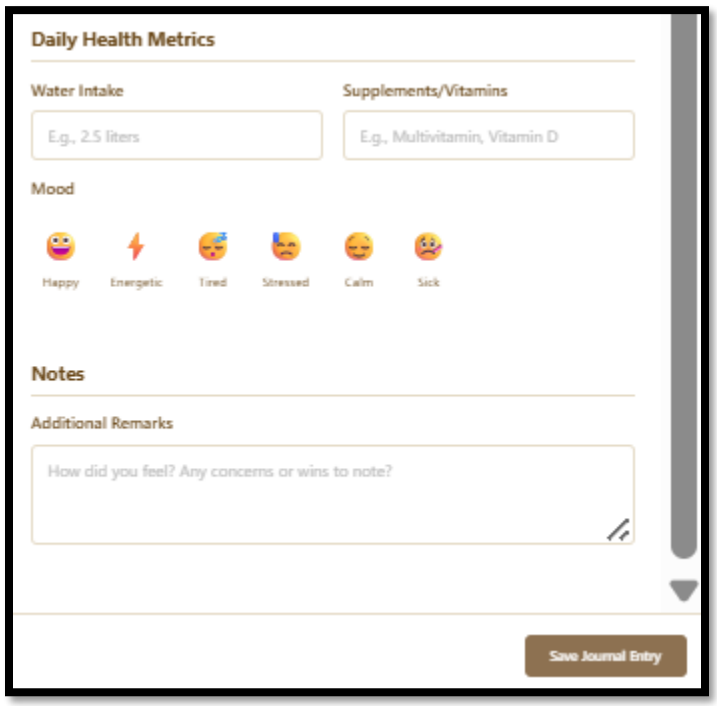


Figure 219: Journal Modal-Part 2

STEP 6:

Select mood

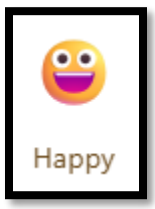


Figure 220: Selected Mood

STEP 7:

Click the “Save Journal Entry” button



Figure 221: "Save Journal Entry" button

STEP 8:

Selected mood will show on the selected date



Figure 222: Selected Mood shown on Selected Date

STEP 9:

Monthly Summary will be updated

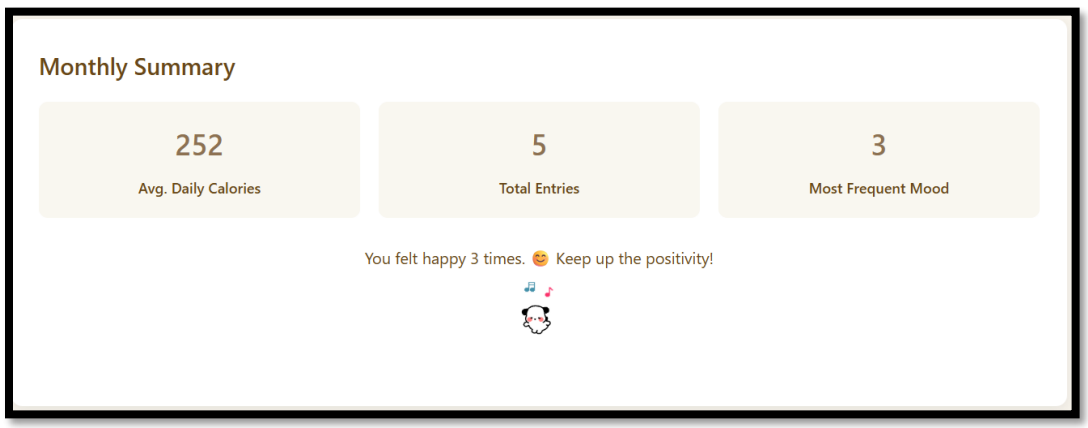


Figure 223: Monthly Summary

10.2.7 How Members Can View Saved Plans in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/DP.MAIN.php

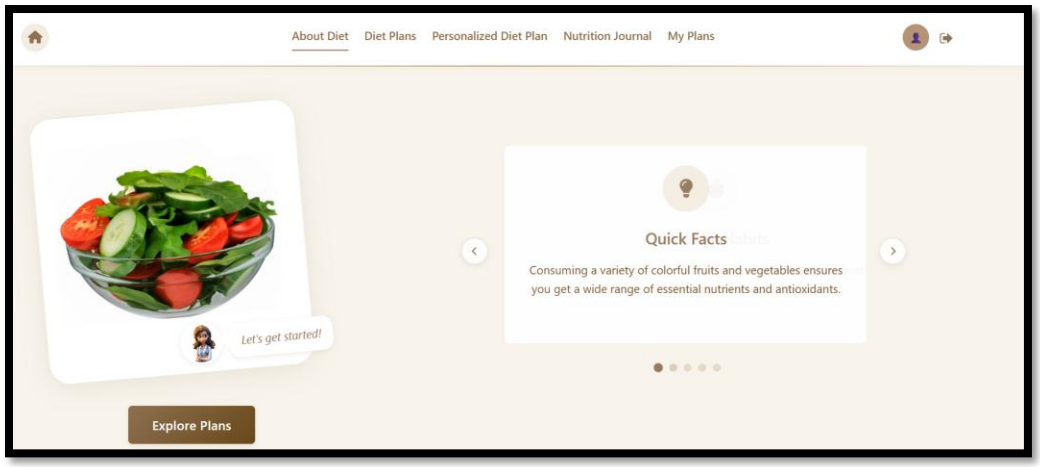


Figure 224: User Manual (DP.MAIN.php)

STEP 2:

Click My Plans



Figure 225: "My Plans" link

STEP 3:

DP.SP.PlansList.php is shown

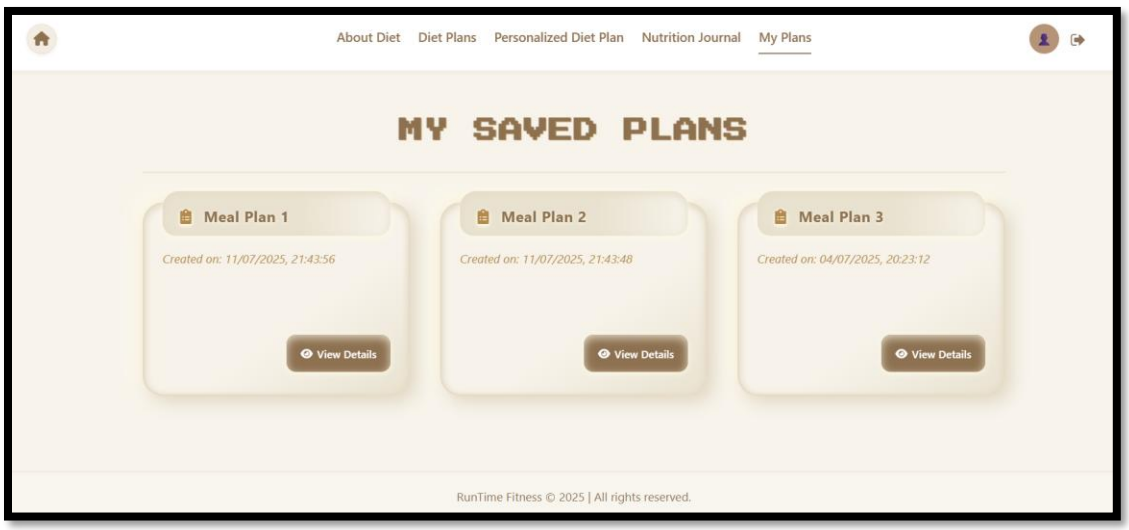


Figure 226: User Manual (DP.SP.PlansList.php)

STEP 4:

Click the “View Details” button

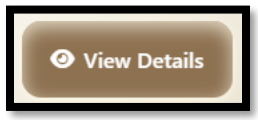


Figure 227: "View Details" button

STEP 5:

DP.SP.ViewDetails.php will be shown

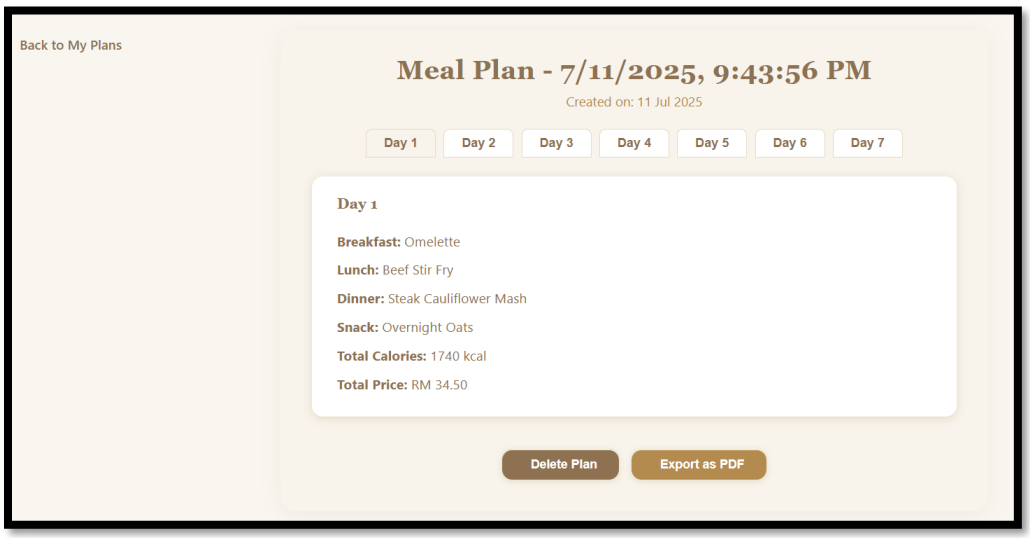


Figure 228: User Manual (DP.SP.ViewDetails.php)

STEP 6:

Click the “Delete Plan” button

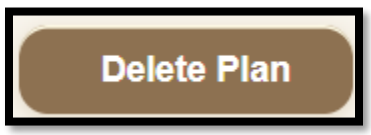


Figure 229: "Delete Plan" button

STEP 7:

Selected meal plan is deleted, remaining saved plans will be display on DP.SP.PlansList.php

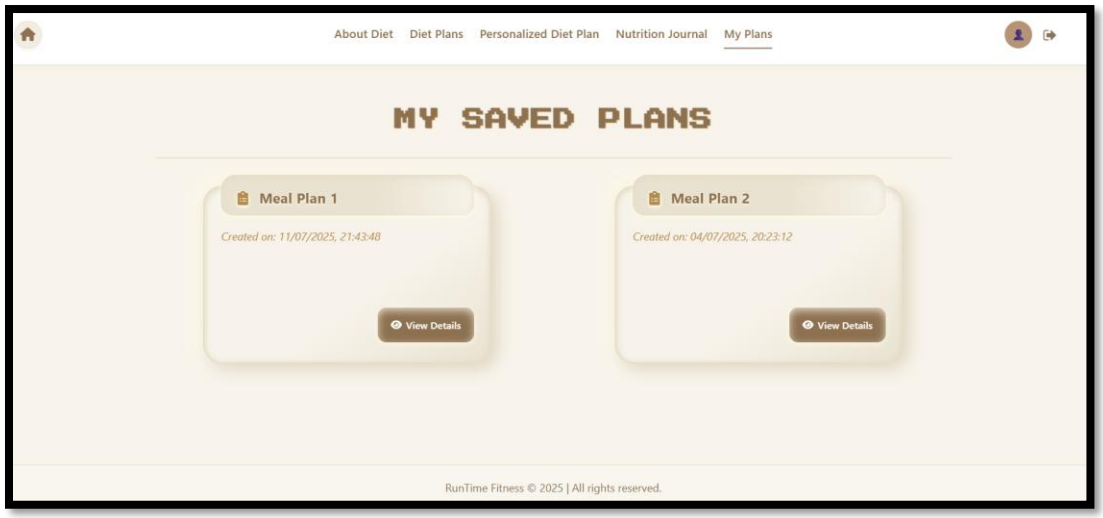


Figure 230: User Manual (DP.SP.PlansList.php)

STEP 8:

Click the “Export as PDF” button

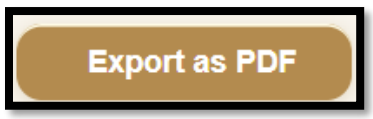


Figure 231: "Export as PDF" button

STEP 9:

Meal plan is exported and save as MyMealPlan(1).pdf



Figure 232: MyMealPlan(1).pdf

10.2.8 How Members Can Access to Workout in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

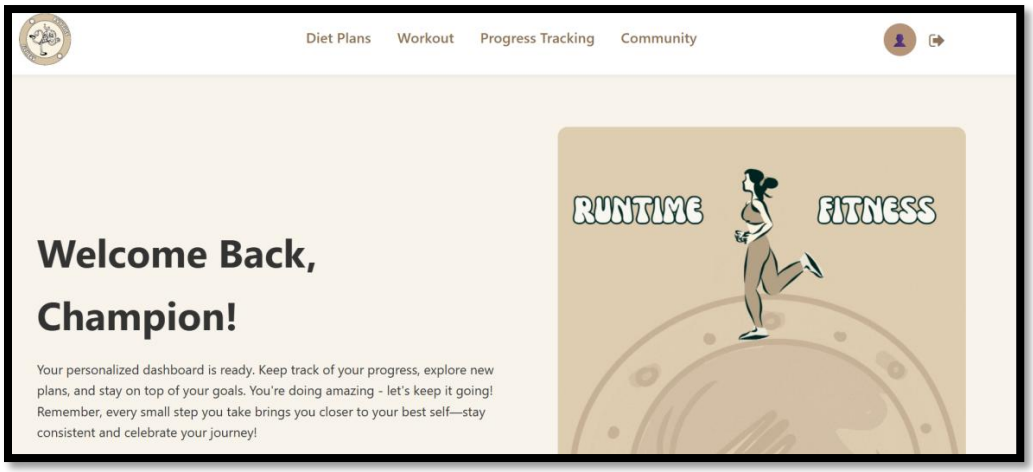


Figure 233: memberHomepage.php

STEP 2:

Click Workout



Figure 234: "Workout" link

10.2.9 How Members Can Access to Workout Library in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/W.Main.php

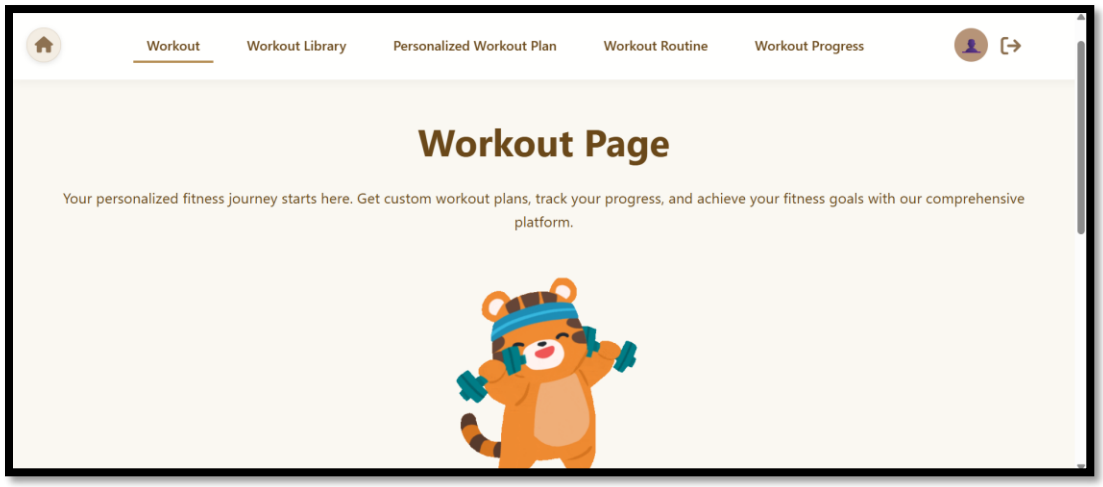


Figure 235: User Manual (W.Main.php)

STEP 2:

Click Workout Library



Figure 236: "Workout Library" link

STEP 3:

W.library.php is shown

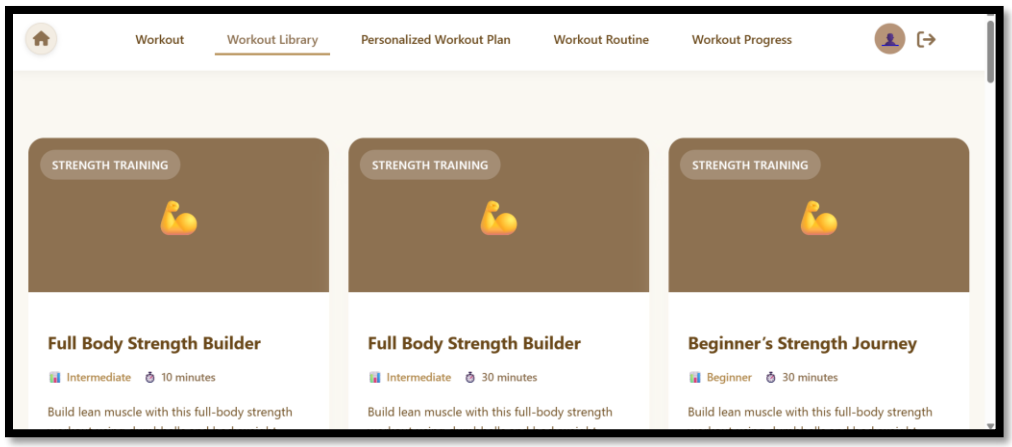


Figure 237: User Manual (W.Main.php)

STEP 4:

Click the “Watch Video” button



Figure 238: "Watch Video" button

STEP 5:

Member will be redirecting to tutorial videos



Figure 239: Tutorial Videos

10.2.10 How Members Can Access to Personalized Workout Plan in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/W.Main.php

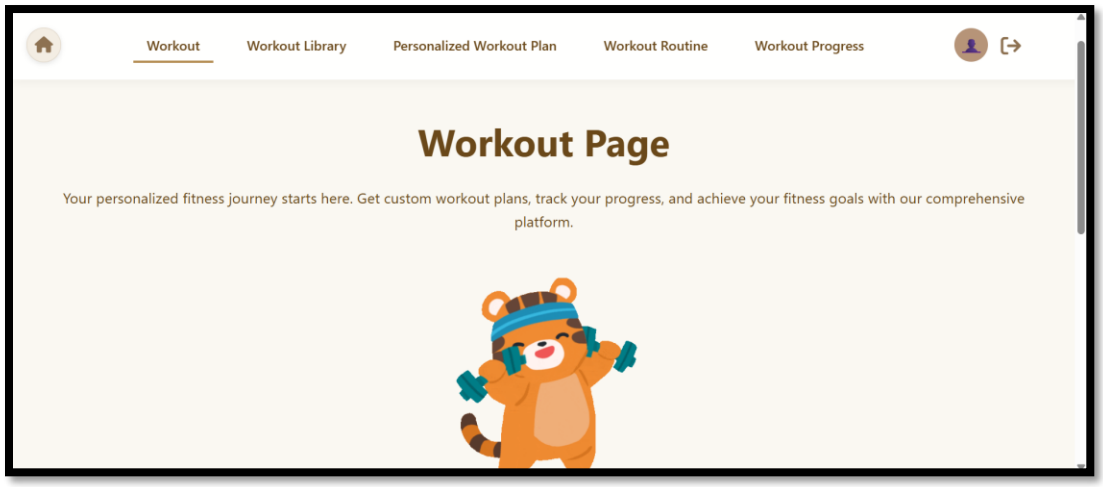


Figure 240: User Manual (W.Main.php)

STEP 2:

Click Personalized Workout Plan



Figure 241: "Personalized Workout Plan" link

STEP 3:

W.P.plan.php is shown

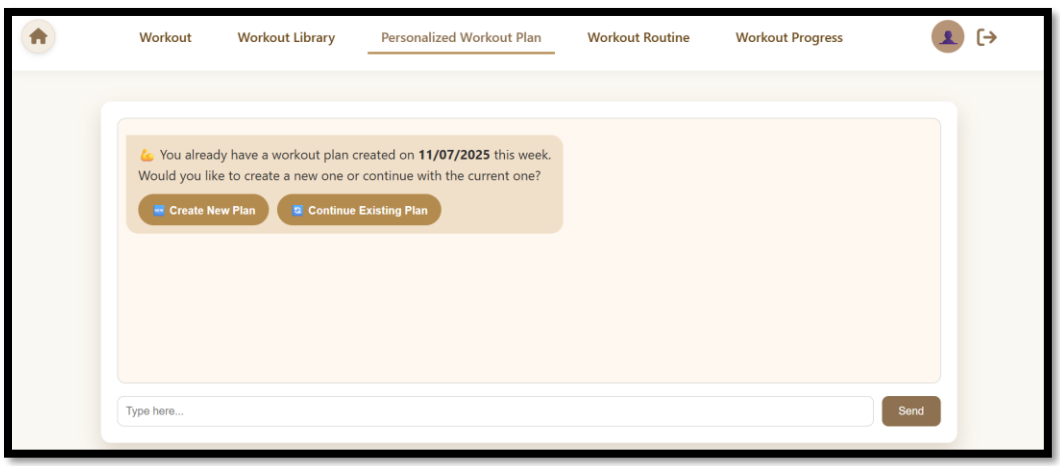


Figure 242: User Manual (W.P.plan.php)

STEP 4:

Type “yes” or “no”



Figure 243: User Manual (W.P.plan.php)

STEP 5:

Click the “Send” button



Figure 244: "Send" button

STEP 6:

Generated Workout Plan will be shown

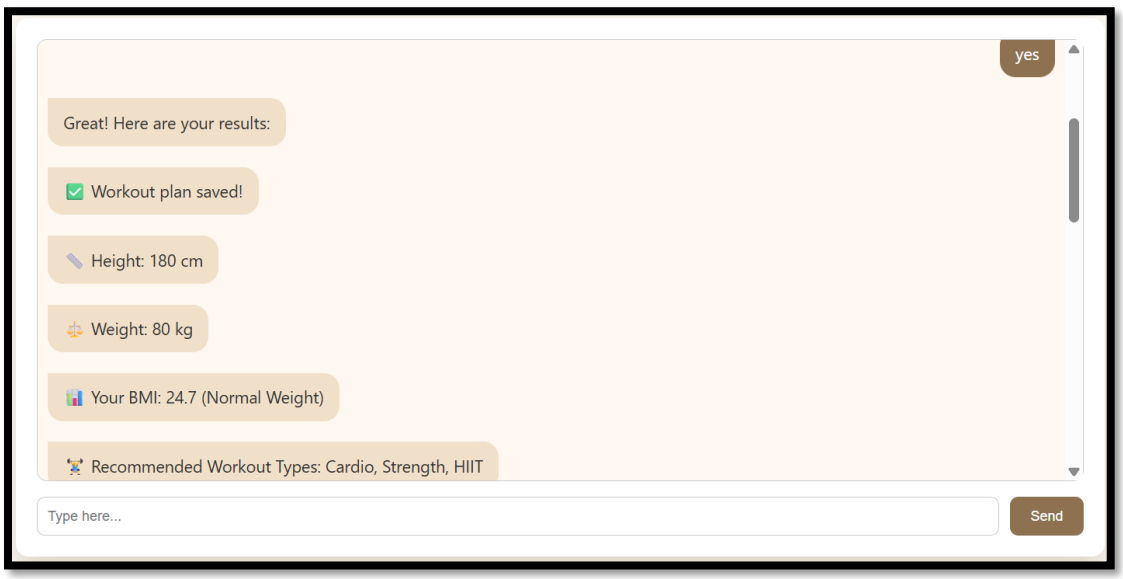


Figure 245: Generating Workout Plan-Part 1



Figure 246: Generating Workout Plan-Part 2

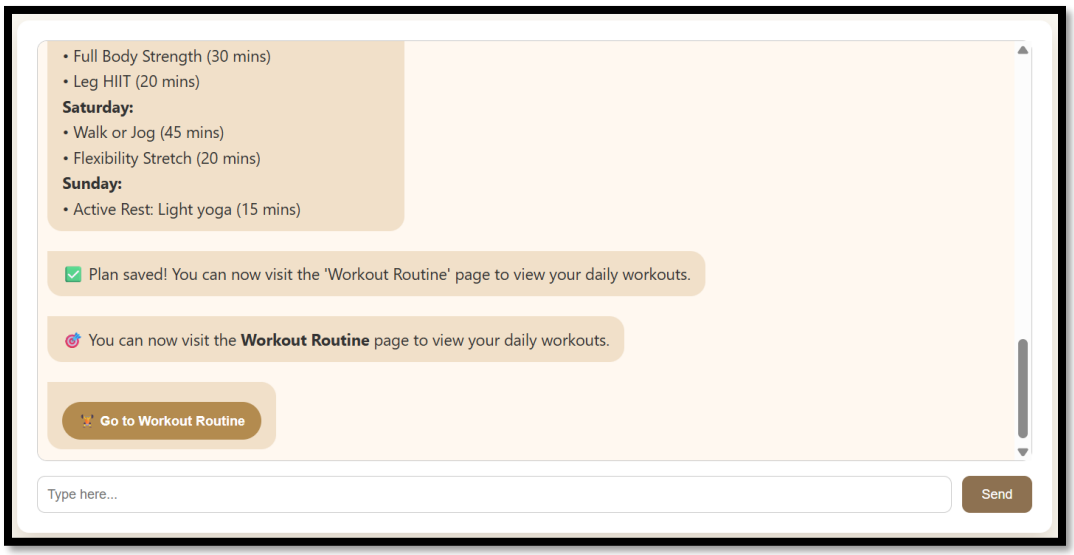


Figure 247: Generating Workout Plan-Part 3

STEP 7:

Click the “Go to Workout Routine” button



Figure 248: "Go to Workout Routine" button

STEP 8:

W.routine.php is shown

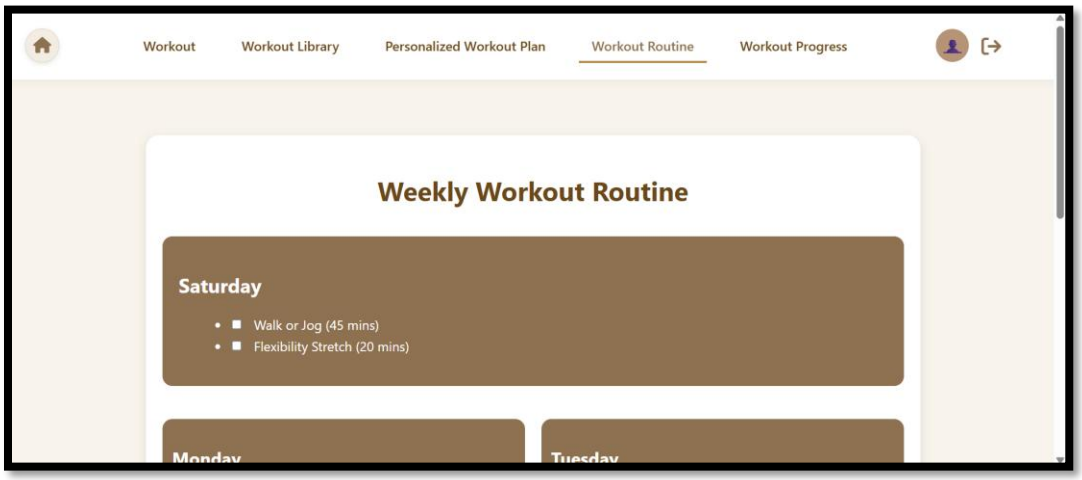


Figure 249: User Manual (W.routine.php)

10.2.11 How Members Can Access to Workout Routine in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/W.Main.php

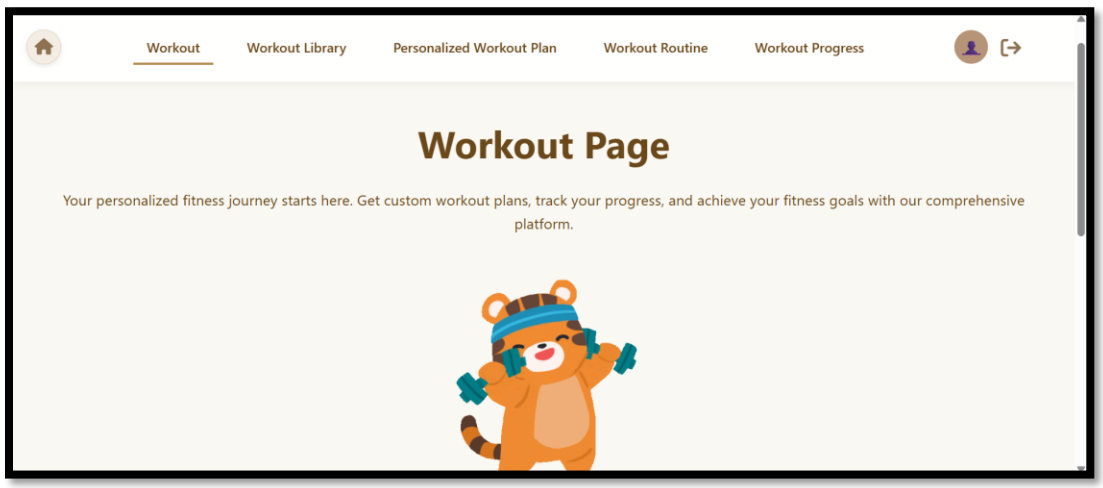


Figure 250: User Manual (W.Main.php)

STEP 2:

Click Workout Routine



Figure 251: "Personalized Workout Plan" link

STEP 3:

W.routine.php is shown

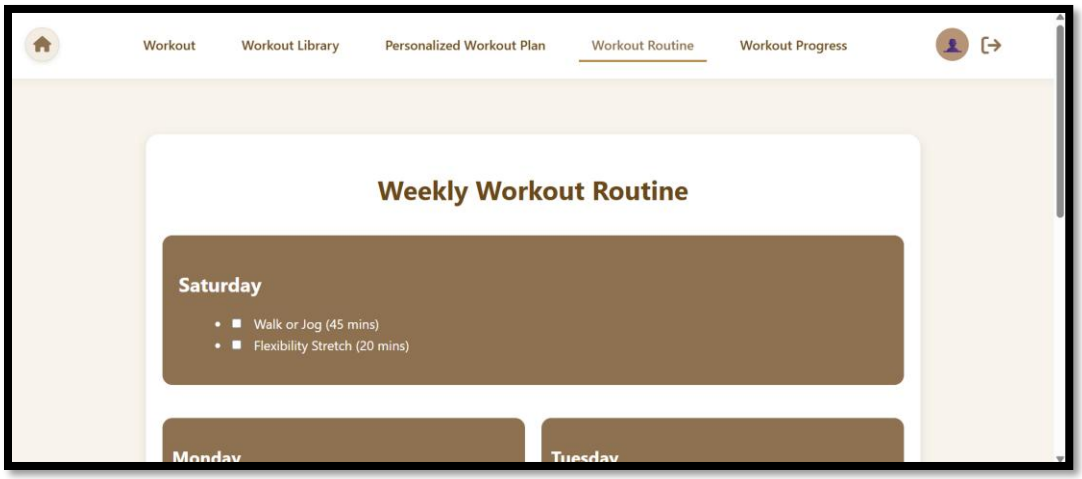


Figure 252: User Manual (W.routine.php)

STEP 4:

Tick the completed activity



Figure 253: Check box for Activity

10.2.12 How Members Can Access to Workout Progress in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/W.Main.php

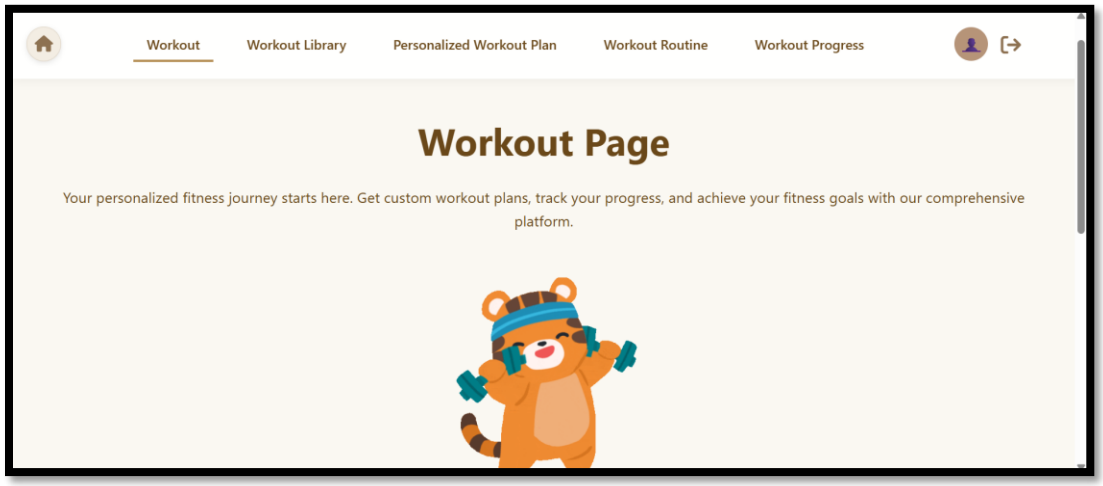


Figure 254: User Manual (W.Main.php)

STEP 2:

Click Workout Progress



Figure 255: "Workout Progress" link

STEP 3:

W.progress.php is shown

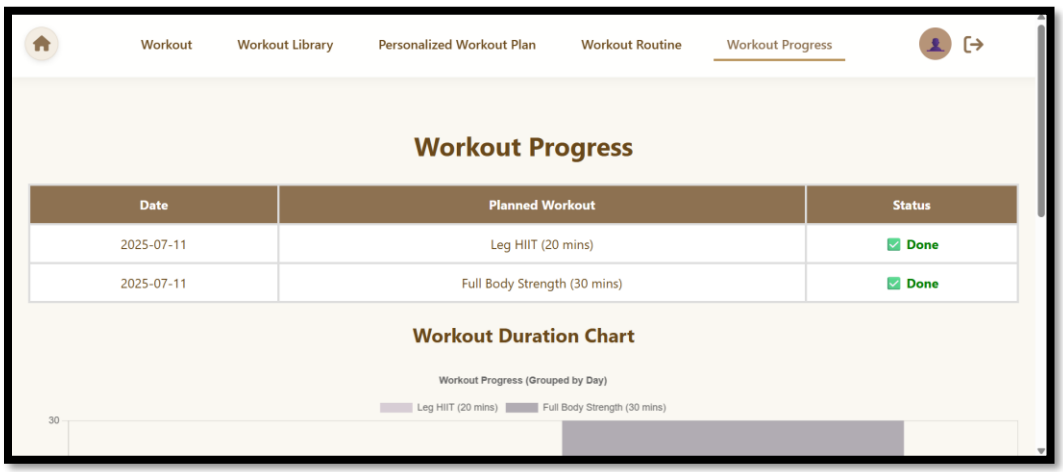


Figure 256: User Manual (W.progress.php)

STEP 4:

The Workout Progress and Workout Duration Chart will be shown

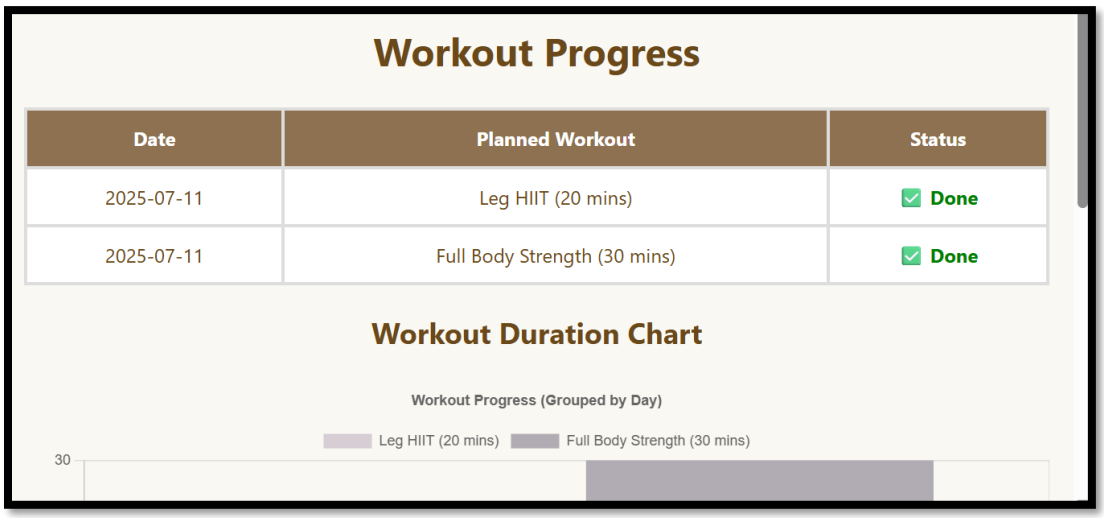


Figure 257: Workout Progress

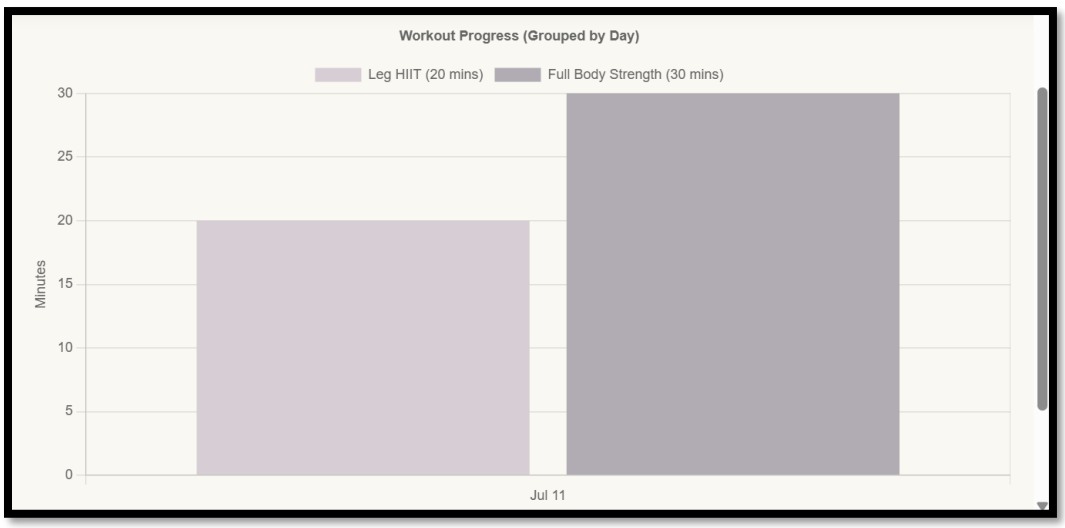


Figure 258 : Workout Duration Chart

10.2.13 How Members Can Update Daily Weight in Progress Tracking of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

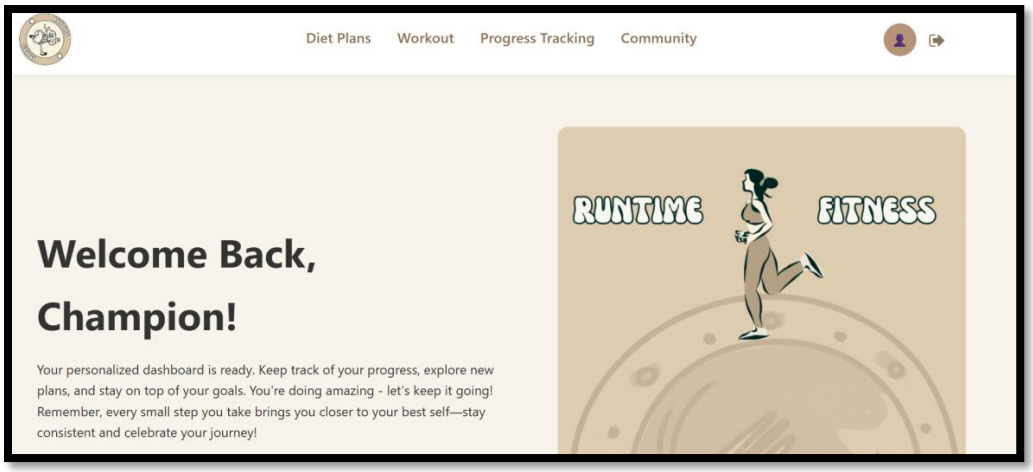


Figure 259: User Manual (memberHomepage.php)

STEP 2:

Click Progress Tracking



Figure 260: "Progress Tracking" link

STEP 3:

Enter daily weight

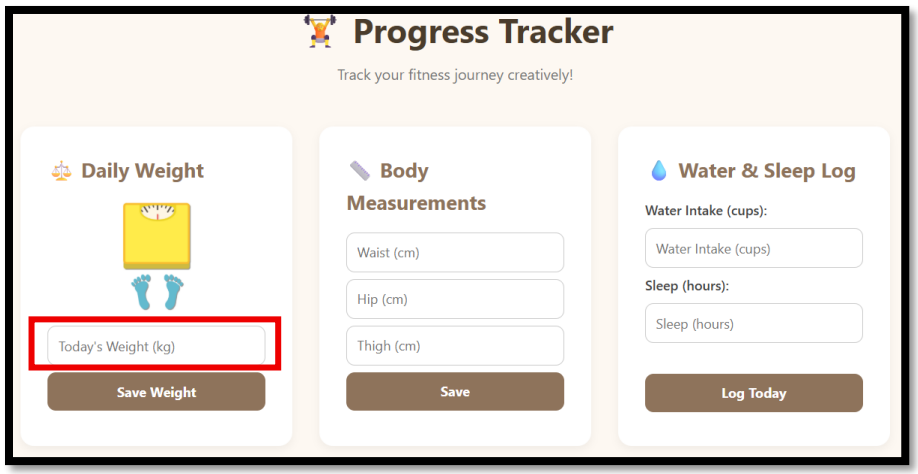


Figure 261: User Manual (tracking.php)

STEP 4:

Click the “Save Weight” button



Figure 262: "Save Weight" button

STEP 5:

The Weight Chart will be updated



Figure 263: Weight Chart

10.2.14 How Members Can Update Body Measurements in Progress Tracking of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

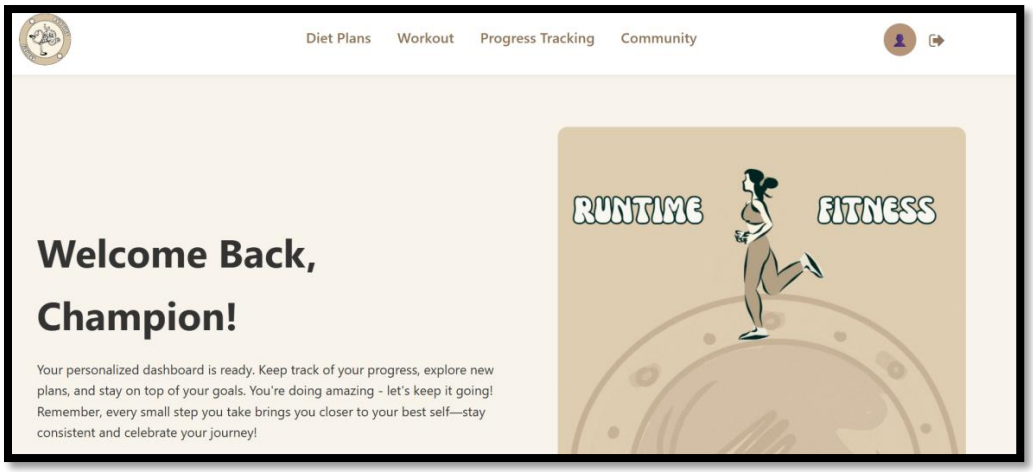


Figure 264: User Manual (memberHomepage.php)

STEP 2:

Click Progress Tracking



Figure 265: "Progress Tracking" link

STEP 3:

Enter waist, hip, thigh

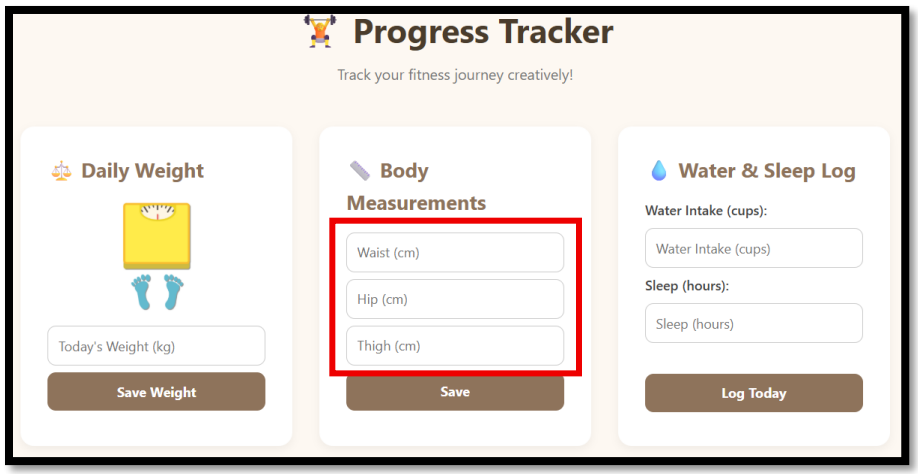


Figure 266: User Manual (tracking.php)

STEP 4:

Click the “Save” button



Figure 267: "Save" button

STEP 5:

The Body Measurement Chart Report will be updated

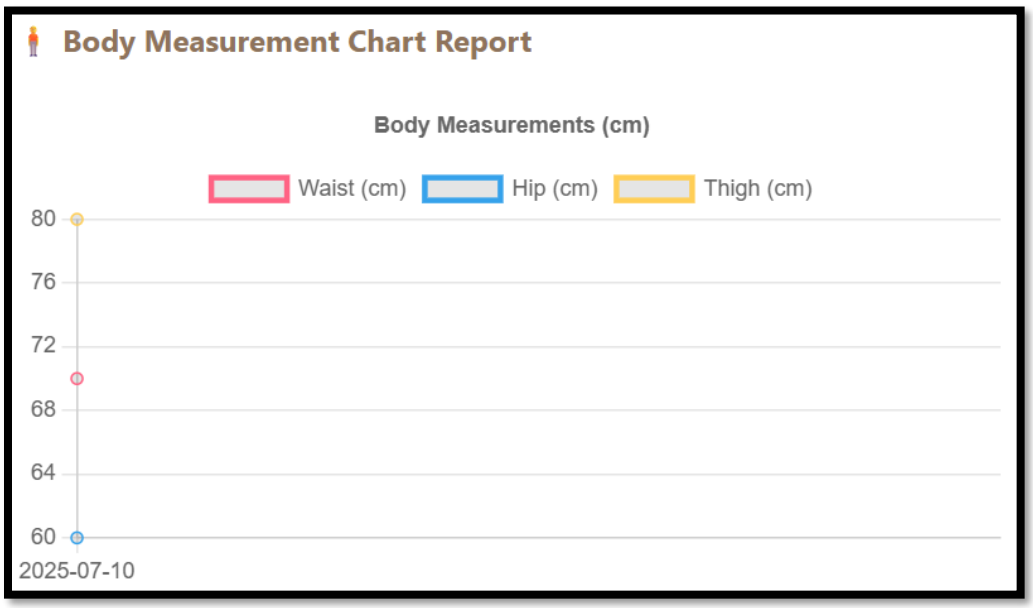


Figure 268: Body Measurement Chart Report

10.2.15 How Members Can Update Water & Sleep Log in Progress Tracking of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

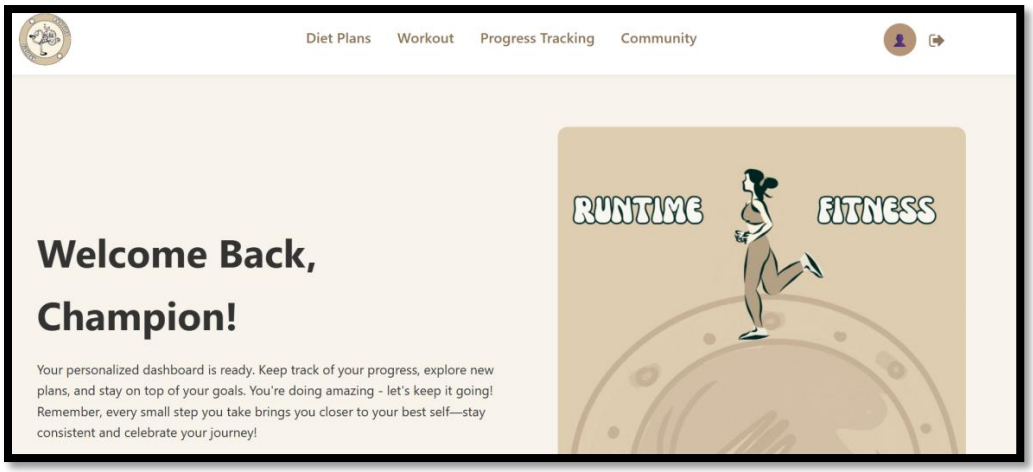


Figure 269: User Manual (memberHomepage.php)

STEP 2:

Click Progress Tracking



Figure 270: "Progress Tracking" link

STEP 3:

Enter water intake and sleep

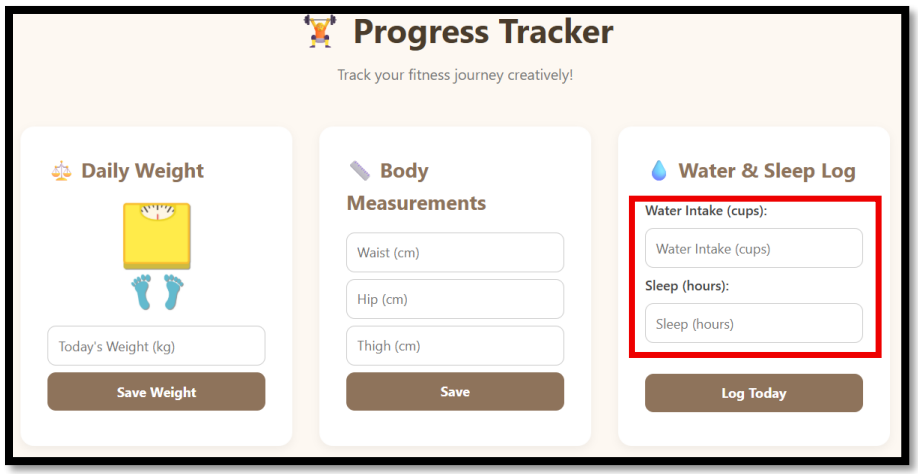


Figure 271: User Manual (tracking.php)

STEP 4:

Click the “Log Today” button



Figure 272: "Log Today" button

STEP 5:

The Water and Sleep Chart will be updated

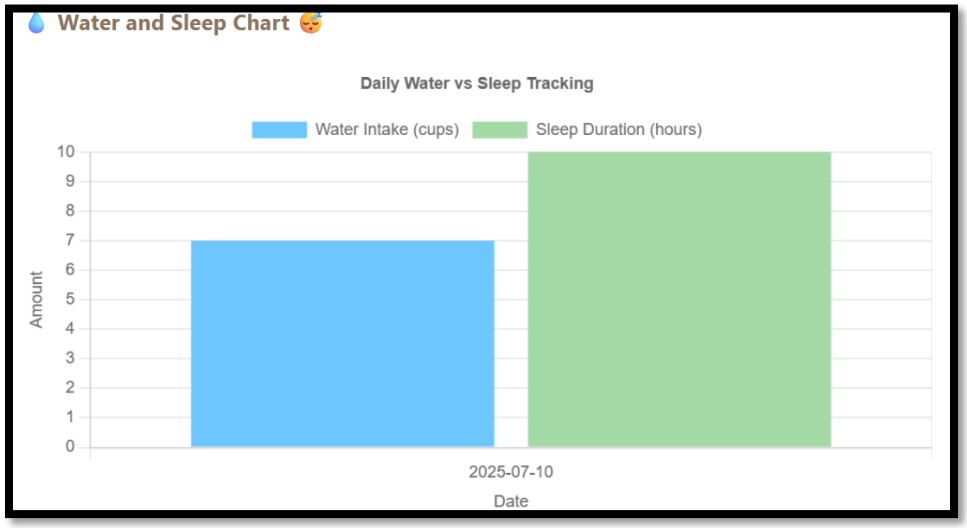


Figure 273: Water and Sleep Chart

10.2.16 How Members Access to Community of RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

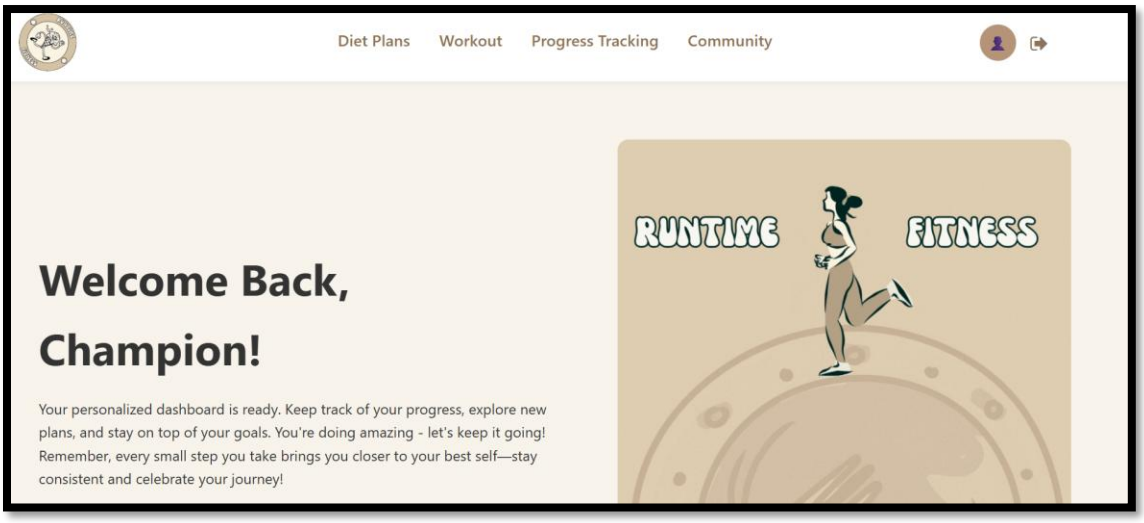


Figure 274: User Manual (memberHomepage.php)

STEP 2:

Click Community



Figure 275: "Community" link

STEP 3:

Enter workout tips, meal ideas, or questions

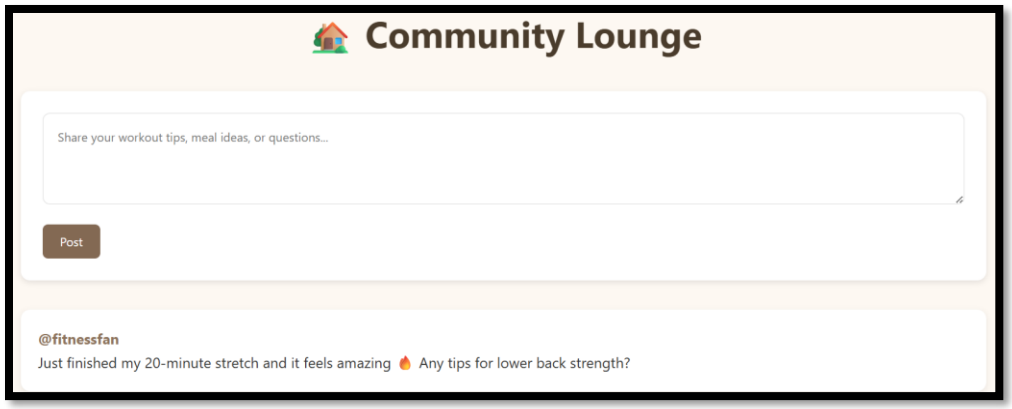


Figure 276: community.php

STEP 4:

Click the “Post” button



Figure 277: "POST" button

STEP 4:

Posted comments is shown.

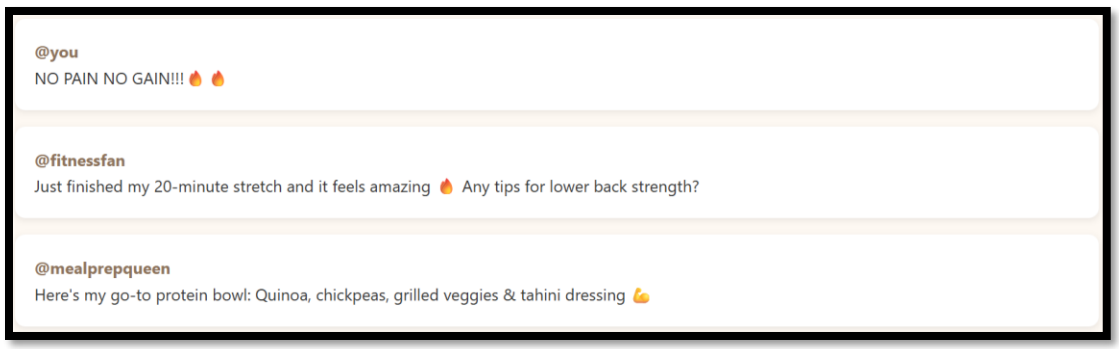


Figure 278: Posted Comments in community.php

10.2.17 How Members Can Edit Profile Details in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

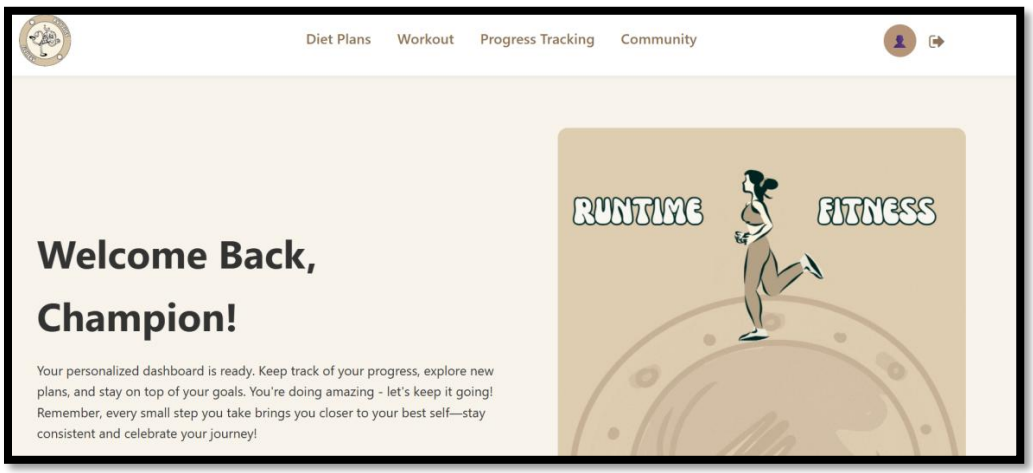


Figure 279: memberHomepage.php

STEP 2:

Click Profile Icon



Figure 280: Profile Icon

STEP 3:

P.viewProfile.php is shown



Figure 281: User Manual (P.viewProfile.php)

STEP 4:

Click the “EDIT” button



Figure 282: "EDIT" button

STEP 5:

Edit profile details

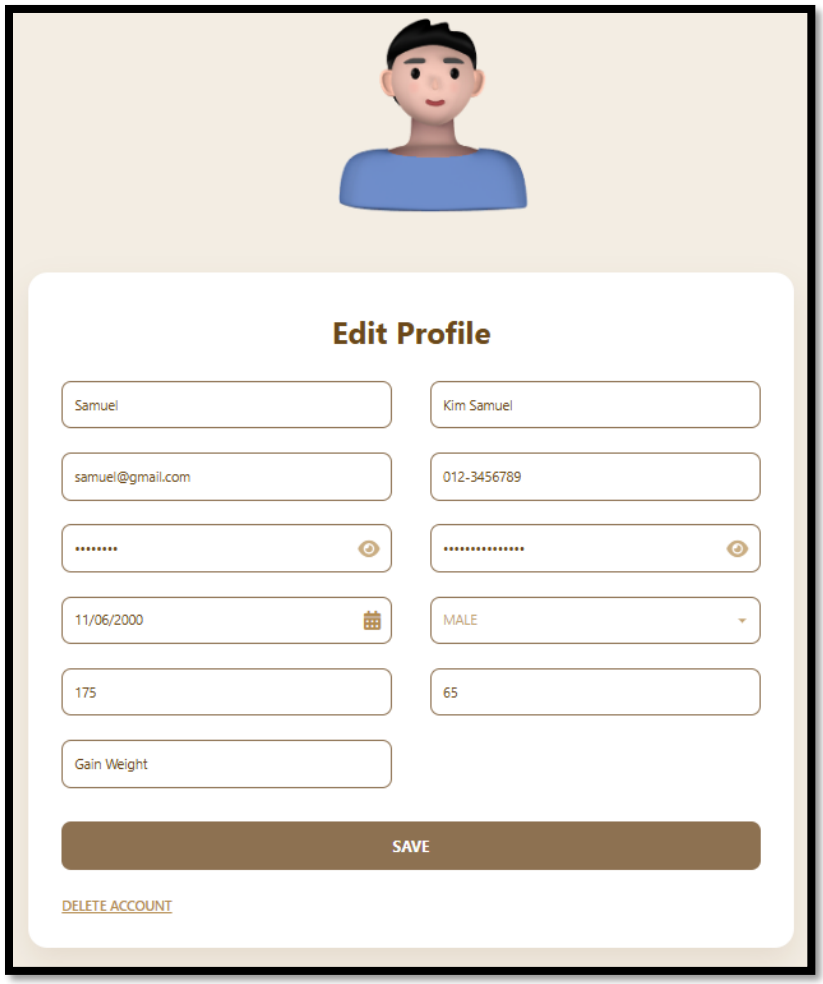


Figure 283: User Manual (P.editProfile.php)

STEP 6:

Click the “SAVE” button



Figure 284: "SAVE" button

10.2.18 How Members Can Delete their Account in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

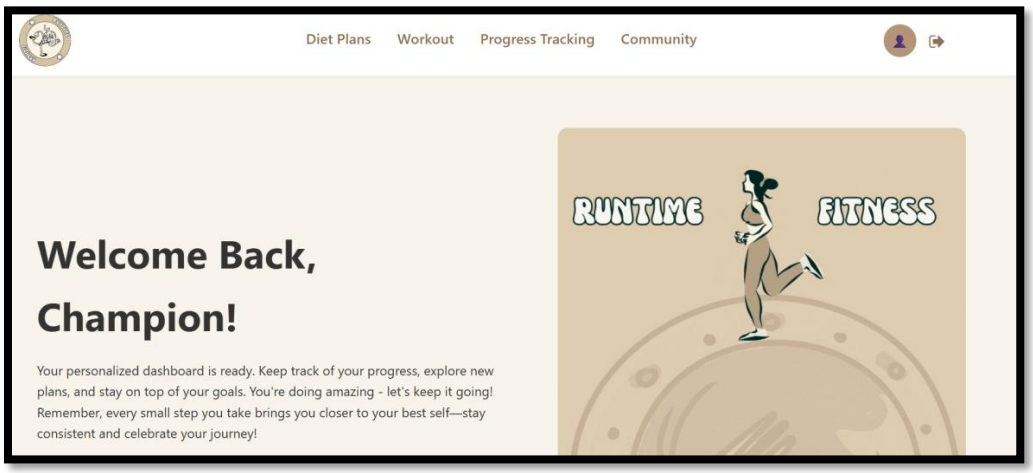


Figure 285: memberHomepage.php

STEP 2:

Click Profile Icon



Figure 286: Profile Icon

STEP 3:

P.viewProfile.php is shown

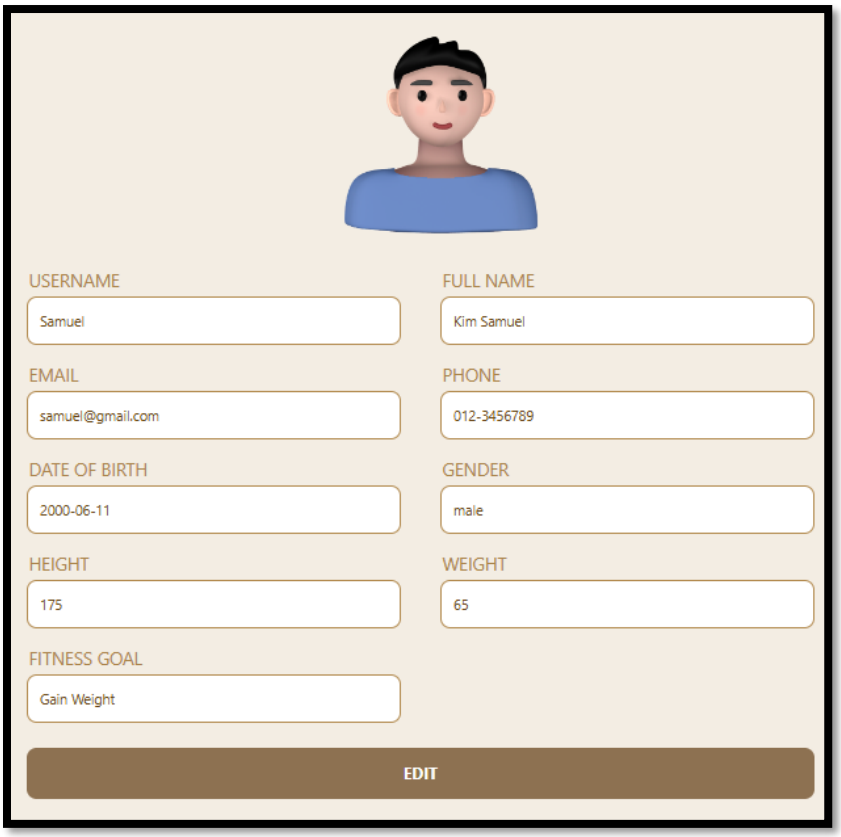


Figure 287: User Manual (P.viewProfile.php)

STEP 4:

Click the “EDIT” button



Figure 288: "EDIT" button

STEP 5:

Edit profile details

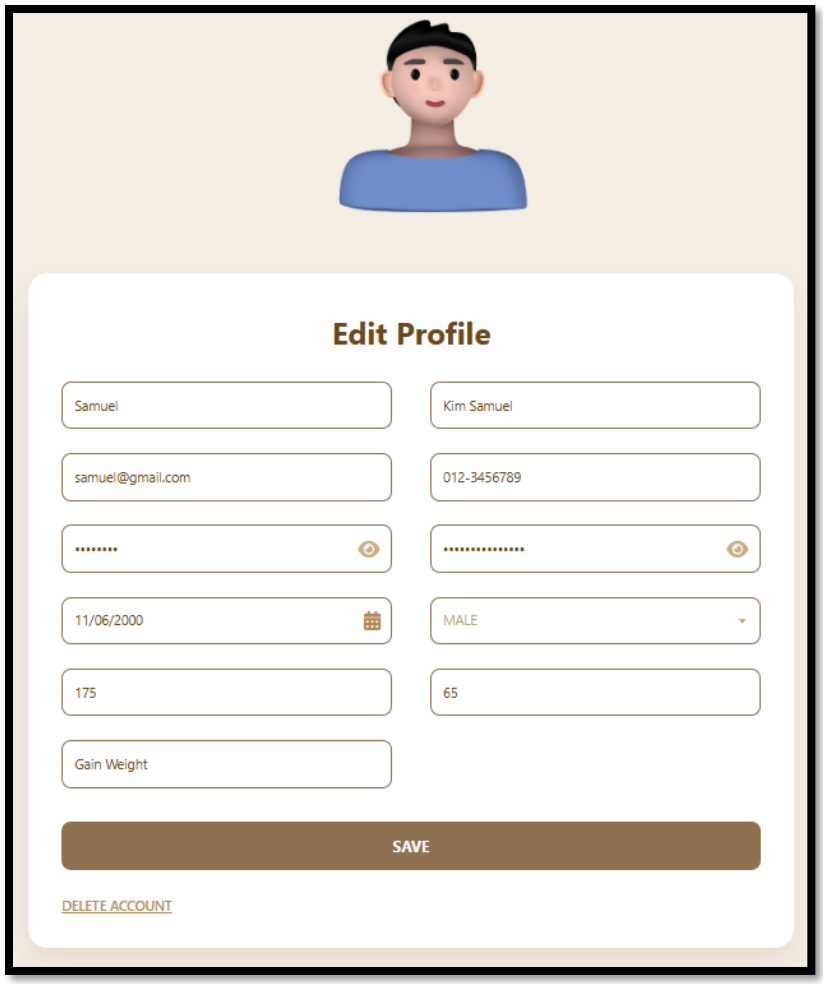


Figure 289: User Manual (P.editProfile.php)

STEP 6:

Click DELETE ACCOUNT



Figure 290: "DELETE ACCOUNT" link

10.2.19 How Members Can Log Out from RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/memberHomepage.php

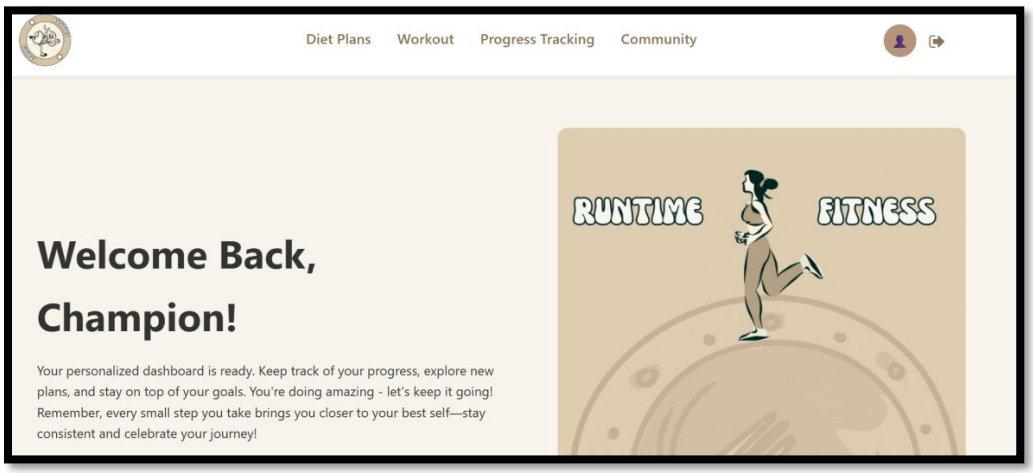


Figure 291: memberHomepage.php

STEP 2:

Click Log Out Icon



Figure 292: Log Out Icon

STEP 3:

The Progress Tracking Reminder is shown.

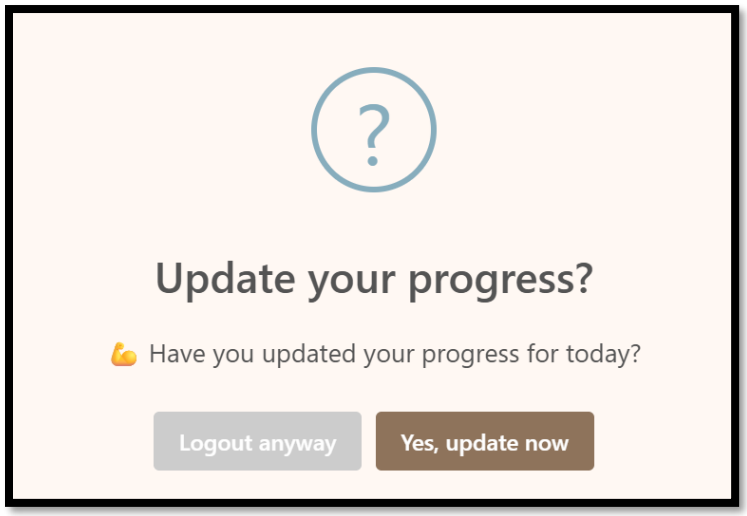


Figure 293: Progress Tracking Reminder

STEP 4:

Click the “Logout anyway” button



Figure 294: "Logout anyway" button

STEP 5:

Logged out successfully message is shown

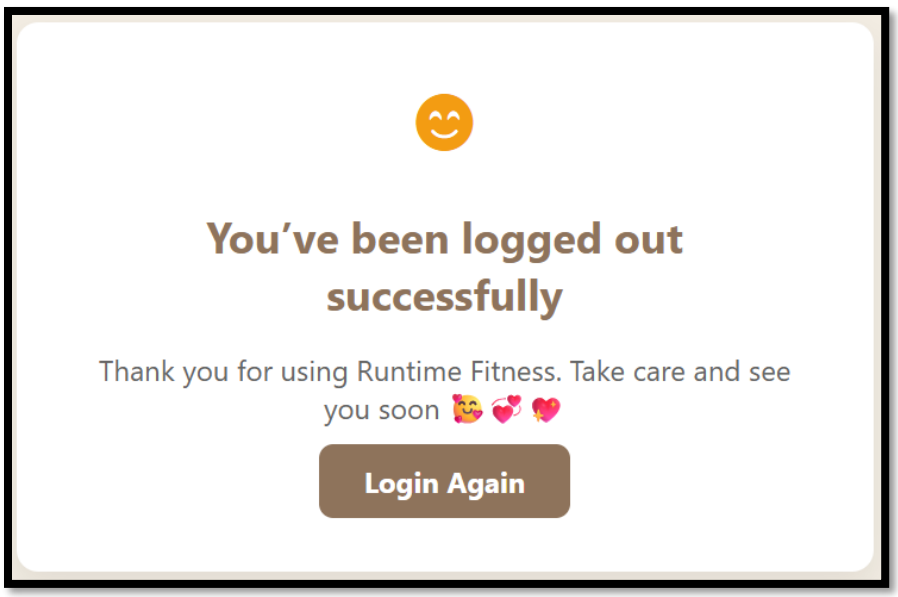


Figure 295: Logged out successfully message

10.3 Admin

10.3.1 How Admin Can Log into RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/homePage.php

STEP 2:

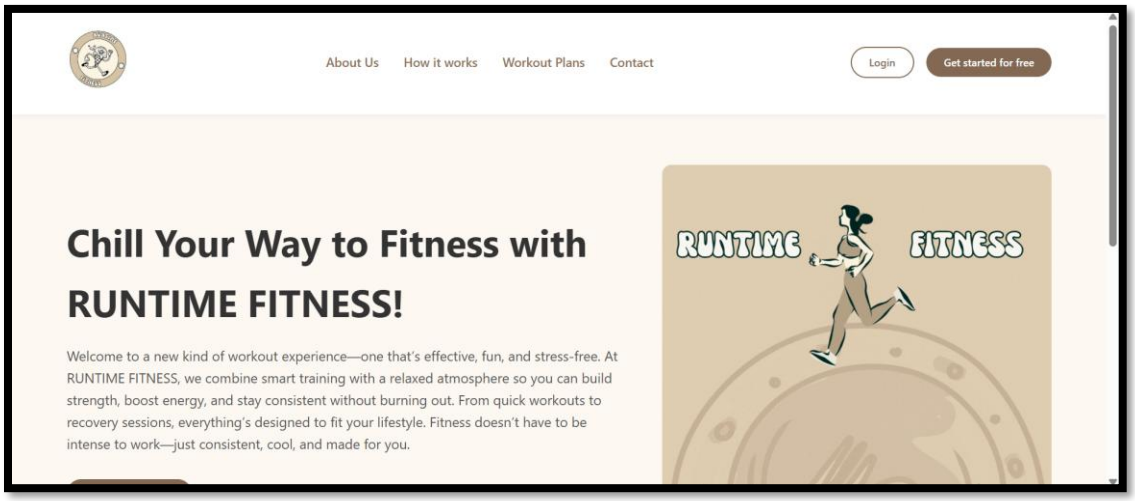


Figure 296: User Manual (homePage.php)

STEP 3:

Click the “Login” button



Figure 297: "Login" button

STEP **4**:

Enter the default username and password for admin:

Username: admin

Password: i_am_admin_123

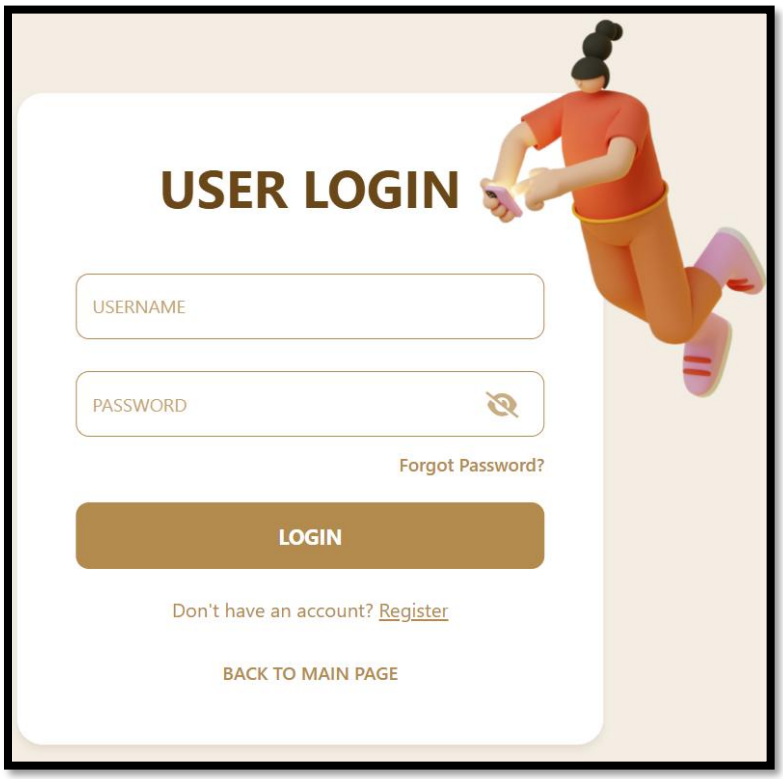


Figure 210: User Manual (login.html)

STEP **5**:

Click the “LOGIN” button



Figure 298: "LOGIN" button

STEP **6**:

adminHomepage.php is shown

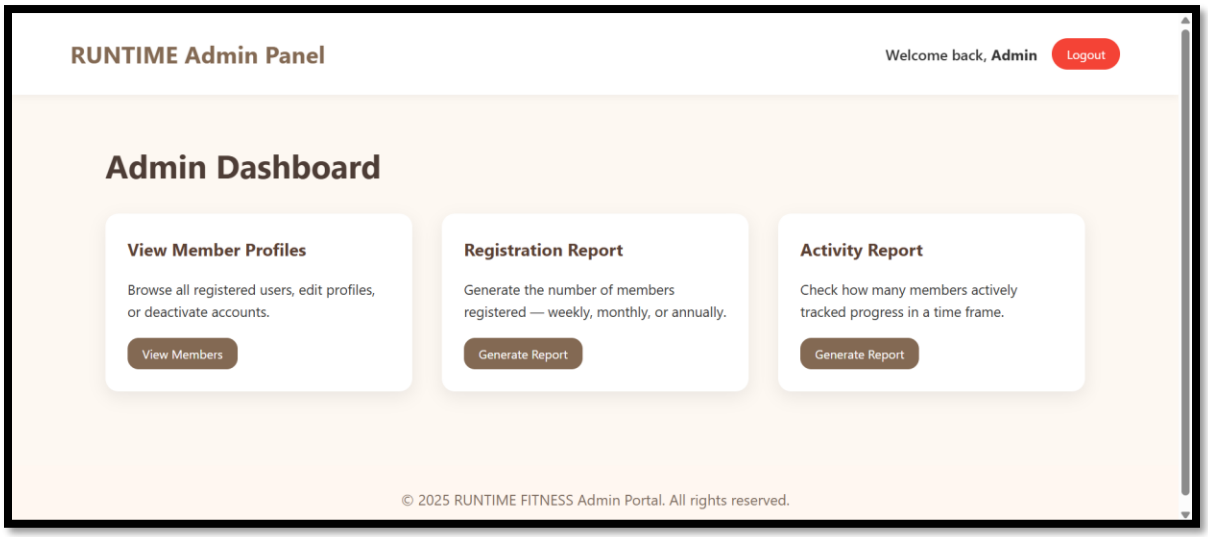


Figure 299: User Manual (adminHomepage.php)

10.3.2 How Admin Can View and Edit Member Reports in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/adminHomepage.php

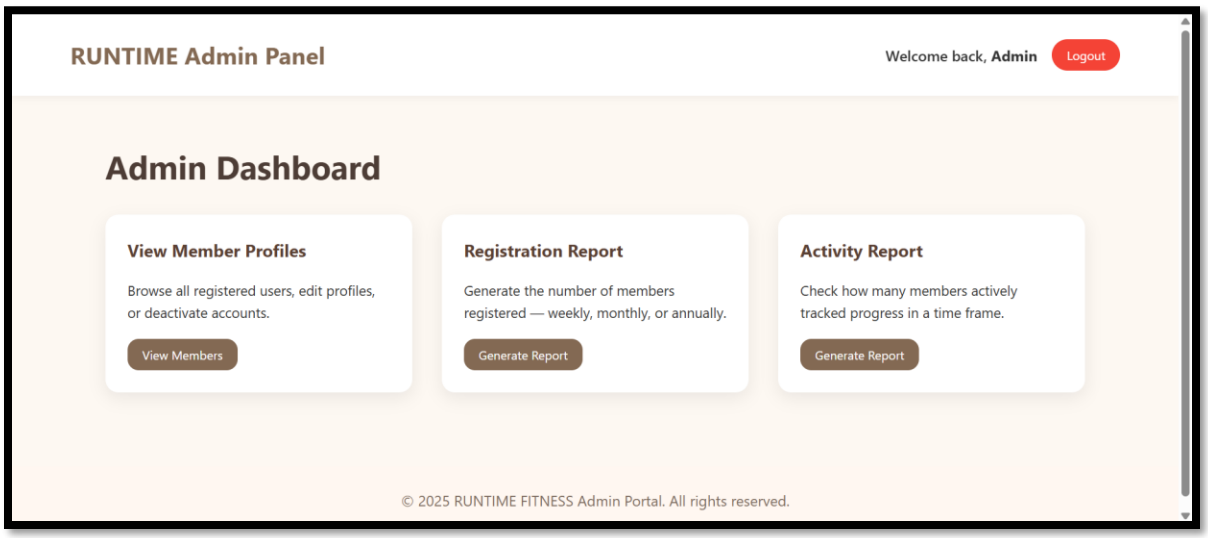


Figure 300: User Manual (adminHomepage.php)

STEP 2:

Click the “View Members” button

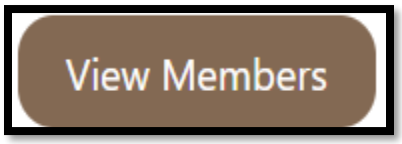


Figure 301: "View Members" button

STEP 3:

view_members.php is shown

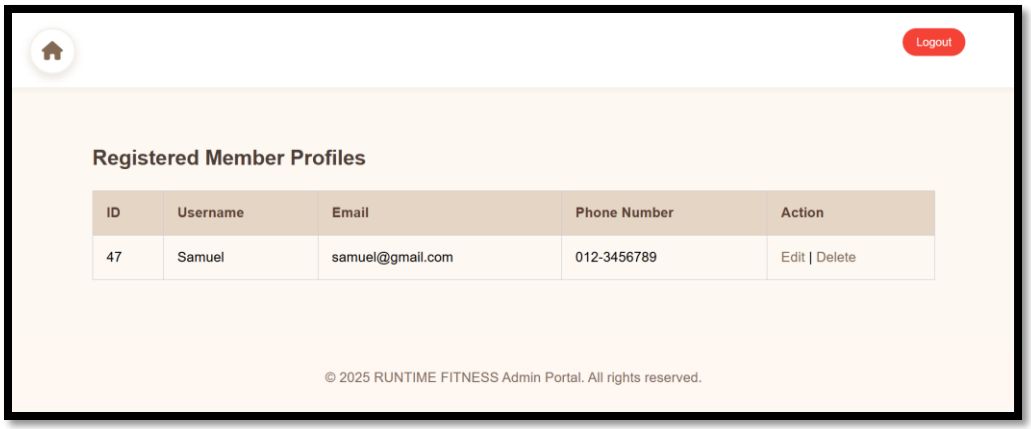


Figure 302: User Manual (view_members.php)

STEP 4:

Click Edit



Figure 303: "Edit" link

STEP 5:

Edit the username, email, phone number of the member.

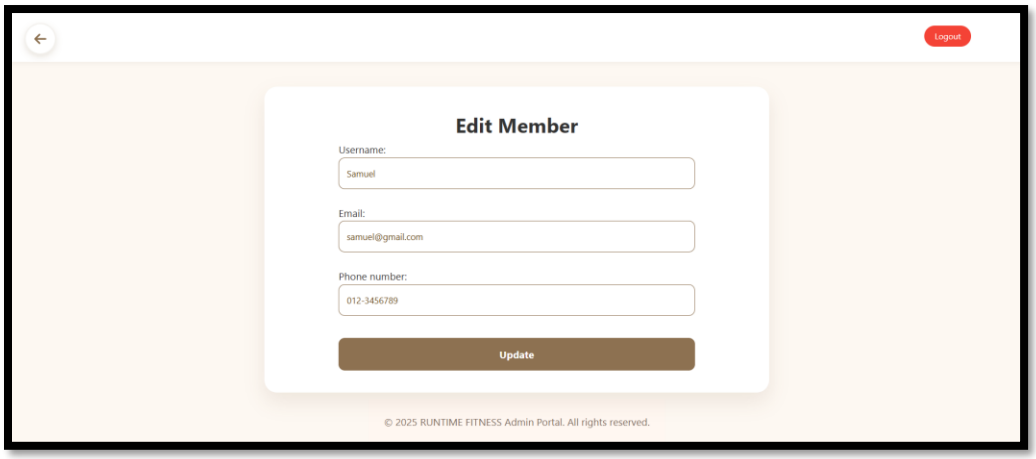


Figure 304: Edit Member

STEP 6:

Click the “Update” button



Figure 305: "Update" button

10.3.3 How Admin Can Delete Members from Member Profiles in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/adminHomepage.php

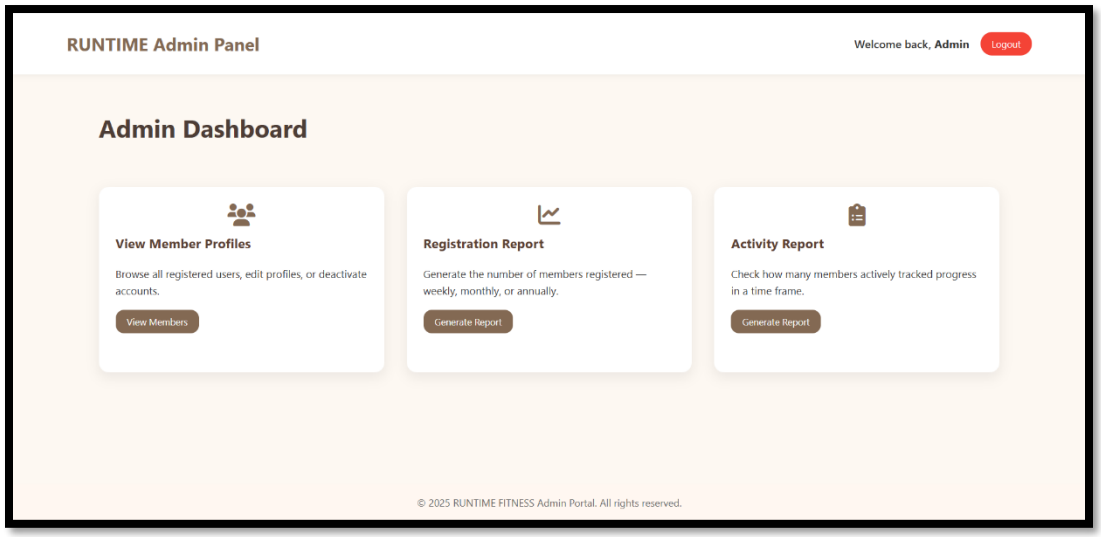


Figure 306: User Manual (adminHomepage.php)

STEP 2:

Click the “View Members” button

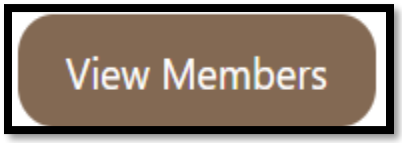


Figure 307: "View Members" button

STEP 3:

view_members.php is shown

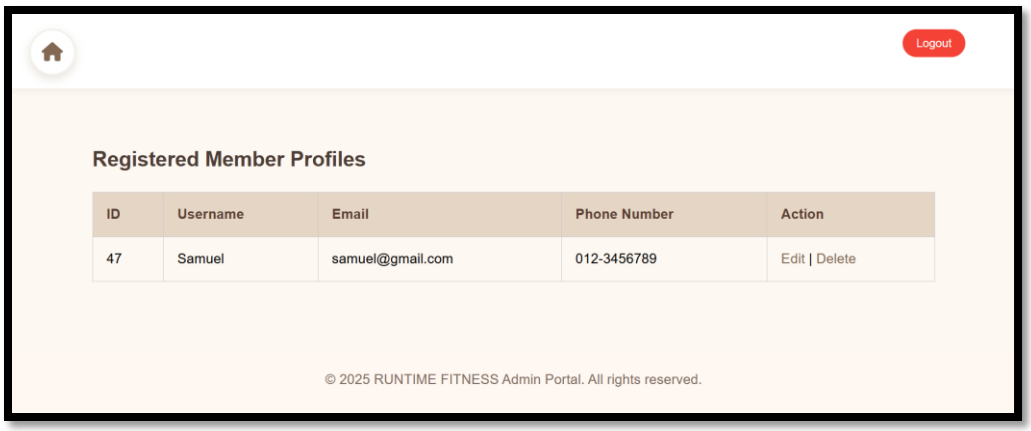


Figure 308: User Manual (view_members.php)

STEP 4:

Click Edit

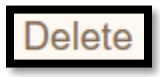


Figure 309: "Delete" link

10.3.4 How Admin Can View Registration Reports in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/adminHomepage.php

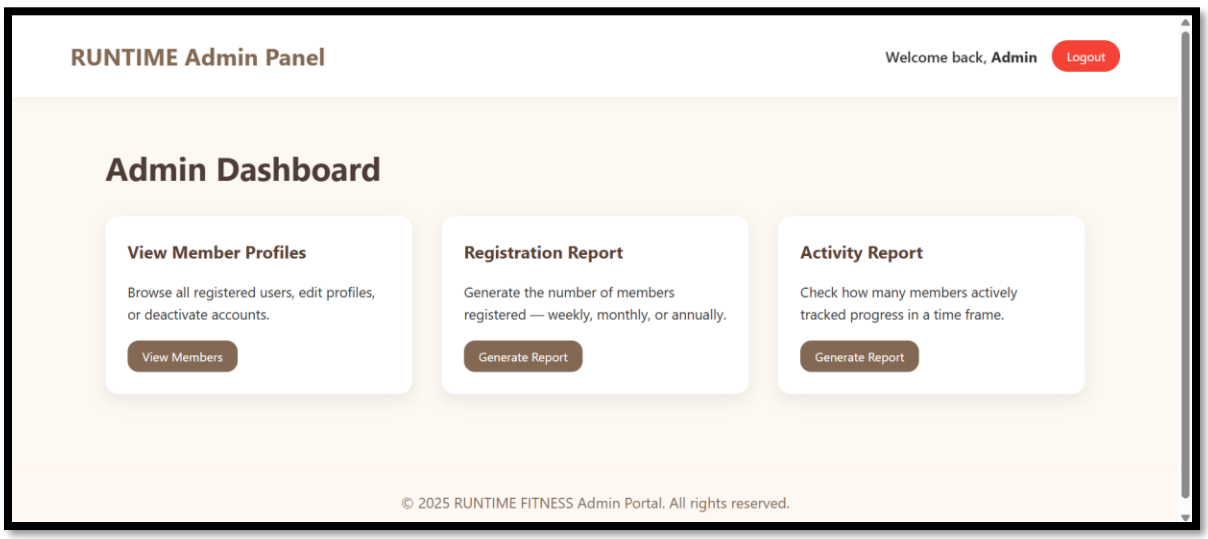


Figure 310: User Manual (adminHomepage.php)

STEP 2:

Click the “Generate Report” button

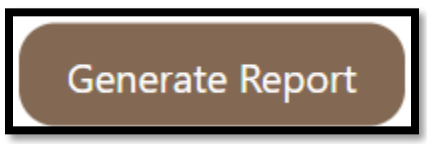


Figure 311: "Generate Report" button

STEP 3:

report_registration.php is shown

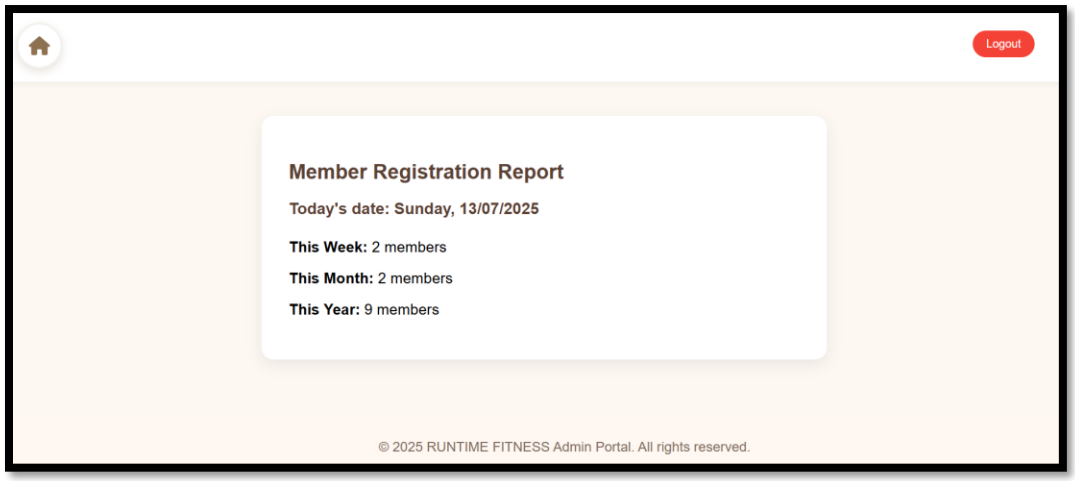


Figure 312: User Manual (report_registration.php)

10.3.5 How Admin Can View Activity Reports in RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/adminHomepage.php

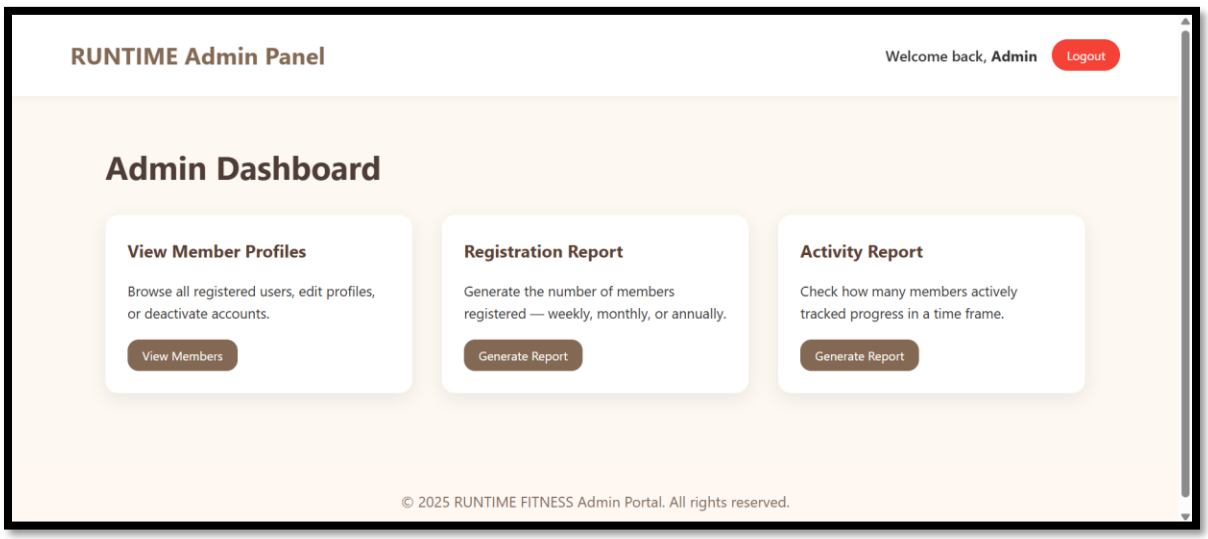


Figure 313: User Manual (adminHomepage.php)

STEP 2:

Click the “Generate Report” button

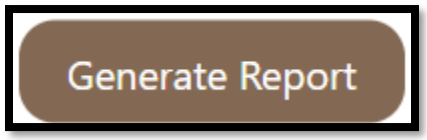


Figure 314: "Generate Report" button

STEP 3:

report_activity.php is shown

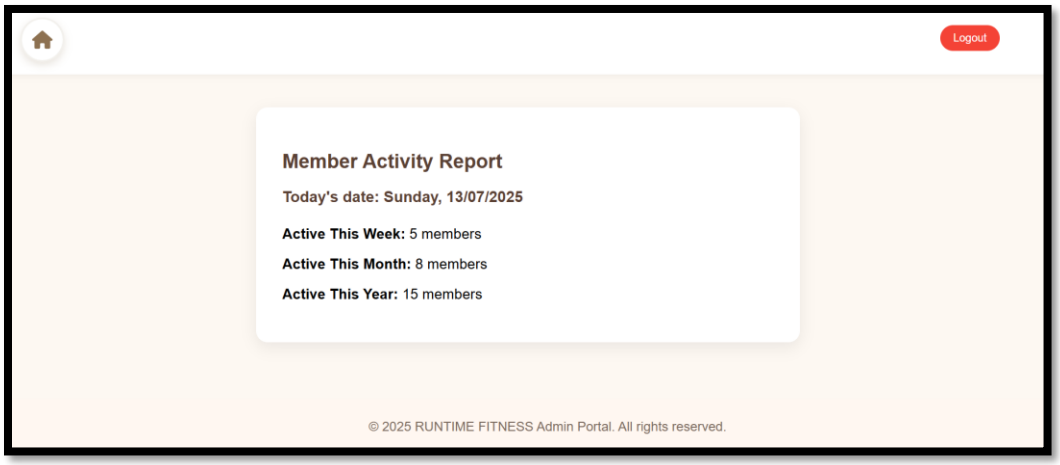


Figure 315: User Manual (report_activity.php)

10.3.6 How Admin Can Log Out from RUNTIME FITNESS

STEP 1:

localhost/RUNTIME_FITNESS_MANAGEMENT_SYSTEM_CP_APR_2025/adminHomepage.php

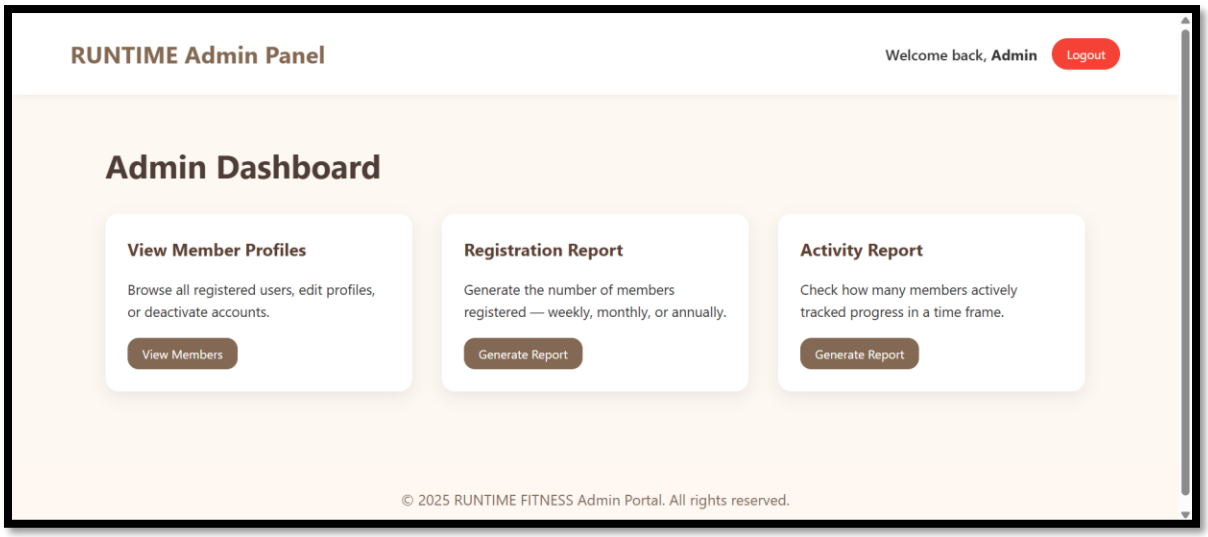


Figure 316: User Manual (adminHomepage.php)

STEP 2:

Click the “Logout” button



Figure 317: "Logout" button

11.0 Conclusion

Throughout the development of this system, our team gained valuable experience in building a project from scratch. In addition to improving our technical skills, we also enhanced our communication and collaboration as a team. Working with GitHub introduced us to the importance of version control and taught us how to troubleshoot effectively through regular testing. Looking forward, we hope this system will help our users to stay consistent with their diets and workouts as they work toward their health goals. Although the current selection of recommended workouts and meals is limited, we aim to enrich the database and diversify the available choices. We realized that manual workout logging can be tiring and potentially discourage user consistency. Therefore, we plan to integrate wearable devices such as Apple Watch, Fitbit, and electronic scales to improve user satisfaction. To streamline the onboarding process, we also intend to allow users to register and log in using their Google accounts. As the system matures, we envision introducing roles such as fitness coaches and nutritionists, enabling members to subscribe to plans. Where paid plans offer personalized, one-on-one guidance to help users reach their goals more effectively.

12.0 Enhancements

Following the initial development and evaluation of the RUNTIME FITNESS Management System, several enhancements were identified to strengthen the system's performance, user experience, and overall features in future versions. Develop a cross-platform mobile application using Flutter or React Native, improving user accessibility on the iOS and Android platforms by offering push notifications and reminders, diet logging, and on-the-go access to workouts. Incorporating an Artificial Intelligence-powered recommendation engine enables the system to analyze user progress and recommend personalized workout routines and dietary adjustments depending on health conditions and performance trends.

Gamification features such as reward badges, activity streaks, and fitness challenges could be added to increase user motivation and engagement. Another important enhancement is the synchronization with wearable fitness devices, allowing logging of data such as step count, heart rate, and sleep duration. Multilingual support allows users to select between multiple languages to accommodate a wider audience.

Implementing email reports and reminders that include weekly progress updates, tailored suggestions, and missed goal alerts can enhance functionality. The administrator dashboard can be improved with advanced analytics and data visualization tools to monitor user activity and system usage patterns, supporting decision-making. Allowing live tutorial videos and live virtual classes within the system would provide members with interactive training content. To improve security and maintain user trust, features such as two-factor authentication (2FA) and activity logging should be introduced to prevent unauthorized access and ensure data protection.

All of these enhancements ensure the RUNTIME FITNESS Management System evolves into a user-centric digital fitness companion that satisfies the users.

13.0 References

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Maryani, Prabowo, H., Gaol, F. L., & Hidayanto, A. N. (2022). Comparison of the System Development Life Cycle and Prototype Model for Software Engineering. International Journal of Emerging Technology and Advanced Engineering, 12(4), 155–162. https://doi.org/10.46338/IJETAE0422_19

Stepanov, D. Y. (2021). Using waterfall, iterative and spiral models in ERP-system implementation projects under uncertainty. Journal of Physics: Conference Series, 2142(1), 012016. https://doi.org/10.1088/1742-6596/2142/1/012016

14.0 Appendix

14.1 Meeting Minutes 1

Minutes of Meeting

Group Runtime	Capstone Project	School of Computing

Date : 28/05/2025
Time : 1045 hrs
Venue : E-09-02, APU Campus

ATTENDANCE

Present:
1. Sim Tian
2. Yap Xin Ling
3. Yeo Pei Wen
4. Yong Xuan Lyn
5. Serene Loh Zi Ting
6. Siew Zhen Lynn

No	Description	Action by	Date
1.0	Distribution of tasks for DFD level 1:		
	Name / **Task**		
	Yap Xin Ling / 1.0 Registration (UserData_T.)		
	Yap Xin Ling / 6.0 Progress (ProgressData_T.)		
	Serene Loh Zi Ting / 2.0 Manage Profile (UserData_T.)		
	Sim Tian / 3.0 Search Preferences (SearchResults_T.)		
	Yeo Pei Wen / 4.0 Class (ClassDetails_T.)		
	Siew Zhen Lynn / 5.0 Diet Plan (DietPlan_T.)		
	Yong Xuan Lyn / 7.0 Chat (ChatHistory_T.)		
	Yong Xuan Lyn / 8.0 Report (ReportData_T.)		
2.0	Clarification of sections required in final documentation: **Process Flowchart** • Each team member will have to prepare process flowcharts for the process included in the system based on DFD level 1. • The process flowchart should include data store. **Interface Design** • Interface design should be a simple wireframe that consists of a minimum amount of 25 interfaces. **Report Design** • There must be at least five (5) reports. • The system must be able to generate the reports stated. • Example case: User Engagement Report = 1 report. But it can be separated into three (3) time frames: - Weekly report - Monthly report - Annual report In this case, there will be a total of **three (3)** reports.		
4.0	Meeting adjourned at 1104hrs.		

Minutes prepared by:
Yong Xuan Lyn
30/05/2025

Figure 318: Meeting Minutes 1

14.2 Progress Sheet 1

Task Name	Start Date	End Date	Priority	Assigned To	Completion Status	Lecturer Comments	Group Leader Comments
1.0 Acknowledgement	30-Apr-25	30-Apr-25	Low	Serene Loh Zi Ting (TP075920)	Completed	On track	Null
3.1 Project Background	14-Apr-25	15-Apr-25	High	Serene Loh Zi Ting (TP075920)	Completed	On track	Null
3.2 Problem Statement	14-Apr-25	15-Apr-25	High	Sim Tian (TP077056)	Completed	On track	Null
3.3 Project Aim	21-Apr-25	22-Apr-25	High	Siew Zhen Lynn (TP076386)	Completed	On track	Null
3.4 Project Objectives	21-Apr-25	22-Apr-25	High	Siew Zhen Lynn (TP076386)	Completed	On track	Null
3.5 Proposed Solutions/Scope	28-Apr-25	29-Apr-25	High	Siew Zhen Lynn (TP076386)	Completed	On track	Null
3.6 Functionalities/Deliverables	28-Apr-25	29-Apr-25	High	Yeo Pei Wen (TP077057)	Completed	On track	Null
4.1 System Development Methodology	28-Apr-25	29-Apr-25	High	Yap Xin Ling (TP077224)	Completed	On track	Null
4.2 Project Gantt Chart	21-Apr-25	22-Apr-25	High	Sim Tian (TP077056)	Completed	On track	Null
5.0 System Hierarchy Chart	14-May-25	21-May-25	High	Yeo Pei Wen (TP077057)	Completed	On track	Null
6.1 Context Diagram	30-Apr-25	1-May-25	High	Yap Xin Ling (TP077224)	Completed	On track	Null
6.3 DFD Level-0	21-May-25	28-May-25	High	Yeo Pei Wen (TP077057)	Completed	Data store names should be in short, upper case, and ends with _T	Null
12.1 Meeting Minutes 1	21-May-25	28-May-25	Low	Yong Xuan Lyn (TP076797)	Completed	On track	Null
13.1 Progress Sheet 1	21-May-25	28-May-25	Low	Yap Xin Ling (TP077224)	Completed	On track	Null

Figure 319: Progress Sheet 1

14.3 Meeting Minutes 2

Minutes of Meeting

Group Runtime	Capstone Project	School of Computing

Date : 19/06/2025
Time : 1852 hrs
Venue : Online, Ms Teams

ATTENDANCE

Present :
1. Sim Tian
2. Yap Xin Ling
3. Yeo Pei Wen
4. Yong Xuan Lyn
5. Serene Loh Zi Ting
6. Siew Zhen Lynn

No	Description	Action	Date
1.0	**Prepare HTML test cases for:** 1. Guest Homepage 2. User Homepage 3. Admin Homepage 4. Diet Plan Page 5. Workout Page 6. Progress Tracking Page		
2.0	Further task delegation in documentation		
3.0	**Announcement:** Removal of the nutritionist and coach role. **Action required:** Modification of DFD level 0 && level 1.		
4.0	Meeting adjourned at 1910 hrs.		

Minutes prepared by:
Yong Xuan Lyn
30/05/2025

Figure 320: Meeting Minutes 2

14.4 Progress Sheet 2

Task Name	Start Date	End Date	Priority	Assigned To	Completion Status	Lecturer Comments	Group Leader Comments
6.2 Data Flow Diagrams & Data Dictionary	11-Jun-25	18-Jun-25	High	All group member	Completed	On track	Null
6.4 DFD Level-1	11-Jun-25	18-Jun-25	High	All group member	Completed	On track	Null
6.5 Program Flowcharts	11-Jun-25	18-Jun-25	High	All group member	Completed	On track	Null
6.6 Database Design	18-Jun-25	25-Jun-25	High	Sim Tian (TP077056) & Yong Xuan Lyn (TP076797)	Completed	On track	Null
6.7 Interface Design	18-Jun-25	25-Jun-25	High	Sim Tian (TP077056)	Completed	On track	Null
6.8 Report Design	18-Jun-25	25-Jun-25	High	Yong Xuan Lyn (TP076797)	Completed	On track	Null
6.9 Test Cases Preparation	11-Jun-25	18-Jun-25	High	Serene Loh Zi Ting (TP075920)	Completed	On track	Null
12.2 Meeting Minutes 2	4-Jun-25	11-Jun-25	Low	Yong Xuan Lyn (TP076797)	Completed	On track	Null
13.2 Progress Sheet 2	4-Jun-25	11-Jun-25	Low	Yap Xin Ling (TP077224)	Completed	On track	Null

Figure 321: Progress Sheet 2

14.5 Meeting Minutes 3

Minutes of Meeting

Group Runtime	Capstone Project	School of Computing

Date : 10/07/2025
Time : 1001hrs
Venue : B-06-10, APU

ATTENDANCE
Present :
1. Sim Tian
2. Yap Xin Ling
3. Yeo Pei Wen
4. Yong Xuan Lyn
5. Serene Loh Zi Ting
6. Siew Zhen Lynn

No	Description	Action by	Date
1.0	**Meeting Agenda:** - To conduct an overall system test. - To list out modules that need improvements.		
2.0	Conduct system testing	Yap Xin Ling	10/07/2025
3.0	Improvements required: - Adding logout buttons to all member accessible pages - Debug "delete user" function under admin features	Siew Zhen Lynn Yong Xuan Lyn	10/07/2025
4.0	Introduction of utilization of basic git commands in Visual Studio Code	Yong Xuan Lyn	10/07/2025
5.0	Meeting adjourned at 1131hrs.		

Minutes prepared by:
Yong Xuan Lyn
14/04/2025

Figure 322:Meeting Minutes 3

14.6 Progress Sheet 3

Task Name	Start Date	End Date	Priority	Assigned To	Completion Status	Lecturer Comments	Group Leader Comments
2.0 Abstract	8-Jul-25	9-Jul-25	Low	Yap Xin Ling (TP077224)	Completed	On track	Null
7.1 Screenshots	1-Jul-25	8-Jul-25	High	Siew Zhen Lynn (TP076386)	Completed	On track	Null
7.2 Significant Source Codes	1-Jul-25	8-Jul-25	High	Siew Zhen Lynn (TP076386)	Completed	On track	Null
8.1 Unit Testing	25-Jun-25	1-Jul-25	High	Serene Loh Zi Ting (TP075920)	Completed	On track	Null
8.2 User Acceptance	25-Jun-25	1-Jul-25	High	Serene Loh Zi Ting (TP075920)	Completed	On track	Null
9.0 User Manual	25-Jun-25	1-Jul-25	High	Sim Tian (TP077056)	Completed	On track	Null
10.0 Conclusion	8-Jul-25	8-Jul-25	Low	Yong Xuan Lyn (TP076797)	Completed	On track	Null
11.0 Enhancements	9-Jul-25	9-Jul-25	Low	Yap Xin Ling (TP077224)	Completed	On track	Null
12.3 Meeting Minutes 3	17-Jun-25	25-Jun-25	Low	Yong Xuan Lyn (TP076797)	Completed	On track	Null
13.3 Progress Sheet 3	17-Jun-25	25-Jun-25	Low	Yap Xin Ling (TP077224)	Completed	On track	Null

Figure 323: Progress Sheet 3